

# IoT

## INTERNET OF THINGS

Principles, Paradigms and Applications of IoT

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# Internet of Things (IoT)

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*Principles, Paradigms and Applications of IoT*

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*by*

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## ***Foreword***

The Internet of Things (IoT) is a pretty interesting technology, it means taking all the physical places and things in the world and connecting them to the internet. I strongly recommend this book for the University and College students and faculty members.

**—Dr. Deepak Khazanchi**

*Ph.D, Professor, University of Nebraska at Omaha, USA*

This is one of the few books carefully and systematically written with deep insights into the principles and core concepts of Internet of Things (IoT). The blend of quick step-by-step practical instructions of IoT use-cases and end of chapter exercises makes it highly suitable to enhance students' understanding. It is also an ideal book for IoT stakeholders, researchers and University professors for use as an instructional guide.

**— Prof. Dharm Singh**

*Namibia University of Science and Technology, Namibia*

This book is enriched with a comprehensive exposition to both the fundamental and advanced topics in modern Internet of Things (IoT) paradigms. It transfers knowledge through an immersive “hands on” practical approach and provides in-depth guidance and understanding coupled with some exercises and multiple-choice questions which makes it ideal for all categories of readers. I strongly recommend it for all IoT enthusiasts.

— **Dr. Ruchi Doshi**

*Adjunct Professor, Jayoti Vidyapeeth Women's University, Jaipur, India*

Internet of Things (IoT) is the dawn of new era in the world of computing that provides a new way to share resources, software & data over web in a highly capable, scalable & secure environment. I recommend this book to anyone who wants to learn the basics of Internet of Things (IoT) and how it works in real world scenarios.

— **Deepak Modi**

*Senior Software Engineer, Microsoft Redmond Woods, USA*

I strongly recommend this book because of the easy language from basic to practical approach.

**Dr. Anders Henten**

*Professor, Aalborg University, Copenhagen, Denmark*

This book is not just about the technology of IoT or more specifically the Industrial Internet of Things (IIoT). It is a book about the digital transformation that is being realized through IoT. The author articulates quite clearly it is not Internet of Things connectivity for its own sake. It is rather the business value, the use cases, and connectivity opportunities that is driving this transformation.

**Trilok Nuwal**

*Software Architect, McAfee, India*

“IoT redefine our industries, organizations, and jobs”. This book plays a vital role in this. Must read

***Prof. Vinesh Jain***

*Govt. Engineering College, Ajmer, India*

IoT is an emerging concept and enabler that has the potential to completely reshape the future of industry. This book is very useful for the students, faculty members and researchers.

***Dr. Do Manh Thai***

*Executive, Govt. of Vietnam*

Internet of Things (IoT) is everywhere in our digital age. Masterminding the ideas and tools behind the many uses of Internet of Things (IoT) is a must. This book will help you grasp and deepen your understanding helped by its clear and concise teaching.

***Prof. Nitesh Chauhan***

*MLVTE Govt. Engineering College, Rajasthan, India*

It's a very useful book, especially for a beginner in IoT learning. The authors used step-by-step approach, which makes it ideal for a beginner. The diagrams are self-explanatory, and the language is easy but evident.

**Prof. Hari Prabhat Gupta**

*IIT (BHU), India*

The two most important technological shifts in the 21st century are the adoption and prevalence of mobile and Internet of Things (IoT) in our day to day lives. It's essential for every business to adopt these technologies, and more importantly for our educational systems to teach and train the next generation of learners on emerging technologies. This book provides a concise, clear and lucid understanding of the fundamentals of Internet of Things (IoT) technologies and concepts.

**Dr. Ricardo R. Saavedra**

*Director, University of Azteca, Mexico*

This book covers almost all the aspects of Internet of Things (IoT) starting from fundamental to practical oriented approach. It incorporates case studies and several advanced topics that may help learners to build strong concepts in the area of Internet of Things (IoT).

**Sujith Jayaprakash**

*Director of Outreach and Collaborations, BlueCrest University College,  
West Africa*

The book is truly comprehensive and can be associated with any university's curriculum.

***Dr. Mukesh Gupta***

*Professor, SKIT, Jaipur, India*

A genuine book for beginners who wants to understand and implement Internet of Things (IoT) concepts.

***Abhishek Maloo***

*Software Engineer, Facebook, San Francisco, USA*

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It is very important to acknowledge the tireless support and endless contributions received from many people in completing this work. Furthermore, this book would not have been a reality without the support of the BPB Publication family, mentors, colleagues and friends. Their unique contributions, advice, time, energy and expertise have improved the quality of this book immensely. Firstly, we would like to thank Manish Jain, Nrip Jain, and BPB Publishing house for giving this wonderful opportunity and the staff members, especially Sourabh, Anugraha and Shruti for their relentless effort to make this book a reality.

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● **Kamal Kant Hiran:** I would like to thank Prof. Anders Henten, Prof. Kund Erik Skouby, Prof. Reza Tadayoni, Prof. LeneTolstrupSørensen, Anette Bysøe, Aalborg University, Center for Communication, Media and Information Technologies (CMI), Copenhagen, Denmark for providing in-depth scientific knowledge in the Internet of Things (IoT) and Cloud Computing research area and support during work. Most importantly, I would like to thank my parents, my wife Dr. Ruchi Doshi and my daughter Bhuvi Jain for their sacrifice during the past years. Special thanks goes to Dr. Mahendra Doshi and Pushpa Doshi for their support in my family during my work. Especially, I would also like to thank Ashok Ghosh Sir, President and Trustee, J K Cement Ltd., and Rinu Ghosh Mam, Vice President, J K Cement Ltd., Prof. P.C. Deka and Sir Padampat Singhanian University (SPSU) family members, Udaipur, Rajasthan, India, for providing a healthy academic environment during my consistent work.

## ***Preface***

Internet of Things (IoT) is presently an emerging technology worldwide. Government, academia, and industry are involved in different aspects of research, implementation, and business with IoT. IoT cuts across different application domain verticals ranging from civilian to defence sectors. These domains include agriculture, space, healthcare, manufacturing, construction, water, and mining, which are presently transitioning their legacy infrastructure to support IoT. Today it is possible to envision pervasive connectivity, storage, and computation, which, in turn, gives rise to building different IoT solutions. IoT-based applications such as innovative shopping system, infrastructure management in both urban and rural areas, remote health monitoring and emergency notification systems, and transportation systems, are gradually relying on IoT based systems. Therefore, it is very important to learn the fundamentals of this emerging technology.

In this book, Principles, Paradigms frameworks and applications of IoT (Internet of Things) in the modern era are presented. It also provides a sound understanding of the IoT concepts, architecture, applications and improves the awareness of readers about IoT technologies, and application areas. The content is written in simple language and suitable for undergraduates, degree and postgraduate students who would like to improve their understanding of IoT. A key objective of this book is to provide a systematic source of reference for all aspects of IoT. This book comprises of nine chapters with close cooperation and

contributions from four different authors spanning four countries and providing a global, broad perspective on major topics on the Internet of Things.

**Chapter** Fundamentals of Internet of Things, discusses the basic concept behind the evaluation of Internet of Things, its role, applications, its physical and logical components, also discusses the connectivity and various communication models by which these devices communicate. At the end of this chapter, the Impact of growing IoT technology on the business and economics is discussed.

**Chapter** IoT Architectures and Protocols, in this chapter various technologies and protocols used in IoT architecture such as Taxonomy of IoT, Three-layer and five-layer architecture, Fog and Cloud-based architecture, Bluetooth, ZigBee, and 6LoWPAN are discussed. Also, at the end of this chapter, it is concluded that currently, this field is in a very nascent stage. The technologies in the core infrastructure layers are showing signs of maturity. However, a lot more needs to happen in the areas of IoT applications and communication technologies.

**Chapter** Programming Framework for IoT, There are various programming languages used for implementing the IoT applications. For developing IoT solution, one would need a programming language which spans the various fields, while being scalable and lightweight at the same time. This chapter discusses the various programming paradigms their interoperability in IoT. This chapter also discusses the topics like Implementation of IoT with Raspberry Pi, Understanding the programming framework for

the Internet of Things (IoT) and various programming paradigms. Readers of this chapter will learn the basic concepts of Arduino Programming, Python Programming and Raspberry Pi also Implementation of IoT with Raspberry Pi.

**Chapter** Virtualization in IoT, Virtualization is a demanding technique in the era of computing. Computing resources are the major components in a wide range of IoT applications. So wide variety of applications can take advantage of virtualization technique. Readers of this chapter will be familiar with the idea behind the virtualization, its Types, Techniques, Benefits, Applications and role of Virtualization Technology in the era of Internet of Things.

**Chapter** Security, Privacy and Challenges in IoT, Businesses can take advantages of IoT for taking accurate and more structured decisions. But before the adoption of the IoT enables infrastructure, it is important to understand about the privacy, security and trust-related threats also about the various challenges associated with the IoT infrastructure. In this chapter, various challenges that are experienced during the development of Internet of Things (IoT) devices is also discussed it analyses various design, development and security issues and challenges associated with the IoT enabled infrastructure.

**Chapter** Emerging Application areas of Internet of Things, reader of this chapter will learn about the varied and the diverse applications of the IoT technology to machines in order to make them smart enough to reduce human labour and make human's life better, safer and sustainable. Also, understand how these Smart devices or "connected devices" are designed in such a way

that they are able to capture and utilize data on human lives and movements on a daily basis which are then subsequently analysed and used to improve the quality and productivity both in our social lives and industries.

Chapter Cloud Computing and Internet of Things, The integration of Cloud Computing and Internet of Things (IoT) speaks to the following huge jump ahead in the Future Internet. The new applications are emerging from this combination called CloudIoT. By reading this chapter, one will understand what is meant by cloud computing and how one can benefit from it. They will also understand the importance of IoT in cloud computing, as well as the types of devices, use in accessing the cloud resources in cloud computing. The integration of both technologies will see the emergence of an almost new way of life and solutions. In addition, the reader will also understand that one utilizing CloudIoT will get to know how easy his/her life will become in terms of having access to things they use on a daily basis.

Chapter Smart City Using IoT Integration, Smart cities use information and innovation to make efficient, improved maintainability, make monetary advancement, and upgrade personal satisfaction factors for individuals living and working in the city. By reading this chapter, one will understand what is meant by Smart City and how the integration of IoT plays an important role to make a city a Smart City. After reading this chapter, one will understand the concept of a smart city, its emergence and definition, dimensions and components of it in details. In this chapter, the adoption of IoT in automation is also explained in detail with design strategies. In the last various issues and

challenges of smart city and different application, areas are also discussed in detail.

Chapter Internet of Things (IoT) Use Cases, IoT is gradually being used in large numbers on a daily basis across the globe giving users (industry and consumers) more options and efficiency. This chapter is intended to present some of the ways that IoT is being deployed and throw more light on the emerging trends and the possible future direction of IoT applications. At the end of this chapter, the reader will be able to identify the various ways, in particular, the three key areas of industry, consumer and governance categories in which IoT concept has been applied and understand how IoT devices work in concert to ensure end-to-end delivery of projects ideas and initiatives.

## ***Errata***

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#### The fetal heart monitor and AnteNatal healthcare

#### Healthcare logistic delivery (drone delivery in Ghana).

#### Smart home and smart buildings

#### Smart education system

#### The medium of instruction (language).

#### Smart classrooms

#### Task-based learning

#### Disability accommodation

#### School security.

#### Challenges of IoT in education

#### Governance use case

#### Smart cities

#### Automatic car parking notification

#### Louisville 'If this then that' (IFTTT) smart city platform

#### District of Songdo new resident integration

#### Copenhagen zero carbon emission agenda

#### Conclusion

#### Multiple choice questions

#### Answers

Fill in the blanks

Answers

Descriptive questions

Reference

Model Question Paper 1

Model Question Paper 2

Bibliography.

*Fundamentals of the Internet of Things*

The **Internet of Things (IoT)** has become a hot topic in both technical and non-technical conversation. Somewhere between 2008 and 2009 the I.T. Company Cisco Systems assessed the things/individuals ratio, and found that it's increasing; in 2003 it was 0.08, and in 2010 it was 1.84. The analyst firm Gartner has forecasted that over 26 billion things (devices and people) will be connected to a giant network (IoT) by the year 2020. But the question is: What exactly is IoT? And how it impacts and will impact on our daily life if any?

To know the answer, let's start reading this chapter.

## Structure

Introduction

Characteristics of IoT

The physical design of IoT

Logical design of IoT

IoT ecosystem (components)

Functional blocks of IoT

Communication models

Application programming interfaces

Evaluating business impact and economics for IoT

Future of IoT

### Objective

On studying this chapter, the student becomes aware of the IoT components, their connectivity to form the IoT altogether, and the future possibilities with IoT.



## Introduction

The IoT can be defined as, *A huge network of interconnected things; things may be small devices, big machines and also includes people. Via this interconnected network communication can occur between things-things, things-people, and people-people.*

Just imagine, while your alarm rings in the morning, the lights of the room lit up, and the coffee maker is notified to start making a coffee for you. The geyser has to automatically set on, warming the water for your shower. While you leave home for the office, the calendar on your smartphone may set the best route for the office and thus instructing your car to drive in the best route. Depending upon the traffic on the route, your office could be informed that you would be late for the meeting.

Moreover, while driving back home; the air conditioners would become operational while you are a few miles to reach home. This scenario is a basic example that can be implemented in practical by connecting the devices around us to form an interconnected network. Such scenarios of interconnected devices are termed as the Internet of Things or the Internet of Everything.

### How does it work?

IoT devices collect, assemble, and share the data by utilizing the environment in which they are working and implant. Collection and transmission of data in IoT devices is done by using various sensors. Nowadays, almost all physical devices have some sensor/s embedded into it. The devices could be mobile phone; home and office electronic appliances, electronic traffic signals, barcode sensors; just about everything that we come across in our daily life. Sensors constantly sense and transmit the data of the surrounding environment and the working condition of connected devices, but the significant query is that how would like these share this much amount of emitted data. What's more, how would we put this data to our advantages?

Moreover, the connected devices use a common language for communication, or perhaps may include an intermediate translator. This emitted data and the data produced by the IoT platform is collectively analyzed and processed to form information which is in turn utilized to implement automation and improve efficiencies of the system. Such a network requires a typically common platform for various electronic sensors to operate on, which thus provide several challenges for the IoT.

An example of IoT working wonders is an approach implemented by an air conditioner manufacturing company, in which, a company designed a belt and machine; both the belt and machine having sensors attached in it. They consistently send data in

regards to the health of the machine, and the production details to the manufacturer to recognize issues in advance.

Each product, while on the belt, is attached a bar code containing special instructions, product information, manufacturer and retailer details, and so on. This barcode can be used by manufacturers to check the distribution of products and the quantity available with the retailers to make the product available if it runs out of stock.

The retailers, in turn, have a barcode reader to keep track of inventory, products coming from manufacturers, and more. The air-conditioner's compressor has a sensor that radiates data regarding the product's temperature and health. This data is consistently monitored by customer care to resolve the issues if any.

There are many other examples that exist like smart cars, smart machines, smart cities, smart homes, where IoT is rethinking our way of life, and changing how we interact with technologies.

## Consumer IoT

The products which are directly used by an individual (consumer) are a part of the consumer IoT. Some consumer products are already widely used, such as fitness tracking devices, smartwatches, and home products (like Nest thermostat or the Apple Home Kit). It's predicted that in 2025, 50% of the grown person will include IoT devices in their life styled.

Rachio lawn-watering system is an example of consumer IoT device and it is explained in below.

### *Rachio lawn-watering system*

With the help of sensors and an active internet connection, Rachio (IoT device) gathers the data like weather prediction data, soil, and plant types, and exposure to the sun, to form a customized watering schedule. An application connected to the physical device (smartphone) assists users to manage the system remotely and check whenever required.

## Industrial IoT

The industrial IoT incorporates the products used by a company to deliver a good or service, for example, manufacturing plant machinery or industrial vehicles. The **Industrial Internet of Things (IIoT)** opens doors for better product designing, improved customer satisfaction, expanded uptime, and increased streams of income.

### Premier deicers

Premier implemented an IoT solution on airplane deicers intending to increase the efficiency, gathering the data on system component performance, fluid pressure, flow, temperature, volumes, and overall usage. With this information, downtime can be minimized by predicting component failure, and ensuring fluid usage does not exceed supply level.

## IoT emergence

The term IoT is not very old, but the actual idea supporting it began to emerge about more than 40 years ago. Kevin Ashton coined the term in 1999 while he was working on exciting RFID technology. The researcher, no doubt, got an attraction for his research, but IoT was not common even for the next ten years. The following [Table 1.1](#) summarizes the emergence of IoT:

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**Table 1.1:** Emergence of IoT

## Characteristics of IoT

The IoT provides an abundance of features enabling heterogeneous devices to dynamically connect for instance via a secure connection, while not being visibly feasible at the other time. Some such features have been elaborated in the following figure:



**Figure 1.1:** Characteristics of IoT

### Interconnectivity

Each connected device in an IoT network has been termed as a node. The IoT enables the connectivity among the devices which are distinct in terms of architecture and operations just by programming the sensors embedded over the devices. The compatibility and accessibility are considered in such connectivity. Compatibility refers to enabling the coherence between the communicating devices such that the data produced or consumed by a device is compatible with the other device.

## Intelligence

The things make themselves recognizable, and they gain intelligence by the context related decisions enabled by the fact that they can convey information about themselves. They can obtain information that has been accumulated by other things.

## Dynamic

The *things* can since change their state from rest to motion and vice versa, so the system (for example, location and speed) cannot be considered static. Moreover, the change in states like sleeping and waking up, connected and disconnected produce the dynamic data. The example of air conditioner manufacturing company depicted in *How does it work?* The section also illustrates the change in the count of the things connected to a network. Thus, dealing with dynamic atmosphere becomes an important property for a network to be called as an IoT network.

## Sensing

The *things* are required to be regularly monitored for any changes in its state or physical environment such as the changes in temperature, humidity, motion, air quality, speed, and pressure, and so on. The sensors can sense such changes and assist in the interconnection of physical devices. For example, the sensor embedded with a bin to make them a part of a smart city. This sensor will consistently generate data about the amount of waste in the bin so that the municipal workers are aware of the requirement of clearance of trash. The workers in this case too have sensors since the nearest ones are only provided notification for the job.

## Expressing

Working on the command lines is sometimes required by the developer. He even writes code in a high-level language. But the IoT system expresses the output to the end-user either in the form of a **Graphical User Interface (GUI)** or in terms of actions performed on things. For example, in the case of the example discussed in *How does it work?* Section, after the sensor, senses the data about the ill health of the air conditioner, a message on the customer's registered mobile number is dropped saying that he can get his product serviced at the nearest service center. This means that in the case of IoT systems, not only the interaction of end user with GUIs, but the real interaction is made as output.



### Extensive scale

An IoT system is required to manage more devices than those directly connected to the IoT system. These too keep on increasing with several factors, for example, an increase in the number of users of the system. Thus, the data generated is also enormous. Moreover, the IoT networks are spread over huge geographic locations which cover almost all the land on the earth in several systems. It is estimation that around 75 Billion devices will be connected to each other by 2025.

### Heterogeneity.

The since belongs to different hardware platform and networks, the compatibility and accessibility are assured, and the connection is made as if each thing on the network possesses homogeneous properties. Any real-world object can be assigned a unique identifier and an embedded system to enable it to share the information over the internet and be a part of the network. IoT has something to offer to everybody. It has reformed the manner we used to socialize, have fun, and do business.

### Endpoint management

It becomes mandatory to include endpoint management in an IoT system as it can otherwise lead to the failure of the entire system. For example, in a smart IoT based coffee machine which orders the coffee beans from the nearest retailer when it is finished, the failure can be faced in case of unavailability of recipient to receive the order at the time of delivery.

## Security

IoT gadgets are normally vulnerable to security dangers. While enjoying the advantages of an IoT system, the safety of endpoints, data, and physical wellbeing of the things cannot be ignored. There is a wide assortment of advances that are related to the Internet of Things that make it secure, however transparent simultaneously. The mechanism of encryption and decryption of generated data, programming sensors to self-monitor the changes, managing security among connections, and many more mechanisms are for example implemented for security.

## The physical design of IoT

The IoT can be seen as a huge network comprising further networks of gadgets and computers connected via a progression of intermediate technologies where enormous technologies like RFIDs and wireless connections may act as allowing of this connectivity. Design of any IoT application must encompass the following task:

**Tagging things:** This includes real-time object detectability and addressability by the RFIDs.

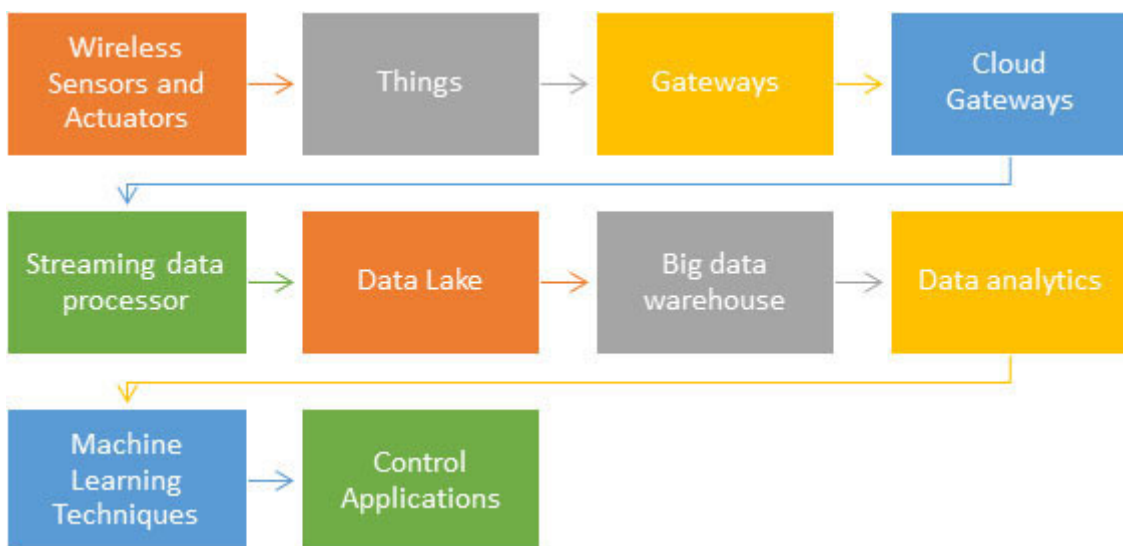
**Feeling things:** The sensors work as essential devices to gather the data from the surrounding environment.

**Shrinking things:** The miniaturizing nanotechnology has incited the capability of smaller things to communicate and connect several *things* or *smart*

**Thinking things:** The intelligence embedded in devices through sensors forms the network interaction to the Internet. It can enable *things* to enhance the intelligence further by analyzing the data collected by sensors.

### IoT architecture and components

IoT architecture comprises several layers of technologies which support IoT. It also outlines how the components of IoT are related to each other to make communication to produce a scalable and compatible system. The following [Figure 1.2](#) depicts the detailed view of the architecture of IoT. The functionalities of architectural components of IoT has been explained as follows:



**Figure 1.2:** *IoT components*

### Wireless sensors and actuators

The sensors are directly connected to physical devices on the IoT network. These gather the data from the things and perform analysis onto it to depict the decisions. Another term for a sensor can be a transducer, produces one form of energy from another. For example, converting sound/vibration energy into electrical energy. In IoT systems *Actuators* are another type of transducers, that performs opposite operation of sensors. In IoT devices Actuators are used to convert the electrical energy into some physical activity.

In a typical IoT system, the information is first collected by the sensor then passed to a control center to perform the decision making; this decision is later sent to the actuator. The efficiency of sensors to depict the changes in the physical environment (lights turned off, reduced room temperature, change in the geographic location of a device, and so on.) determines the efficiency of the overall system. The sensors can be classified based on the task performed by them, for example, light sensor, temperature sensor, infrared sensor, and smoke sensor, and so on.

## Things

In the Internet of Things, the term Things represents any physical object which is capable of sense and communicates with other objects. These objects can be small devices, big machines, and even sometimes people. For example, refrigerators, street-lights, smart-homes, manufacturing-machinery, vehicles, laboratory-equipment, and everything else imaginable. Some of the cases, the sensors are not attached physically to things. Such objects need to be monitored, for instance, monitoring what happens in the surroundings of a thing.



## Gateway.

Data produced by things is required to be sent to cloud, and cloud to things. This is done through gateways. Gateways are used to provide the interconnectivity between the cloud, and things associated with the IoT. Moreover, it enables pre-processing and filtering of data before transferring it to the cloud. Gateways are also used to transfer the control commands from the cloud to the things. These commands are then executed by actuators present in the things.

### Cloud gateway

Data used in the communication between the IoT cloud server and field gateway must be encrypted and compressed. Cloud gateway encourages data compression and encryption and guarantees communication. It also guarantees the compatibility between different protocols supported by different gateways.

### *Streaming-data processor*

This component ensures the effective flow of input-data into control application and a data lake.

### Data lake

A data lake stores the data generated by associated things in its natural format. The Big Data comes in streams or in batches. At the point when the data is required for significant insights, it is removed from the data lake and is at the point loaded to big data warehouse when the information is required.

### Big data warehouse

The data lake contains all kinds of data that is generated by the sensors, but the warehouse holds only structured, processed and matched data. When pre-processed and filtered is required, it can be fetched from a big-data-warehouse. Additionally, data warehouses are responsible for storing the context details about sensors and things like location of the installed sensors; control applications and commands sent to things.

## Data analytics

To figure out the current trends, and to find the actionable solutions; data analysts utilize the data stored in big-data-warehouse. Such data can be utilized by data analysts to find the relationship and patterns between the data. Furthermore, such data patterns can be used to create useful algorithms and different control applications. Moreover, during the data analytics, various tasks could be performed like: drawn as schemes, plans, diagrams, info-graphics, big data derives, for instance: the presentation of things, assisting in identifying inefficiencies and working out the methods to make customer-oriented and reliable IoT system.

### *Machine learning and the models that ML generates*

Machine learning has provided a great prospect to create more effective and more precise ML models. These models can be used for control applications. Based on the collected historical data from the big data warehouse (for instance, in a week or a month), these ML models can be updated. Once the effectiveness and appropriateness of new updated models are checked by data analysts, new updated models can be used by the control applications.

### Control applications

The control applications automatically give directions and alerts to actuators, for instance:

On a weather prediction, the windows of a smart home can be directed by an actuator to open or close.

The smart watering systems can be directed to water the plants, and the sensors detect that the soil is getting dry.

The equipment in an industry can be connected to an IoT system to be continuously monitored by sensors, in the case of the situations reflecting the risk of failures, the notification messages to the engineers can be sent.



## Logical design of IoT

The logical design illustrates the abstract representation of processes and entities without going into the details of lower-level implementation. The following terms are required to be understood for a complete understanding of logical design:

Functional blocks

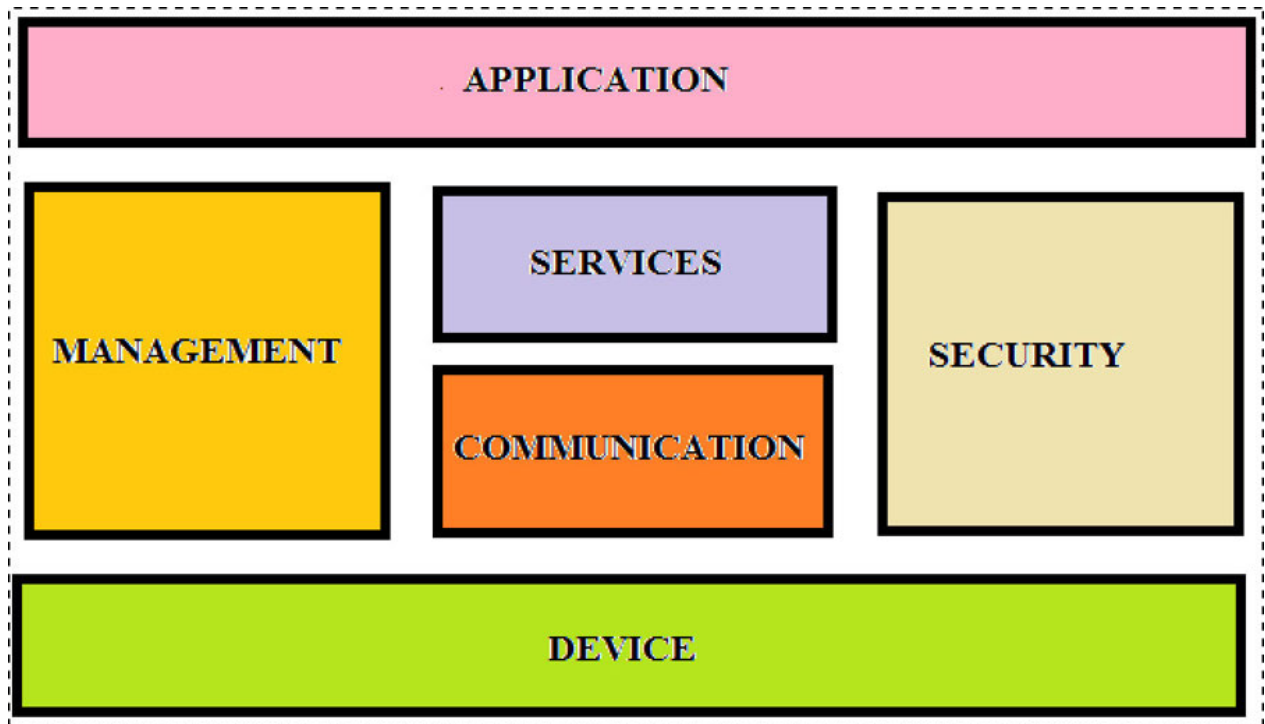
Communication models

Communication APIs

Functional models

An IoT framework contains several functional blocks that provide the system, the capabilities for identification, sensing, actuation, communication, and management. These are depicted in the following

[Figure](#)



*Figure 1.3: IoT functional models*

## Devices

The physical connected things are devices. These share data among each other. This data is handled either on the same device on which it is produced or is processed on the remotely located central server. Several interfaces like those for sensors, network connectivity, sound/video, memory, and storage interfaces assist communication to other different devices in case of both wired as well as wireless connections.

For instance, the data generated by a sensor embedded into soil moisture monitoring equipment in a lawn, when processed, can assist in finding the optimal watering schedules. According to the moisture level, calculated by the sensor, the watering schedule can be calculated on the remoteserver.

## Communication

The connected devices and remote servers communicate among each other through communication devices. The communication protocols include the set of rules to assist in communication. These protocols usually work in the application layer, transport layer, network layer, and the data link layer.

## Services

An IoT framework serves different types of functionalities, for example, services for device modelling, device control, information distribution, information analysis, and device recovery.

## Management

Management includes several functions to govern an IoT system, like information processing, processing of relevant tasks, processing of requestor information, decision making, and so on.

## Security.

The authentication and authorization assure privacy, data integrity, and message security. Such services have been provided by the security functional block.

## Application

This layer provides the user interface. Applications allow the users to picture, and analyze the status of the system at the current stage of action, and in several cases can also predict the future perspectives.

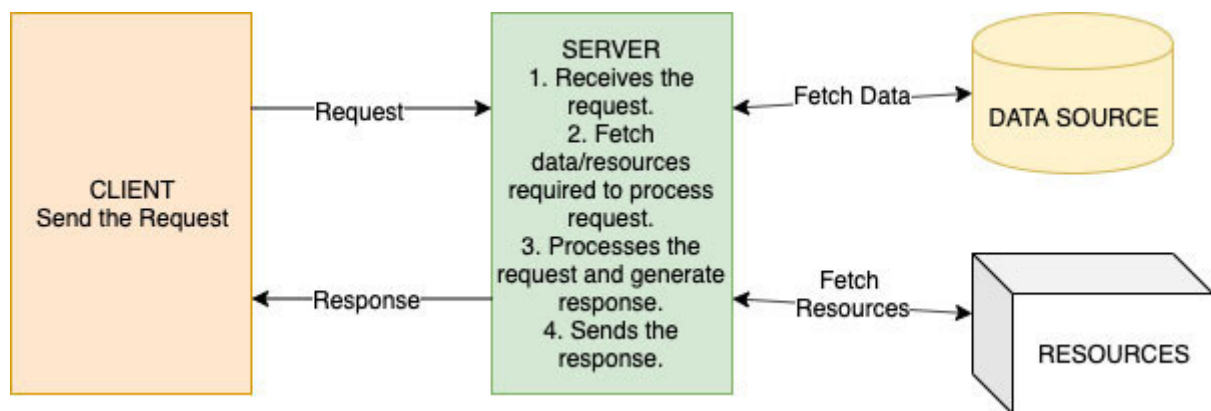


### Communication models

For operational perception, it is important and useful to understand how various IoT devices communicate with each other. In terms of technical communication models, some major IoT communication models are as described in the following sections.

### Request and response model

This model follows client-server architecture. The client, when required, requests for the information from the server. This request is usually in encoded format. The server categorizes the request, and fetches the data from the database and its resource representation. This data is converted to response and is transferred in an encoded format to the client. The client, in turn, receives the response. The request-response is a stateless model since the data between the requests is not retained. The working process of request response model is shown in [Figure](#)

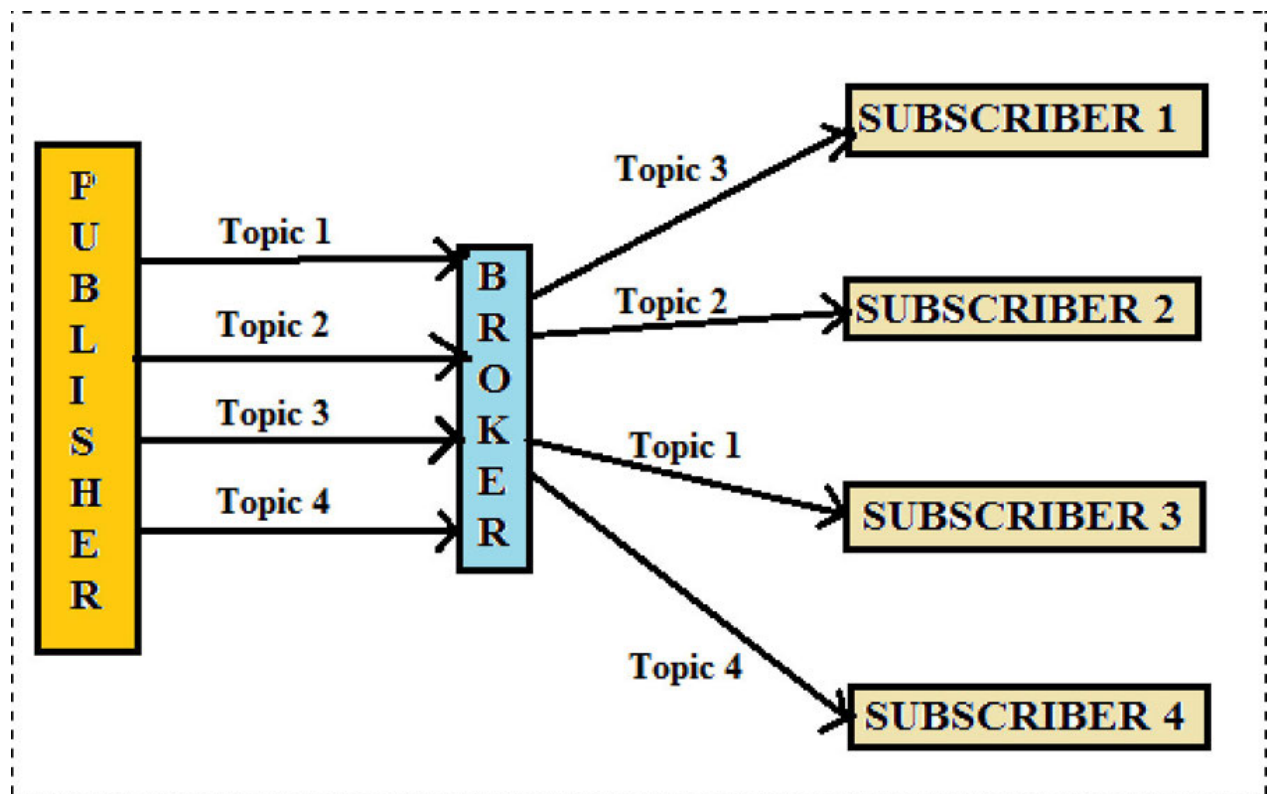


**Figure 1.4:** Request response model

### Publisher - Subscriber model

The model comprises publishers, brokers, and consumers. Publishers are the source of data, which is sent in the form of topics. The topics are sent to the intermediate broker, which in turn transfers them to subscriber/consumer who has subscribed for the particular topic. The broker only has the information regarding the consumer to which a particular topic belongs to, which the publisher is unaware of. The working process of Publisher-Subscriber model is shown in

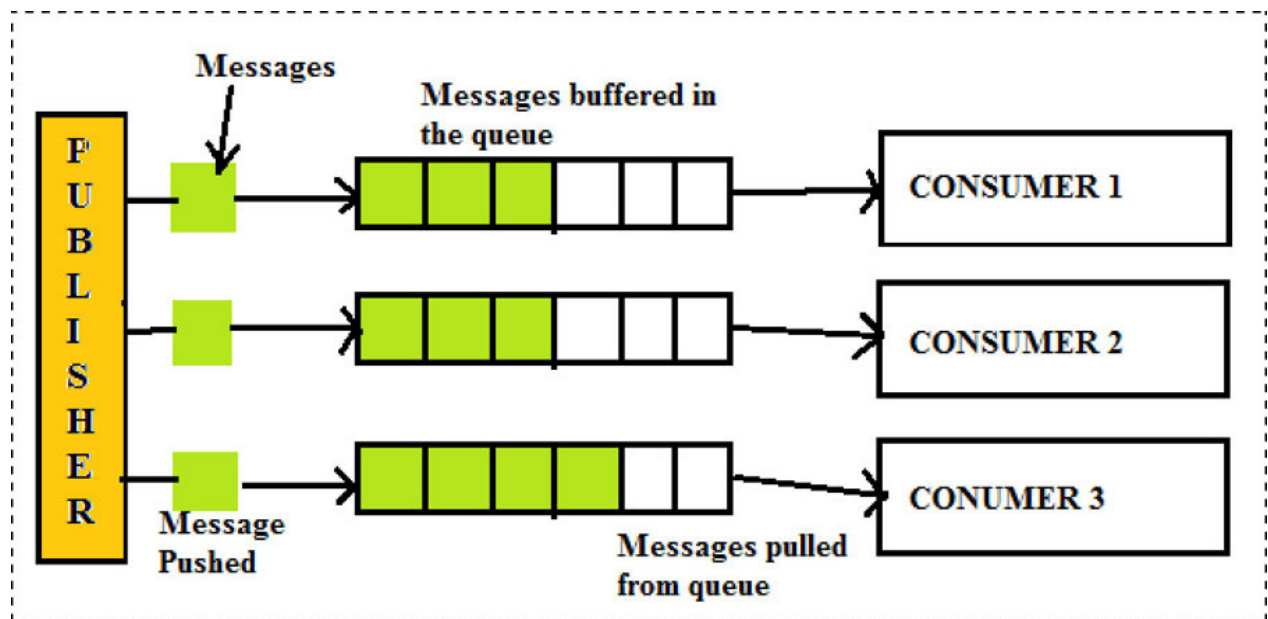
Figure



**Figure 1.5:** Publisher-Subscriber model

### Push-Pullmodel

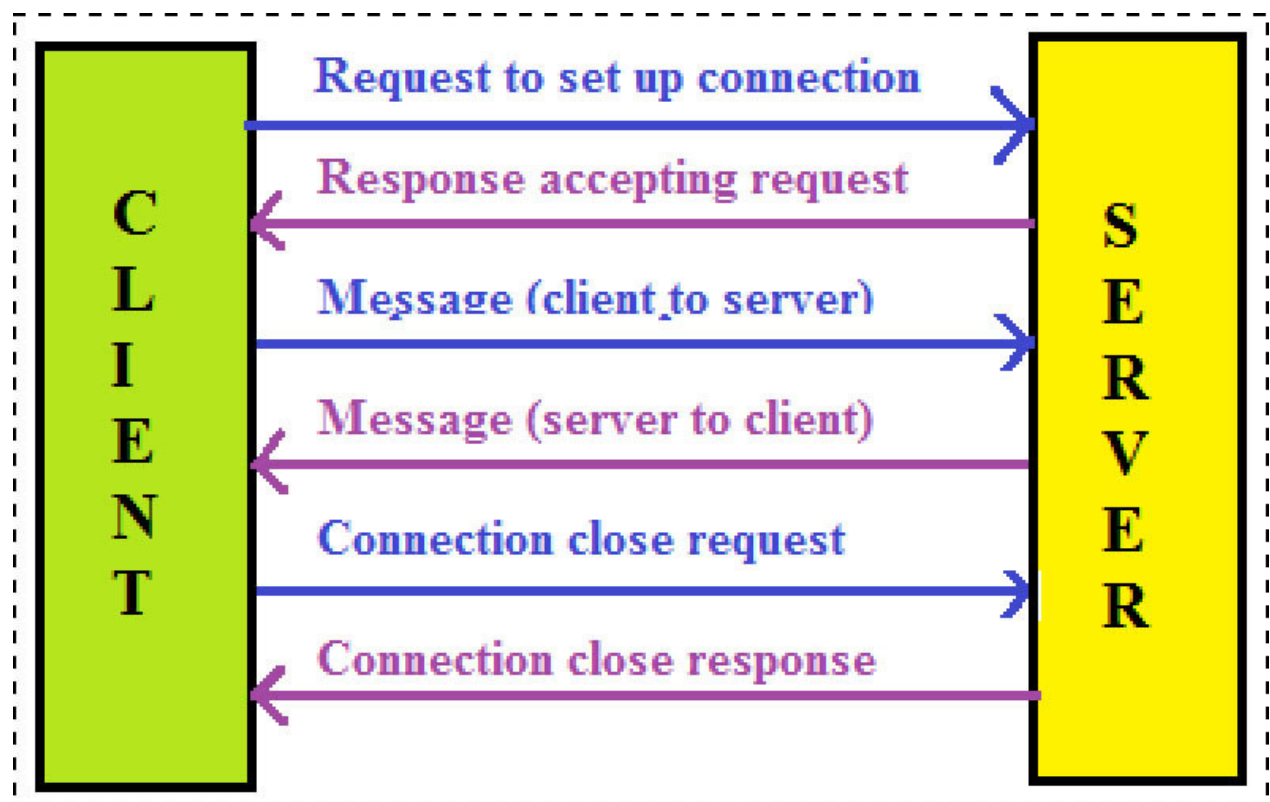
The push-pull model constitutes data publishers, data consumers and data queues. The working process of Push-Pull model is shown below in [Figure](#) whereas publishers publish the messages/data and push it into the queue. The consumer, present on the other side, pulls the data out of the queue. Thus the queue acts as the buffer for the messages when the difference occurs in the rate of push or pull of data on the side of a publisher and consumer:



**Figure 1.6:** Push-Pull model

### Exclusive pair model

Exclusive pair is a bi-directional model, including full duplex communication among client and server. After the connection is set up between client and server, both can share messages with each other. The once opened connection, will not be closed until the client requests to close the connection. The server has the record of all the connections which has been opened. The working process of Exclusive Pair model is shown in [Figure](#)



**Figure 1.7:** Exclusive pair model

## IoT communication APIs

The IoT communication generally uses two APIs. These are:

REST-based communication APIs

Web socket-based communication APIs

## Representational State Transfer (REST).

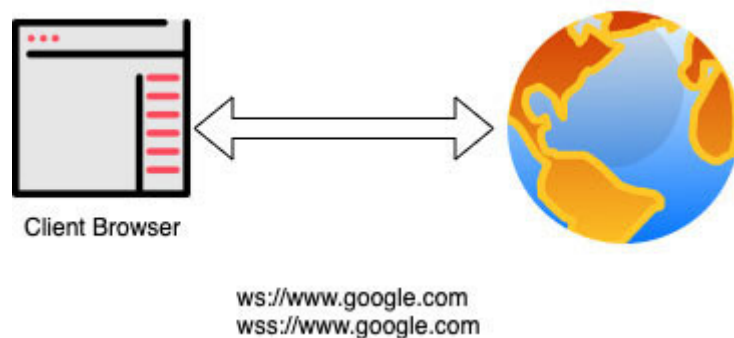
In case of **Representational State Transfer (REST)**, every time the client wants to retrieve data from a server a connection needs to be established. A client sends a request, which is received by server. The server then processes the request and sends the response/data back to the client. In this way, REST is unidirectional, therefore there is more overhead in this case. The applications demanding real time responses, or those requiring to display the streams of data could not be made to work in this case. An example of REST based communication is shown in [Figure](#) whereas client send the request to the server using HTTP or HTTPS Protocol:



**Figure 1.8:** REST based communication APIs

### Web Sockets Based Communication API

In web sockets-based communication API, over TCP connection: the connection with the server is established only once which is done by initial handshake with the server. The server can send data at any time and the client can handle receiving that data, the client can also send other requests to the server. Such kind of API is really useful when you need to have low latency with data interactions on your web application. An example of Web-Sockets based communication is shown in [Figure](#) whereas client and server are communicating through ws: or wss: (Web-Socket based communication APIs):

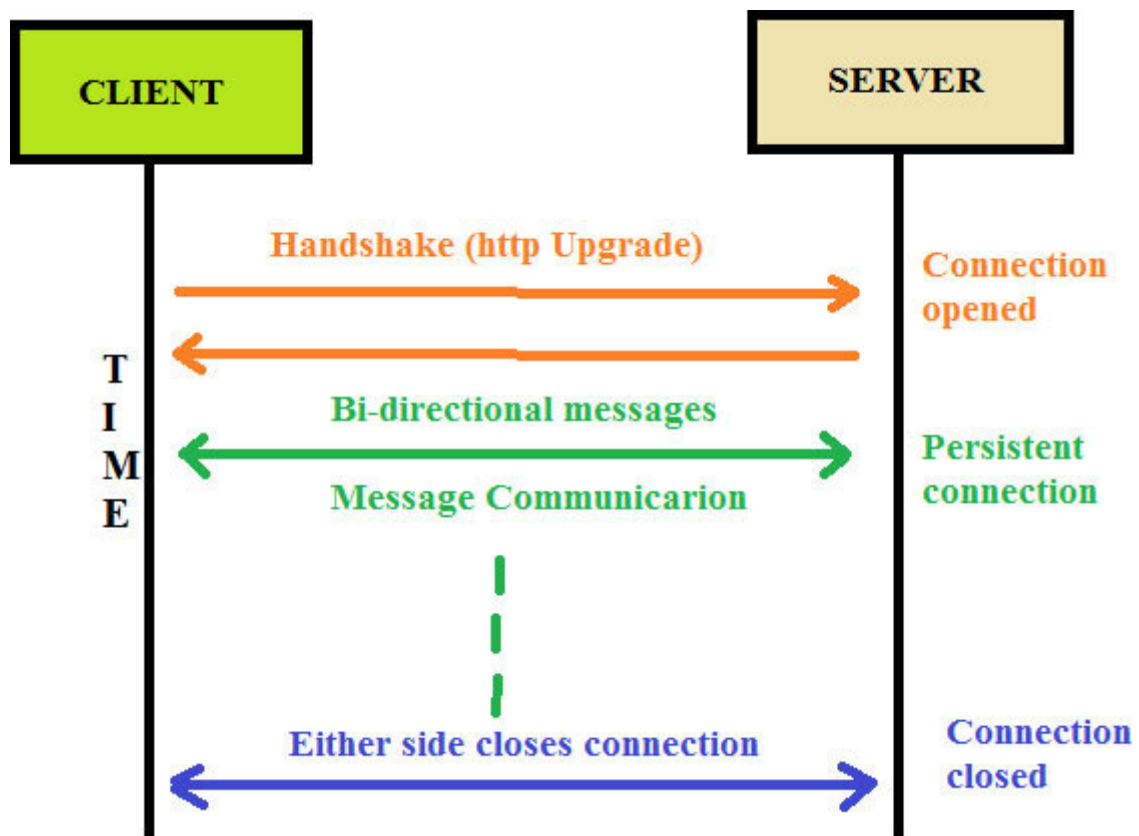


**Figure 1.9:** Web-Socket based communication APIs

Using web-socket based API; messages can be transferred in both the direction client to server and server to client. [Figure 1.10](#) shows that the connection with the server is established only once which is done by initial handshake with the server. After successful establishment of the connection, server can send data any time to



the client; client can also send a new service request to the server. The established connection can be closed by any device either client or server at any time of instance:



**Figure 1.10:** Bi-directional message transfer Web-Socket based APIs

Difference between web-socket and REST API

The difference between web-socket and REST based API is shown in [Table](#)

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**Table 1.2:** *Differences between web-socket and REST API*

### Evaluating business impact and economics for IoT

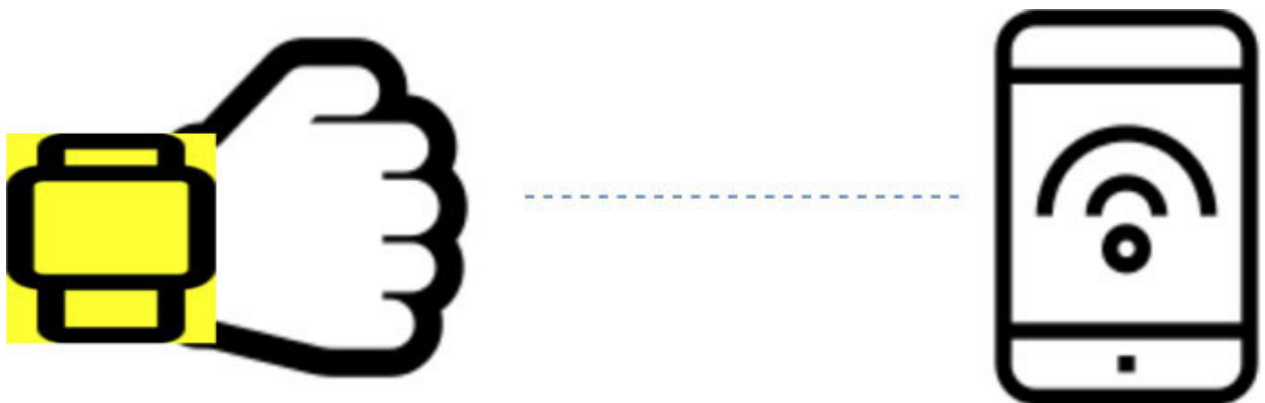
With the emergence of IoT, some people believe that the implementation of IoT would reduce the employment opportunities for humans, while others opine that the tasks causing boredom can be allocated to be assigned to machines while humans can work on the other important tasks to be accomplished. The time would determine this impact. But it can be said that the impact would be profitable. It can be calculated that the 10 percent increase in the IoT systems would raise the GDP of *Germany* by \$370 billion and that of US by \$2.26trillion over 15 years (2018-2032).

## Some emerging fields of IoT which can impact on business and economics

IoT have several applications in various fields, which will put a big and growing impact on business and economics, some of the IoT applications in various fields are discussed below.

## Wearable

Wearables are among the earliest devices those deploying IoT as an application. The wearables can be used for several purposes; health monitoring is one amongst them. Guardian glucose monitoring device is one of the applications to assist people suffering from the diabetes by monitoring the sugar level in blood. A tiny electrode with embedded glucose sensor is simply kept over the skin to provide the information by using radio frequency.



**Figure 1.11:** Wearable device

A wearable device like a smart-watch shown in [Figure 1.11](#) can be used for monitoring the footsteps required to reduce obesity based on the current weight and fitness target of an individual is an important application including sensors to calculate the footsteps, pulse rate and the distance covered and that information will be sent to user mobile for analysis.

## Smart Home

A smart home may, for instance, include the connectivity among all the home appliances, even the cooking utensils. Turing the air conditioners on while you are few meters away from the home, feeding the pet through smart pet feeder, Turning the room lights on after detecting that the owner has woke up just by the sensing done by wearables are some of the examples of IoT. Another example of smart home technology is the Amazon The Echo is a Bluetooth speaker which is powered by Amazon's handy voice assistant. You can use Alexa to control most of the gadgets in your house by the sound of your voice. Also, the smart home surveillance cameras to keep a watch through a camera over the surroundings. A notification is sent to the owner's mobile whenever someone tries to breach the security and pass the secure area.

## Smart Cities

Smart traffic handling is one of the IoT based application of a smart city. As more people are advancing to cities, the problem of traffic congestion is becoming worse fortunately, the Internet of Things is working in the area to benefit the individuals in a quickest possible way. For instance, the smart traffic signal signs can modify their timings to involve commutes and manage traffic and keep cars moving. The city officials can collect and analyze data from traffic cameras, smart phones, and road sensors to detect traffic incidents in real-time. Drivers can also be made aware of the accidents and routed to roads that are less congested. The possibilities are never ending and the effect will be considerable. [Figure 1.12](#) depicts the various services and facilities of a city, building, transportation, home, agriculture and many more are interconnected to share real time information that makes a city a



**Figure 1.12:** Smart city



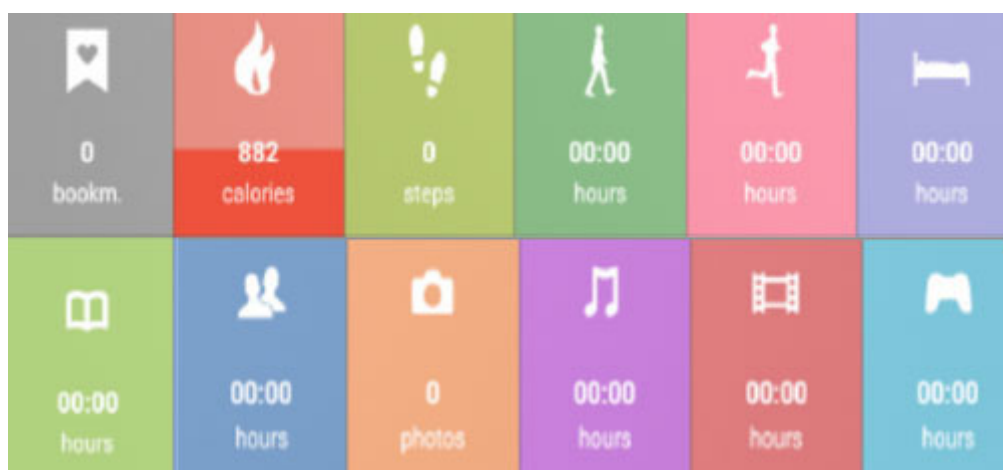
### Smart retail

The shopper can be sent advertisements reflecting the fresh arrivals by the store. Based on the clicks by the user, the retailer is sent the detailed information regarding the interests of the shoppers. Moreover, the less attracted products can be made unavailable based on the smart feedback and making the demanded products available. While shopping in a smart mall, the products can be made to introduce themselves. The customer can be provided with the option of paying through his credit/debit card or smart phone. The employee of the mall can be informed to add more items to the rack of sold items.

## Health care

IoT includes applications in healthcare too, those assisting patients, physicians, hospitals, and healthcare insurance companies. By using wearables and other personal health monitoring equipment embedded with the IoT, the patient can monitor his health and judge if a visit to a doctor is required or not.

On the other hand, the physicians too can regularly get the report regarding patient adherence to the treatment. The real time analysis of staff deployment in the hospital can too be monitored through IoT. The health insurance companies benefit from IoT by capturing the data (heart beat, walking habits, reading habits and many more as shown in [Figure](#) of clients from health monitoring applications and analyzing it for organizing the insurance plans:



**Figure 1.13:** IoT in health care

## Agriculture

The sensors can well monitor weather conditions, soil type and humidity level, and crop growth. Automating processes like fertilization, irrigation, pest control, weed control, and prediction of the harvest using IoT has assisted farmers a lot. For instance, **GreenIQ** is an interesting application using smart agriculture sensors. It has an embedded smart sprinklers controller that enables a farmer to manage the irrigation and lighting systems from a remote location. Another application of IoT included sensors for informing the farmers if some animal like cows from outside cross the boundary of his field.

Besides the above discussed IoT fields, there are many other fields are also exist which can make a big impact on business and economics, because the use and demand of such IoT applications and devices are growing rapidly in all over the world, and it makes a positive impact on production, development, and the business growth. Some of emerging and growing fields of IoT has been summarized in the [Table](#)

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**Table 1.3:** Emerging fields of IoT

## Conclusion

The **Internet of Things(IoT)** is a giant network of connected devices, assembling and sharing data about how they are utilized and the environment in which they are operated. In this chapter we have discussed the basic concept behind the evaluation of Internet of Things, its role, applications, it's physical and logical components, also discussed the connectivity and various communication models by which these devices communicate. At the end of this chapter, we spread light on the Impact of growing IoT technology on the business and economics. With the emergence of IoT, some people believe that the implementation of IoT would reduce the employment opportunities for humans, while others opine that the tasks causing boredom can be allocated to be assigned to machines while humans can work on the other important tasks to be accomplished.

### Points to remember

The devices connected in an IoT network are termed as

Any IoT device directly utilized by a consumer is part of the consumer IoT. For example, wearables.

Any IoT device used by the industry to assist in the development of consumer products is an example of Industrial IoT. For example, industry equipment diagnosis with IoT device.

Ability to access information from anywhere at any time on any device.

Sensors gather the data from the things and perform analysis onto it to depict the decisions.

Actuators perform reverse action to that of sensors. These perform physical actions in response to the sensed input.

The gateway is responsible for sending and receiving data to and from the cloud.

The industrial IoT includes any product used by a company to deliver good or service, such as factory machinery or industrial vehicles.

Kevin Ashton coined the term in 1999 while he was working on an exciting RFID technology. Although the advancements in the field had started for more than 40 years ago.

The harmonious blend of network, sensor positioning, and complex data analysis methods currently empower applications to aggregate and follow up on a lot of data produced by IoT gadgets in homes, open spaces, industry and the common world.

### Multiple choice questions

\_\_\_\_\_ coined the term Internet of Things.

Tim Berner's Lee

Steve Jobs

Glen Macaughty

Kevin Ashton

IoT is an acronym for:

Idiocy of technology

Internet of Things

Internet of Thing

Internet of Technology

IoT can be defined as?

The day that computer take control of the world



The intelligent connection of people, process, data, and things

Both of these

None of these

IoT is built on\_\_\_\_\_.

Networks of data gathering sensors

Cloud computing

Both of these

None of these

IIOT stands for what?

Informational Internet of Thing

Industrial Internet of Thing

Industrial Internet of Things

Innovative Idiocy of Technology

What challenges should be kept in mind in IOT?

Network Configuration

Security

Energy Consumption

All of the above

### Answer

*d*

*b*

*c*

*c*

*c*

*d*

### True or false

IoT gadgets are normally vulnerable to security dangers.

A network allows the things to connect to each other and transmit the data.

The device that senses the physical activity performed by a device is called as actuator.

Sensors perform the energy conversion on taking the input sensed by actuators.

Response and Request model include publishers and subscribers.

WebSocket is bi-directional API.

The stateless http is used in REST APIs.

The data from things to cloud is sent by gateways.

Steve Jobs coined the term *Internet of Things*.

Bidirectional communication is used in REST APIs.

*Answer*

True

True

False

False

False

True

True

True

False

False

### Descriptive questions

Define the term Internet of Things; and how it works? Explain with the help of an example?

What is the basic difference between consumer IoT and industrial IoT? Support your answer with the help of example.

What are the various characteristics of IoT? Explain in detail.

Briefly describe about the various components of IoT architecture.

Explain the difference between Request-Response and Publisher-Subscriber communication model. Draw a neat and clean diagram to explain the working of both of the model.

Show the working of a Push-Pull communication model also explains how it differs from exclusive pair model of communication.

Explain the difference between REST-based and Web Socket Based API.

On the basis of your understanding about IoT evolution and its exponential growth, what is your opinion about this statement:

*Implementation of IoT would reduce the employment opportunities for humans?*

### IoT Architectures and Protocols

The **Internet of Things (IoT)** is the name of multiple electronic devices that are equipped with a unique IP address and communicated over the Internet. We can describe IoT in many ways, but ultimately, with the help of the Internet, we can control the number of electronic devices in this development through a single device. Here, the sensors should be provided by the electronic devices in IoT technology, and the signal should be sensed electrically and functions accordingly. And the sensed data is transferred via the Internet to the other computer. For seamless and smooth connectivity between the devices, there is the need for common protocol or architecture that will ensure that devices can *communicate* to and *understand* each other across space and time. In this chapter, we will review and present various IoT based architecture and protocols.



## Structure

Taxonomy

Three-layer and five-layer architecture

Cloud and fog based architecture

Representative architecture

Near Field Communication (NFC)

Wireless Sensor Networks (WSN)

IoT network protocol stack

Bluetooth, ZigBee, and 6LowPAN

## Objectives

After reading this chapter, you will be able to:

Understand how the various components of cloud computing work together to form the basic architecture of cloud computing.

Examine the relationship between the various layers in the IoT architecture.

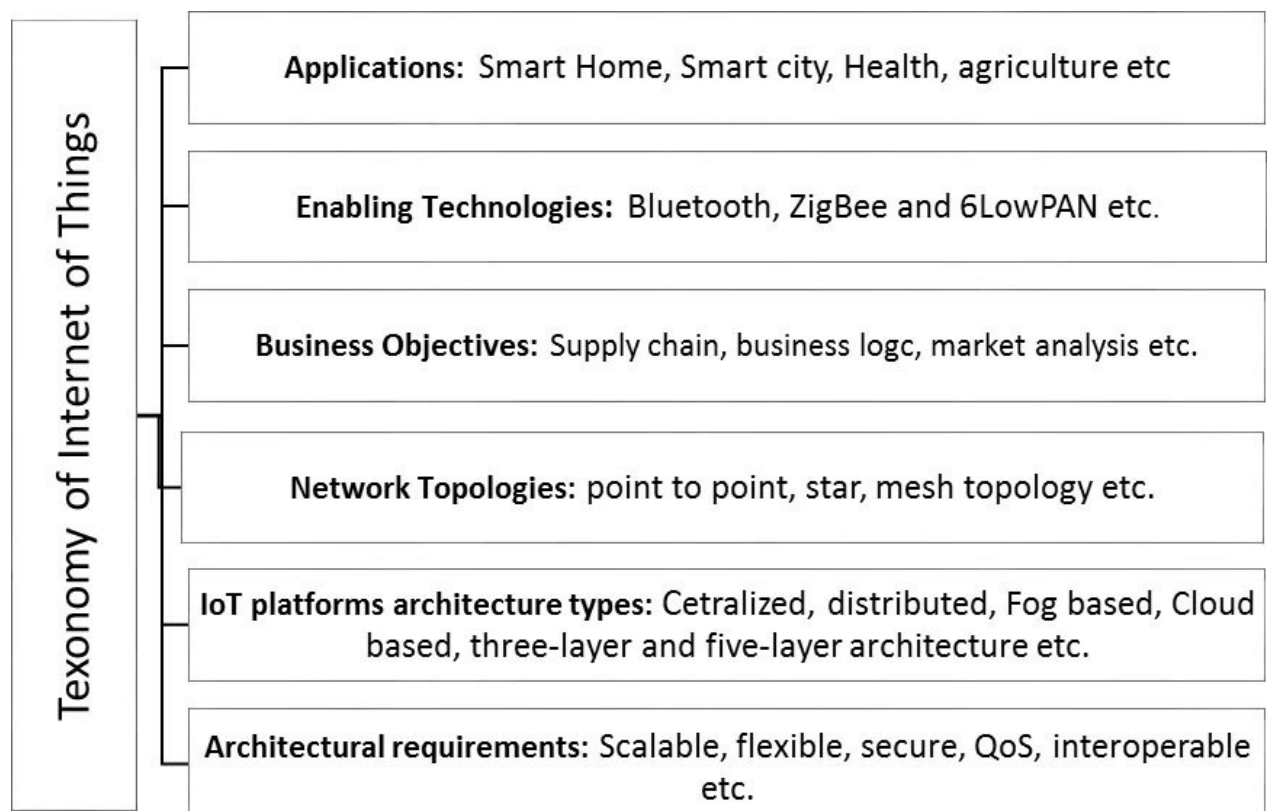
Appreciate the role of fog and cloud computing architecture in deploying IoT infrastructure and services in a smart applications environment.

## Introduction

In today's world, the Internet has become deeply embedded in our day-to-day lives that we can hardly ignore anymore. Yet still, it is even the beginning, and in not too distant future, we are going to see massive intrusion and use of the Internet in unprecedented levels and across-wide varieties of devices. In this respect, sensors will be required to connect devices at various places and locations to generate data for analysis and action. For seamless and smooth connectivity, there is the need for common protocol or architecture that will ensure that devices can *talk* to and *understand* each other across space and time.

## **Taxonomy**

Taxonomy refers to a process whereby items are named and classified according to their similarities and differences. In the area of IoT, taxonomy is based on the architectural elements and the protocols of IoT. [Figure 2.1](#) depicts the taxonomy of internet of things based on the parameters such as IoT applications, business objectives, enable technologies, platforms and architecture types, networking topologies, and architecture requirements:



**Figure 2.1:** *Taxonomy of IoT*

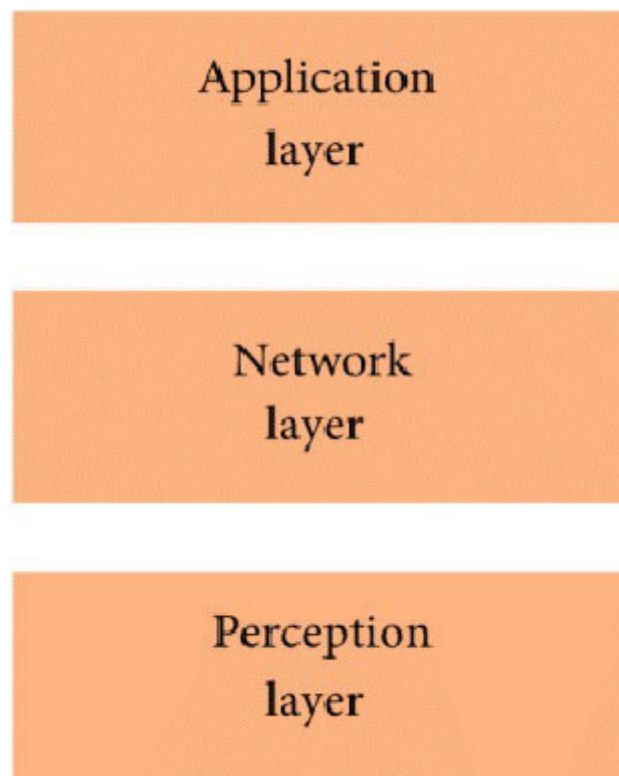
By far, the most commonly used or forms of IoT architecture are the fog and the cloud-based. Fog based architecture is that part of

computing which consists of layered approach useful for processing and filtering data.

The cloud-based architecture consists of temporary storage and has a small processing unit with some security features. Entities or components communicate over the network using varieties of a set of protocols and standards. For the short-range, low power communication protocols, the most often used are the **Radio Frequency Identification (RFID)** and **Near Field Communication (NFC)**. Bluetooth, Wi-Fi, and ZigBee are examples of short-range communication protocols.

### Three-layer and five-layer architecture of IoT

IoT should be able to connect and transfer data among billions and trillions of devices. For that, it is necessary to have a well-structured and organized architecture. This architecture is supposed to be capable of accommodating a wide range of components and technologies that form a part of the IoT ecosystem. The first and most basic three-layer architecture is shown in [Figure](#). It was established in the initial phases of study in this area. The architecture comprises of three layers, namely, the perception, network, and application layers:



**Figure 2.2:** *Three-layer architecture*

**The perception layer (or object layer/devices layer):** The perception is the external layer with sensors for sensing and collecting environmental data. In the environment, it detects those physical parameters or recognizes certain smart objects. The physical devices include different types of sensors like the ones based on **micro-electromechanical systems (MEMS)** technology.

Many different types of Sensors are available for this use like proximity sensors, light sensors, gesture and optical sensors, touch and fingerprint sensors, pressure sensors, and many more. Standardized plug and play mechanisms are used by the physical layer to consolidate and construct the heterogeneous types of sensors that belong to the IoT device ecosystem. The data collected at this layer is passed on to the next layer using different network channels.

**The network layer:** The network layer provides the link between network devices and servers to create smart devices or IoT. Its features are also used for transmitting and processing sensor data. The data transmission can happen using any of the following technologies:

RFID

3G

GSM

UMTS

Wi-Fi

Bluetooth low energy

Infrared

ZigBee

Specialized processes for handling functions such as cloud computing and data management are also present in this layer.

**The application layer:** The application layer makes it possible to provide specific services to users. It is that part of the IoT architecture that outlines the various applications for IoT to be deployed, for example, in smart homes and smart health.

This layer is responsible for meeting the various kinds of services requested by users (customers). The type of service requested by the customer depends on the specific use case that is adopted by the customer. For example, if a smart home is the use case under consideration, then the customer may request for specific parameters such as **heating, ventilation, and air conditioning (HVAC)** measurements or temperature and humidity values. This layer provides the various types of smart services, which are



offered by various IoT verticals. Some of the prominent IoT verticals are as follows:

Smart cities

Smart energy

Smart health care

Smart buildings or homes

Smart living

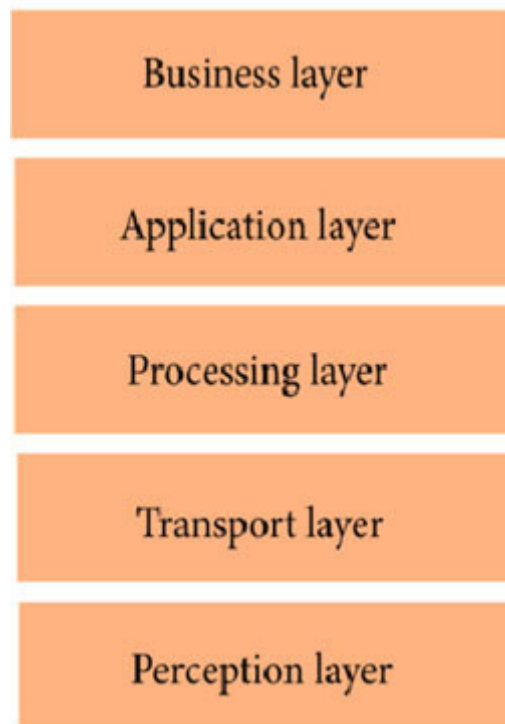
Smart transportation

Smart industry

The three-layer architecture defines the main idea of the IoT, but it is not sufficient for research on IoT because research often focuses on finer aspects of the IoT. That is why we have many more layered architectures proposed in the literature.

### The five-layer architecture

The five-layer architecture includes the two layers of perception and application, which have been described above. The functions of the other three layers, namely the transport, processing and the business layers are at this moment described below. [Figure 2.3](#) shows the five-layer architecture of IoT:



**Figure 2.3:** 5-layer architecture

**The transport layer:** The transport layer is responsible for transferring sensor's data from the perception layer to the

processing layer and vice versa by means of a variety of connectivity such as RFID, NFC, LAN, Bluetooth, and many more.

**The processing layer:** This is where the middleware layer which stores, analysis, and processes a large amount of data in the transport layer architecture. It manages the different services to the lower layers and uses technologies such as databases, cloud computing, and Big Data processing modules to achieve its functions.

**The business layer:** This part of the architecture manages and responsible for the whole IoT system. This layer includes the applications, business and the profit models as well as user privacy. It performs the overall management of all IoT activities and services. This layer uses the data that are received from the network layer to build various components such as business models, graphs, and flowcharts. This layer also has the responsibility to design, analyze, implement, evaluate, and monitor the requirements of the IoT system. This layer can use big data analysis to support decision-making activities. This layer also performs a comparison of obtained versus expected outputs to enhance the quality of services.

### Cloud and fog based architecture of IoT

Fog based architecture or also known as fog networking uses edge devices to perform a significant amount of computation, storage, and communicate locally over the Internet backbone. Fog in IoT utilizes a decentralized computing infrastructure in which data storage, computing, and applications are located somewhere between the data source and the cloud. Thus, it makes it possible to bring the benefits of the cloud closer to where data is created or produced and acted upon.

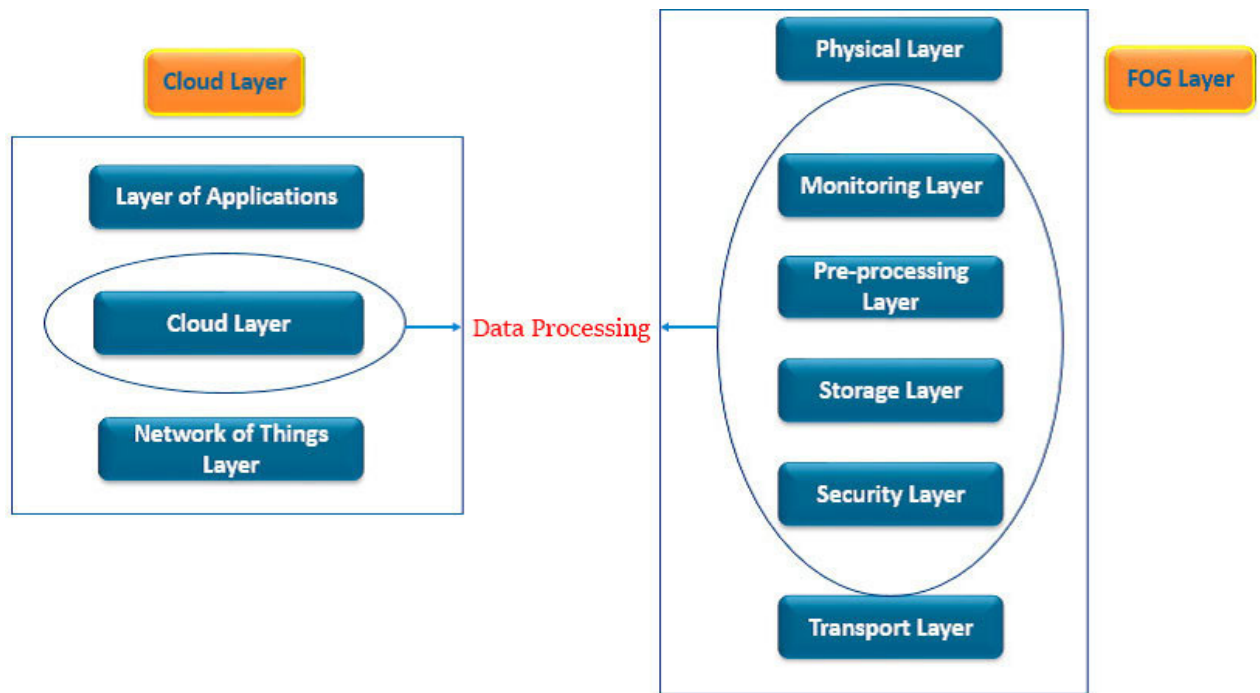
In recent times, the trend is to advocate for Fog computing as system architecture. In this respect, sensors and network gateways are used in data processing and analytics. This type of architecture offers a layered approach that covers areas of control, storage, and protection between the physical and transport layers, while the monitoring layer takes control of the power, resources and services.

### Cloud-based architecture of IoT

Cloud computing provides scalable and flexible services which include information storage options, software tools and analytics, suitable platform, and core infrastructure for the development. It is an extension of cluster and grid computing used to collect resources at one place and utilize them to high-performance computing.

It provides three types of services namely; **Software as a Service (SaaS)**, **Platform as a Service (PaaS)** and **Infrastructure as a Service (IaaS)**. It also provides mobility features for information handling and storage is reloadable from nearby clients. We can also have big data, machine learning as well as data analytics along with cloud computing for more information. The nature of information sensed as well as produced in the form of data by an IoT device.

The cloud platform receives and aggregate data summaries from many fog nodes, and it also performs analysis on IoT data and data from other sources to gain business insight. Cloud computing architecture can send new application rules to the fog nodes:



**Figure 2.4:** Data processing in the cloud and fog-based architectures of IoT

[Figure 2.4](#) describes how data is being processed between fog and cloud-based architecture. The left part of the figure shows the cloud layer which consists of three layers, at the center is cloud layer, above the cloud layer is an application of layers and below the cloud layer is a network of things layer. The right side of the figure describes the fog-based architecture of IoT. It consists of six layers- physical, transport, security, monitoring, pre-processing, and storage.

## Fog based architecture

Fog based architecture presents a layered approach which inserts monitoring, pre-processing, storage and security layers between the physical and transport layers. Fog based architecture is also known by another name that is edge-based architecture. Fog based architecture is an advanced version of cloud-based architecture. Fog computing acts on IoT data in milliseconds, based on policy and sends selected data to the cloud for long-term storage. Fog computing performance is better than cloud computing for handling user request. Fog computing has flexible infrastructure.

In fog-based structure, response time is low, Unlimited number of users as well as resources. Besides sending a vast amount of data to the cloud, it analyzes most time-sensitive data at the network edge. Mainly fog architecture considered only the four layers, that is, the following layers of fog architecture are:

**Monitoring layer:** It performs the monitoring functions like check the availability of resources and requirement of services by the clients and various responses.

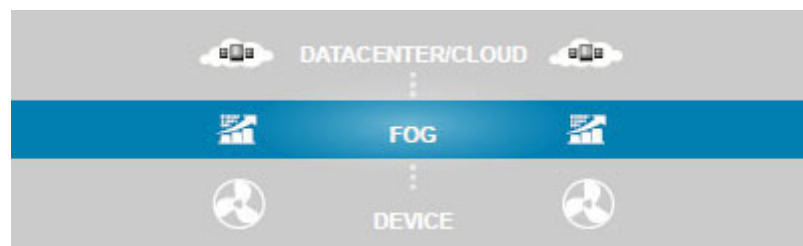
**Pre-processing layer:** It helps in analyzing the data by doing filtering processes.

**Storage layer:** The data from the pre-processing layer is sent to the storage layer. It helps in storage in a different format as per

requirement and needs with suitable protocols.

**Security layer:** It helps in offering privacy status to the data flow as well as helpful in the encryption and decryption of data.

[Figure 2.5](#) shows that the fog lies at the center, cloud lies above the fog layer where the data is stored whereas device lies below the fog:



**Figure 2.5:** *The Fog Extends the Cloud closer to the devices producing data*

The most time-sensitive data are analyzed on the fog node closest to the things generating the data. In a *Cisco Smart Grid* distribution network, for example, the most time-sensitive requirement is to verify that protection and control loops are operating properly. Therefore, the fog nodes closest to the grid sensors can look for signs of problems and then prevent them by sending control commands to actuators.

Data that can wait seconds or minutes for action is passed along to an aggregation node for analysis and action. In the Smart Grid example, each substation might have its aggregation node that



reports the operational status of each downstream feeder and lateral.

Data that is less time-sensitive is sent to the cloud for historical analysis, big data analytics, and long-term storage (see sidebar). For example, each of thousands or hundreds of thousands of fog nodes might send periodic summaries of grid data to the cloud for historical analysis and storage.

### Advantages of fog computing

**Greater business agility:** Developers with proper skills and tools develop applications and market them according to the requirement or demand. Fog applications help in operating the machinery and tools in the way of customer requirement.

**Better security:** With the proper privacy and along with the cybersecurity solutions, protect fog nodes for better security by using the proper procedures as well as using the same policy.

**Deeper insights, with privacy control:** Fog computing analyze sensitive data locally instead of sending it to the cloud for analysis.

**Lower operating expense:** It Conserves network bandwidth by processing selected data locally instead of sending it to the cloud for analysis.

### *Representative architecture*

**Social Internet of Things (SIoT)** is that part of an IoT which is capable of establishing social relationships with other objects with respect to humans. SIoT, attempts to moderate the challenges of IoT in the areas of scalability, trust, and resource discovery by taking a cue from social computing. Representative architecture is a part of the social Internet of things that provides such as navigation where one device is initialized and through it, we can navigate to other connected devices and linked back to the start device thereby creating autonomous relationships between objects and humans.

## Basic components of SIoT

IoT requires many devices to be interoperable for the model to function well. The following are some of the major components that enable SIoT applications to be successful:

**ID:** This refers to the unique method of object identification which is assigned to objects in a typical system. Examples of an ID include MACID, IPv6ID, universal product and other custom methods.

**Meta-information:** This describes the form and operations of a device in a system. It is needed to establish relationships with other devices as placing them appropriately within the universe of IoT devices.

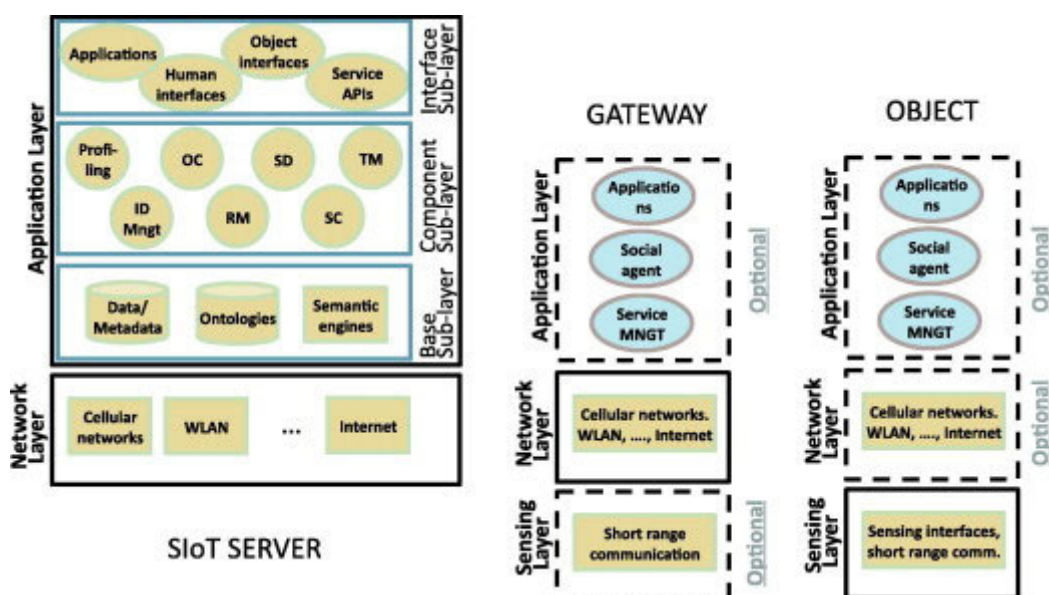
**Security controls:** This is synonymous to *friend list* on Facebook, where an owner of the device puts some restrictions regarding how some devices can connect to them. It is sometimes also known as owner controls.

**Service discovery:** Similar to a system like a service cloud, dedicated directories are created to store details of devices that provide certain kind of services. Keeping the directories up to date make it possible for devices to learn about other devices.

**Relationship management:** This refers to the relationship between devices and how they are managed. For example, storing the relationship between the light controller and a light sensor.

**Service composition:** This part of the module in the SLoT provides better-integrated services to users. It allows the system to establish a relationship with an analytics engine where large data that are generated are analyzed to learn about the usage pattern for further improved output or services. Thus it is possible to identify users based on say three categories of heavy, medium and low energy consumers in their community or among their Facebook friends.

[Figure 2.6](#) describes the social IoT architecture, which consists of server-side along with the client, in SLoT server, there are two layers network layer and application layer. The other side consists of a gateway and object, the gateway and object are further divided into three layers that are sensing layer, network layer, and application layer:



**Figure 2.6:** *Social Internet of thing architecture*

Representative architectures come under social internet of things have a server-side architecture which consist of mainly two-layered approach first one is network layer which constitutes cellular networks, WLAN, internet, and many more, and the second layer is further divided into three parts, that is, base sublayer, component sublayer, and interface sublayer, where base sub-layer constitutes data/metadata and semantic engines, component sublayer consist of profiling, TM, and more, and interface sublayer constitutes applications, human interfaces, object interfaces, and service API's. The server connects to all the interconnected components, composes the services, and acts as a single point of service for users.

The server-side architecture primarily features three main levels. There is the base layer that stores the database of all devices including their attributes, meta-information and relationships, then the second layer that contains the code to interact with the other devices, querying their status as well as using some of the subset to affect their service and finally the topmost layer which the application layer providing services to users.

The object-side has two layers namely a first layer which allows devices to connect through standardized protocols and exchange information and the social layer which manages the execution of applications and interacts with the application layer of the server.

## Near Field Communication (NFC)

**Near Field Communication (NFC)** is a short-range wireless connectivity standard (Ecma-340, ISO/IEC 18092) that uses magnetic field induction to enable communication between devices when they're touched together or brought within a few centimeters of each other. Jointly developed by Philips and Sony, the standard specifies a way for the devices to establish a **peer-to-peer (P2P)** network to exchange data. After the P2P network has been configured, another wireless communication technology, such as Bluetooth or Wi-Fi, can be used for longer-range communication or for transferring larger amounts of data.

NFC is used in the following ways:

You could take pictures with a cell phone with a built-in camera, and touch an enabled computer or television set to transmit the images for display.

You could download applications or games to a handheld device by touching the computer.

In conjunction with another wireless technology, you could transfer large files between two devices, such as a laptop and a desktop, simply by touching the two together.

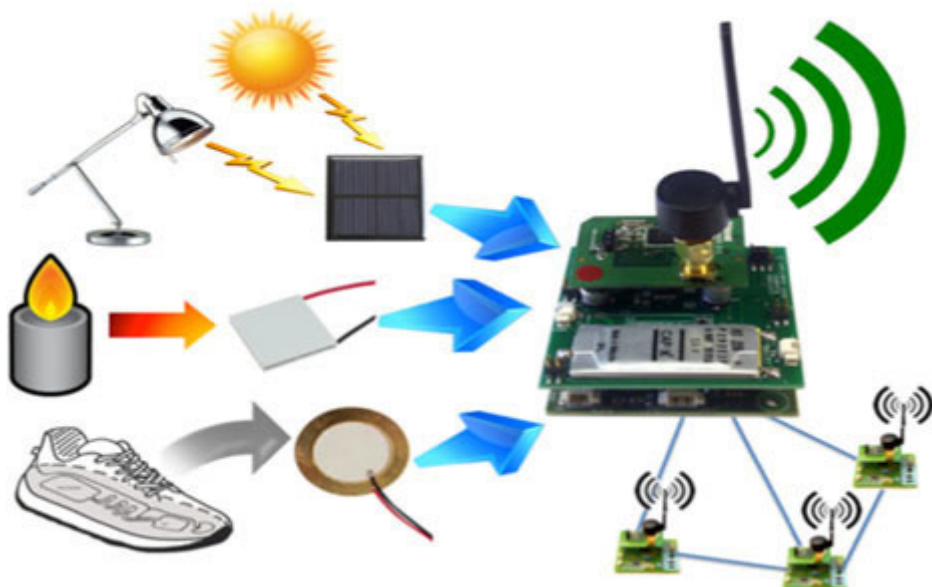
Google launched Google wallet sometime back that could support MasterCard PayPass, PayPal, and offers money transfers between smartphones. As the technology grows, more NFC compatible smartphones will be available and more stores will offer NFC card readers for customer convenience.



## Wireless sensor network (WSN).

**Wireless sensor network (WSN)** is the combination of a large small-sized sensor combined wirelessly. They are connected to increase the processing power. There is a base station which controls all the devices. They as an individual, have very less processing power and consume less power. They are used in the field of habit monitoring, security and military, collect real-time data, and many more.

In [Figure](#) there are many devices which are connected by various sensors. These sensors are connected to the base station. The base station receives the information from the sensors and processes it for achieving a particular task. Various models or processes are running on the base station, which provides the useful information:



***Figure 2.7: Wireless sensors in devices***

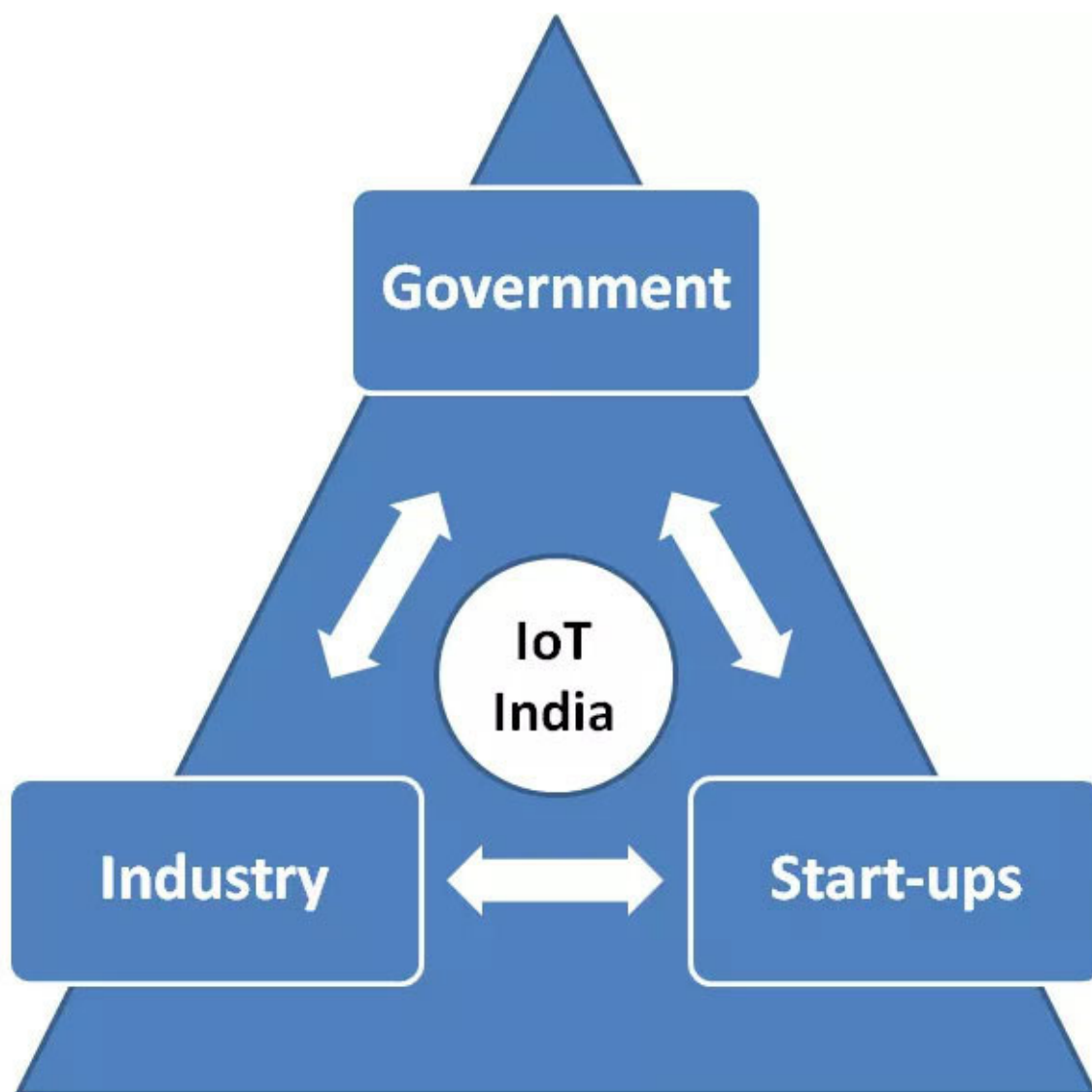
In the above [Figure](#) there are many devices which are connected by various sensors. These sensors are connected to the base station. The base station receives the information from the sensors and processes it for achieving a particular task. Various models or processing are running on the base station, which provides the useful information.

Now it comes to mind how these can be utilized by connecting to the cloud? The major protocol used for connecting the base station to the AWS cloud is MQTT protocol. This is a lightweight protocol. The data collected from the base station is transferred to the cloud for processing. There are various services which are used by the cloud. The data is processed after storing it in s3 or glaciers.

There are various benefits for connecting the base station to the cloud as there are numerous options available for processing the data. The data can be sent to various areas across the globe for various purposes.

### IoT network protocol stack

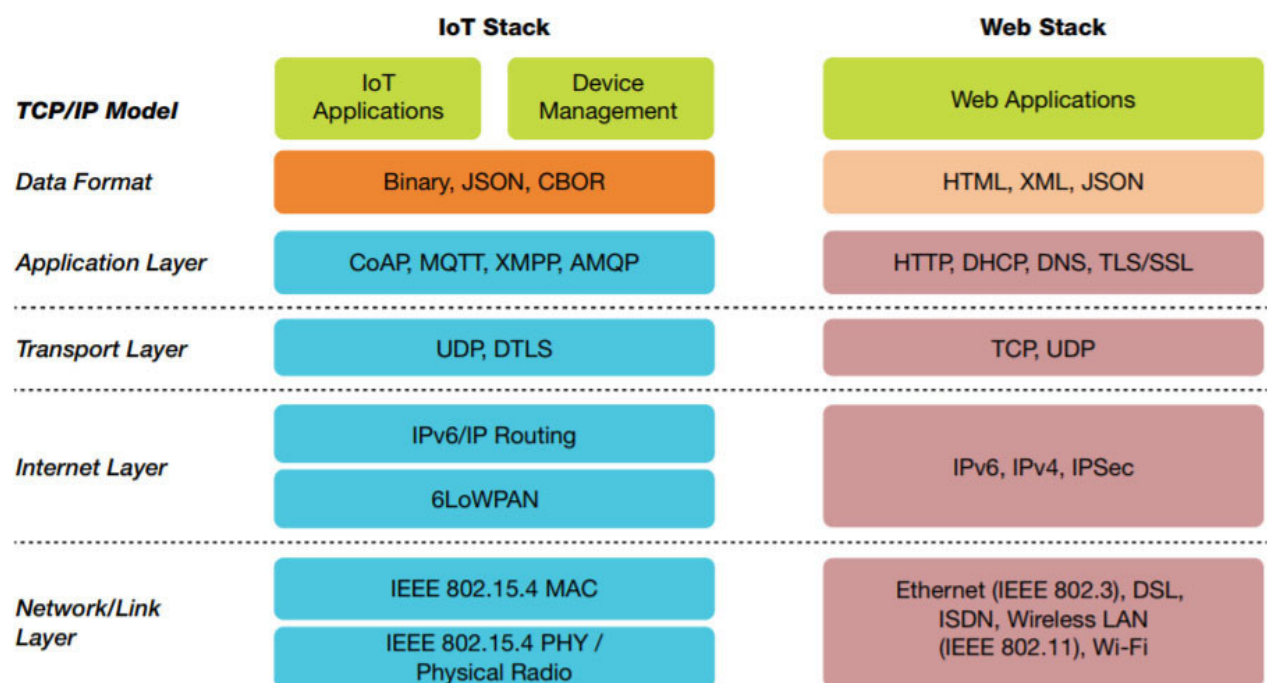
IoT stands for things on the internet. IoT is rising at a very rapid rate in the current scenario. IoT is an emerging tool with the advancement of technology that provides a connection between industry, start-ups, and government. The relationship between government, start-ups, and industry is shown in [Figure](#)



**Figure 2.8:** Relationship between government, start-ups, and industry

When we talk about the relationship between the industries, then there come various problems which the industry could face while using the IoT stack. There are many security issues which come when we are dealing with the IoT.

There are various protocols on various layers when we compare them based on architecture. The main and most important layers are transport layer, and internet layer as only these two layers are majorly responsible for the transfer of information from one location to another. Most of the bugs and attacks are also made on this layer. For example, if we are transferring the data between two machines then while transferring the data, an attacker can intercept the data in between if the connection is not secure by using man in the middle attack. Let's discuss the function of each protocol in detail:

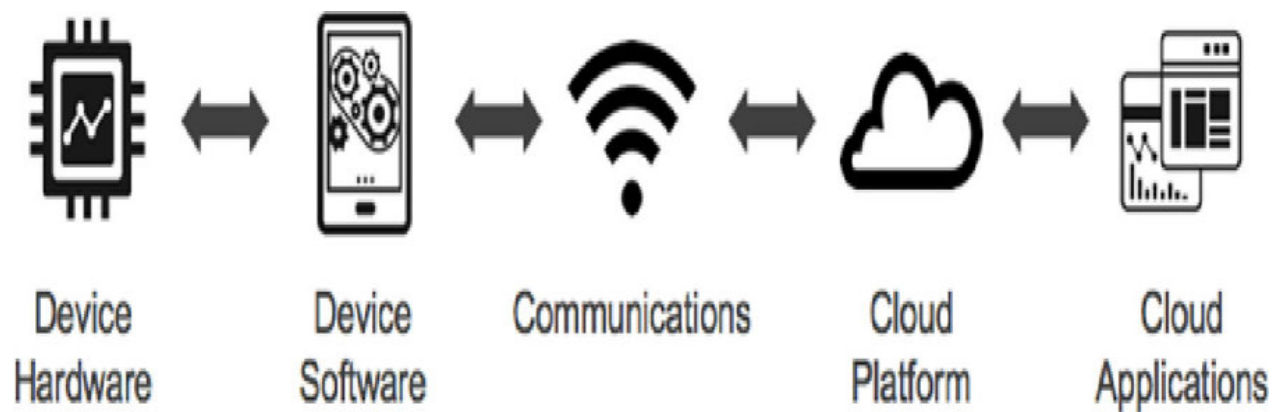


**Figure 2.9:** Arrangement of the protocols in the network.

The connection is made with the help of all the layers shown in the above. The network layer comes at the action at first. After the allocation of MAC address of the machine, then the packet is sent to the internet layer. At internet layer, an IP address is routed to an appropriate channel with the help of IP routing, once routing is done, Transport layer transport the packet from one location to the destination with the help of UDP as well as TCP protocols. TCP provides safe transfer of data while UDP provides fast transfer of data, the protocol is selected based on the requirement of the system. while the data is reached the destination, there are few application layer protocols which are designed for specific tasks, for example, MQTT protocol transfers the data to the Amazon cloud. This protocol is specifically made by Amazon for its cloud service. After the data is reached in the application layer, it can be used by the various application used in the field of IoT.

### IoT technology stack

Every IoT project is having set of interconnected components like hardware, software, communication, cloud and applications. Various components of an IoT project and their interconnection are shown below in [Figure](#)



**Figure 2.10:** *The IoT technology stack*

## Device hardware

The device hardware is the most basic part of the IoT project. We have to choose the IoT device according to the needs of the type of work we are doing. If we are doing any complex task we have to use more complex hardware as if we are using raspberry-pi we have sufficient memory, but if we replace Raspberry-Pi with the Arduino, the Arduino may not work properly. More importantly, we have to take a look at our needs before deciding the hardware we need.

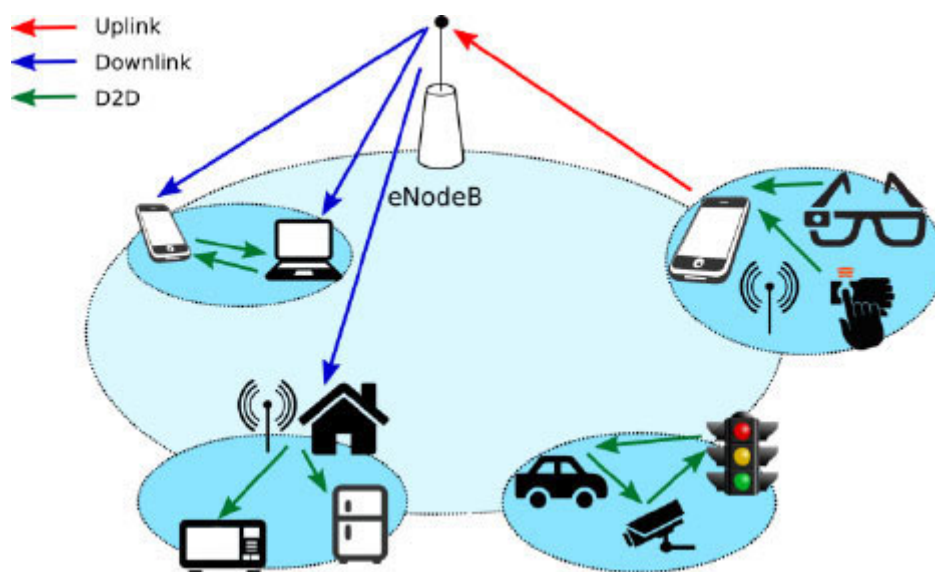
### Device software

Device software is also a very important part of the IoT technology stack. The software provides a mind to the hardware. There are various types of operating system like Linux, Windows, Solaris, Brillo, and more, which can be used as a base for running different softwares on it. For example, you want to use the hardware to the maximum then you should run Linux as Linux is very close to the machine it can talk to the machine in a more effective way as the other operating systems. Thus, this solely depends upon the type of function performed.



## Communication

Communication is a method of exchanging information. This is most important to choose it carefully. Communication should be decided according to the topology of the system, such as number of sensors in the IoT project, how these sensors should communicate with each other in a sink or they have to connect with single server it completely depends upon the architecture of the IoT project.:



**Figure 2.11:** Communication across devices

## Cloud platform

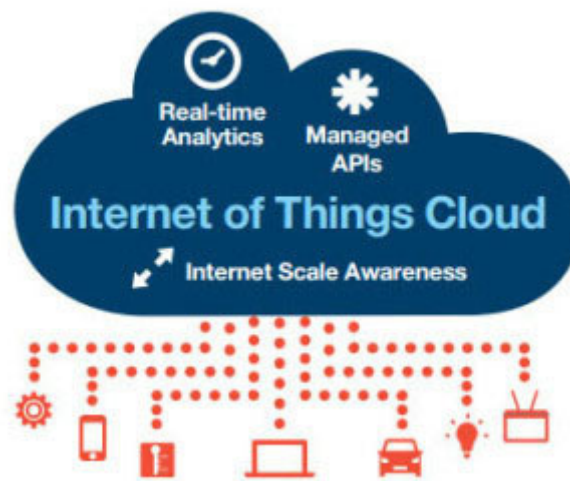
This platform is the backbone of the IoT platform. This is further divided into various parts:

**Analytics:** Analytics refers to search the data from the large chunk of data and finding the patterns from the data. This is the most crucial part as if we are not able to find the appropriate data from the large chunk of data, then it will be difficult to process the data.

**Cloud API's:** API stands for application program interface. These are the interfaces between the service which we want to use and the device. They control the usage. This also provides a facility to process the data in the real-time.

### Cloud applications

There are various applications of the cloud. It provides services like storage to the device. Applications can be developed using cloud-based or web-based. [Figure 2.12](#) depicts the devices such as mobile phone, laptops, connected cars etc. that are communicating through cloud based applications. These depend upon the requirements of the user client. These are of internal-facing or customer-centric:



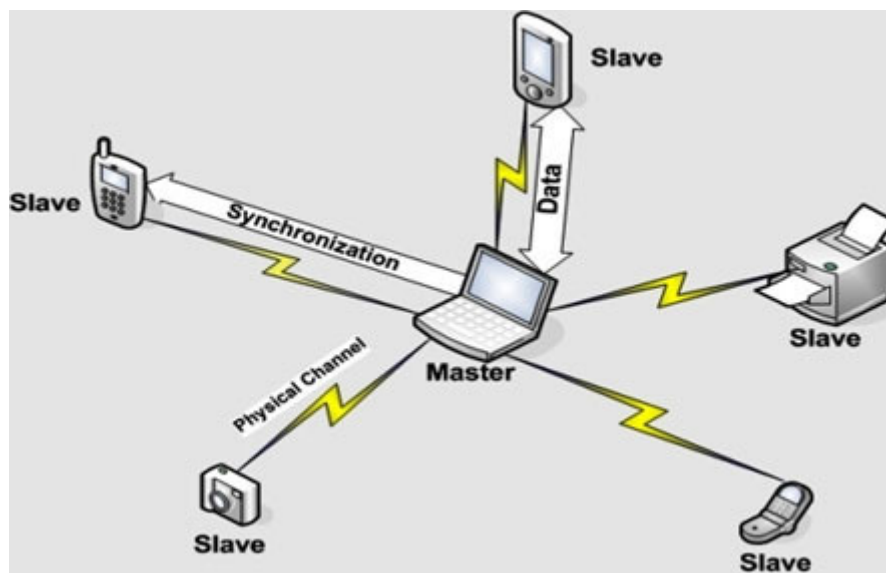
**Figure 2.12:** *Devices connected to the cloud*

### Bluetooth, ZigBee, and 6LowPAN

Communication between IoT devices is a big challenge, because of the mobility and low power capacity of the IoT nodes. Therefore most of the IoT devices communicate through low power wireless technologies for communication. Most of the IoT devices communicate through either a Bluetooth, ZigBee or 6LowPAN.

## Bluetooth

Bluetooth is a wireless technology that covers the small distance to link different devices like mobiles, laptops and other network devices. We use it for transfer files or to transfer a small amount of data. We use this technology over short-range that are, 50-150m and device must have Bluetooth 4.2 core specification with a basic data rate of 1Mbps. It uses Ultra High-Frequency radio waves from 2.4-2.485GHz. We find Bluetooth technology in smartphones, smartwatches, laptops, wireless speakers and wireless headsets. A master-slave interconnection between Bluetooth devices are shown below in [Figure](#)

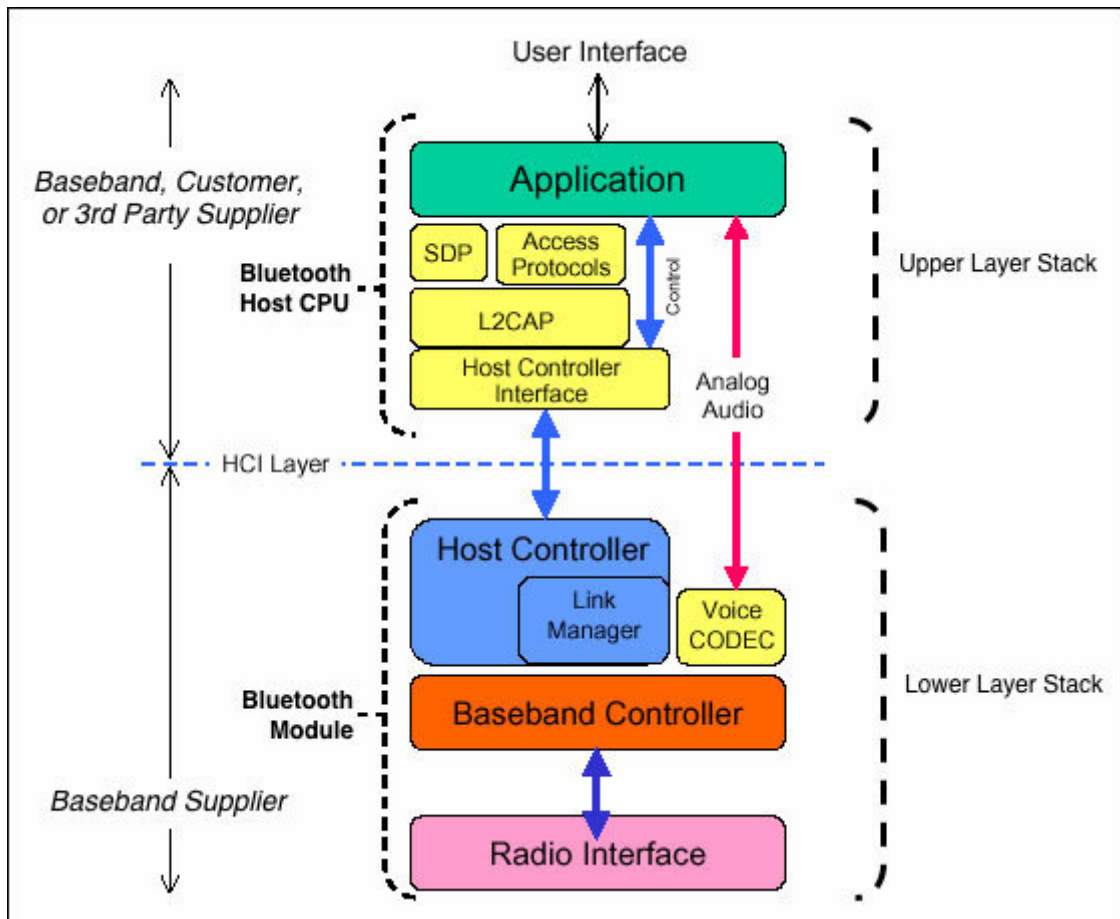


**Figure 2.13:** Bluetooth interconnection

In the above figure, we see the master and slave devices. Bluetooth mainly used to connect a minimum of two Devices and maximum eight devices and one device is considered as master (starts communication) while other seven as a slave (give a response to master). When we power on the Bluetooth, the master device start searching for all the active devices. Once it finds the suitable device from the listed devices, it sends a request in the form of a radio signal to that device, and once the slave responds to that request, these devices synchronize over the frequency hopping sequence. There are also paired devices in the list, and we easily transfer files between them.

Bluetooth has core and profile specification. Core specification explains the stack protocol of Bluetooth while profile specification gives information to use Bluetooth protocol.

Core specification consists of layers, as shown in the figure given below:



**Figure 2.14:** Stack protocol of Bluetooth

As we see figure, low stack layer consists of radio layer, baseband controller and link manager protocol (LMP) while upper stack layer consists of **Logical Link Control and Adaption Protocol (L2CAP)** and **Service Discovery Protocol (SDP)**.

Radio is a base layer of Bluetooth stack protocol. This layer describes characteristics of a transceiver and is responsible for modulation/demodulation for transfer data. Baseband controller defines the packet format, timing for transmission and receiving, control on the channel. LMP establishes and maintains the link between the upper and lower stack layers. L2CAP is above the **HCI (host controller interface)** and check the communication

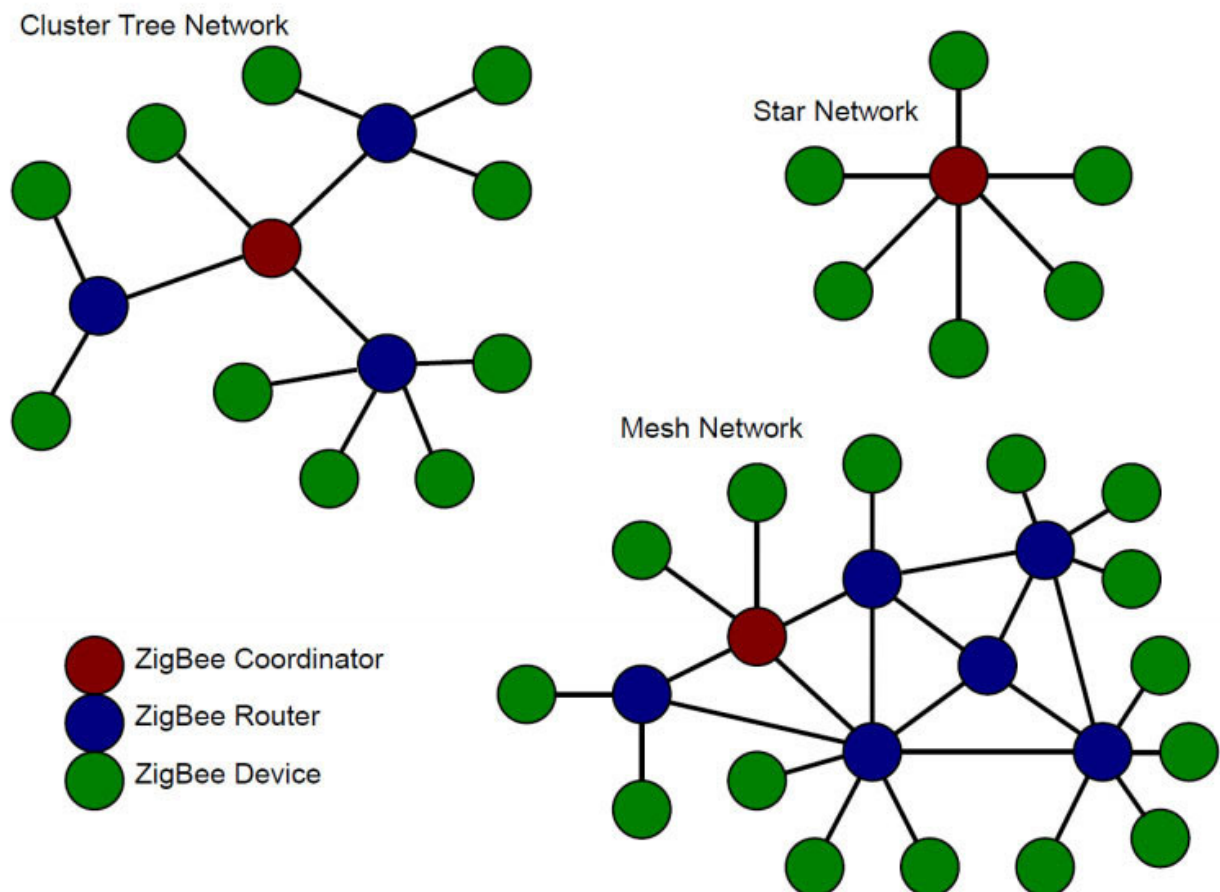
between the upper and lower layer. SDP gives device information, service provided, and more information to other Bluetooth devices.



## ZigBee

ZigBee is a wireless technology used for small scale projects. ZigBee is based on the IEEE 802.15.2 protocol. We use this technology over a small range, that is, 10-100m. This technology is operating at 2.4GHz and exchanges data at low data rates, that is, 250kbps over a short area. This technology has various applications in agriculture, automotive sensing, smart homes, remote control, and many more.

ZigBee network has various types of topologies-star, mesh (peer to peer) and cluster tree topology, as shown in the figure below:



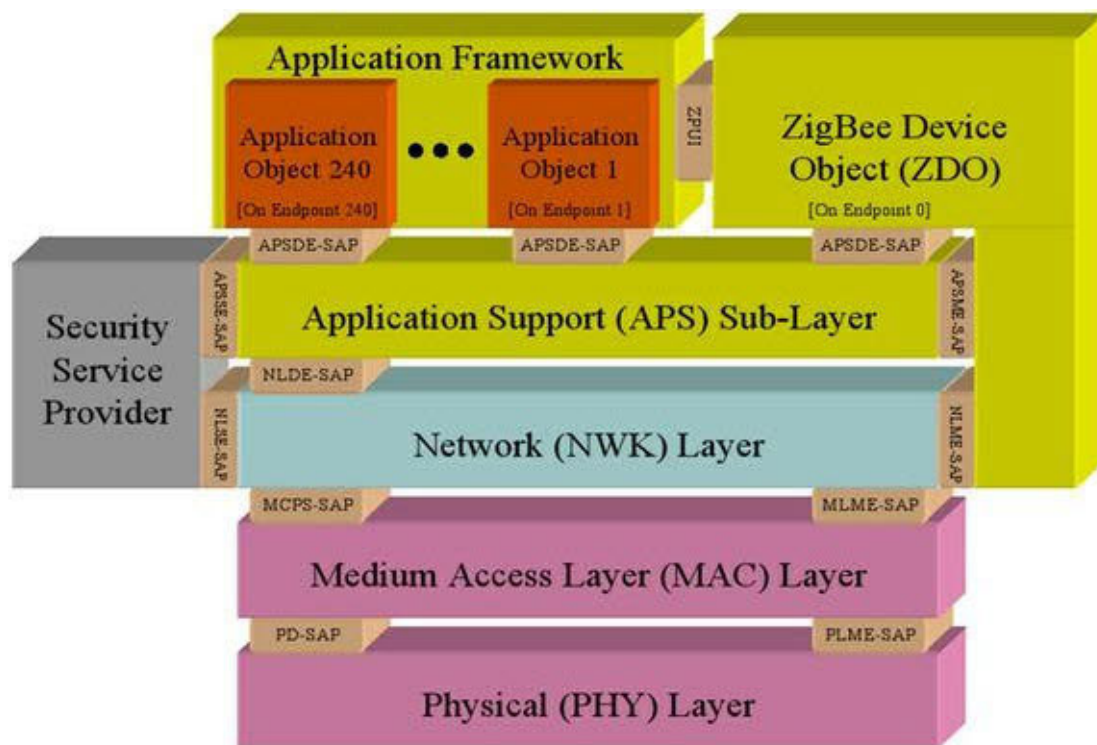
**Figure 2.15:** *Different types of networks*

In star network we are having ZigBee coordinator at the center from where the communication starts and all the devices are connected to it. Each ZigBee device communicates with the coordinator. If any device wants to send any packet to another device, then it must go through the coordinator, two devices can connect only through the coordinator. We use this topology in personal computers, toys, home automation etc.

In peer to peer or mesh topology, we have a single ZigBee coordinator while more than one number of routers and the end devices. In this number of devices helps to increase the size and one device connect to others if they are in the range of one another. In this message or packet passes through multiple hops to reach the destination. If any device fails, then this topology uses another path to reach the destination. We use this topology in industrial control and monitoring, WSN, and more.

In a cluster tree network, we have central coordinate, and nodes that are connected to this center node are known as children. Parent with children makes cluster, and there is number of clusters in this topology. Communication is only possible if two nodes are directly connected; no two nearby nodes communicate directly. Each cluster must have their ID, that is, Cluster ID. This topology covers a large area.

ZigBee consist of various layers that are shown in the following diagram:



*Figure 2.16: Layers of ZigBee*

In the figure shown above, we see the **physical layer (PHY)** that is base layer use for modulation and modulation of incoming/outgoing signals. **Medium Access Layer (MAC)** is above the PHY layer. This layer use carrier senses **multiple access collision avoidance (CSMA)** to access the network to transfer data. **Network layer (NEW)** is above the MAC layer, and it starts a network, route discovery, check connection and disconnection. **Application Support (APS)** sublayer use for data managing services. Application Framework provides a key-value pair and generic message services.

## 6LoWPAN

6LoWPAN, an IPv6 acronym for low-power wireless personal area networks, is a popular wireless communication standard.

6LoWPAN makes interaction over the protocol of IEEE 802.15.4 using IPv6. This standard establishes a layer of adaptation between the 802.15.4 layer of communication and the layer of transport. 6LoWPAN devices can communicate on the Internet with any other IP-based device. IPv6's choice is due to the large addressing space in IPv6. 6LoWPAN networks connect to the Internet via a gateway (WIFI or Ethernet), which also supports conversion protocols between IPv4 and IPv6 since today's Internet is predominantly IPv4. IPv6 headers are not small enough to fit into the 802.15.4 standard small 127-byte MTU.

The adaptation layer directly performs the following three modifications to minimize overhead communication:

**Header** compression 6LoWPAN defines IPv6 packet header compression to reduce overhead IPv6. Some of the fields are omitted as they can be extracted from data on the level of links or can be spread across packets.

**Fragmentation:** The minimum size of the IPv6 MTU (total transmission unit) is 1280 bytes. On the other hand, in IEEE 802.15.4, the maximum size of a frame is 127 bytes. So, we need

to fragment the packet of IPv6 and it is done through the layer of adaptation.

**Network layer forwarding 6LoWPAN:** This facilitates routing mesh conducted on the network layer using short addresses of the link tier instead of the network layer. You can use this feature to communicate within a network of 6LoWPAN.

### Points to remember

In this chapter, we have discussed various technologies used in the IoT domain. Currently, this field is in a very nascent stage. The technologies in the core infrastructure layers are showing signs of maturity. However, a lot more needs to happen in the areas of IoT applications and communication technologies:

These fields will mature and impact human life in inconceivable ways over the next decade. Fog computing provides awareness and gives the response to events by eliminating a round trip to the cloud for analysis.

It enables cloud computing to handle two Exabyte of data generated from IoT. It avoids the expensive bandwidth by offloading gigabytes of network traffic from the core network of things.

Finally, fog computing gains deeper and faster insights, leading to higher service levels and improved security and privacy.

### Key terms

PHY-physical layer

MAC-Medium Access layer

CSMA-Carrier sense multiple access

APS-Application Support Sub-layer

LMP-link manager Protocol

L2CAP-Logical Link Control and Adaption Protocol

SIOT-Social Architecture of IoT

SDP-Service Discovery Protocol

HCI-Host Controller Interface

### Multiple-choice questions

What are the important components in IoT?

Hardware

Software

Verbal exchange infrastructure

a and b both

Which protocol is most common for short-range, low power communication

Bluetooth

RFID (Radio Frequency Identification)

ZigBee

Wi-Fi

Which layer is not present in the three-layer structure of IoT?



Application Layer

Perception Layer

Business Layer

Network Layer

ZigBee is based on which protocol?

IEEE 802.15.2

IEEE 802.15.3

IEEE 802.15.4

IEEE 802.15.5

What is the minimum size of the IPv6 MTU (total transmission)?

1280 bytes

1680 bytes

2460 bytes

3620 bytes

## Answers

*d*

*b*

*c*

*a*

*a*

### Fill in the blanks

Physical devices include different types of sensors like the ones based on ..... technology.

The five layers are perception, transport, processing, application, and .....

Near Field Communication (NFC) is a ..... wireless connectivity standard that uses magnetic field induction to enable communication between devices.

Bluetooth is ..... technology that covers the small distance to different link devices like mobile, laptops and other network devices.

ZigBee network has various types of topologies such as star, mesh, and ..... topology.

## Answers

electromechanical systems

business layer

short-range

wireless

cluster tree

### Descriptive questions

Explain ZigBee technology.

Explain the different components that form the ZigBee network.

What are different types of topology in the ZigBee network?  
Explain any 2 of them.

What are the different layers in ZigBee architecture? Explain all layers.

Explain data services of application framework?

Who's currently in on the action with NFC technology and mobile payments?

What do you mean by the 3-layer architecture of IoT?

What are the different layers present in the five-layer structure of IoT?

Explain the three common layers and their roles in IoT architecture.

Explain the basics of Bluetooth technology.

What are the applications of Bluetooth technology?

Explain the working of Bluetooth technology with the help of a diagram.

Explain the Bluetooth specifications.

Explain Bluetooth architecture in detail.

Explain diagrammatically low stack layer and upper stack layer.

What is the cloud-based architecture?

What is fog based architecture?

Explain the working of fog-based architecture?

Explain the benefits of fog-based architecture?

When to consider fog computing?

Describe the characteristics of Fog based architecture?

What is the social architecture of IoT?

Describe the three basic components of SIOT and justify with example?

What is representative architecture?

Explain 6LowPAN.

### Programming Framework for IoT

**Internet of Things(IoT)** is finished utilizing various sorts of gadgets, made by various sellers and having various details. So we need to study the interoperability in IoT. Their assurance and usage are essential to be logically unavoidable, which make them basic parts of the future internet. An impressive point of view where IoT is implemented with Raspberry Pi are consolidated is predicted as perilous and as a drawing in authority of perpetual application conditions. There are various programming languages used for implementing IoT applications. For developing IoT solution, one would need a programming language which spans the various fields, while being scalable and lightweight at the same time.



## Structure

Interoperability in IoT

Programming paradigms

Introduction to Arduino programming

Introduction to Pythonprogramming

Introduction to Raspberry Pi

Implementation of IoT with Raspberry Pi

Conclusion

## Objectives

After studying this chapter, you should be able to:

Understand the programming framework for the IoT and various programming paradigms.

Learn basic concepts of Arduino programming, Python programming, and Raspberry Pi. And also implementation of IoT with Raspberry Pi.

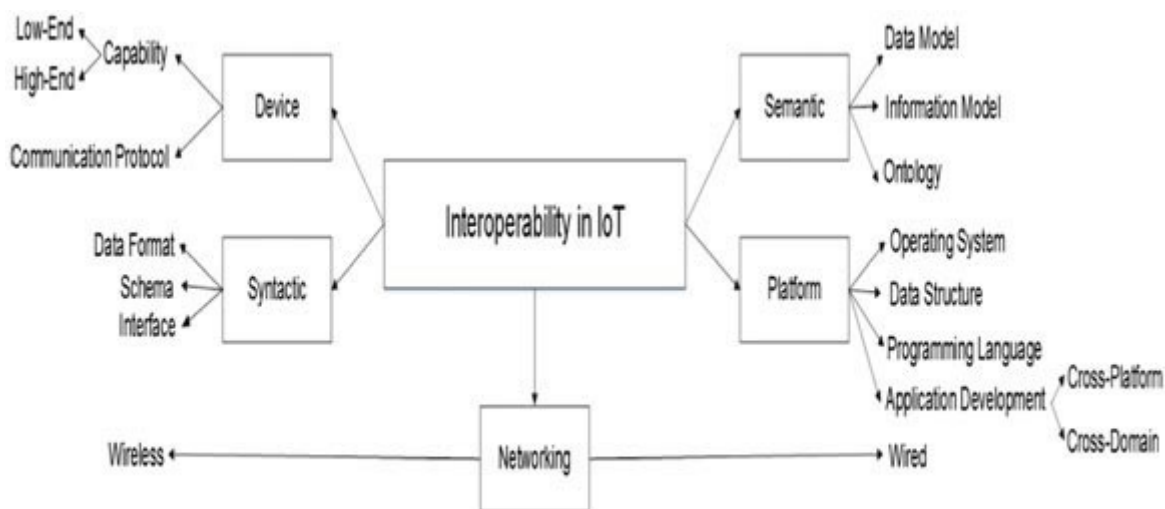
Understand interoperability in IoT with some examples.

### Interoperability in IoT

IoT is finished utilizing various sorts of gadgets, made by various sellers and having various details. IoT doesn't have one given standard so thus what happens is, for various things distinctive IoT gadgets are made by various sellers, following various determinations. Again these various gadgets by various merchants, they pursue various conventions, not really that they all pursue a similar convention. Indeed, even the sort of clients their clients and client profiles, these can likewise be extraordinary. So there is a lot of assorted variety that is characteristic to these frameworks is the motivation behind why it is imperative to address this specific issue.

In IoT, one of the central issues or issues being examined altogether is the assorted variety of conventions, gadgets, client gatherings, and numerous different perspectives from various points. So there have been studies, and one of the fundamental prerequisites to deal with this decent variety issue is to have a kind of interoperability between these various perspectives. Interoperability implies what, that let us state that one specific gadget is following a specific convention, another gadget pursues another convention. So how would they converse with one another? They communicate in various languages. So also, at various physical levels, there are various determinations and various gadgets. How would they converse with one another? They all have been made in various manners because there is nobody

standard that has been followed in building up these frameworks. So when you need to fabricate a particular IoT framework involving all these distinctive heterogeneous items, gadgets, conventions, norms, and so forth., you have to have some handshaking and that handshaking is the place these conventions have been concocted, which can be some agent, a middleware rather which can help these two distinctive different gatherings to have the option to converse with one another. So let us attempt to comprehend the interoperability issue in further detail so when we talk about IoT we are discussing enormous-scale networks.



**Figure 3.1:** Interoperability in IoT

Enormous scale networks require the utilization of a huge number of various gadgets, millions and billions of them. These gadgets are dispersed all over in the web or all over on the planet and what is required is to have some participation between the various gadgets some coordination component to be implemented between these various gadgets to have the option to converse with one

another. So this is one issue. The second issue is that the gadgets have been planned with various particulars, heterogeneous in all regards, in the physical gadget level, at the convention level and the client level. So heterogeneous IoT gadgets and their subnets is a test that must be chipped away at with regards to the web of things. Another exceptionally regular worry that is explicit of IoT is the gadget arrangement. Commonly, for these IoT gadgets their arrangement is obscure all over as illustrated the following figure.



**Figure 3.2:** *Value in IoT interoperability*

The distinctive setup modes for IoT gadgets originate from obscure proprietors and that naturally acquires a parcel of intricacy and that must be taken care. Another intriguing unpredictability how you handle contrasts in semantics. Not just so there could be clashing semantics also. In this way, various rationales of handling are applied to the equivalent IoT network gadgets or applications by various designers and diverse client gatherings and more. So how would you handle these distinctions these contentions in the semantics? That is the place interoperability

comes in. Interoperability is a component of a framework or item whose interfaces are perfect to work with different items or frameworks now or later in either execution or unhindered access.

It's been comprehended in the scenery of what we have talked about up until now. So what is required is to have all these various units, gadgets, conventions, and so on that we have talked about before to have the option to convey seriously so that there is trade of information and trade of administrations and the interoperability must be dealt with in such a way, that to the client it should feel that the person is gaining admittance to the administrations of the IoT framework in a consistent way. The client ought not to need to get into how these are executed, what is the interpretation that is going and things like that. So why interoperability is significant with regards to IoT on the grounds that it is required to satisfy diverse IoT targets.

Physical articles would need to associate with other physical items for sharing data; any gadget can speak with some other gadget whenever and at anyplace. So gadgets imparting whenever, anyplace and any sort of gadget, that implies anything whenever anyplace communication, so in the event that we must have anything whenever anyplace communication, at that point clearly we have to deal with this issue. That is the motivation behind why heterogeneity and interoperability are the center issues of IoT that must be tended to before huge scale IoTs are made practical. Issues, for example, **Device to Machine communication (D2M)**, **Device to Device communication (D2D)**, **Machine to Machine communication (M2M)** are likewise one of the goals in IoT. We have just experienced M2M communication and afterward we have

D2D communication. This communication is fundamentally a standard in the LTE propelled where essentially one cell phone talking straightforwardly to another cell phone is tended to. And after that you have the gadget to machine communication.

Presently we are going to discuss various kinds of Interoperability. It is of two sorts: user interoperability and device interoperability. Client interoperability is done in the middle of a client and a gadget though device interoperability is between two unique gadgets. So the issues are extraordinary. Suppose that there are two gadgets (CCTV cameras) one is situated in Delhi and the other gadget which is another CCTV is situated in Tokyo and their gadget level semantics are taken care of in Hindi and Japanese language separately and after that there is an IoT client who is situated in America. So now this American client needs to work with both of these gadgets remotely from America. So utilizing IoT both **A** and **B** give continuous security administration where in this specific model **a** is set at Delhi while **b** is in Tokyo and **u** is a client from America. In this way, **a**, and **u** all utilization various dialects, that is, Hindi, Japanese, and English separately. Presently the client **u** in America needs continuous assistance of CCTV camera from the gadgets **a** and **b**, so what all issues can happen? Number one, **u** doesn't have the foggiest idea about the gadgets **a** and **b**, number two issue, gadgets **a** and **b** are diverse in the part of semantics and linguistic uses and thusly it is hard to discover the CCTV gadget. Client can't comprehend the administration gave by **a** and **b** in light of the language distinction and comparatively **a** and **b** don't see each other commonly for a similar explanation. So it is presently clear those distinctions in grammar, contrasts in semantics, and contrasts in the client details all these are

acquiring parcel of multifaceted nature for a basic fundamental issue that is code to IoT.



## Programming paradigm

To understand programming paradigms, we have to understand what a paradigm means. A paradigm is basically a way or a pattern in which we accomplish a certain task. It's just a style so if you have a given way of doing a task, somebody else may have another way of doing that, in a different style, a different paradigm, a different pattern or model in which they get the job done. Whether it be let's say if I'm a financial guy trying to come up with a model or a pattern to get me super rich on the stock market however frivolous that maybe or maybe, there's different paradigms with religions for example, different moralities, different ways and different styles. So you've got different patterns or paradigms of morality if you will. Well we have programming paradigms, we have a way in which we can write our program and we have different styles to write our programs. So let's take a look at a few major paradigms.

## Assembly.

Assembly is one of the most basic paradigms. Now we started out with writing assembly programs in 1's and 0's and don't forget this is a paradigm it's a model it's a way of doing things. It's the lowest level writing in ones and zeros and we don't come across this style anymore but this is assembling. Nevertheless, there is one more, higher, level to the assembly paradigm. The higher level is in fact the first generation of language to come out that allows us to write in human readable syntax and what we did was we created instructions, line upon line and precept upon precept. You can look at this style like a to-do list of instructions, just one pretty long to-do list of instructions and it is line upon line and precept upon precept and instruction after instruction. This is an assembly type paradigm and of course you can use this paradigm in modern languages.

## Procedural

Modern languages allow you to write a command after a command, but when we're trying to create large-scale applications that require a lot of user interaction while we need something that's a bit more modular. And this one is called procedural programming. Now procedural programming is where we have subroutines. What are subroutines? They're functions. They are simply a set of instructions that we run. Subroutines, you know what you do when you do a routine. While let's say when you wake up in the morning you make a cup of coffee, well you have a subroutine inside of your mind that's make a cup of coffee for example. And you run through that subroutine in your head and of course you get a coffee at the end of it. So the whole point is that is more modular. If we break down our instructions into smaller chunks instead of having line upon line and precept upon precept, in that case we can execute those subroutines at a later stage. This is procedural programming.

## Functional

Now procedural programming can alter variables and the state of the program and so forth. But then also what we have is functional programming, the functional paradigm. Now this is a little bit different. The key value of functional programming is that functional programming believes all computation as the execution of a series of mathematical functions. It's a programming paradigm rooted well in mathematics, it is language independent and it's all driven by this one key ideal that all computations are shown as the execution of a series of mathematical functions and we say that this mathematical function maps inputs to outputs. Also, mathematical functions can't change their inputs so as a direct consequence of everything in our programs being represented by functions or the execution of functions, we can't change any variables and this concept is called immutability. And immutability rules out loops. It rules out counters, it rules out all variable reassignment and as a result we lose some express ability. Functional programming is a little bit different we treat the functions as if they were values. It's purely functional meaning we don't really try to change the state of a program so a good example of this is a mathematical equation. Let's say we have a math's application and all we're doing is we're going from one subroutine to maybe another subroutine to another subroutine and eventually after all these subroutines run it just gives us a value out. So for example, a good calculator application, where we're treating the function as if it's a value, and we're just doing

function calls after function calls and this is a nice robust way of writing programs. But don't assume that procedural is the same as functional. That's a common misconception and it's actually not true.

## Object-oriented programming

And finally we have one that's called **object-oriented programming (OOP)** or you could say object-oriented paradigm. This is a little bit different and is a very famous programming paradigm. OOP allows us to create and obviously build objects, but what it is, is that these programs are orientated around objects, and the objects are created either manually or dynamically. The programs orientate, modify and work with objects and other object like structures. Needless to say, the programs must resemble the real world. Our programs may need to resemble stock in a warehouse or products that have been produced. So we do need object orientation.

But there are certain individuals who tend to go one way or the other. This should not be the case. No one paradigm suits every problem. So for example you need to look at what your application is doing. If what it is doing is something very simple that can go without subroutines or functions, and then just use the assembly. And if you wanted something that's a bit more modular, you need certain subroutines instruction sets for example you want to mirror somebody waking up in the morning and opening the fridge and making breakfast and making a cup of coffee and so forth, we have all these different subroutines, procedural, different sets of instructions. The state of the application can be changed and variables can be modified and so on. And then also you have more functional style programming which is more for analysis and mathematics whereby we're treating the function is

the value in a way. And then finally we have object orientation. If I'm writing a program for a warehouse to keep an inventory of all the products in the warehouse, then I will go with OOP.

## Multi-paradigm programming

But also you have something called multi-paradigm programming languages. Now these allow you to get a combination of different paradigms. You can choose any style you like and your program doesn't have to be all OOP or all procedural. For example, this warehouse let's say they're selling stock. Well I need to do OOP for the stock and so forth. But let's say they want to do analysis, they want mathematical formulas and algorithms to find out what products are doing well and you'll find in big companies they want to do this, like Amazon, they look they're logistics and all the rest of it. So they need to do lots of mathematical equations, so at this point I would go with functional programming. And then let's say we have a user interface. Well there's only one user interface and let's say I go with procedural because the user is going to click a button over here or click a button over there different subroutine for this button and a difference of routine for that button, so maybe I'll go for procedural in that case. So you see how you can combine these paradigms to come up with the optimal multi-paradigm to handle the development of your application.



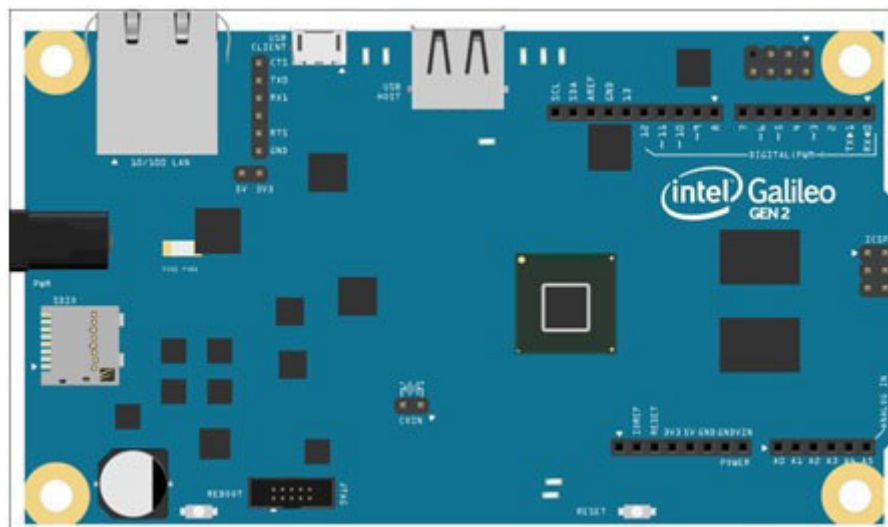
## *Introduction to Arduino programming*

Arduino is an open-source gadgets stage with the point of simple to-utilize equipment and programming. Arduino sheets can understand inputs-light on a sensor, a finger on a catch, or a Twitter message - and afterward transforms it into a yield - initiating an actuator. We can offer directions to the board about what to do by sending a lot of guidelines to the microcontroller on the board. For that we utilize the Arduino programming language, and the Arduino Software (IDE), in view of processing.

For long time, Arduino has been the cerebrum of thousands of ventures, from ordinary gadgets to complex logical instruments. An overall network of producers like specialists, craftsmen, software engineers, understudies, and experts has assembled around this open-source stage, and their commitments have signified an immense measure of available learning that can be of extraordinary assistance to tenderfoots and specialists the same.

### Arduino Intel Galileo Gen 2 board

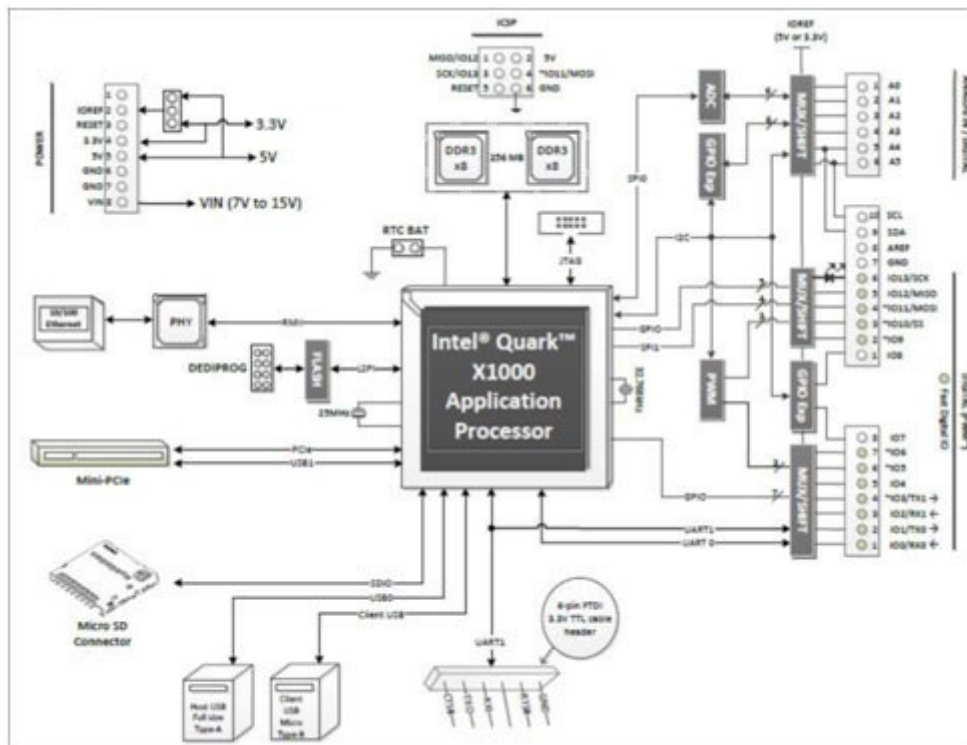
The Intel Galileo Gen 2 board is an Arduino embedded computer that can be used to develop and prototype the IoT projects. The board incorporates Intel architecture and uses an Intel Quark SoC X1000 system on a chip, known as SoC or application processor. The SoC is a single-core and single-threaded application processor that is compatible with the Intel Pentium 32-bit instruction set architecture (ISA). The operating speed of the board is up to 400 MHz.



**Figure 3.3:** Intel Galileo Gen 2 board

On the right-hand side of the CPU, the board consists of two integrated circuits that provide 256 MB of DDR3 RAM memory. The operating system and Python will be able to work with this RAM memory. Like in any other computer, RAM memory loses its

information after we turn off the power supply of the board. Therefore, we say that RAM is volatile.



**Figure 3.4:** Pin diagram of board

### **Boardpins**

The board provides the following I/O pins:

14 digital I/O pins

Six PWM (Pulse Width Modulation) output pins

Six analog input pins

## Buttons and LED

There is a button that is labeled REBOOT that resets the Intel Quark SoC X1000 application processor. The sketch and any attached shield are reset by a button labeled as RESET.

5 rectangular LEDs are present next to the USB 2.0 host connector: 3 LEDs on the right-hand side and two on the left-hand side of the connector. The following and the LEDs:

**OC:** LED flag over-current when the board is fueled through the smaller scale USB connector. Notwithstanding, the above element isn't empowered on Intel Galileo Gen 2 sheets, and that is the reason, we simply have the LED off. On the off chance that the LED is turned on, it implies that the board isn't working appropriately or the power supply is coming up short. This LED normally turns on when the board isn't working any longer.

**USB:** It is the small scale USB prepared LED. The LED turns on after the board has completed the boot procedure and enables us to interface the small scale USB link to the smaller scale USB association named USB CLIENT. We should never associate a link to the small scale USB association before this LED turns on the grounds that we can harm the board.

**L:** This LED is associated with stick 13 of the computerized I/O pins, and that is the reason, an elevated level sent to stick 13 will

turn on this LED and a low level will turn it off.

**ON:** It is a power LED and shows that the board is associated with the power supply.

**SD:** The LED shows I/O action with the micro-SD card connector, marked SDIO, and subsequently, this LED will flicker at whatever point the board is perusing or composing on the micro-SD card.

## Getting started with Arduino programming

Arduino programming is done in the Arduino **Integrated Development Environment (IDE)**. Arduino IDE is system run software that allows users to write programs for different Arduino boards, in Arduino language.

Arduino board programming's first step is downloading and installing the Arduino IDE. It is an open source and runs on Windows, Linux, and macOS X.

**Example:** Setup input and output pins for Arduino:

```
void setup ()
{
  pinMode (pin-number, INPUT); // set the 'pin-number' as input
  pinMode (pin-number, OUTPUT); // set the 'pin-number' as output
}
```

### **EXPLANATION:**

Here, setup() is the readiness square. To start with, the setup capacity is executed when the program is executed, and it is called just once. This capacity is utilized to instate the stick modes and start sequential communication. This capacity is required and is incorporated regardless of whether there are no different proclamations to be executed.

After the execution of `setup()` work, the execution square keeps running straightaway. This square has articulations, for example, activating yields, understanding sources of info, checking conditions, and so on.

The `loop()` is an execution block. The `loop()` is a function that executes the given set of statements (enclosed in curly braces) iteratively.

Example: How to write a **`loop()`** function?

```
Void loop ()
{
digitalWrite (pin-number,HIGH);// the component connected to
'pin-number' is turned ON
delay (1000); // wait for 1 sec
digitalWrite (pin-number, LOW); // the component connected to
'pin-number' is turned OFF
delay (1000); //wait for 1sec
}
```

**In Arduino, the time duration is measured in milliseconds.  
Therefore, whenever we mention the delay, it is kept in milliseconds.**



## Introduction to Python programming

For developing IoT solution, one would need a programming language which spans the various fields, while being scalable and lightweight at the same time. We will now begin the path towards creating IoT applications with Python. Python is one of the most adaptable programming languages. Python can be used to create multi-platform desktop and Web, mobile, and other science-related applications. We can work with huge amounts of data and build the complex algorithms. Python has thousands of packages, which allows us to interpolate its capabilities to any type of field.

We can use our current learning of Python and the majority of its packages to code the various pieces of our IoT ecosystem. One can also utilize the object-oriented features, which we love, from Python. We can utilize every one of the packages that we definitely know, to communicate with Web services, databases and different APIs. Python makes it simple for us to hop into the IoT world.

In the course of the most recent ten years, Python has turned out to be widely prevalent as a standard programming language. A portion of the significant favorable circumstances of Python over different languages include:

It is a straightforward language to learn and simple to execute and convey, so there is no compelling reason to exhaust extended

periods of time in adapting bunches of gathering choices and organizing principles.

It is compact, embeddable and expandable, so it is free of the framework, and along these lines underpins a great deal of single board PCs present in the market nowadays, paying little mind to working framework and the design.

One of the most significant points of interest is Python has a huge network which causes us and furnishes with help and libraries for the language.

## Getting started with Python for IoT

**The Jupyter Notebook:** The most effortless approach to compose Python code is the Jupyter Notebook. In Jupyter note pad, it is anything but difficult to blend code with remarks and discourses, and furthermore blend the code with its yield; including diagrams and other data representations.

Jupyter note pad comprises of cells for dialog, known as *markdown cells*, and *code cells*, containing Python.

**Python at the order line:** Specialists of data science for the most part utilize an *order line condition* for working.

**Python contents:** When one improves at programming, they'll need to be able to compose their very own full projects in Python, which works in a way simply like another program on your PC.

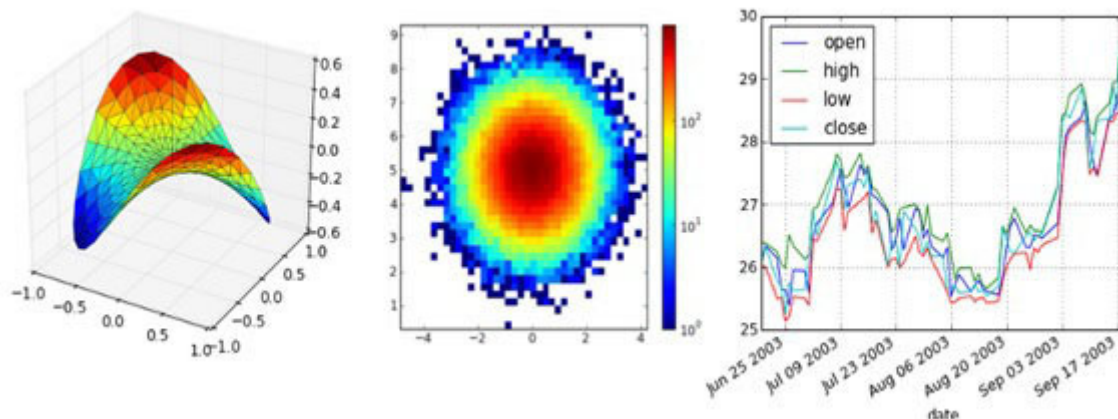
**Python libraries:** We can likewise make our own python libraries. These are called modules and can be brought into the note pad and conjured.

**Pandas:** Pandas is a package for data investigation. It is an option in contrast to utilizing SQL databases. It has a great deal of favorable circumstances over SQL, for example, a progressively streamlined way to deal with data taking care of and investigation,

direct activities on datasets and the capacity to deal with unordered and heterogeneous data.

**Numpy:** Python has records rather than an exhibit. Numpy is really a package for logical processing utilizing Python, which is a lot lighter. One of the highlights is to peruse sensor data from databases in mass and afterward do handling on them utilizing the inbuilt capacities.

**Matplotlib:** Matplotlib has various different styles of charts which can be plotted by utilizing data from databases. It's a useful instrument to quickly have an understanding of the data.



**Figure 3.5:** *Matplotlib different styles of charts*

**OpenCV:** OpenCV is a package that is a Python port of the C library utilized for picture preparing. OpenCV comprises of significant level variations of elements of picture preparing which make picture examination a lot simpler.

**Tensorflow:** Tensor stream is a package utilized for numerical calculations for AI. It utilizes an alternate numerical portrayal called data stream diagrams which use edges as data exhibits and hubs as scientific activities. It manages a great deal of non-straight datasets or work widely with choice trees and neural networks.

**Mraa:** It is a skeleton GPIO library for most Python supporting SBCs. Since it is a significant level library, writing to and perusing from pins is a solitary line errand, and it additionally gives assistance to communication rules.

**Sockets:** Attachments are a TCP/IP and UDP networking encouraging package in Python. Both TCP/IP and UDP being transport layer conventions are ideal for communication with same Wi-Fi network gadgets.

**Mysqldb:** MySQL is the go-to social database for generally engineers. It is an extremely simple and straightforward device which expels the requirement for executing shell directions inside Python content for perusing and keeping in touch with a database.

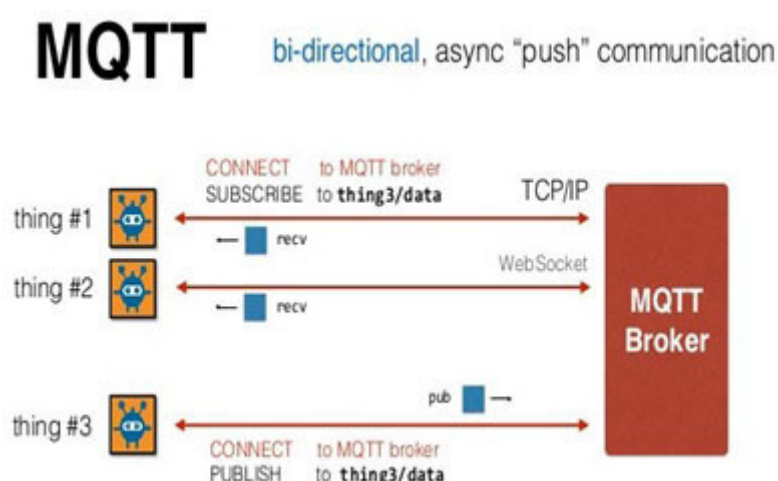
**Tkinter:** Loads of GUI improvement libraries are given by Python and Tkinter is one of the most generally utilized libraries out of them that come limited in with all Python conveyances. Each area of Python content can be taken care of by means of a totally impromptu GUI. It is extremely useful in circumstances like usefulness testing or over and over executing a similar code.

|                     |        |
|---------------------|--------|
| Annual Rate:        | 4      |
| Number of Payments: | 48     |
| Loan Principle:     | 10000  |
| Monthly Payment:    | 225.79 |
| Remaining Loan:     | 0      |

Final Balance    Monthly Payment    Quit

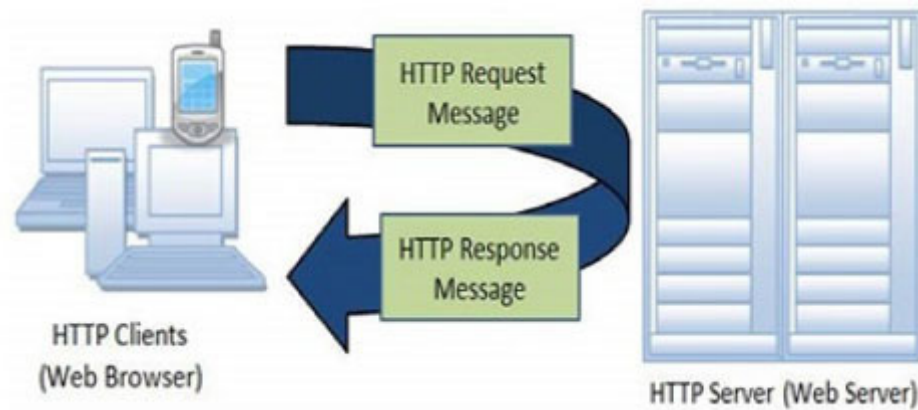
**Figure 3.6**

**paho-mqtt:** MQTT is a convention grown exclusively for reason for the IoT worldview. Its emphasis on fast communication for low payload communication between assets obliged gadgets. The mqtt library gives a very easy to use rendition of the convention to be utilized with installed frameworks. MQTT solicitations can be made straightforwardly inside Python, with no extra setup to be finished. It is particularly valuable in the prototyping stage.



**Figure 3.7: MQTT**

**Requests:** HTTP is one of the major protocols used in internet based resource exchange that is more suited towards large data exchanges. The requests package is used in Python to make HTTP calls and parse responses. This package is useful when dealing with third party cloud services based on HTTP.



**Figure 3.8:** HTTP protocol

## Introduction to Raspberry Pi

Raspberry Pi is a compact, fully featured single board mini-computer. We can connect external devices like mouse, keyboard, camera, display to Raspberry Pi (via a USB port provided) and it will behave as a usual version of computer. Mostly Raspberry Pi is used for processing of real time image and video, for applications based on IoT, robotics, and more. The Raspberry Pi may be used for surfing the web, developing documents and spreadsheets, playing video games, looking motion pictures and listening to music.

Performance wise Raspberry Pi is slower than computers and laptops but it is still a mini computer that can provide various expected features and abilities at low power consumption.

Raspberry Pi Foundation officially provides Debian (open-source) based Raspbian OS. Also, they provide NOOBS OS for Raspberry Pi. We can install several Third-Party versions of OS like Arch-Linux, Ubuntu, RISC OS, Windows 10 IoT Core, and more.

Raspbian OS is available free to use for everyone as the official OS. This OS is efficiently optimized and contains all the necessary applications pre-installed for IoT development. Raspbian can be used for browsing, Python programming, office, games, and more, Raspbian has GUI that has many useful tools.

We should use SD card (minimum 8 GB recommended) to store the OS (operating system). Raspberry Pi is greater than computer as it gives entry to the on-chip hardware, that is, GPIOs for



growing an software. Via having access to GPIO, we will connect gadgets like LED, motors, sensors, and so on and may manage them too.

It additionally provides exceptional surroundings for studying programming and digital making. We could additionally join up hardware to the Pi's GPIO (general reason input/output) pins and learn how to program the use of electronics components.

The Raspberry Pi also can be built into custom tasks together with interactive museum reveals or domestic automation answers.

Raspberry Pi's are a practical, portable and inexpensive option to use. The operating system in Raspberry Pi comes pre-installed with lots of options which make it easy to get started with coding.

Hardware requirements for Raspberry Pi are:

A Raspberry Pi computer with an SD card or micro SD card

A monitor with a cable (and, if needed, an HDMI adaptor)

A USB keyboard and mouse

A power supply

Headphones or speakers (optional)

An Ethernet cable (optional)

Software requirements for Raspberry Pi is:

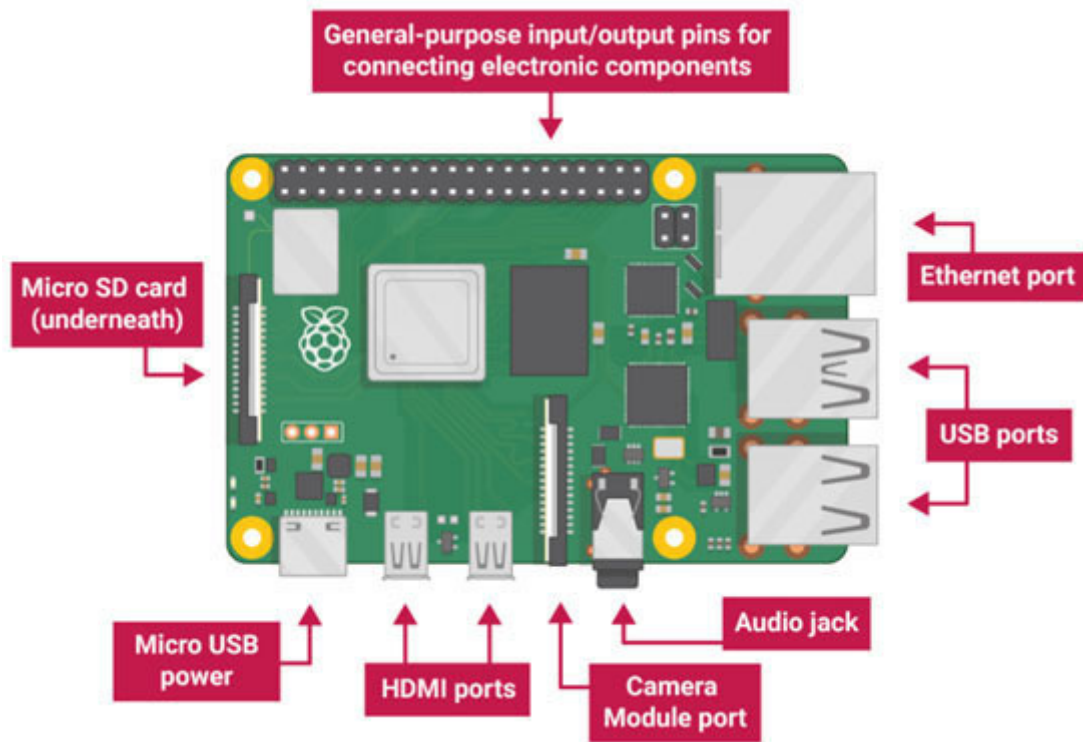
Raspbian, installed via NOOBS

## Why Raspberry Pi?

The Raspberry Pi is a microcontroller and a popular preference while developing IoT products. It offers a Linux server with a tiny platform at very reasonable price. The Raspberry Pi has featured several IoT projects on their website. For an instance, the world's first cloud texting enabled coffee device was powered by Raspberry Pi.

Raspberry Pi can be connected to a TV, laptop display, and it uses a general keyboard and mouse that can be connected using a USB port. It is consumer friendly. It does all the entirety we expect a computer or laptop to do like surfing the internet spreadsheets, phrase-processing, video games, and also watching HD movies and listening to music. It is used in many applications like wide array of digital maker projects, music machines, and parent detectors to the weather station and tweeting birdhouses with infrared cameras.

## Raspberry Pi and its components



**Figure 3.9:** *Raspberry pi and its components*

**USB ports:** These are used to connect a mouse or a keyboard. We can also connect other, which include a USB drive.

**SD card slot:** The SD card slot is where the SD card goes in. That is wherein the running device software and your files are saved.

**Ethernet port:** It is used to connect the Raspberry Pi to a network with the help of a cable. Raspberry Pi can also connect to a

network through Wi-Fi and LAN.

**Audio jack:** It is a port to connect headphones or speakers.

**HDMI port:** This is the port that we connect or reveal (or projector) the output to display from the Raspberry Pi. If your display has speakers, you can also use them to hear sound.

**Micro USB energy connector:** It is the place where you connect the power supply. We must always do this last, after we have connected all of the different components together.

**GPIO ports:** These pins permit the user to connect digital components together with LEDs and buttons to the Raspberry Pi board.

## Implementation of IoT with Raspberry Pi

The IoT is interfacing gadgets, creatures or individuals furnished with single identifiers and the capacity to consequently move and the ability of independent exchange of data to a network without requiring human intercession or human-to-gadget communication. IoT has developed with the introduction of remote advances, MEMS and the internet.

### Internet Gateway Device

An Internet Gateway Device has the adaptability to course data drawing closer from the WSN network onto the web and after that sends the data and data originating from the internet to WSN network. It is a greater amount of like Wi-Fi switch for IoT. In the internet portal gadget we utilize the model B of Raspberry pie, which incorporates a quad-center ARM Cortex-A7 CPU that keeps running at 900MHz and 1GB of LPDDR2 SDRAM (for a 2x memory increase). There is complete similarity with Raspberry Pi we are verified. Broadcom's new SoC, the BCM2836, is a significant factor.

## Five stages we are utilizing Internet Gateway Device

Embed Linux working framework on Raspberry Pi

Modify Linux to work with the prototype

Develop Python library for communication of RPI with Xbee ZB

Wrote program from sensors and device controlling

Create WI-FI usefulness on RPI for internet connection



### WSN nodes

A **remote sensor network (WSN)** for the most part comprises of 3 segments: entryways, hubs, and programming framework. The spatially disseminated estimating hubs communicate with the sensors to watch resources or their condition. The data procured is then remotely transmitted to the door that gives an association with the wired globe where you can gather, process, dissect, and present your estimation data utilizing the product framework. Switches are an individual kind of measurement hub that you can use to grow the separation and constancy in a WSN. Sensors can be available on the vehicles, streets, structures, emergency clinics, and individuals permitting various applications, for example, front line tasks, restorative administrations, calamity reaction the executives, debacle help. and condition perception.

## IoT applications

Weather security and temperature camera

Working doctor who props with Raspberry Pi

Sensing an air quality monitoring hat

Beer and wine fridge

Raspberry Pi internetdoorbell

IoT washroom

Training a rat behavior science at home

Smart doorbell

Smart microwave using Raspberry Pi

The above mentioned is IoT using Raspberry Pi. Currently, IoT is made up of a collection of different, purpose-built networks. Today's cars, example, have multiple networks to control safety features, engine function, communication systems, and many more things. Commercial and residential buildings also have many

different management systems for venting, heating and air condition (HVAC), telephone service, lighting, and security.

### Case studies of IoT using Raspberry Pi

**Smart home automation:** IoT is essentially changing the gadgets at home with a specific goal in mind so as to work on them all in all. With the assistance of IoT different home gadgets are mechanized which can connect among themselves and with the clients and condition. Sensors alongside actuators are every now and again utilized together for mechanization in home gadgets. A couple of home mechanization IoT assignments for building undergrads incorporate computerization with the assistance of keen carport entryway, hand motions, facial acknowledgment entryway, robotized blinds and shrewd morning timer.

**Brilliant car navigation:** Industrial IoT Automation has been fundamentally utilized in the enterprises so as to decrease the human mistakes inside the procedure of creation. Utilizing IoT gadgets, businesses could be equipped for controlling and checking hardware, different gear forms and different applications with considerably less or no human intercession. Computerization encourages us to lessen and maintain a strategic distance from the slip-ups and improving the proficiency of generation. Principally consequently, numerous businesses are looking to embrace IoT framework into their tasks. A portion of the businesses additionally give IoT instruction and preparing to their representatives to get mastery on an approach to oversee IoT structures and contraptions. A portion of the mechanical robotization IoT ventures for definite year understudies are vehicle

reproduction, shrewd leaving framework, keen structure undertaking, biometrics and brilliant security model. IoT can play out any ideal activity like controlling a gadget or observing from a remote area. IoT can play out any favored activity like controlling a gadget or checking from a faraway area. This innovation makes gear and hardware more digitized and associated. This innovation is supported even by methods for the administration to accomplish better power execution, a cleaner city and better profitability. A couple IoT ventures for keen urban communities are stuff tracker, brilliant waste gatherer, shrewd vitality meter perusing, and savvy fluid level checking.

## Conclusion

Programming models offer an approach to order the programming languages as indicated by the style of PC programming. The various highlights of different programming languages figure out which programming models or worldview they have a place with. Therefore, a few languages may fall into just one kind of worldview while others may fall into various models or standards. A portion of realize ideal models are concerned for the most part with suggestions for the execution model of the language which incorporates permitting symptoms or if the succession of activities is characterized by the execution model. Different models are principally worried about the manner in which the code is being sorted out like gathering code into units alongside the express that is adjusted by the code. Others are centered altogether with this style of sentence structure and syntax.

### Points to remember

Interoperability takes care of perhaps the most serious issue of IoT and is the explanation that IoT is such huge in size today. It makes the communication between heterogeneous gadgets utilizing distinctive innovation and conventions conceivable and has empowered IoT to hold its differing nature that makes it so helpful.

It makes the stage come to fruition and aides the middleware innovations work consistently. It is of two kinds: user and device interoperability. Various kinds of cosmology help to assemble the information base about gadgets so this is required so if two gadgets need to converse with one another, the philosophy the relating learning premise of these various gadgets must be framed.

Gadget interoperability comprise of two sections: **Universal Middleware Bridge Adapter** and **Universal Middleware Bridge** This works in a state of harmony and makes everything conceivable.

Programming models offer an approach to order the programming languages as indicated by the style of PC programming. The various highlights of different programming languages figure out which programming models or worldview they have a place with.

Arduino, an open-Source programming stage which is broadly utilized to make and sending IoT applications. We learned about the Arduino installed PC for example Intel Galileo Gen 2 board which comprises of an Intel Quark SoC X1000 framework on a chip.

Python is a lightweight, adaptable and flexible programming language to create multi-stage work areas and Web, versatile, and logical applications. We can utilize our present existing learning of Python and its packages to code the diverse part of our IoT biological system.

The idea of Raspberry Pi was started by the need to consolidate PC's little size and moderate cost. Numerous devotees imagined utilizing a little PC as a modest home theater (HTPC), or auxiliary low-control utilization work area.



### Keywords

**Handshake:** the steps of verifying the connection, the speed, or the authorization.

**IEEE:** Institute of Electrical and Electronics Engineers

**XML:** Extensible Markup Language

**USB:** It stands for Universal Serial Bus is a common interface which facilitates communication between devices and a host controller like a computer. Many peripheral devices such as keyboards, cameras, printers, media devices, scanners etc.

**Ethernet:** It is a system of computer network technology which is used to form local area networks (LAN), wide area network (WAN) along with various protocols to transfer the data and to avoid transmission of two or more systems at a given point of time.

**OS (Operating system):** It is a system that controls the software and hardware of a device.

**SD card:** It is a card which stores information in devices like camera, mobiles, and more.

**IDE (Integrated Development Environment):** It is a software application which provides very useful and comprehensive utilities and functions to computer programmers for development of software.

**PWD (Pulse Width Modulation):** It is a technique used to control the analog circuits with the help of microprocessor's digital outputs.

**RAM (Random Access Memory):** It is the hardware in a computing device where the operating system (OS), data and application programs in current use are stored so that they can be easily and quickly accessed by the device's processor.

**Python libraries:** It is a collection of methods and functions that helps us to perform various actions without writing the whole code.

**GPIO (General Purpose Input Output):** There are two rows of GPIO pins in a Raspberry Pi that are the connections between the Raspberry Pi and the real world.

**MQTT (Message Queuing Telemetry Transport):** It is a messaging protocol which is lightweight used for mobile devices and small sensors and is optimized for unreliable networks or high-latency.

### Multiple choice questions

What are the important components in IoT?

Hardware

Software

Verbal Exchange Infrastructure

a and bboth

MQTT stands for \_\_\_\_\_

MQ Telemetry Things

MQ Transport Telemetry

MQ Transport Things

MQ Telemetry Transport

MQTT is \_\_\_\_\_ protocol.

Machine to Machine

Internet of Things

Machine to Machine and Internet of Things

Machine Things

\_\_\_\_\_ method saves the received arguments in three attributes.

\_\_Init

Init\_\_

\_\_Init\_\_

\_init\_

Which one out of these is not a data link layer technology?

Bluetooth

UART

WiFi

HTTP

Which of the following language is preferred for IoT analytics?

Python

S

R

All of the mentioned

Which one is simplest form of analytics?

Predictive

Descriptive

All of the mentioned

Prescriptive

MQTT is:

Based on client-server architecture

Based on publish-subscribe architecture

Based on both of the above

Based on none of the above

## Answers

*d*

*d*

*c*

*c*

*d*

*a*

*b*

*b*

### Descriptive questions

What OS are often used or installed in Raspberry Pi?

Name some useful applications of IoT.

Which ports are used for connecting mouse and keyboard?

Name Raspberry Pi's components.

Which type of RAM is used in the Arduino board?

What are the ways in which one can write a Python program?

How are the various Python libraries useful for IoT?



*Virtualization in IoT*

Virtualization is a demanding technique in the era of computing. Computing resources are the major components in a wide range of IoT applications. So wide variety of applications can take advantage of virtualization technique. In this chapter, you will be familiar with the idea behind the virtualization, its types, techniques, benefits, applications, and role of virtualization technology in the era of the IoT.

## Structure

Introduction of virtualization technology

Types of virtualization

Virtualization and IoT

Embedded virtualization

IoT and embedded virtualization

Conclusion

Review questions

## Objective

After studying this unit, you should be able to:

Understand the concept of virtualization technology, its types and application areas, and the benefits of virtualization.

Analyze the requirement of virtualization in implementation of IoT based applications.

Understand the concept of embedded virtualization and role of embedded virtualization in IoT virtualization.

## *Introduction of virtualization technology.*

A process of creating a virtual version of an actual thing is called virtualization. In information technology and computing, virtualization refers to create a virtual version of actual resources, like virtual machines, virtual networks, virtual applications, virtual software, and more.

In the year of 1960, the concept of virtualization came into existence. That time a method was used for logically dividing computing and storage resources of the mainframe computer and allocating to different applications. Since then the meaning of that term has been extended.

### Definition

With the help of the virtualization technique, a software version of actual hardware can be created.

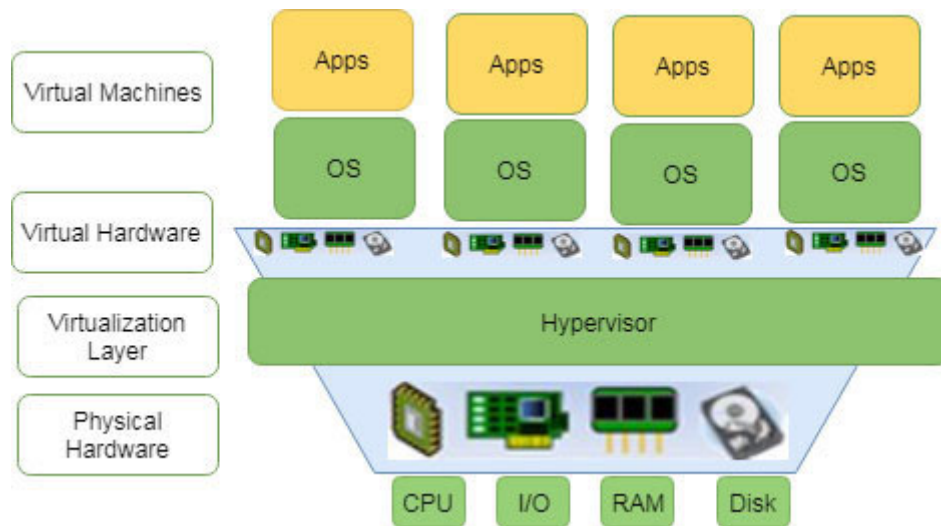
In other words, virtualization can be defined as: *Creation of new virtual version of any product or service.*

#### **For example:**

Disk partition (Partitioning of one hard disk into multiple segments like C:/, D:/ or E:/ drive, mostly we are using in our computer system).

Virtual memory.

As one hard disk can be partitioned in multiple logical partitions in our system in the same way, with the help of virtualization technique multiple logical partitions of actual or physical hardware can be created, and that logical portion is called virtual version of the actual machine. [Figure 4.1](#) shows the block diagram of the virtualization process:



**Figure 4.1:** Conversion of the 1-Physical machine to multiple virtual machines

The combination of virtual infrastructure offers layers of abstraction between various resources like storage, networking hardware, computing, and various applications that are running on it.

## Virtual machine

A closely detached software device (software version of actual hardware) that could run its own operating systems and application as if it is running on a physical computer is called a virtual machine. In [\*Figure\*](#) the box having name VM1, VM2, and VM3 are three virtual machines. The virtual machine contains its VRAM (Virtual RAM), virtual hard disk, virtual CPU, and virtual network interface card. Therefore an individual operating system can be installed on each VM.

### Guest OS (Operating System).

An Operating System that is running in a VM environment or on a different physical system is called the Guest operating system.

In [Figure 4.3](#) the box having name Guest OS1, Guest OS2, and Guest OS3 are three guest operating systems that are running on VM1, VM2, and VM3 respectively.



## Host OS

An operating system that is running in actual hardware or on an actual physical system is called the Host operating system. In [Figure](#) the second layer from the bottom shows the Host OS, which runs on physical hardware.

## Hypervisor

A hypervisor is software; it is used to create and run virtual machines. It is also called a **virtual machine monitor (VMM)**. The third layer from the bottom in [Figure 4.3](#) shows the placement of Hypervisor software. The host machine is a computer on which a hypervisor can run virtual machines. Examples of the hypervisor are listed below:

VMware workstation, VMware ESXi

Oracle Virtual Box

Hyper V

Virtual PC

Citrix Xen-Server

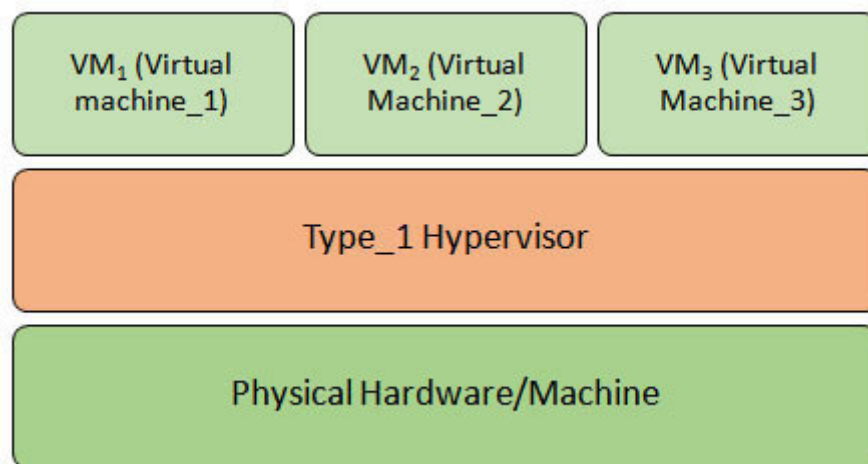
KVM

### Types of hypervisor

Hypervisors are of two types called type\_1 and type\_2 hypervisor. The difference between type\_1 and type\_2 hypervisor is based on its working characteristics.

### Type 1 hypervisor (bare metal)

A hypervisor which can directly be installed over the physical hardware, without the use of any operating system comes in the category of type-1 hypervisor. After installation of the type-1 hypervisor, we can create multiple virtual machines of required configuration. [Figure 4.2](#) shows the virtualization using type-1 hypervisor:



**Figure 4.2:** Virtualization using type-1hypervisor

#### **Example:**

VMware EsXi is an example of type\_1 hypervisor.

### Type 2 hypervisor

A hypervisor that cannot be installed directly on the physical hardware and requires a host operating, over which it can be installed comes in the category of type-2 hypervisor. A hypervisor shown in [Figure 4.3](#) is a category of type\_2 hypervisor:

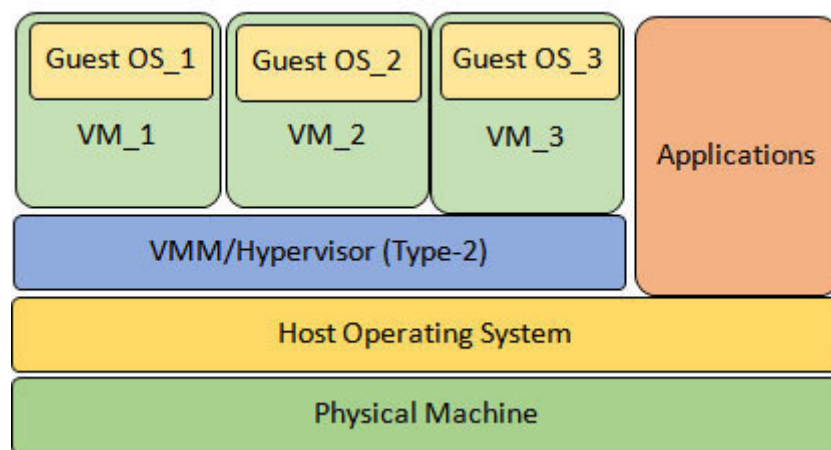
#### **Example:**

VMware-Workstation

Oracle-Virtual-Box and more.

### Hosted virtualization

A virtualization method where virtualization and partitioning services run on top of an host operating system is called hosted virtualization. The type-2 hypervisor provides the hosted virtualization. Block diagram of hosted virtualization shown below in [Figure](#)



**Figure 4.3:** Hosted virtualization

### Benefits of virtualization

In information technology and computing, the use of virtualization can increase the scalability, agility, and flexibility of IT resources. It also enhances the resource availability, workload mobility, and performance of automated operations. These benefits of virtualization make IT less costly and simpler to operate. Some additional benefits of virtualization are:

Availability of SDDC (Software-defined datacenter).

Reduced operating and capital cost.

Improved IT agility, productivity, efficiency, and responsiveness.

Fast provisioning of resources and applications.

Minimized resource downtime.

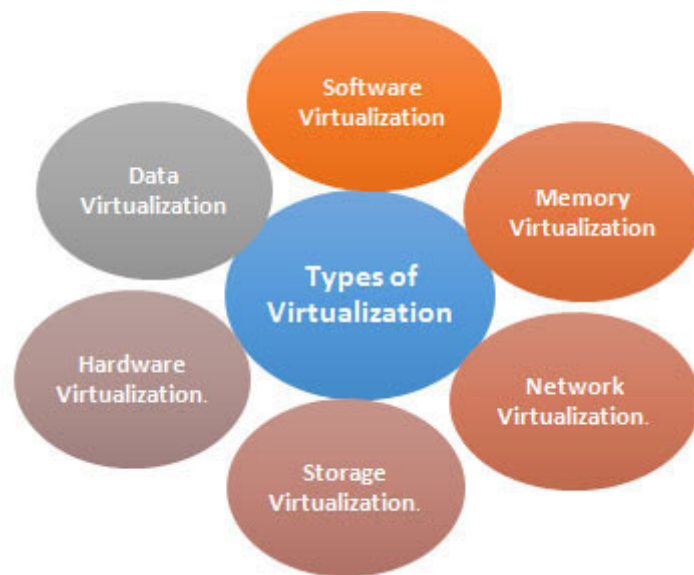
Improved disaster recovery mechanism.

Simplified management of datacenter.

Improved business opportunities.

### Types of virtualization

Based on the characteristics of the computing resources virtualization can be categorized into 6 types as shown in [Figure](#)



**Figure 4.4:** *Types of virtualization*



## Hardware virtualization

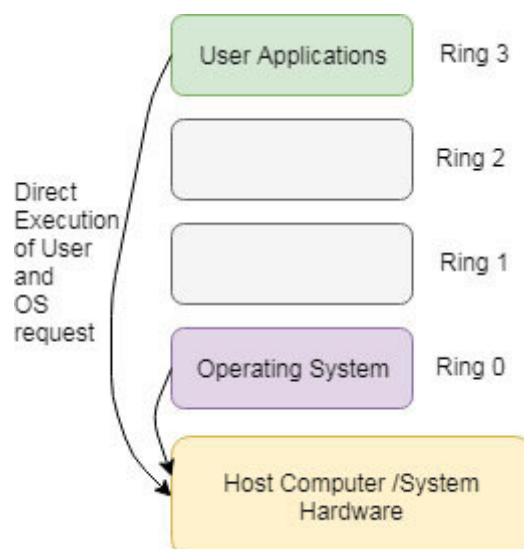
Hardware virtualization is the most important type of virtualization. It relates to resource utilization and application uptime. The main motivation behind this type of virtualization is server consolidation and optimal utilization of the processor. Server consolidation means, bringing together many physical servers into one large physical server. Server consolidation improves the utilization of the processor and other resources.

Hardware virtualization is not possible without a hypervisor. The hypervisor manages the processor, memory and other resources and allows multiple operating systems to run on the same machine. There are three basic techniques, by which a CPU/Processor of x86 architecture handles and executes a variety of instructions. These instructions can be initiated by different Guest OS and different software programs of different virtual machines.

## x86 Architecture

x86 architecture is a series of instruction set architecture (ISA) for computer processors. Intel Corporation develops it. The character x in x86 architecture indicates the version of ISA. ISA defines how a CPU will execute and handle the operating system and software level instruction.

In the x86 instruction set architecture, there are four levels of privilege. This four-level of privilege are categorized as Ring 0, 1, 2, and 3. According to these privileges, computer hardware can be accessed by Operating systems and other software applications. Typically user-level applications and instruction execute in Ring 3. OS level instruction requires direct access of memory and other hardware; therefore OS executes its privilege instructions in Ring-0. Privilege levels of x86 architecture (without virtualization) are shown in [Figure](#)



**Figure 4.5:** *Privilege levels of x86 architecture (without virtualization)*

## x86 hardware virtualization

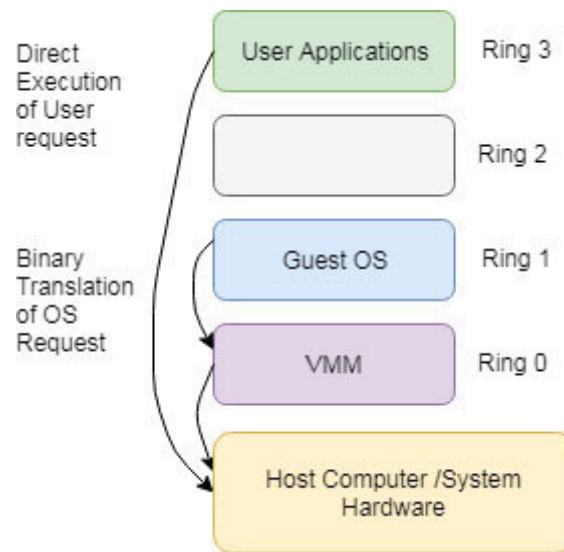
Based on the privilege level of x86 hardware, there are three basic techniques of x86 hardware virtualization, they are:

**Full-virtualization (binary translation with direct execution):** With the help of full virtualization, an isolated and virtual version of a complete computer can be created, which includes memory, CPU, and I/O devices. A virtual machine created by full virtualization technique looks like a real computer. An important characteristic of full virtualization is, it allows us to run an arbitrary guest OS.

Binary translation is an approach that is used in full virtualization techniques. It involves three steps, the first examination of executable instruction of Guest OS (before it runs), second if it found *unsafe/sensitive* instruction translate theses unsafe or sensitive instruction into equivalent *safe* instruction using binary translation and in the third step it runs the translated code.

User-level applications run in Ring-3 (least privilege level) where it is not possible to execute unsafe and sensitive instruction. This instruction can be executed directly by the hardware. This blending of direct execution of the user-level request and binary translation of OS requests delivers the functionality of full virtualization.

**Examples:** Microsoft virtual server and VMware products like VMware ESXi are based on full virtualization technique.



**Figure 4.6:** Binary translation approach to x86 virtualization

Some important characteristics of full virtualization are:

Run time binary translation of sensitive instruction is a time-consuming process. Therefore the performance of full virtualization suffers.

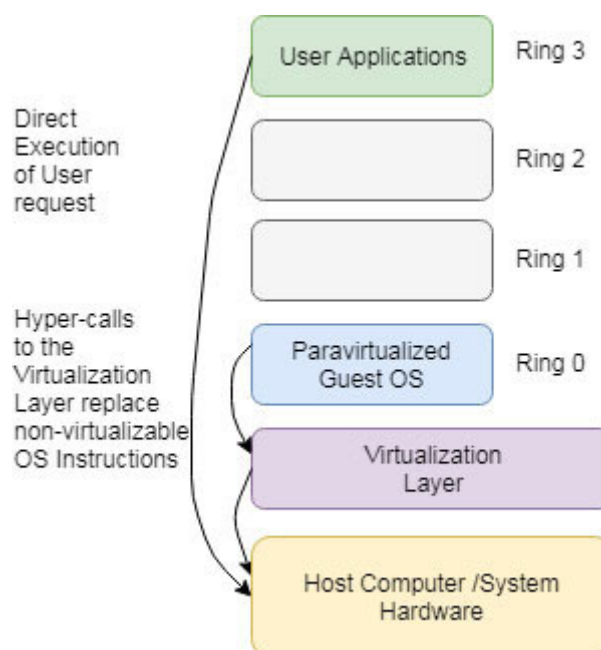
To improve the performance of full virtualization, a code cache can be used to store binary translated instruction, but it will increase the cost of memory usage.

On the x86 architecture, the performance of a full virtualize solution is usually 80% to 97%.

Input-output demanding applications can be a challenge for full virtualization technique.

**OS-assisted virtualization/Para-virtualization (PV):** PV or OS-assisted virtualization includes the process of OS Kernel modification. During the modification of Guest OS kernel, non-virtualizable, and sensitive instruction of Guest OS is replaced with hypercalls (modified OS-instruction according to hypervisor). Interrupt handling, memory management, and timekeeping are some other critical kernel operations. In PV, to handle such critical operation, the hypervisor provides a hypercall interface.

**Example: Xen** is an open-source project based on Para-virtualization technique. Xen virtualizes memory and processor by means of a modified Linux kernel. It virtualizes the input-output by means of convention guest OS device drivers.



**Figure 4.7:** Para-virtualization approach to x86 virtualization

The benefits of PV are as follows:

No-need of runtime binary translation of OS-level instruction.

Easier backups.

Fast migrations.

Improved system utilization.

The disadvantages of PV are as follows:

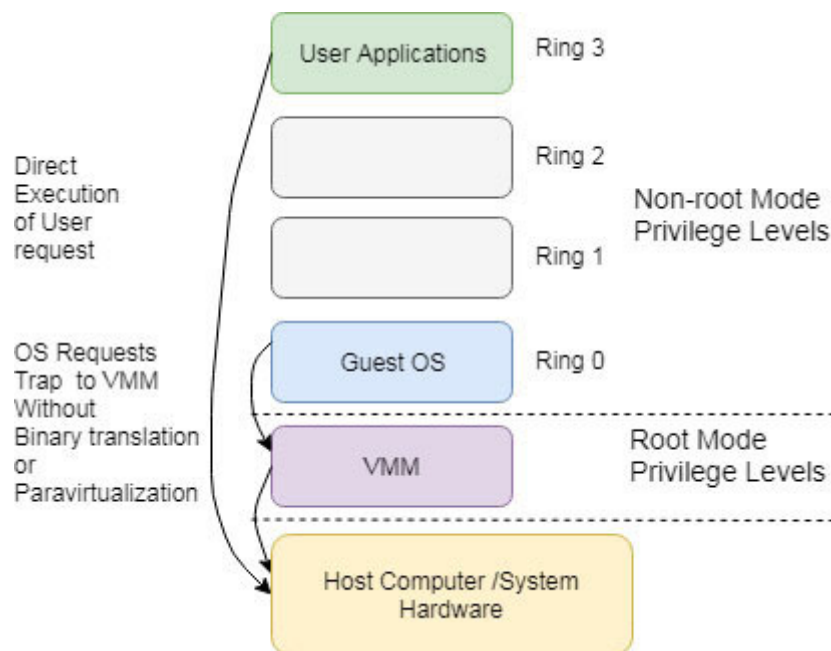
The actual performance benefits of PV vary with the workload some Para-virtualized applications perform better than others.

PV also demands a suitable hypervisor and a modified OS capable of interacting with the hypervisor through hypercalls.

**Hardware-Assisted Virtualization (HVM):** AMD's AMD-V and Intel's VT-x, both organization targets a new privileged execution mode of CPU. This new privileged level is called Ring-oP (privileged root-mode). Hardware-assisted virtualization technique allows to run VMM(Virtual machine monitor) in Ring-oP privilege mode. While Guest OS runs into a de-privileged non-root mode, also called Ring-oD (see [Figure](#)

Sensitive and unsafe instructions calls of Guest OS are set to trap to the hypervisor and handled by hardware automatically. It

removes the need for binary translation and/or Para virtualization. VMware is the first organization that takes advantage of this hardware feature of the Intel process, but that is limited to 64-bit Guest OS support:



**Figure 4.8:** Hardware assist approach to x86 virtualization

From the software perspective, the VT-x instructions work by providing additional instructions that make it easier to write a **VMM** (virtual machine monitor). The VT-x instructions provide a mechanism to allow the root-mode VMM (the actual hypervisor) to launch a new VM context, with a set of operations that will trap, and restore control to the VMM. It is done in a manner similar to the working of page tables and page fault handler. Whereas in VT-x, with hardware support, a structure of instructions defined by the ISA is specified and the location of this structure is given to CPU by switching it into appropriate



mode and the whole process can be done in hardware rather in software.

### Full virtualization versus Para-virtualization

Para-virtualization is different from full-virtualization. In full virtualization, Guest-OS is unaware that it is virtualized and its sensitive instruction/calls are trapped and binary translated during run time. While in Para-virtualization, Guest OS is aware of it that it is virtualized and it is sensitive, and non-virtualizable instruction/calls are replaced with Hypercalls at compile time. The main advantage of Para-virtualization is low virtualization-overhead. The performance benefits of Para-virtualization over full-virtualization can vary and depends on the workload.

The Para-virtualization technique cannot be applied on operating systems that don't support kernel modification. For example, operating systems like Windows 2000 and XP don't allow to modify its kernel, so it shows poor compatibility and portability. In the production environment, Para-virtualization signifies some support and maintainability issues because it requires deep kernel modification. The deep kernel modification creates data structure dependency between Guest OS and Hypervisor it tightly couples the hypervisor and guest OS. Data structure dependency and tight coupling prevent the guest OS from running on native hardware or another Hypervisor.

## Summary and comparison of x86-virtualization techniques

The comparisons between all x86virtualization techniques are summarized in a tabulated form (see [Table](#)

|                                   | Full Virtualization                         | Hardware Assisted Virtualization                                                                                  | Para-Virtualization                                                                                                                                   |
|-----------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technique                         | Binary Translation and Direct Execution     | Exit to Root mode to Privileged Instruction                                                                       | <u>Hypercalls</u>                                                                                                                                     |
| Guest Modification/ Compatibility | Unmodified Guest OS Excellent Compatibility | Unmodified Guest OS Excellent Compatibility                                                                       | Guest OS Modified to issue <u>Hypercalls</u> so it can't run on Native Hardware or other Hypervisors, Poor Compatibility, not available on Windows OS |
| Performance                       | Good                                        | Fair. Current performance lags Binary translation virtualization on various workloads but will improve over time. | Better in certain cases.                                                                                                                              |
| Used By                           | VMware, Microsoft, Parallels                | VMware, Microsoft, Parallels, <u>Xen</u>                                                                          | VMware, <u>Xen</u>                                                                                                                                    |
| Guest OS Hypervisor Independent?  | Yes                                         | Yes                                                                                                               | <u>Xen</u> Linux runs only on <u>Xen</u> Hypervisor<br>VMI-Linux is an Hypervisor agnostic                                                            |

**Table 4.1:** Summary and comparison of x86-virtualization-techniques

## Network virtualization

In the field of computing and information technology, network virtualization, or network virtualisation is the process by which hardware, software, and network functionality of network resources can be combined and converted into a single software-based administrative entity.

Network virtualization can be categorized into two types:

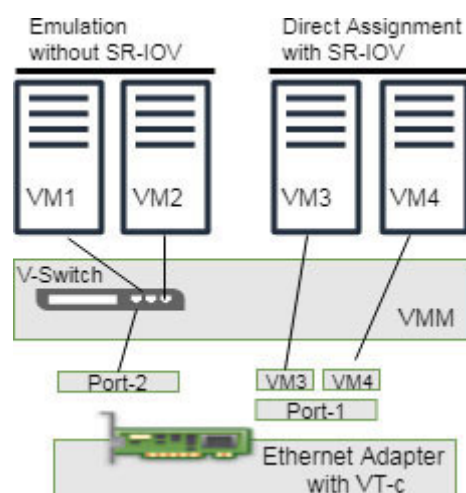
**External virtualization:** It is based on the external component of networks. In this virtualization, many networks or parts of networks can be combined to create a virtual unit of the network., An example of this category, is the creation of a **Virtual Local Area Network**

**Internal virtualization:** This category of virtualization provides network-like functionality to a software container that can run on a single network server.

For example, Single **PCIe (Peripheral Component Interconnect Express)** shared among multiple VMs. [\*Figure 4.9\*](#) shows that how a single network adapter can be shared between multiple virtual machines using with and without SR-IOV technique.

**Single root Input-Output virtualization (SR-IOV)** is a PCI Express technology that makes one physical device computer's physical function into multiple virtual functions. In networking a single adaptor port is a physical function, and its virtual functions can be used to handle the traffic of virtual machines. Different virtual machines running in a common virtual environment are allowed to access a single PCI Express hardware through SR-IOV technique. In contrast, **Multi-Root IOV(MR-IOV)** allows input-output PCI Express hardware to share its resources and functionality among different VMs that may run on different physical machines.

To use the functionality of SR-IOV-enabled PCIe devices, appropriate hardware and BIOS support are required. Its functionality also depends on drivers (Guest OS drivers) or Hypervisor instance. The number of virtual functions supported by PCIe hardware depends on the device. SR-IOV enabled PCI expressed device presentsits multiple instances like instance to the Guest OS and hypervisor (see [Figure](#)



**Figure 4.9:** Single PCIe shared among VMs with and without SR-IOV

## Storage-virtualization

Storage virtualization is a logical process, by which various physical storages available on different network storage, can be grouped into a single logical unit. This single logical storage unit acts as a single physical storage device. Storage virtualization has many advantages like it reduces system downtime, convenient in load balancing, and speed and performance enhancement.

Storage virtualization having two subtypes:

**Block-level virtualization:** It is commonly known as storage virtualization. It is generally used in server applications such as database server, where the software needs to block-level access to data. In block storage data will be stored into a disk in the form of memory blocks, these disks typically but not always reside in **Storage Area Network Arrays (SANs)**.

**File virtualization:** It is something completely different. Some applications need to access data in the form of a complete file instead of Block-by-block. File virtualization serves these applications. In file virtualization files resides in a file system and file system is located on **NAS (Network-attached storage)** device.

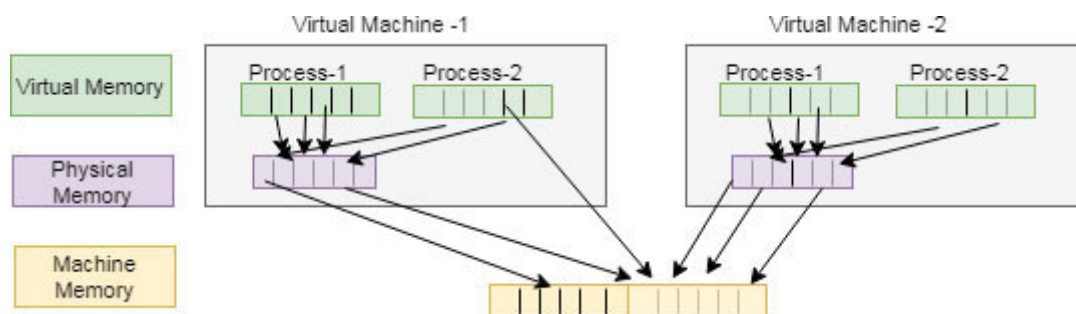
## Memory virtualization

Memory virtualization is a process of combining the physical memory of different servers to make a pool of single virtual memory. By doing this process, a system can get the advantage of bigger working memory. Sometimes windows operating system of Microsoft is doing the same process and allows a portion of secondary storage to work as an extension of primary memory (RAM).

In a VM (created through hypervisor), the process of memory virtualization is very similar to the virtual memory management system in a modern operating system. The modern operating system uses a page table to map a virtual page number with the actual/physical page number. To optimize the performance of virtual memory, the modern x86 CPUs comprises a separate **MMU (memory management unit)** and a **TLB (Translation look-aside buffer)**. To support a guest operation in a virtual machine, the MMU of the system has to be virtualized because the guest OS cannot access actual physical memory directly.

A single physical system can be shared between multiple virtual machines. To manage the physical memory between multiple virtual machines, an extra level of memory virtualization is required. In this step memory of the physical system is shared and allocated among different VMs dynamically. It is the

responsibility of VMM to Map physical memories of Guest OS to actual machine memory. To accelerate the mapping, shadow page table technique is used by VMM. In [Figure](#) it can be seen if the process wants to access the actual memory, two levels of translation are required on every memory access. To avoid these two-level translation and to map virtual memory directly to the machine memory, a TLB hardware is used by the VMM:



**Figure 4.10:** *Memory virtualization*



## Software-virtualization

Software virtualization is a technique to create virtual environments on the primary system so that the Guest OS and its applications can be install and run on a native or primary OS. For example, working of an Android OS and its supporting apps on a Host Windows OS. Software virtualization has the following sub types:

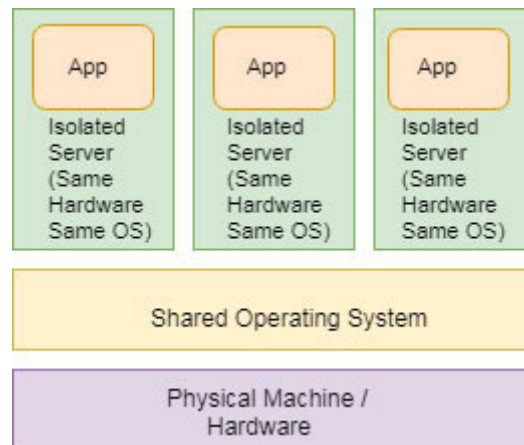
**Server virtualization:** In server virtualization, a software application is used by the network administrator to create multiple and isolated virtual servers for hosting specific services or processes to serve specific applications. Advantages of server virtualization are:

It reduces the requirement of physical servers.

It reduces the amount of energy required to Power and Cools the server.

Due to the fewer number of server maintenance costs will also be reduced, optimal utilization of hardware resources.

**OS level virtualization:** Operating system-level virtualization (OS virtualization) is a type of server virtualization technology. It involves altering an operating system, making it possible to run diverse applications that can be operated by different users working on a single computer at one time see [Figure](#)



**Figure 4.11:** *OS-level/ server virtualization*

**Application virtualization:** Application virtualization is a software technology that refers to run a computer program on a thin client, like a network server or terminal (on a virtual machine in another location), but it gives the illusion that its running on an operating system of the local machine. This type of virtualization is also known as process virtualization.

**Desktop virtualization:** Desktop virtualization is a process to create a virtual version of the actual desktop with its associated applications so that it can be accessed from a remote computer via a connected network. Desktop virtualization software works on a client-server model.

**Data virtualization:** The process of data virtualization is used to manage and manipulate the data by creating an independent abstract layer of actual data. Creating a view in a database is an

example of data virtualization. Data virtualization allows to retrieve and manipulate the data without requiring the technical details and physical location of stored data. It helps reduce data inputs and data formatting errors.

## Summary comparison of various types of virtualization

Various types of virtualization are summarized in a tabulated form (see [Figure](#))

| Types of Virtualization                                                                                                                            |                                                                                                                                            |                                                                                                                   |                                                                                                                              |                                                                        |                                                                                                                             |                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b> <ul style="list-style-type: none"><li>•Full</li><li>•Bare-Metal</li><li>•Hosted</li><li>•Para</li><li>•Hardware-Assisted</li></ul> | <b>Network</b> <ul style="list-style-type: none"><li>• Internal Network Virtualization</li><li>• External Network Virtualization</li></ul> | <b>Storage</b> <ul style="list-style-type: none"><li>•Block Virtualization</li><li>•File Virtualization</li></ul> | <b>Memory</b> <ul style="list-style-type: none"><li>• Application Level Integration</li><li>• OS Level Integration</li></ul> | <b>Data</b> <ul style="list-style-type: none"><li>• Database</li></ul> | <b>Software</b> <ul style="list-style-type: none"><li>•OS level</li><li>•Application Level</li><li>•Service Level</li></ul> | <b>Desktop</b> <ul style="list-style-type: none"><li>• Virtual desktop Infrastructure</li><li>• Hosted Virtual Desktop</li></ul> |

**Figure 4.12:** Summarized types of virtualization

## Virtualization and IoT

The demand for the IoT has been increased in recent years. Its access to the common man has also increased. With this potential of the use of its applications has also picked the demand. Due to it, a large amount of data is generated and processed.

To process the application of IoT, there is the requirement of hardware like processing power, storage, and network capabilities. But such hardware is present in a limited amount. The data generated in intelligent applications are huge, so there is a necessity of more hardware. Adding more hardware resources will increase set up cost and has some environmental issues. Due to this there is strong need to provide management to the use of existing resources efficiently and optimally.

In order to resolve these issue virtualization can be considered as a solution. Virtualization means running more number of the machine on a given hardware. This means running multiple operating systems together on the underlying hardware. This technique helps in saving more hardware requirements without compromising the performance. In addition to this the other idea is to integrate the real world with the virtualization to increase access to new possibilities in Internet of Things that could forecast past, present and future.

## *Virtualization for IoT resource management*

IoT has gained momentum in the past few years. This concept describes that the devices are connected to each other through internet connectivity. The devices pass on the information which is further processed by the processing unit. The processing unit will decide the action to be taken in intelligent way. The various applications of IoT are:

Smart Cities

Smart Agriculture

Smart Transportation System

Smart Garbage System

Smart Parking System

Smart Healthcare System

The devices in IoT can be classified into resourceful and resource-constrained devices depending on the resources available in a particular device.

The data collected in the IoT environment has different characteristics as it is not collected from a single device but from multiple devices. To manage such data there is requirement of managing the resources. As IoT, are not confined on a limited scale, but on a commercial scale, so this poses another challenge.

Now virtualization plays a dominant role here in managing the resources. The virtualization evolution has also gone through various phases, and this technique is matured now. There are various types of virtualization, like:

Server virtualization

Memory virtualization

Network virtualization

I/O virtualization

OS virtualization

Thus, virtualization can be used to provide a good solution for networks in order to deliver the resources necessary for IoT so that the need for setting up network infrastructure can be reduced.

Through open flow technology and open architecture, there has been standardization of setting up the Virtualized IoT devices.

Nowadays there are various applications like Docker and LXC containers which can be used to set up the IoT environment. Docker is a cloud-based system that allows you to build a secure application that can be supported on multiple platforms.



## IoT resource management through container-based virtualization

A container is unit of software that combines the code and its dependencies (system files, bin and library files) in order to run the application quickly, reliably and safely on cross platforms. The difference between the architecture of container based virtualization and hypervisor based virtualization is shown below in [\*figure\*](#). Containers are adopted quickly for IoT application development as container based application includes:

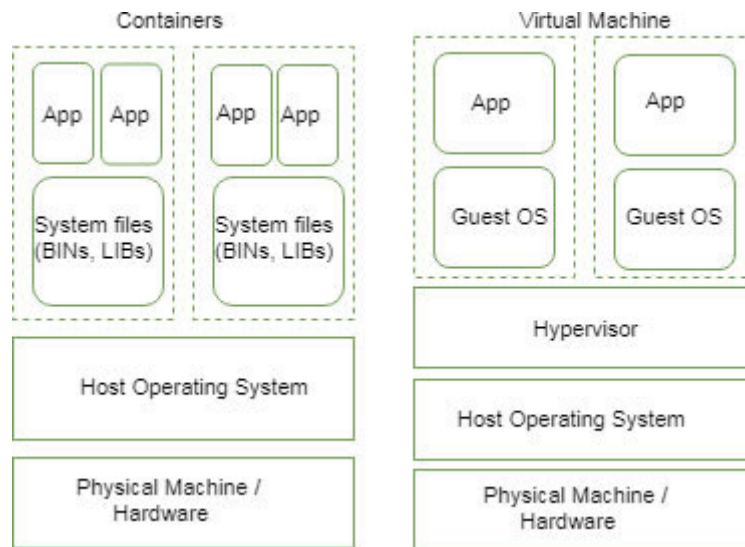
Fast creation and initiation of virtualized instances

Reduced Software and hardware costs.

Secure application platform

Decrease application design time

Reduce the overhead of hardware and software



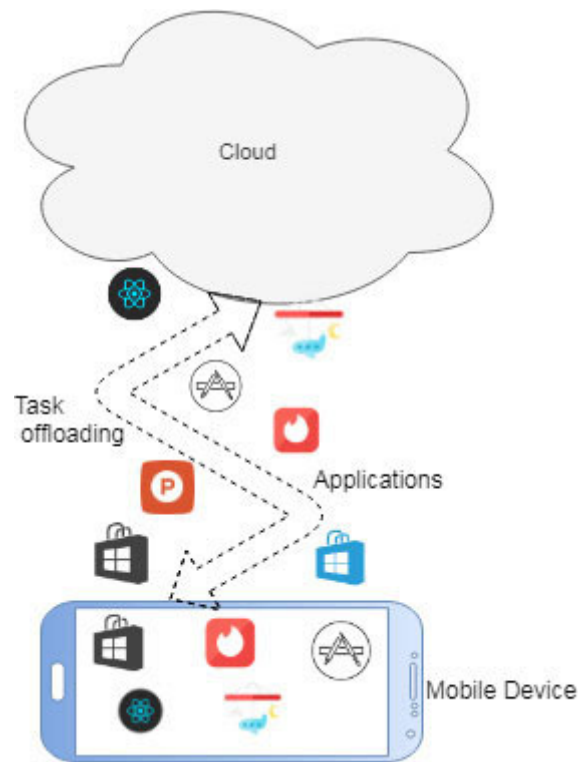
**Figure 4.13:** *Container-based and Hypervisor based virtualization architecture*

IoT applications which have a complex structure consisting of heterogeneous devices. These devices are interconnected and range from servers to desktops to small devices such as fog devices, mobile-devices, wireless sensor nodes and more. The task performed by WSN nodes, mobile-devices, and edge/fog devices are resource-constrained intensive applications that demands deployment of a huge number of hardware devices and generate huge amounts of data. Therefore, there is a need to pay attention to resource management of these devices.

### IoT resource management through task offloading by mobile devices

Mobile devices like mobile phone are used as an essential component in many IoT applications. Such mobile devices are used in data transmission although they do not have a fixed physical connection to link to the network. These mobile devices allow massive data generation. These devices have limited applications and resources to process and analyze the enormous data, due to mobility in nature. To balance the resource-constrained nature of mobile devices an offloading mechanism is used to send the data over to cloud or remote server for data processing computation and analysis. (see [Figure](#)

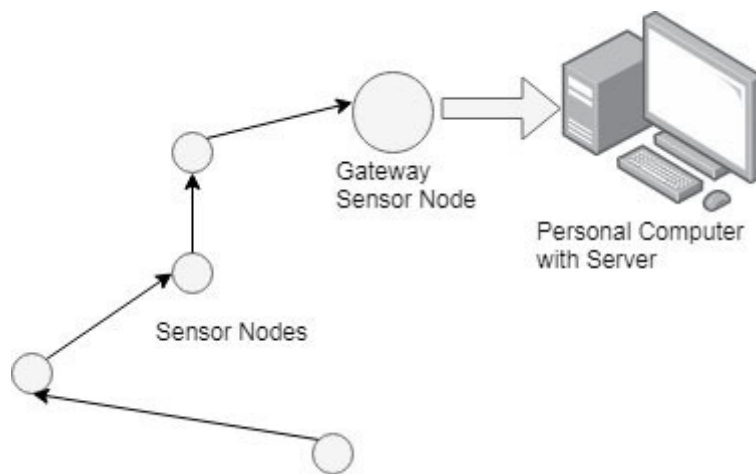
Think Air, Cloudlet, Cuckoo, and more, are some virtualization approaches used in offloading tasks in mobile devices and compensate for the resource-constrained nature of mobile devices:



**Figure 4.14:** *Offloading in mobile devices*

### IoT resource management through virtualization technology in WSN devices

In IoT applications, WSN devices play a significant role. WSN devices are used to sense, compute, and communicate the data to monitor the environment in which it is working. (See [Figure](#)



**Figure 4.15:** WSN architecture

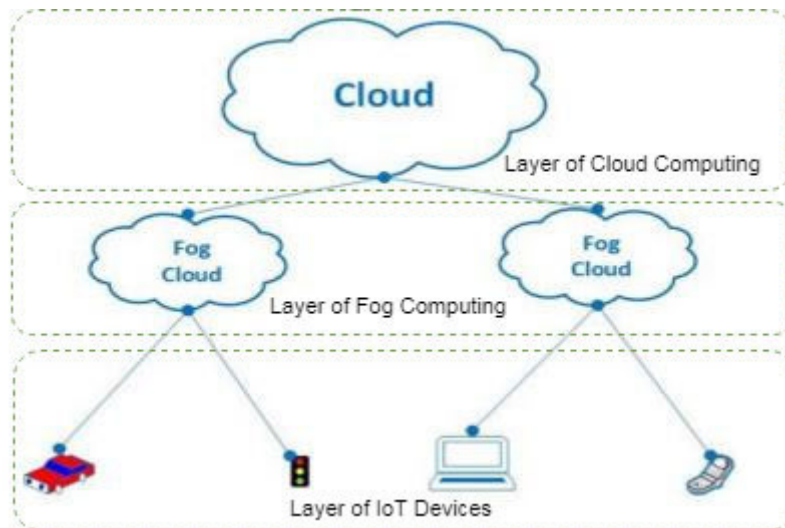
WSN systems have certain drawbacks, such as the small amount of memory, limited storage capacity and limited range of communication. Such limitations can reduce the performance of IoT applications. These drawbacks of WSN devices can be overcome by adopting virtualization technology in the wireless sensor network. Virtualization is used to create multiple logical instances of physical computing resources. WSN networks and devices can also be virtualized. Using WSN virtualization, a single sensor and a single network can be used to run multiple applications and multiple tasks concurrently.

**Software-Defined Networks (SDNs)** is a platform for network virtualization in WSN. It offers many advantages in resource management such as scalability, reduced cost of deployment, versatility and many more.

### *IoT resource management through virtualization in fog platforms*

Fog computing is an extension of cloud-based computing services towards the network edge. Due to the mobility of IoT devices, they are having limited resources for computing and analysis. To manage the limitation of such resources in IoT applications, the use of fog platforms and virtualization in fog platforms becomes a trend. An architecture of fog computing in [Figure](#)

To manage constrained resources in IoT, the computational offloading technique is being used. Using this technique, the resource-constrained computational task of IoT applications (node and network-level) is pushed towards the fog devices. There are various benefits of computational offloading such as reduced latency, high data accuracy, improved quality of service, and more. Moreover the benefits there are various issues in fog computing like service migration in Fog, privacy, trust, and security. To determination of some of the above mention issues in fog computing virtualization can play a leading role:



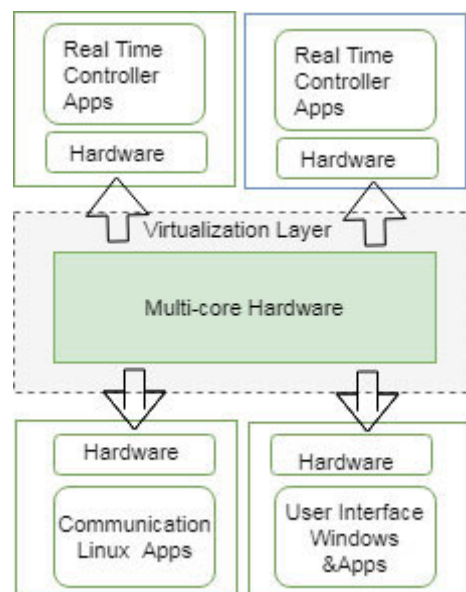
**Figure 4.16:** *Fog computing architecture*

Fog computing helps in managing resources of IoT applications. It performs various operations like storing-and-processing of data between the cloud, fog, and end-devices. In fog computing, there are many techniques that can be used to perform offloading, one of which is virtualization. Using virtualization in fog computing makes it possible to perform multiple functions into a single device by creating multiple virtual copies of a single resource. Virtualization in fog computing thus assists in the management of resources in IoT.



### Embedded virtualization

An industrial process requires many different systems to accomplish the job. For example, it may require a system for signal processing, one system that collects data, one system for vision and one system for human-machine-interface. Each individual system requires hardware and it cost money, electric power, space, maintenance, spare parts, update-and-replacement strategy, due to this it increases capital and the operational cost of the complete process. But using *embedded* virtualization, multiple different workloads can be consolidated into a single multicore-chip. A model of embedded virtualization is shown in [Figure](#) whereas four different systems are sharing the single multicore hardware through virtualization.



***Figure 4.17: Embedded virtualization***

Combining multiple functionalities into a single piece of hardware through embedded virtualization technique reduces the complexity of end-system. This new piece of hardware may be more expensive and more powerful than the older one uni-functional hardware. Hardware designed through embedded virtualization simplifies the process of maintenance-and-replacement and reduces the capital and operational cost of end-system.

### *Benefits of virtualization for embedded systems*

Virtualization is now getting attention from well-established enterprises, for developing a range of embedded solutions. To allowing multiple systems workload/function to run simultaneously on a single hardware, embedded virtualization empowers OEMs (original equipment manufacturers) to:

Reduced system cost, power consumption, and size by consolidating multiple separate functions into one embedded system.

Enrich the market with new innovations although preserving legacy code.

Use of software redundancy and application isolation to enhance safety, security, and availability of services.

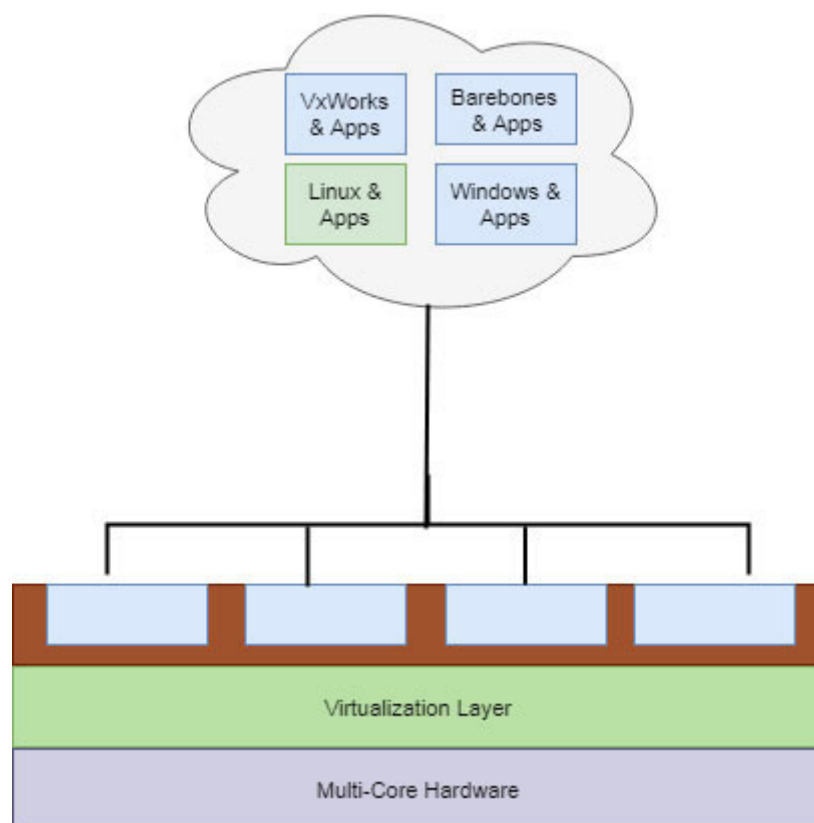
## IoT and embedded virtualization

Traditionally, industrial control devices were manufactured with unique functionality. These devices were capable of doing a fixed function. But now the time has changed. Manufacturers are developing such devices that are more generic in nature. Initially, these devices are comparatively bare. A personality or functionality to these devices can be given during the deployment by the system integrator through programming or downloading software from flashcards, USB or using some sort of interconnection. Responding to such innovation, technology used in industrial control devices now has changed dramatically from microcontrollers to multi-core, 8-bit to 64-bit processors, and now internet of things IoT is shifting the scenario again. Now manufacturers are taking advantage of the various opportunities created by internet of things.

A new technology, that is, embedded virtualization is a boon for manufacturers. The revolution and progress of IoT are fueling the embedded virtualization on fire.

Embedded virtualization provides an additional level of abstraction to the devices. During the manufacturing of devices using embedded virtualization, devices are provisioned with a generic management layer and few empty slots. These empty slots will work as virtual machines and can be provisioned with different functionality during deployment. During the deployment phase, these empty slots can be provisioned with logic that may consist of an operating system and/or with business logic.

After the production in the factory, the device is just hardware, which consists of two layers a generic management layer and a virtualization layer. [Figure 4.18](#) depicts the provisioning functionality into the virtual machines on a device with a generic management layer. During the deployment the generic management layer interconnects with external storage like USB or cloud and downloads contents from it, these downloaded contents can be a business logic that will be installed into the virtual layer of the device and allow the device to perform a specific function.



**Figure 4.18:** Provisioning functionality into the virtual machines on a device with a generic management layer

### How is embedded virtualization different?

In IT enterprise and data centers virtualization is used to create virtual machines using hypervisors so that different servers and operating systems (like Windows and Linux on Intel architecture) can be installed and executed on a single piece of hardware to fulfill the consumer's requirement.

Characteristics of embedded systems are to have real-time components that are used to handle and process the time-critical task. Embedded systems consist of a real-time component, whereas task has to complete in a guaranteed time period. Embedded systems are also used to process non-real-time components that may include processing real-time information, managing or configuring the system.

Often non-real-time applications run on a different processor. These non-real time components of an application can compromise the real-time nature of the system. Using the virtualization these components can be consolidated on a single physical hardware and real-time integrity of the system can be ensured.

Using embedded virtualization, the hypervisor of the system is a *to the metal* software layer. This hypervisor provides abstraction and creates virtual machines by partitioning physical memory and I/O resources between multiple virtual machines. Embedded

virtualization provides isolation between multiple virtual machines which provides better security and higher performance within each virtual machine.

## Conclusion

The use of IoT devices is going to be increased in our day to day life. It leads to a situation of data explosion. Therefore the demand for improved connectivity, improved processing power and the requirement of data analysis are increasing day-by-day. To deal with such increasing demands of computational resources, different forms of virtualization can be applied.

To manage, analyze, and process voluminous data generated by IoT, computational offloading technique is being used. Using this technique, the resource-constrained computational task of IoT applications (node and network-level) is pushed towards the fog and cloud environment. There are various benefits of computational offloading such as reduced latency, high data accuracy, improved quality of service, and more.

Network virtualization also plays a key role in dealing with such inherent complexity in IoT. SDN is a platform for network virtualization in WSN. It provides many resource management benefits like scalability, reduced deployment cost, versatility, and many more.

The revolution and progress of IoT are fueling the embedded virtualization on fire. Embedded virtualization provides an additional level of abstraction to the devices. During the



manufacturing of devices using embedded virtualization, devices are provisioned with a generic management layer and few empty slots. These empty slots will work as virtual machines and can be provisioned with different functionality during deployment. During the deployment phase these empty slots can be provisioned with logic, that may consist an operating system and/or with business logic. Such feature of embedded virtualization provides valuable flexibility to IoT system.

### *The future is virtualized*

One thing is very clear, now IT companies and hardware manufacturers are adapting infrastructure virtualization, network virtualization, and embedded virtualization to meet the growing demanding requirement of IoT infrastructure. The use of the virtualization technique can increase the scalability, agility, and flexibility of IoT resources. It also enhances the resource availability, workload mobility, and performance of automated operations. These benefits of virtualization make IoT less costly and simpler to operate. Some additional benefits of virtualization are: it reduces the CAPEX and OPEX for the manufacturing floor, and also significant savings in development time for end devices.

### Points to remember

Virtualization is a technique to create software version of actual hardware.

A closely detached software device (software version of actual hardware) that could run its own operating systems and application as if it is running on a physical computer, is called virtual machine.

An OS that is running in a VM environment is called a Guest operating system.

An OS that is running in actual hardware or on an actual physical system is called the Host operating system.

A hypervisor is a software, it is used to create and run virtual machines. It is also called virtual machine monitor (VMM).

A Hypervisor which can directly be installed over the physical hardware, called type -1 hypervisor.

A Hypervisor that cannot be installed directly on the physical hardware and requires a Host operating, over which it can be installed comes in the category of type-2 Hypervisor.

In information technology and computing, the use of virtualization can increase the scalability, agility, and flexibility of IT resources.

Virtualization can be categorized into 6 types such as: hardware virtualization, network-virtualization, storage-virtualization, memory virtualization, software virtualization, and data virtualization.

This blending of direct execution of the user-level request and binary translation of OS requests delivers the functionality of full virtualization.

In Para-virtualization or OS-assisted virtualization, non-virtualizable and sensitive instructions of Guest OS are replaced with Hypercalls (modified OS-instruction according to Hypervisor).

In hardware-assisted virtualization, sensitive and unsafe instructions calls of Guest OS are set to automatically trap to the hypervisor and handled by hardware. It removes the need for binary translation and/or Para virtualization.

Network virtualization is the process by which hardware, software and network functionality of network resources can be combined and converted into a single software-based administrative entity.

Storage virtualization is a logical process, by which various physical storages available on different network storage, can be grouped into a single logical unit.

Memory virtualization is a process of combining the physical memory of different servers to make a pool of single virtual memory.

Software virtualization is a technique to create virtual environments on the primary system so that the Guest OS and its applications can be install and run on a native or primary OS.

In server virtualization, a software application is used by the network administrator to create multiple and isolated virtual servers for hosting specific services or processes to serve specific applications.

Operating system-level virtualization (OS virtualization) is a type of server virtualization technology.

Application virtualization is a software technology that refers to run a computer program on a network terminal, but it gives the illusion that it's running on an operating system of local machine.

The process of data virtualization is basically used to manage and manipulate the data, by creating an independent abstract layer of actual data.

### Multiple choice questions

A closely detached software device (software version of actual hardware) that could run its own operating systems and application as if it is running on a physical computer, is called\_\_\_\_\_.

Guest OS

Host OS

Virtual machine

Physical machine

An OS that is running in a VM environment or on different physical system is called\_\_\_\_\_.

Guest OS

Host OS

Virtual Machine

Physical Machine

An OS that is running in actual hardware or on an actual physical system is called\_\_\_\_\_.

Guest OS

Host OS

Virtual Machine

Physical Machine

A software, that is used to create and run virtual machines called \_\_\_\_\_.

Guest OS

Hypervisor

Virtual Machine

Physical Machine

A Hypervisor which cannot be installed directly on the physical hardware and require a Host operating, over which it can be installed comes in the category of\_\_\_\_\_.

Guest OS

Type-1 Hypervisor

Type-2 Hypervisor

Virtual Machine.

A Hypervisor which can directly be installed over the physical hardware, without the use of any operating system comes in the category of:

Guest OS

Type-1 Hypervisor

Type-2 Hypervisor

Virtual Machine.



## Answers

*c*

*a*

*b*

*b*

*c*

*b*

### Fill in the blanks

A virtualization method where virtualization and partitioning services run on top of a Host OS is called\_\_\_\_\_.

In the x86 instruction set architecture, there are\_\_\_\_\_ of privilege.

Based on the privilege level of x86 hardware, there are \_\_\_\_\_basic techniques of x86 hardware virtualization,

Binary translation is an approach that is used in \_\_\_\_\_technique.

\_\_\_\_\_virtualization includes the process of OS Kernel modification.

Hardware assisted virtualization technique allows to run VMM (Virtual machine monitor) in \_\_\_\_\_privilege mode.

In \_\_\_\_\_network virtualization many networks or parts of networks can be combined to create a virtual unit of network.

\_\_\_\_\_is a process of combining the physical memory of different servers to make a pool of single virtual memory.



## Answers

Hosted Virtualization

Four Levels

Three

full virtualization

Para or OS-assisted

Ring-oP

External

Memory virtualization

### Descriptive questions

Define virtualization technique. What are various types of virtualization? Explain in brief.

Draw and explain privilege levels of x86 hardware architecture, also explore the various techniques of x86 hardware virtualization.

How Para-virtualization is different from full virtualization?

Draw and explain the process of memory virtualization.

What are the various types of software virtualization explain in detail?

How virtualization technique is useful for IoT resource management?

Explain the difference between Hypervisor based virtualization and container-based virtualization.

What do you mean by embedded virtualization? What are the various benefits of embedded virtualization?

How embedded virtualization is different from other types of virtualization?

Explain the difference between external and internal network virtualization techniques in detail.

*Security, Privacy, and Challenges in IoT*

**Internet of Things (IoT)** application development is ceaselessly proving itself to be revolutionary and foundation for the alternative contemporary technologies like AI. IoT may be transformational forces which will enhance performance through IoT analytics and security to yield maximum outcome. Businesses can take advantages of IoT for taking accurate and more structured decisions. But before the adoption of the IoT enables infrastructure, we need to understand about the privacy, security and trust-related threats and also about the various challenges that are associated with the IoT infrastructure.

## Structure

In this chapter, we will discuss the following topics:

Challenges in IoT

Design challenges

Development challenges

Security challenges

Privacy threats

Other challenges

Trust management



## Objectives

After studying this unit, you should be able to:

Understand about the various challenges that are experienced during the development of IoT devices.

Analyze various design, development and security issues, and challenges associate with the IoT enabled infrastructure.

## Introduction

IoT can be stated as *Intelligent interactivity between human and things to exchange data, information, and knowledge for innovative worth creation.*

The term IoT-Internet of Things was first coined by *Kevin* Executive Director of MIT's Auto-ID Lab in the 1990s. Most of the leading merchants and technology leaders are taking steps ahead to leverage the opportunities related to web of things, and outline IoT, according to their specialty.

IoT will attach gadget simple anted in numerous systems directly to the web. Once device/object characterizes it-self on a digital platform, it can be handled or controlled from any location in the present network. With the help of this connected web, each entity - good vehicles, smart devices, smart homes, good appliances; unbreakable connected bond is the biggest development of recent time. This connectivity web enables, capturing of data from different locations, ensuring ample amount of additional techniques of stimulating efficiency, rising protection and obviously, IoT security and privacy-related challenges.

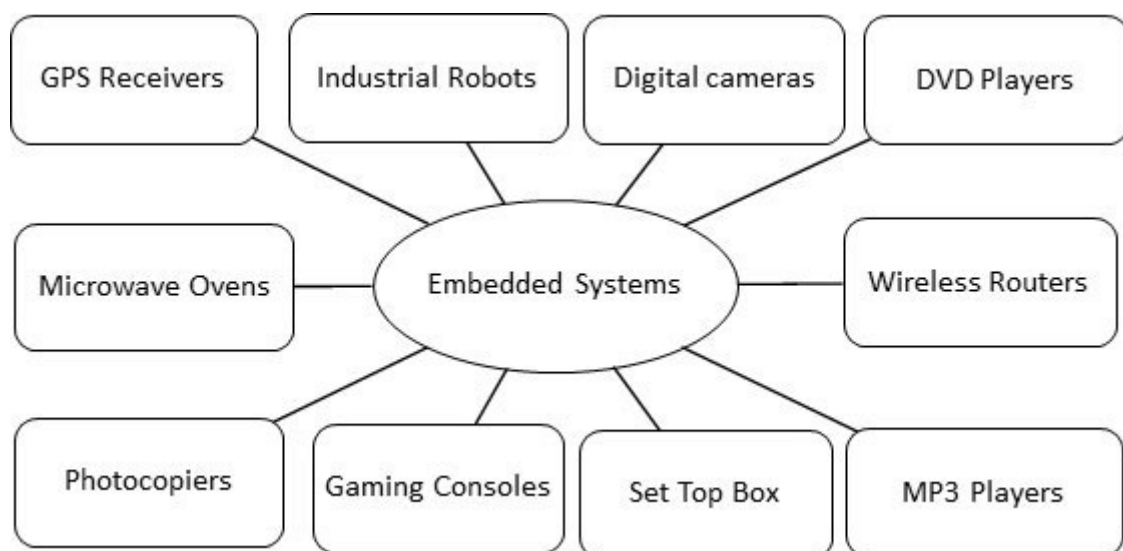
In the sequent paragraphs, we'll be introducing about the concerning challenges in IoT development.

### Design challenges

The design phase of IoT is a very vast field. This phase is having countless challenges. A diverse variety of IoT devices having different types of design challenges. The design phase of an embedded system also has many challenges. There are a variety of common challenges in designing embedded and IoT devices with some specification. Some of the design challenges are as follows.

### Absence of essential flexibility for running applications over embedded systems

As the demand of connected devices is increasing day by day, embedded systems are proving efficient to work with totally different devices, due to its flexible nature; it is also able to adapt to different networking architectures so that new functionalities can be copied in the real-time environment. [Figure 5.1](#) shows that flexible embedded systems can work with totally different devices:



**Figure 5.1:** *Flexibility of embedded systems*

Developing flexible, embedded IoT systems is a challenge for designers. It is very difficult to design flexible IoT systems and devices that are having the capability to adapt the latest applications. Difficulties during developing flexible embedded IoT systems are:

Problems in making certain smooth integration of the latest services.

It's hard to adopt a new environment.

Frequent hardware and package changes are difficult.

There are issues in packaging tiny chip and the difficulty in integrating these tiny chips with low weight and a lesser amount of power consumption.

Running different energy awareness operations becomes difficult.

## Security issues

IoT systems and devices must be secured robust and trustworthy. But Unstable performance in real-time embedded environment can be seen in entire IoT hardware products. Engineers face issues in ensuring the security of embedded parts as they operate in an extremely resources constrained and physically insecure environment.

But nowadays due to the technological advancement, firm performance in real-time embedded environment can be seen in the latest IoT hardware products. The systems are designed to be robust and trustworthy. They are also made secure with cryptographic algorithms and various security procedures. Various types of security approaches are included to protect all parts of the embedded system from model to deployment.

### High power dissipation

High power dissipation of hardware components like microprocessor and embedded chips is another challenge of the embedded and IoT system design. Hardware design uses more power to obtain the most effective performance from application and devices in a given period.

Deployment of an embedded system with a greater ratio of transistors and an adequate or sufficient amount of power consumption quantitative ratio is also one of the persistent challenges. High power dissipation coming up with low power embedded system is due to two reasons:

There is an increase in the capacity distribution per semiconductor. Thus, to reduce the power consumption of overall embedded systems, engineers should use economic system design with technology alone.

The main concentration of engineers consumes low power and delivers high performance, thus increasing the frequency of the system, resulting in the burning of additional power. They should pay more attention to choices of styles.

### Difficulties of testing

To ensure that the product design is reliable, thorough testing, verification, and validation are other problems. Limitations related to the software updates make the testing process very challenging in finding bugs as would be prudent for a given software version. It also increases the importance of build and deployment procedure:



**Figure 5.2:** *Testing of system for bugs*

The tasks that can be performed during the hardware testing, verification, and validation are as follows:

**Hardware testing:** Wherever hardware tools are used by embedded developers, it is comparable to all types of testing. The system's performance, consistency, and validations are tested as per merchandise demand, which refers to the embedded hardware.



**Verification:** It is done to ensure that the verification has been carried out properly or not.

**Validation:** In this, it is validated whether the merchandise matches the necessary quality standard and passes them.

### Insufficient practical safety of safety-critical embedded systems

Safety-critical embedded systems are very sensitive and special embedded systems. Failure of such systems or applications based on such systems can lead to serious injury, loss of life, significant property damage, or damage to the environment. Railways, automatic weapons, medical care devices, nuclear are the examples of fields where safety-critical systems are being used.

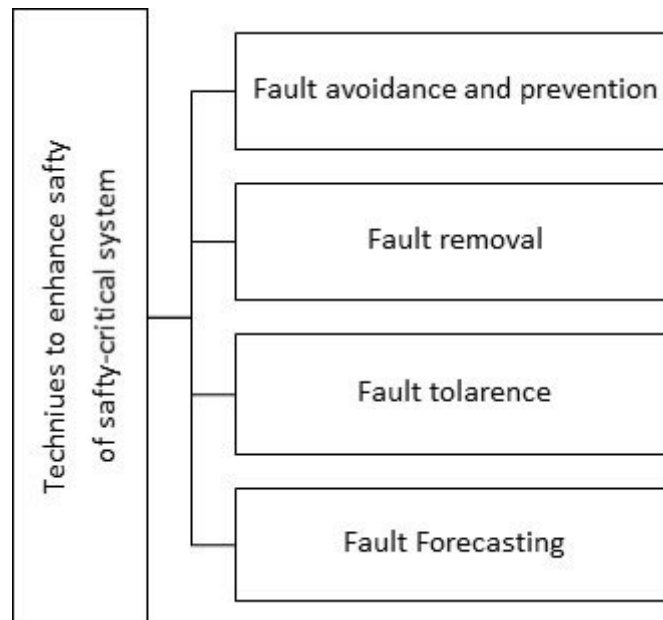
It is highly considered that designing such systems is a very complex process because the designing process includes the combination of renowned design approaches and techniques in software and hardware to fulfill two different types of requirements, non-functional requirements, and functional requirements:

**Functional requirements:** are set of activities or operations are expected to perform by an embedded system.

**Non-functional requirements:** are the set of required attributes like small size, safety, low cost, ease of maintainability, high reliability, and availability.

Insufficient practical and functional safety of the safety-critical system can increase the chances of its failures. Therefore, some general techniques can be used to enhance the safety of safety-critical systems. [Figure 5.3](#) shows the classifications of these

techniques, which can be used according to the requirement of different stages:



**Figure 5.3:** *Techniques to enhance the safety of safety-critical system*

**Fault prevention/avoidance:** How to prevent or how to avoid fault occurrence. These techniques are used to reduce the occurrence of faults during the phase of system development.

**Fault-removal techniques:** How to remove and reduce the quantity of faults. These techniques are used in the second step. It consists of inspection and testing.

**Fault-tolerance:** How to avoid system failures after fault occurrence. During the development phase these fault-tolerance techniques are employed. These techniques empower the system to endure

remaining faults so that system can deliver accurate service in the existence of faults.

**Fault-forecasting:** How to make available evaluation of system, via approximating the existing number of faults, the future occurrence, and significances of faults. The robustness of fault tolerance capability can be access by using these techniques.

An effective combination of these techniques can improve the degree of success. The fault-forecasting technique is used to evaluate the degree of success.

### Increased cost and time-to-market

Embedded systems are tightly restricted by cost, in addition to flexibility and safety. The design cost of an embedded system is depends on it's built in hardware style, digital electronic components, functional requirement, development process, and manufacturing quantity.

The cost of an innovative product and time-to-market is directly proportional to the product development cycle. [Figure 5.4](#) depicts, if innovations and products are not carried and deployed according to the product development life cycle then the cost of product will rise exponentially. It can happen because of the growing system complexity. With the time complexity of the system is growing exponentially. Growing system complexity is the result of innovation of advanced protocols, advanced radio frequencies, and growing number of applications, among others:



**Figure 5.4:** Increased costs

Using cutting edge technology during the design and development phase of embedded system may lead to reduce the development cost of embedded system.

Hardware/software code-designers together ought to solve the design time issues and convey embedded devices at the proper time to the market.

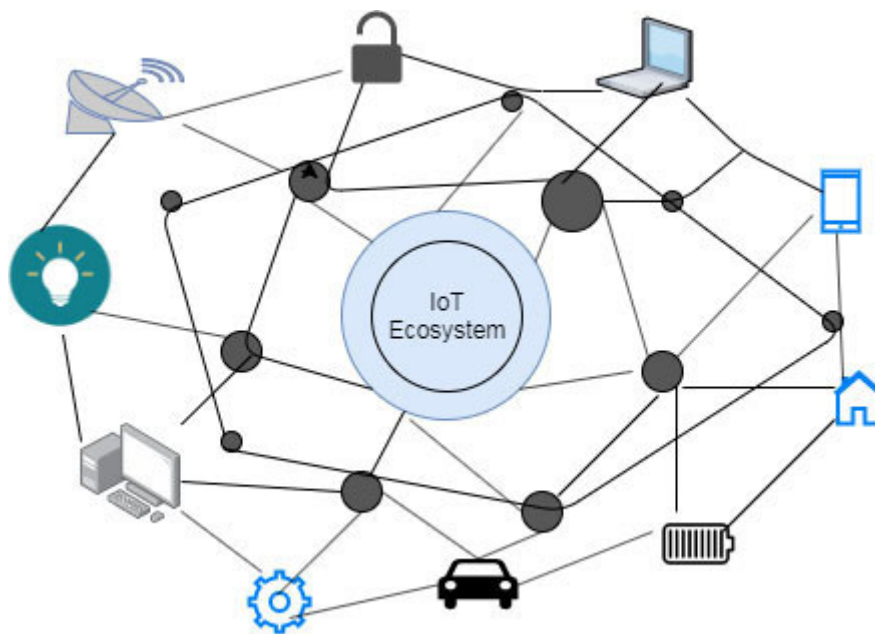
### Development challenges

The difficulty with IoT is that many companies focus exclusively on the growth of IoT while not assessing the first difficulties they face. Many of these companies have no background in IT trade or package growth, but most are committed to offering internet-connected equipment. Even enterprises that have package and hardware style expertise usually take IoT as alternative ancient computing technologies and build huge mistakes once developing IoT devices.

## Connectivity

Connectivity is the original problem, that is, a way to connect computers to the internet as well as the platform for cloud computing. However, this can be determined to an excellent extent by the setting of the device application and also by the type of communication infrastructure that these devices provide.

Connectivity of the various devices in an IoT ecosystem is shown in [Figure](#)



**Figure 5.5:** *IoTecosystem connectivity*

IoT utilizes a centralize client-server model in its current form to produce assorted servers, workstations and system resources. This can be quite cost-effective for current systems as the IoT remains



in its infancy; however, what happens once many billions of devices concurrently allude to the network?

According to Gartner's updated accounts, by 2020, more than 20 billion units will be connected to IoT. It is just a matter of your moment before consumers start to experience major IoT connectivity, effectiveness, and general performance bottlenecks.

For instance, if you want to create a smart home computer, such as an internet toaster, you will have access to a home Wi-Fi router or IoT router ZigBee/Z-Wave. Therefore, one or more transmitting media should be supported by your device. However, access to the Wi-Fi network is not available in some settings, such as IoT farming or smart cars, and the mobile network is also your first option.

Therefore, if you want to balance your alternative side and make style decisions, the prospects are endorsed, providing all options, and investment. For example, transmitting information to the cloud service through a cellular network would be expensive, but you will confirm picking up and running original mode or blockchain mode to create IoT system that is relatively less addicted to cloud computing.

You should also know, of course, that IoT remains an early technology undergoing major changes and modifications. There are many uncertainties and trends in this area. Consequently, techniques in use today may deteriorate in the longer term.

On the contrary, IoT devices have an expanded life cycle compared to personal computers and intelligent phones that are simple to replace with fresh products. A smart refrigerator, for instance, should operate for at least 5 to 10 years. Therefore, you want to create a notion to verify that your device will retain its property and adapt to new techniques once IoT starts to take shape in the future.

### Flexibility and compatibility

Since the IoT pattern is constantly dynamic, therefore an IoT product must also support future technologies. However, it is compulsory to keep equilibrium between package and hardware once an IoT item is introduced.

Developing your device's dedicated hardware helps to achieve optimum performance, although product updates may also be prohibited. On the contrary, selecting relevant storage and computing resources and IoT-customized operating system may cause performance degradation, but it allows you to expand your computer, use fresh features, and solve bug exploitation patches.

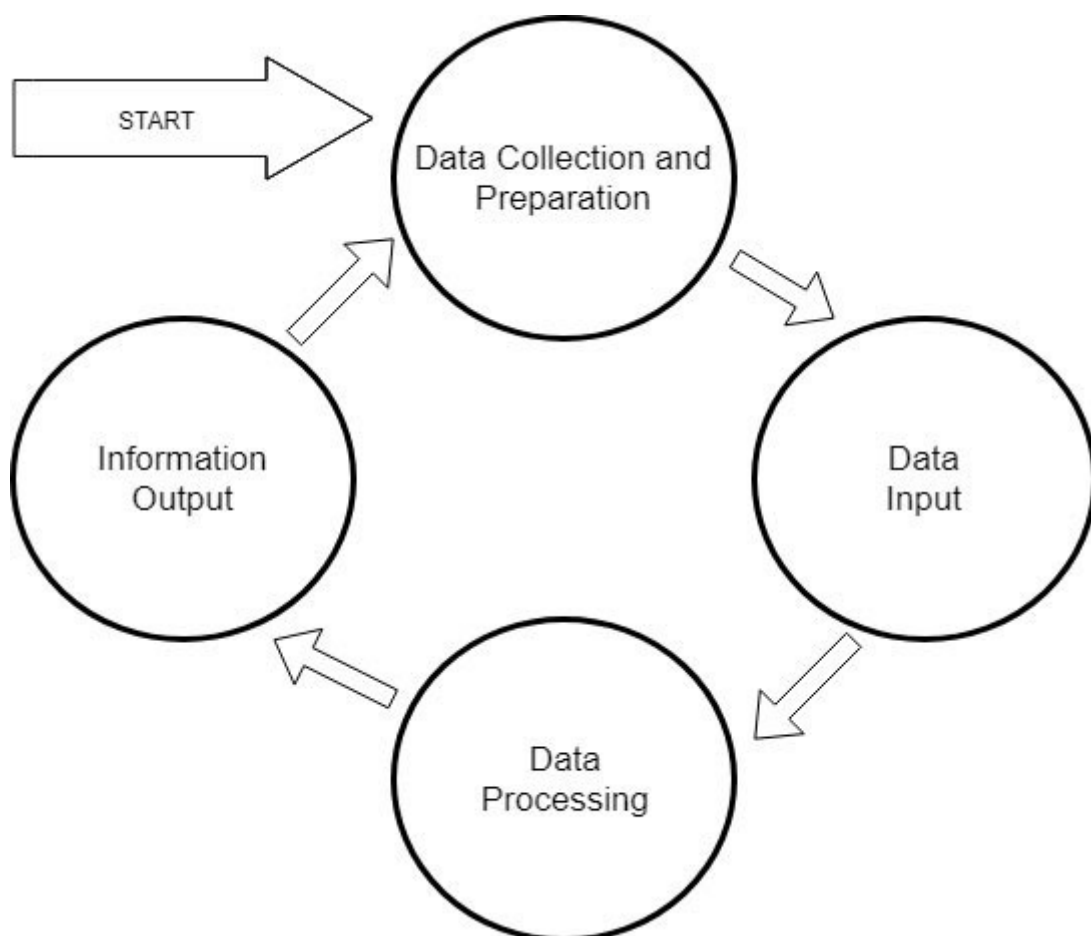
Some suppliers may attempt to give relevant APIs and SDKs whenever possible to allow development staff to add characteristics or functions to their IoT systems. An instance of honesty is Amazon Echo. This IoT instrument will execute the programming of the expansion in a thousand completely distinct directions.

Once the IoT product has been developed, you must ensure the compatibility and that your IoT device is embedded seamlessly with the IoT scheme of customers, without increasing the complexity or transportation of any setbacks to current knowledge. For this purpose, you want each package and hardware to be considered.

An optimal situation shows that customers do not have to be pressured into a brand-new implementation merely because they buy a smart device for their homes. The main examples are the Apple Home Kit and Samsung Smart-Things. Each allows personnel development to provide users with fresh IoT features in settings known to users.

### Data collection and processing

In relation to safety and privacy, you would like to discover a joint technique for processing all the information gathered. To handle the scale of your cloud storage and satisfy your platform requirements, you want to initially evaluate the amount of processed and picked up information. Data collection and processing is a continuous process, [Figure 5.6](#) illustrate the data collection and processing cycle:



**Figure 5.6:** Data collection and processing cycle

Data collection and preparation is the first step in this contentious process. Prepared IoT information is as valuable as gold, but it is ineffective if it is stored on your server and not processed correctly. Therefore it is more essential that you are advancing in processing the information gathered.

To the advantage of information processing step, you need to explore the skills and instruments that will make the greatest use of your expertise. These instruments include the recruitment of knowledge advisors and the adoption of relevant assessment and additional machine learning knowledge will complete multiple sensible functions. Following are the functions:

**Supplement existing data:** Before migrating their services to IoT, most companies already have extensive understanding of their clients. Integrating current understanding with information gathered from IoT devices will bring fresh company ideas and extra revenue generation possibilities.

**Analysis and more user division:** Knowledge such as data and information gathered from IoT devices can also inform you a good deal of knowledge about customer preferences and features. Analysis and classification of IoT information will make it easier for companies to learn more about the requirements and preferences of their clients and change them to solve linked problems in a very wiser way.

### Security challenges

IoT security always has a questionable one. The main challenge to be considered is that IoT's security and privacy are fundamentally distinct from the network security:



***Figure 5.7: Security of IoT devices***

Before using and installing the IoT devices following safety style points which are significant, are explained in the below sections.

### Physical security

IoT devices are generally settled in open areas and physically unprotected. You want to make sure that a vicious organization, broken by hackers, will not maliciously manipulate them.



## Encryption

You also want to secure understanding or protection that holds equipment of any kind. While entering a safety protection portion on each IoT device is costly, encrypting information is still essential.

### Knowledge security exchange

In addition, data protection is vital as a result of which data should be transmitted from IoT sensors and devices to the entrance and then to the cloud. Therefore, it must be mandatory to use encrypted transfer protocols. In addition to cryptography, to verify IoT safety, you must consider authentication and permission.

### Cloud storage security

The data placed in the cloud is as brittle as the IoT system's alternative components. The knowledge or data stored in the cloud should be protected by your platform. Applicable cryptography, access management, and so on is protective measures.

## Update

There are perpetual security vulnerabilities despite the amount of effort you make to strengthen your product code and equipment. In this situation, you want to initially have an idea for repairing mistakes and rapidly unlocking patches instead of for a prolonged quantity of time leaving mistakes unfixed. Next, you would like to give a smart and safe method to clients to repair mistakes. Currently, updating on-line devices over the air is well-liked, but you want to make sure that top-of - the-line techniques themselves do not become vulnerable to security.

## Data security issues

Recently, enterprises and clients are concerned about data safety with the occurrence of numerous ransom-ware assaults.

Additionally, there is a business spying chance to achieve material ownership. IoT service providers should therefore ensure that their data is secure.

Using an extensive mode of governance that offers safe access to sensitive records and data can address these problems of data safety. This topic describes various safety related problems that are essential to understand for the effective execution of IoT. Some of the problems are:

### **Brute-forcing as well as the problem of default-passwords:**

Escorting transportation devices using default passwords and without informing customers to alter or change them when they receive them, is one of the most effective example of this problem. In the most troublesome DDoS attacks Mirai-Botnet were used. Almost all IoT devices are susceptible to weak credentials and login information. Mirai malware only succeeded in identifying vulnerable IoT devices through using default password and default usernames to login and infect them. Accordingly, any business that has used default credentials on its appliances will position both its companies and property and the custom.

**IoT malware and ransomware:** As the range of IoT linked machines continues to increase in subsequent years, various types of malware and ransomware can be used to exploit them. Conventional ransomware relies on encoding to fully lockout the users from distinct systems and platforms. There is continuing hybridization of each strain of malware and ransomware directed at combining distinct kinds of assault. Ransomware attacks could certainly aim at limiting device functionality and simultaneously stealing user data. For instance, an IP-Camera is suitable for capturing subtle and sensitive information from a wide-ranging spectrum of places.

**IoT botnets targeting cryptocurrency:** Cryptocurrency is the latest and hottest virtual currency. Popularity and valuation of cryptocurrency is increasing day by day. It's valuations, proves to be too appealing for hackers. Therefore it is hot target of many hackers. They are trying to decode and redeem it on the crypto craze. Somewhat app development using blockchain is the main vulnerability instead of blockchain itself. Hackers are practicing social engineering to get usernames-passwords and personal keys, and the same can be used by the hackers, in the future and blockchain based app could be hacked. Monero Open Source is presently deep-rooted in each of the distinct digital currencies with IoT devices. Manipulation of information integrity presents a high risk of open crypto market flooding and cryptocurrencies 'already volatile value and structure disruption.

**Untrustworthy communication:** Several IoT devices send emails without encoding to the network. This can be one of the biggest out there IoT safety challenges. It is time for each company to

ensure that their cloud services and devices are encoded at the highest level. The most efficient way to prevent this danger is to use transportation encoding and standards such as TLS. Another way of using distinct networks is to isolate distinct equipment. Private communication can also be used to guarantee secure and confidential transmission of information.

**Remote car access:** Using connected IoT equipment, smart vehicles are on the brink of becoming reality. However, it possesses a greater danger of an automotive hijack due to its IoT association. An experienced hacker can hijack your delicate vehicle by remote access. This may be an alarming scenario as someone has the capacity to handle your vehicle and can leave you at risk of crime.

**Home invasions:** Maybe one of the most appalling threats IoT will face is a home invasion. Nowadays, in homes and offices, IoT devices are used in an oversized variety. Therefore a rise in home automation market can be seen. Security of these IoT technologies is a serious concern. It may expose your IP-address to know your home address. Hackers may sell these very important data to criminal outfit underground websites. Furthermore, if you are victimizing IoT devices in your safety systems, there may be a possibility that they will also compromise as they leave your home at enormous future risk.

**Artificial intelligence and automation:** As IoT devices are capturing data from our daily life activities, companies are eventually managing millions of IoT devices and systems. These devices are producing a large amount of data. It is quite difficult to manage such a large amount of user data from data collection and

networking point of view. Therefore AI enabled automation instruments are used to scan large quantities of information, and one day IoT executives and network safety administrators will impose data-specific guidelines and detect unusual traffic patterns and data. However, it may be too hazardous to use autonomous systems to produce autonomous choices influencing various duties across gigantic infrastructures such as healthcare, power and transportation, particularly if you consider a code error or algorithmic misconduct program needed to decrease the entire infrastructure. Mirai botnet was the biggest active botnet based on IoT for 2 years. In 2017, it was the Reaper, a botnet that was much more hazardous than the renowned Mirai. Over the next couple of years we might see many micro-breaches through Internet security.



## Privacy challenges

As far as privacy is concerned, you want to understand that information gathered by IoT devices is simply subject to legislation and regulations constraints.

For example: A fitness tracker, for instance, will obtain a lot of user data, which is protected within the US by HIPAA. If you store this kind of knowledge on the cloud server, this indicates that the data should be adjusted to the legislation and rules that are attached.

As a rule of thumb, to prevent storing identity information within the cloud, you would more anonymized client understanding. This rule, once incidents happen, defends you against legal punishment.

Figure 5.8 shows a pictorial representation various objects associated with the cloud that can face privacy challenges:



**Figure 5.8:** *Privacy of data on cloud*

### Types of privacy challenges

The IoT creates distinctive privacy challenges, many of which transcend current info privacy issues. Unauthorized tracking can be used to reveal the privacy of an object. Object could be a data, database, person, and many more. Two important aspects of privacy challenges are:

**Object privacy:** Stealing data, eavesdropping, and tracking come under the category of object privacy.

**Securing location privacy:** Revealing data, tracking and monitoring in the category of location privacy. Following are the list of some IoT applications and scenarios in which to handle the privacy challenges is an important and necessary concern:

Home appliances

Transport

E-Governance

Social networking

Defense

### Scenario to understand privacy challenges

Consider a scenario of a sensitive and smart television, in which an integrated voice and vision recognition device is embedded. It will endlessly hear in-home conversations and watch in-home activities and selectively transmit that information to a third party cloud storage that can process to reveal the privacy of many lives.

In addition, multiple IoT scenarios require device deployments and multinational or global data assortment operations that cross social and cultural borders. What this means for the case of the IoT's widely applied model of privacy security.

In order to appreciate the possibilities offered by IoT, it is suggested that ways of respecting individual choices on privacy should be established across a wide range of expectations, while innovation in fresh technologies and services should be encouraged.

### Other challenges

There are many securities and privacy related issues in IoT design. These issues must be address during the development, and adopting IoT infrastructure and devices. But other than the privacy and security, there are some other challenges which must be taken in consideration. In the subsequent topic, we will explore some other challenges that are having equal importance.

### Meeting customer expectations

Success of IoT implementation is directly proportional to the how clearly and effectively problems statements are defined. However most of the IoT service providers are lacking in it. Customer satisfaction, effectiveness of service and productivity are the alternatives which will influence the customer to adapt IoT services:



**Figure 5.9:** *Consumer ratings to products*

It is a challenging task to understand customer requirement and act accordingly. This complete process is contentious cycle that requires a nice retroactive. IoT implementation challenge cannot be even greater than this customer issue statement understanding gap. Therefore,

figuring out the key performance indicators is crucial for IoT consultants to measure or improve through an IoT solution.

### *Keeping IoT-hardware updated*

Irrespective of by what means the IoT or cloud is used by a company, the integrity of data may be a prevalent task. Therefore it is essential to attune and regulate IoT sensors, detectors, and other electrical devices on a regular basis. With such information coming in from multiple sources, separating helpful, unjust data from inapplicable chatter is robust. Even as you would have any other electrical device. Many smart devices are equipped with the next-generation sensors as well as panel meters, chart recorders, current clamps, and power monitors. Synchronizing the data flow between all these hardware without the assistance of a knowledgeable or professional team is difficult.



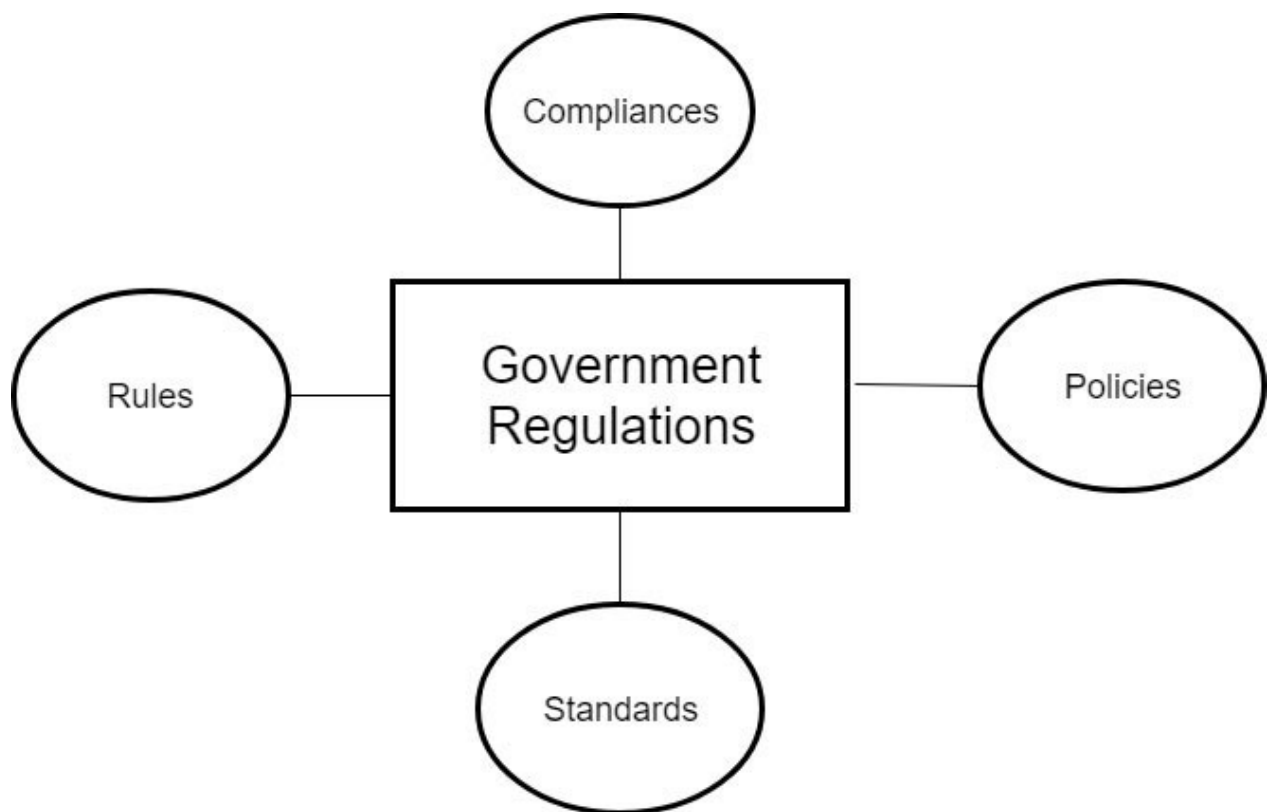
### *Analytics challenges*

The real value of the IoT solution is achieved using actionable insights derived from the IoT information collected. This requires a superior analytics platform that can handle the ginormous amount of data that needs to be added to the solution.

Data analytics partners must realize that they must involve data processing, cleaning and representation while designing the IoT application. Therefore, leaving enough space for real-time extensibility to an IoT solution that will help solve the crucial challenge of implementing IoT.

### Waiting for governmental regulation

While some companies are embracing IoT in actual time, others are reluctant. In several instances, these companies are waiting with fresh norms and guidelines for public authorities to intervene. Some of the government regulations are shown in [Figure](#)



**Figure 5.10:** Some of the government regulations

Since IoT, cloud and even a popular network are not connected to a specific town, country or area, who is accountable for establishing these laws and regulations? Some of the government regulation includes constraints and regulation based on product standards, policies, compliance, and other rules.

For an instance the sheer variety of devices connected to IoT makes matters more complicated. Since these instruments come from various sources, as well as from global partners and vendors, the quality of incoming shipments will be monitored by a localized administrative agency.

## Trust management

In context of information technology and information system, trust management system can be defined as an abstract system that can be used to process and evaluate the social trust of a system. To evaluate social trust of automated device symbolic representations of social trust can be used. Usually it is used to aid decision making process in automated devices.

It can be understood with a real life such as, an experience of a movie tickets. A person can buy a movie ticket that permits him to enter the cinema hall. The printed movie ticket can be seen as a symbol of trust that means the person having the printed ticket has paid for his seat and is permitted to enter in the cinema hall. However, after purchasing the movie ticket, it can be given to some another person, thus it can be seen as transferring of such trust in a symbolic way. Though at the gate of cinema hall printed ticket will be checked only, not the identity of a bearer.

Trust management having an important role in IoT systems and devices. Some of its important roles are: Enhancing user privacy and its information security, reliable data mining and data fusion, providing context-aware and quality services. Trust management helps IoT users to overcome the insights of risk and uncertainty, also it enhances the trust in people to accept and consume IoT devices, applications and services.

### Trust properties

Trust can be influenced by many non-measurable and measurable properties. Therefore it is a very complicated concept. Trust is directly related to the security. User safety and system security are the necessary components to gain the system trust. But, trust is more than security. Other than the security, it relates on numerous factors, such as:

Goodness

Strength

Reliability

Availability

Ability

Other characters of an entity

Another important concept related to trust is privacy. Managing trust in privacy means the capability of an object to determine to whom, whether, and when, the information about itself is to be disclosed or released. An IoT system can gain the user trust by

preserving its user's privacy. Such systems can be considered as trustworthy IoT systems.

In the emerging information system and an autonomous system like IoT, Trust is an important and critical issue. Although the generalize concept of trust is very rich still it can be summarized in the following five properties:

**Objective property of trustee:** Reliability and security of trustee.

**Subjective property of trustee:** Goodness, kindness, and honesty.

**Objective property of trustor:** Criteria and policies for the trust decision

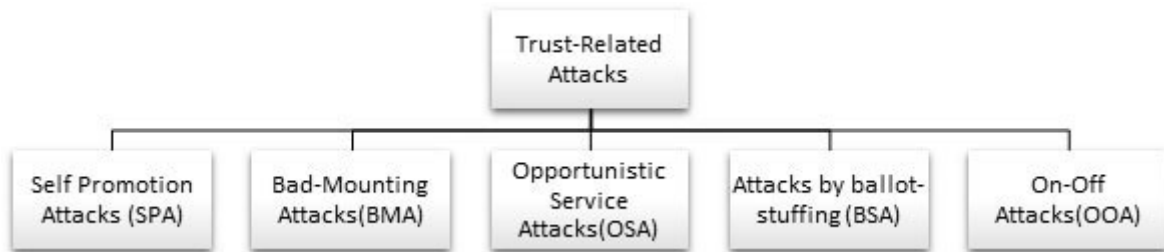
**Subjective property of trustor:** Willingness and trustworthiness

**The context in which trust relation lies:** like environment, location, time device is being used, and its operation mode, and many more.

IoT trust management systems should assist detect malicious nodes by supporting other security mechanisms and protocols like system for intrusion detection, protocols and mechanism for authentication check, mechanisms to check and preserve privacy, and other important management systems.

### Trust issues and trust-related attacks

Indeed, most trust management schemes assume collaboration between distributed entities. Cooperation will merely be broken by inconsiderate behaviors and exploited by malicious aggressors to trigger later trust-related attacks. Following are the various types of trust-related attacks:



**Figure 5.11:** Trust related attacks.

**Self-promotion-attacks (SPA):** In such types of attacks, malicious nodes manipulate their name by offering sensitive recommendations.

**Bad-mouthing attacks (BMA):** In these attacks, malicious node manipulates the name of another trusted node creating unhealthy suggestions.

**Attacks by ballot-stuffing (BSA):** These kinds of attacks are also known as attacks by *Good*. Some malicious nodes will operate these attacks to trigger the attack. In reality, by offering reasonable

suggestions for it, a malicious node manipulates the name of another malicious node.

**Opportunistic-service attacks (OSA):** in this type of attack, a malicious node tries to become timely by offering authentic services to keep its name high. The basic objective is to cheat various malicious nodes to maintain *unhealthy-mouthing and sensitive-mouthing*

**On - off attacks (OOA):** In this type of attack, honest and unhealthy service is provided by the malicious node, or else. The goal is to remain sensitive to its name, and it will compromise the network by offering an honest suggestion for malicious nodes or an unhealthy-recommendation for trusted node. Finding such attacks seems like a much difficult task.



## Conclusion

There are many problems in the method of producing IoT products. In this chapter, we have listed some important issues. If you don't believe about these problems properly, without a torch, you'll walk into a deep channel. IoT devices are having a wide variety of security issues and challenges. Under this scenario, you will have to feel and understand that designing and adopting IoT devices and system, it is very important to consider the privacy, security and trust-related parameters. The challenges faced by IoT's development are wider and more complex.

### Points to remember

While for centuries the idea of combining computers, sensors and network for tracking and controlling facilities, the latest confluence has been important technologies and market trends for IoT is entering a fresh reality.

IoT can be stated as, *Intelligent interactivity between human and things to exchange data, information and knowledge for innovative worth creation.*

Absence of essential flexibility for running applications over embedded systems, security, power dissipation, difficulties of testing, the safety of safety-critical systems, and increasing cost and time-to-market are some important design challenges.

Connectivity, flexibility, compatibility, data collection, and pre-processing of data are some development challenges in IoT implementation.

Physical security, encryption, knowledge security exchange, cloud storage security, update, and data security issues are some important security challenges in IoT development.

Unauthorized tracking, securing object privacy, securing location privacy, are some privacy challenges in IoT enabled environment.

There are some other challenges which must be considered during the development and adoption of an IoT system. These challenges are Meeting customer expectations, keeping IoT hardware update, some analytics challenges, and governmental regulations.

Trust management system can be defined as an abstract system that can be used to process and evaluate the social trust of a system.

Trust can be influenced by many non-measurable and measurable properties like security, safety, goodness, reliability, strength, ability, availability, and some other characters of an entity.

Self-promotion, bad-mouthing, ballot-stuffing, opportunistic service, and on-off attacks are some important trust-related attacks.

### Multiple choice questions

What are the important components in IoT?

Hardware

Software

Verbal Exchange Infrastructure

*a* and *b* Both

Data collection presents which obstacles to the protection of information and data protection legislation?

Maintenance

Cost

Legal and restrictive

Which open source is currently deep-mined with IoT devices in each of the different digital currencies?

Monero

OpenAI

Mirai

In which attack the malicious node manipulates its name by offering sensitive recommendations?

On - off attacks (OOA)

Attacks by ballot-stuffing (BSA)

Self-promotion attacks (SPA)

Which security challenge is faced despite the amount of effort you make to strengthen your product code and equipment.

Update

Cloud Storage Security

Knowledge Security Exchange

## Answer

*d*

*c*

*a*

*c*

*a*

### Fill in the blanks

The term IoT-Internet of Things was first coined by ....., Executive Director of MIT's Auto-ID lab in the 1990s.

It is necessary to ..... whether the merchandise matches the necessary quality standard and passes them.

..... is one of the major problems, that is, a way to connect computers to the internet as well as the platform for cloud computing.

The ..... escorted transport equipment with default passwords and not telling clients to modify them when they receive them.

The data identified should be accurately processed and analyzed in a trustworthy way, while preserving confidentiality and privacy is called .....

## Answers

Kevin Ashton

Validation

Connectivity

Mirai Botnet

Data fusion and mining trust (DFMT)



### Descriptive questions

What is the Internet of Things (IoT) and what are the challenges faced in IoT devices?

Explain different types of data security issues?

Draw a diagram to show data collection and processing and the challenges of it?

Describe in detail design challenges?

Explain trust issues and trust-related attacks?

### IoT Applications Area: Emerging Application Areas of IoT

The, application of **Internet of Things (IoT)** seems to permeate in all aspect of our lives and is therefore pervasive across both our private and public life as well as in the industries and business. IoT is now going beyond traditional devices such as phones, computers and laptops to hitherto unimaginable areas such as wearables, implants, politics, and more.

In the business world, the **Industry Internet of Things (IIoT)** which is empowering industrial engineering in areas such as agriculture, health, military, constructions, and many more, holds great potential for quality control, high productivity, efficiency, and sustainability. The emerging application of IoT in our lives is now unpredictable and only subject to one's imagination.

In this chapter, we will explore the varied and the diverse applications of the IoT technology to machines to make them smart enough to reduce human labour and make our life better, safer and sustainable. We will also understand how these Smart devices or *connected devices* are designed in such a way that they can capture and utilize data on our lives and movements daily which are then subsequently analyzed and used to improve the quality and productivity both in our social lives and industries.

## Structure

In this chapter, we will discuss the following important topics of IoT applications in emerging areas:

Smart home

Health care

Agriculture

Military

Politics

Construction

Other application areas

Overview of key challenges of IoT implementation

## Objectives

At the end of this chapter, you will be able to:

Anticipate the varied emerging applications of IoT in our everyday activities and business.

Understand the extent to which the adoption of IoT technology can disruptive our day-to-day activities and in businesses.

Appreciate how IoT has been or being incorporated in some selected work areas of our lives.

## Introduction

The IoT can be defined and characterized as the application of network and computing functionalities to sensors and devices that are not designated as computers but enable machine-to-machine communication without or with minimum human input.

The application of IoT technology can, for example, make it possible for you to switch on the heating and cooling appliances in your house from a remote location either through your mobile phone or some form of computer device, most probably connected to the Internet. In this respect, you can envisage a situation where smart software installed on your phone sends and receives notifications from devices and sensors in the house that activates or disables them within pre-determined times and even warn you when things are out of range.

When objects or machines are IoT-enabled, it creates the capability to generate data for analysis and enhance the decision-making process. IoT analytics are therefore important, especially when objects that are IoT enabled also become Internet-enabled and interact via some form of network environments such as a combination of edge and cloud computing.

From a business or commercial perspective, a medical doctor could diagnose a patient remotely if the medical equipment or medical device is connected to IoT device and hooked up to a

private cloud that makes accessibility possible. It can trigger a technical support notification either for servicing or consumable replacement.

The IoT paradigm has now generated compelling economic incentives for businesses and even individuals to minimize cost, increase energy savings as well as add value and efficiency to our daily operations and routine activities through semi or full automation. The possibilities are, therefore, endless, and the only limit is your imagination.

Socially, this is already creating new services that were not possible before. It is important to note that the use of IoT technology in our every-day life activities are not merely for the fun of it but are to make life better.

At the moment, it can be said that many of us are already participating in the IoT ecosystem without knowing or being active user when we purchase items or devices that have the capability of connectivity. Examples of such equipment or devices include smartphones, smart TV sets, Internet-enable devices, and more. Once configured, these interact daily with cloud services and service providers' servers without any intervention from us. Essentially, living a connected life!

Several companies are currently involved in the application of IoT in all our spheres of life. Notable amongst these companies are Amazon, Bosch, Axeda, Google, Siemens, Honeywell, Cisco, GE, Dell, Hitachi Data Systems, Huawei, Apple, Intel, IBM, Microsoft, Oracle, and Samsung.

Finally, the IIoT which is empowering industrial engineering in areas such as health, military, agriculture, construction, entertainment, and many more, that hold great potential for quality control and sustainability will be examined.

The following topics regarding the application areas will now be presented to highlight the diverse ways that IoT are being adopted and implemented in all our sphere of lives: homes, health care, agriculture, activity monitoring, military, politics, constructions, and other applications.

### Homes (the smart home).

The implementation of IoT in our homes is by far the most popular application of IoT in the emerging areas in our lives, which is considered as *smart* home. In simple terms, it is the extension and application of IoT devices on physical items which hitherto stand alone and connect them to a central point (usually a server) for remote control or access. Thus, such connectivity makes it possible for the homeowner or user to automate both simple and complex tasks that create convenience, comfort, efficiency, and to some extent, lower cost of operations or consumption.

The number of people looking for smart homes devices keeps soaring up every day, just as the many companies going into the production and provision of IoT devices for Smart Homes. The Smart Home concept is made possible as a result of the ability of the connected devices to communicate with each other or a central node courtesy Internet connectivity. The continuous improvement in the speed of the Internet (now talking about 5G connectivity) is giving the impetus to people to create their smart home ecosystem as long as they are electronic and digital.

There are also smart home systems that have artificial intelligence and learn from your home routine. It makes your home look like something out of a sci-fi movie. Examples of smart home devices are smart thermostats, smart lights, smart plugs, smart security cameras, and smart doorbells. These are just a few.





### Understanding the smart home concept

The smart home concept is a new way of lifestyle where people to seek to control the things in their homes, most often through a smart device from a remote location (when away from home). It is all about convenience, comfort and safety at one's home.

For instance, in a situation where you leave the house forgetting to switch off your lights or even turn off the heaters/air conditioners; it could result in an expensive electric bill at the end of the month for you. The smart home concept or installation will relieve you of the burden of such an expensive mistake by automating such as tasks without your intervention once you set the right parameters of operations.

Thus, day-to-day routine tasks of turning on lights when it's dark and off when its daytime will all be done automatically. In a smart home ecosystem, your doorbell will send you a notification when there is someone at the gate, and your refrigerators can detect the type of items and their expiry dates and send you a notification for replenishment.

Above all, you can send commands to or pre-program several of your home devices to undertake a task at a certain time when you are not in the house; thus making your home comfortable and convenient for your busy working schedule.

At the heart of the smart home, the concept is your ability to connect or adapt the various appliances or devices in a stand-alone working condition to network connectivity where central control is possible. The bottom line is that once a device can be made digital or electronic, then the IoT application is possible.

### Internet of Things (IoT) in healthcare

The adoption of IoT in healthcare provides several benefits both to patients and the health care institution. To the patient, data help to improve their lives while doctors, specialists, nurses and the general staff are empowered to make better decisions and faster by leveraging their access to IoT embedded devices. Hence, reduction in both errors and costs to the benefits of the citizens and humanity in general.

In general, health care services are beginning to attract the attention of IoT developers and users. The health care services stand to benefit from the improvements in health care by way of remote monitoring and telemonitoring of patients and assets. With the adoption of IoT in health care, medical professionals can remotely diagnose, monitor and even administer treatment or conduct surgery on patients.

A recent paradigm shift in patient care where hospitals and emergency care environments are being moved to private, or patient's home is gradually gaining prominence. Citizens' awareness and demands for home and remote access to their medical needs and conditions (especially the upscale private health care services) are fast becoming popular. Such an approach is only possible through a more integrated and IoT enabled environment.

Tagging and monitoring sensitive medical equipment such as predictive maintenance of health care equipment like smart beds and medical wearable devices are also gaining attention within the health care management system. All of these are geared towards the good of patients, health care institutions, governments, and the citizenry in general.

Amongst the countries currently spearheading the adoption of IoT in the healthcare sector and leveraging health-related data from IoT devices are USA, Canada, UK, Germany, and most developed countries. As of 2017, about 30% of health organizations in the USA were using IoT for sensitive data (healthcare and cybersecurity). Research shows that by 2019, 87 percent of healthcare organizations *will have adopted IoT technology and 76 percent believe it will transform the healthcare industry* (research from Aruba March 2017). The application of IoT in healthcare is endless and sits in a vast ecosystem.

### Internet of Things (IoT) in agriculture

The IoT adoption in agriculture revolves around a system that is built for monitoring crop field with the help of sensors capturing data on light, humidity, temperature, soil moisture, and more, and automating the irrigation system.

It is thus a system of smart farming which is a capital intensive and hi-tech system geared towards growing cleaner and sustainable food for the masses. It is basically the application of modern **ICT (Information and Communication Technologies)** into agriculture.

IoT based technology of farming such as smart farming creates opportunities for farmers to improve productivity and at the same time, help reduce waste on the farm. For example, the number of farm trips (by farm vehicles) are significantly reduced as some activities are pre-programmed and continuous feedback to farmers sent automatically.

The focus of adoption of IoT-based farming is not only on the traditional, large farm operations but also strategically towards approaches that tend to lift other emerging techniques and trends in agriculture such as organic farming, family farming as well as both complex and simple small spaces farming of cattle culture, preservation of high-quality seeds and subsequently enhancing transparent farming.

The IoT-based agricultural environment positive impacts are enormous in terms of ensuring food for all for the ever-growing population of the world. For example, IoT-based smart farming can provide great benefits, including more efficient water usage or the optimization of inputs and treatments.

Some of the major applications of IoT-based smart farming include the points explained in the following sections.

### IoT and precision farming

Precision farming or agriculture is when farming practices are enabled to be more controlled and accurate with respect to growing crops and managing livestock. This is emerging farm management where IoT plays a major role in ensuring optimized output and long-term sustainability.

IoT components such as robotics, autonomous vehicles and hardware, sensors, and many more, are embedded and incorporated in the farm management ecosystem to provide data for the timely decision-making process. For example, access to high-speed internet and mobile devices by manufacturers have enabled some of the key technologies now being used in precision agriculture.

Precision agriculture is currently becoming important in agriculture with an increasing number of firms leveraging this technique around the world. For example, Cropetrix is a precision agriculture organization focusing on ultra-modern agronomic solutions while specializing in the management of precision irrigation.



### *IoT and livestock monitoring*

Monitoring the movements and the health of livestock has never been better than with the adoption of IoT in agriculture. By utilizing wireless IoT applications, farm owners can collect data regarding the location, well-being, and health of their livestock. The health status of animals can be ascertained and remedial action taken to prevent the spread of diseases amongst the flock/stock. Such an approach of incorporating IoT in agriculture has the potential to lead to lower labour cost and optimize farmer time and as well as financial resources in managing their ranches.

In North America, one company that is aggressively adopting IoT in agriculture is JNB. The company is using IoT to provide cow monitoring services where cattle owners can observe their animals that are pregnant and about to deliver.

### *IoT and agricultural drones*

Drones fitted with IoT devices are now being deployed to boost and enhance agriculture management practices. Some of the benefits of both ground and aerial-based drones irrigation, crop and livestock monitoring, crop health assessment, crop spraying and planting as well as soil field analysis. Thus, the use of drones provides real-time data that allows timely decision-making process leading to potential increases in yields.

One organization that uses drones to gather data through IoT sensors is Precision Hawk. The drones conduct in-flight observations and provide monitoring services that are valuable in mapping and surveying of farmland.

### IoT and smart greenhouses

Greenhouse farming, a technique which has the potential to enhance farm yields such as vegetables, fruits, and crops through the use of IoT embedded devices is yet another novelty in agricultural farm practices. With the help of IoT, a smart greenhouse can monitor and control the climate around the greenhouse farm, thereby eliminating the need for human attention or intervention. It is because the smart greenhouse technique employs sensors which measure different environmental indicators in response to different plant needs as these sensors are connected to cloud servers and are accessible remotely.

Because the IoT in the agriculture market is still developing, there is a lot of opportunities for businesses to innovate and set themselves apart as early adopters, and as such become very successful.

Thus, the use of IoT in greenhouse farming provides a mechanism where manual intervention is reduced to the barest minimum due to semi-automation or full automation of certain key control activities. There is a reduction of production and energy losses as well as labor costs. This design provides cost-effective and optimal solutions to farmers with minimal manual intervention.

A company that has taken the lead in smart greenhouse farming is Illuminum Greenhouses. Based on a drip installation and agri-technology, the company uses sensors to monitor the greenhouse water consumption state and sends reports (alerts) to the farmer electronically on a continuous basis as feedback for action. The installed or connected sensors can also send messages regarding actuators that open valves, doors or windows, and many more.

Notwithstanding the numerous benefits that smart greenhouse offers, there are certain challenges that one needs to be aware when considering investing in a smart greenhouse. For example, choosing or identifying the hardware requirements for your IoT network is a major constraint.

Smart farming applications should be able to give farm owners access to information on the farm wherever and whenever (mobility). Ensuring a robust infrastructure is therefore important, failing of which invites and increases security breaches.

## IoT in military application

The adoption and implementation of IoT in the military is now an area of battle readiness capability assets or competitive advantage. In the not too far future, battles will be fought from a central command room with virtually no human (soldiers) on the battleground. There is rapid adoption of wide varieties of IoT embedded devices in military equipment with both long and short-range connectivity to a central network for control purposes. Thus, the integration of sensors and actuators into existing military infrastructure, thus turning an old and near outmoded military equipment into a state-of-the-art battle-ready device or equipment is now creating several global warmongers and military superpowers.

The IoT in military applications is giving opportunities to allies and adversaries to update their warfare capabilities and thus pushing the boundaries of military exploits to their wit ends. Warfare is more complex, multi-faceted and highly unpredictable. Battles will be won not on the field but from an arm-chair.

The following sections explain some of the ways that the military can improve their efficiency and capabilities.

### Military smart bases

The adoption of IoT devices into military installations and equipment are now giving a new dimension to warfare and troops safety. Connecting certain military devices or equipment with sensors or actuators can generate both data and network access that can be used or analyzed to improve troop safety, readiness and efficiency. Thus, from a remote location (headquarters), access to military base security cameras about their movements and activities can improve safety while at the same time minimizes workforce. It further improves the capacity, output and the well-being of personnel within the base.

### Military logistics

One of the great benefits of IoT adoption in modern times is its impact on logistics and supply chain management. From logistics in agriculture through various socio-economic transactions and activities to military, healthcare, and many more, the connected sensors and the digital analytics of IoT technology provide effective and efficient management by way of tracking and near real-time consumption of materials/goods or services. Thus, the deployment of military equipment or arsenals can be tracked from the point of origin to destination with minimum human intervention.

## Military data warfare

One of the important benefits of IoT to the military is its ability to generate large data when connected to platforms that manage aircraft, weapon system, troops and ground vehicles, especially at one central point or server. Analysis of such data will equip the military or the armed forces to review or identify any potential threats ahead of time and with more accuracy for the timely decision-making process, especially during a war or actual battle.

**Adelphi, a US army Research laboratory (ARL)** firm based in *Maryland, USA*, is currently conducting studies on how smart city connectivity and infrastructure can be used on the battlefield. The firm is testing **long-range wide-area network (LoRaWAN)**, the protocol often deployed in smart cities to bring together the IoT devices and sensors in dense, urban areas.

*As reported by GCN, the overall aim is to see if smart city technologies can improve the capabilities of "Internet of Battlefield Things" (IoBT) and military applications where buildings, clustered objects, and walls might otherwise interfere with signals -- and, therefore, operations.*

IoT devices such as televisions, smart lights, mobile phones, and household items and appliances generate vast amounts of data that become available for predictive analysis and monitoring. On



the battlefield, the same principles can apply to bases, mobile gadgets, drones, and vehicles, including ships and cars.

The principle of IoBT is to harness the information generated via sensors and networks for military use, in what IEEE calls *the full realization of pervasive sensing, pervasive computing, and pervasive communication*.

Thus, warfare in cities is not the only potential use case for these experiments. By identifying and dealing with any connectivity issues between IoT and mobile devices, this may also help law enforcement, threat detection and response, as well as disaster recovery efforts. Military or army installations can be viewed or considered as cities in their own right and hence can be akin to a smart city project.

## Internet of Things (IoT) and politics

The political campaign landscape is changing rapidly, all due to the rapid intrusion and infusion of IoT. the game of politics has been played on the altar of massive political infrastructure based on polling, survey, door-to-door campaigning, and cold calls. These approaches historically have been instrumental, but in recent times, depending on these strategies to woo voters are becoming obsolete, and waste of resources as IoT technology is set on changing the dynamics between government and voters.

One of the key advantages of IoT in politics had been within the political campaign space where it provides significant opportunity to gather lots of data

In a Politico article about IoT and politics (cited by *Nathan Rockershausen*, 2016), author Phil Howard states:

*Instead of small survey samples with noticeable error margins and carefully worded questions, the device networks will generate many details about our lives — all the time and 'the room for error in data collection is significantly reduced, and the efficiency is vastly improved using big data analysis.*

-- *Phil Howard, 2015*

In 2008, the political landscape of USA was dramatically changed when *Barack Obama* took full advantage of the technological advancements in social media within his campaign bid for the Whitehouse. Even though he did not use IoT directly, the use of the social media platform was, in a way, a pointer to how IoT can be adopted. The success of his campaign with the use of online data analysis has caused the IoT to become a valuable political tool.

The emergence of IoT promises to improve many aspects of life (both public and private). This includes activities that are associated with the citizens' governance collectively termed as politics. The probing question on everyone's mind is, how can IoT change the dynamics in government-citizenry interaction or relations? And how can IoT technology impact or change the voting process in elections? And to what extent should IoT be allowed to influence the political decision-making process bearing in mind the high emotional attachment associated with politics globally and certainly more pronounced in developing countries?

*Mark* professor of practice in information systems management at the University of Warwick's, says (cited in a blogger by:

*Mobile devices and social media is already disrupting politics as seen in the 2016 U.S. presidential election and politicians are often left struggling to try to understand what the technology is doing to the national markets, its impact on taxation (and) income, and how to respond in legal and in education investment to attract and grow nations and GDP.*

-- Mark Skilton

Politicians are often worried about how to generate jobs for their constituents and significant of them are becoming wary of the effects or the growing disruptive impact of IoT by way of automation which has the potential of making some jobs obsolete resulting in how jobs are distributed and people lay-off. There is also the fear of data being misused or abused.

As experts have warned; poor planning and implementation of IoT technologies in politics could have a devastating effect. The danger of IoT in politics is further corroborated by a senior advisor at regarding voting:

*There are a variety of ways to use the Internet for good in elections, but transmitting votes is not yet one of those.*

—Pamela Smith

However, from a positive perspective, the data generated via IoT will present an opportunity for politicians to provide better services and engagement with the citizen. From the citizen's perspective, IoT that allows civic expression, create more visibility in the political process and behaviour could be beneficial to the general populace.

The IoT is diminishing the role of direct interaction between political campaigners and the electorate. Increasingly, some of the old campaign methods are being rendered useless with the Internet-enabled devices collecting massive behavioural data about

constituents. Political activists have now been transformed from an *act then observe approach*, to an *observe then act approach* (Nathan Rockershousen,

### Internet of Things (IoT) and constructions

The construction job sites and locations are becoming yet another avenue for the deployment of IoT to improve safety, efficiency, and productivity. Varied activities related to construction can be connected with sensors to generate data or provide automation to achieve optimal performance.

These may include but not limited to remote operation of construction sites, replenishment of supplies, construction tools and equipment tracking, equipment servicing and repair, remote usage monitoring, power and fuel management and savings, **Augmented Reality (AR)**, **Building Information Modeling (BIM)**, and many more.

In terms of remote operation, if machines are hooked up to the web, either physically or wirelessly, one can give instructions remotely to operators. It can operate alone in hazardous areas or environment for humans. Wearables computing such as Google Glass can help construction workers and engineers on-site to access instruction and operational manuals in hands-free mode benefit from remote support mode that makes it possible to see what is being done.

Ensuring Supply replenishment timely and in **Just-in-Time (JIT)** delivery is becoming more readily available and efficient when construction items are labelled with RFID tags or sensors that are

connected, especially to a central node. It ensures that idle is reduced significantly and thus projects have better chances of being completed on schedule. IoT makes automation possible!

Through the adoption of IoT in construction, tools and equipment are tagged and/or attached with sensors that can help project managers and supervisors to track the location and usage of such items which tend to prevent theft and pilfering, misplacement, and misuse/mislaidd.

Machines embedded with IoT devices announce their state of conditions or status for continuous remedial action. This is possible as the IoT devices allow them to transmit information for attention. This makes the Equipment and Servicing unit of any construction company to be pro-active as they can obtain advance information of the state of machines before breaking down.

According to Murphy's Law (1940) that says *Fixing machines before they break makes more sense than waiting for failure... which is all too likely to happen just at the wrong time.*

Remote Usage Monitoring is also one of the key benefits of the adoption of IoT in construction. When construction equipment is linked or connected to or with IoT device, it makes it possible for the machine to generate data when the users or machines are logged in (automatically). This ensures that machines limits and working hours are monitored (remotely), and workers or machine overuse are prevented due to fatigue and possible accidents. Opportunities for immediate correction or remedial action can be taken.

Power and Fuel savings via IoT adoption in construction industry are also important in ensuring a well-managed construction project and companies. Such embedded devices send information like electrical power usage and idling time (which uses fuel) so that lighting hours and off periods adjusted without impacting negatively on project time and delivery of construction machines.

IoT devices in construction also contribute towards BIM. This is a model used to direct real-life construction when sensors are placed strategically to provide update on continuous basis with feedback on the way materials on site are affected by climates and time. BIM can provide data which can lead to a good understanding of issues such as changes in energy efficiency in say roofing, how structures behave under earth tremors and earthquakes or how bridges react under the might weight of passing traffic.



### Internet of Things (IoT) and other application areas

The adoption of IoT technology is now permeating in all aspects of our life to the extent that, people are now talking or even experimenting embedding chips under our skin or brains to generate data for all sorts of analysis.

In addition to the various areas mentioned above, IoT adoption are being contemplated in areas such as wearables, smart grids, IIoT, connected car, connected health (digital health, telehealth, telemedicine), smart retail, smart supply chain, and many more.

### *Brief overview of key challenges of IoT implementation*

In as much as the IoT technology provides huge benefits and probably limitless possibilities of its application in our day to day lives and activities, so does it also come along with its extra baggage of negative externalities, disruptions and complex legalities both at individual (private) and group levels as well as business and commercial/industrial limits. Below sections are some of the key areas that continue to present challenges to the wide-scale, effective and rapid adoption of the IoT technology globally.

## Security

The main purpose of IoT is to connect devices to each other via the Internet. However, as more devices are connected, they tend to pose potential risk and become subject to digital burglary, data breaches and malware attacks. The insatiable drive of hackers to penetrate the network either to disrupt service or steal data has become the bane of IoT (Internet) enthusiasts and business chief executives. This is therefore forcing the corporate world to spend significant amount of money in combating the security threat and so industries leveraging modern technologies, such as biometrics or cryptography to overcome the threat. The challenge with IoT devices, of course, is that they're part of a network. If an attacker gets a foothold, IoT architectures tend to make it easy for them to penetrate other connected devices.

### Network connectivity

For a successful IoT deployment, there is the need for strong and reliable network connectivity. Unfortunately, there are several places around the globe without Internet connectivity, even in USA. The situation is dire in developing countries, more especially in Africa. No or poor Internet connectivity is one of the many key barriers faced by industries in IoT deployment.

## Infrastructure

IoT is significantly associated with massive data collection and storage and in view of the fact that most users require near real-time data for effective decision-making process. This means that firms must find and incorporate a sustainable data storage solution for the daily massive data generation by the IoT devices deployed. Companies should switch from centralized data storage to a distributed cloud platform. Besides, companies should ensure that they have the right infrastructure that manages IoT connected devices, sensors, and the data from it.

### IoT suppliers/vendors

Currently, most firms outsource their IoT solutions to third parties. Therefore, identifying and choosing the right vendor for your IoT solutions is very important to the successful adoption. In the light of this, there is the need to choose a robust IoT platform that is capable of high scalability and optimization and as well as having disaster recovery without compromising efficiency and precision.

Every IoT application is built on an ecosystem of hardware, software, network connectivity, and services that turns data into intelligence. No single technology or service provider can supply all the necessary building blocks.

### IT professionals and experts

The key characteristics of IoT are that it gathers large amount of data on daily basis. This means that, for industries to derive meaningful and actionable insights from IoT data for informed decisions, there is the need for the right analytics (tools and skills). However, the presence of large pool of skilled analysts is difficult and expensive to come by.

Like any other IoT embedded devices system, there is the need for high quality data analysts to make sense of the huge data collected. One needs a powerful data analytic capabilities and application of predictive algorithms and machine learning to derive good insights on the data collected. Hiring the right talent with the right niche skills is important for IoT adoption and success.

## Conclusion

It is now evident that the application or the adoption of IoT in our day-to-day activities is taking new dimensions which had never been thought of before. IoT is becoming pervasive in all aspects of our life and the only limit to its applicability is our imagination. Thus, emerging areas such as agriculture, health, military, and many more are becoming game changers.

However, the application and adoption of IoT come with its own cost and disadvantages. These challenges which range from the cost of hiring skilled IT professionals, issues of choosing the right IoT providers or suppliers (vendors), privacy and security breaches to ensuring robust network connectivity, companies are unperturbed and undaunted in their efforts to adopt IoT.

In conclusion, it can be said that IoT has been around for some time now albeit slow, it has come to stay. Despite the teething problems in its massive deployment, most of the technical problems will be resolved within the next few years and will thus pave the way for its massive uptake and adoption. The world must get ready for what will be known as the IoET!



### Case studies

The following case study of Smart Home written by Kelvin Wireko, (a blogger and technology consumer guide advocate) is presented below to illustrate how IoT is being used in our home setup for comfort, convenience and safety.

### What is smart home technology? How does it work?

Now that you understand the concept behind smart homes. The next thing is to understand how smart home technology works. A basic understanding is important if you plan on turning your home into a smart home. Even so, it is still not enough. Because in some cases you might have to call an expert to do the installation for you. Smart home technology is all about linking the smart devices together where they can be controlled and monitored from a centralized point which can be your smartphone, computer or tablet.

Think of it as a hive mind. Where a queen ant sends out commands to other ants for them to carry it out. In this case, the queen is the centralized point and the ants are the smart home devices. Smart home devices are connected to a smart hub which is then connected to your smartphone. A command is sent from your smartphone to the smart hub. The smart hub then receives the command and sends it to the smart home devices.

And all that makes it possible is the internet. It is through the internet that the smart devices, smart hub and smartphone can communicate with each other.

And for it to work flawlessly, you should always have a fast internet connection at all times. Plus, you should make sure that every one of the devices is connected to the same internet

network. The best part about smart home technology is that you can control and monitor the smart devices remotely.

### What is a smart home system?

Without a smart home system, you can't make use of smart home technology. Because you need a point where all the devices are monitored and controlled.

An army is useless without its general. And so the smart home system or smart hub serves as the general of the army of smart home devices or you could say the brain.

A command centre of sorts where you send commands from a centralized point, that is, your smartphone as explained earlier. It's thanks to the smart home system that you're able to enjoy the convenience of owning a smart home.

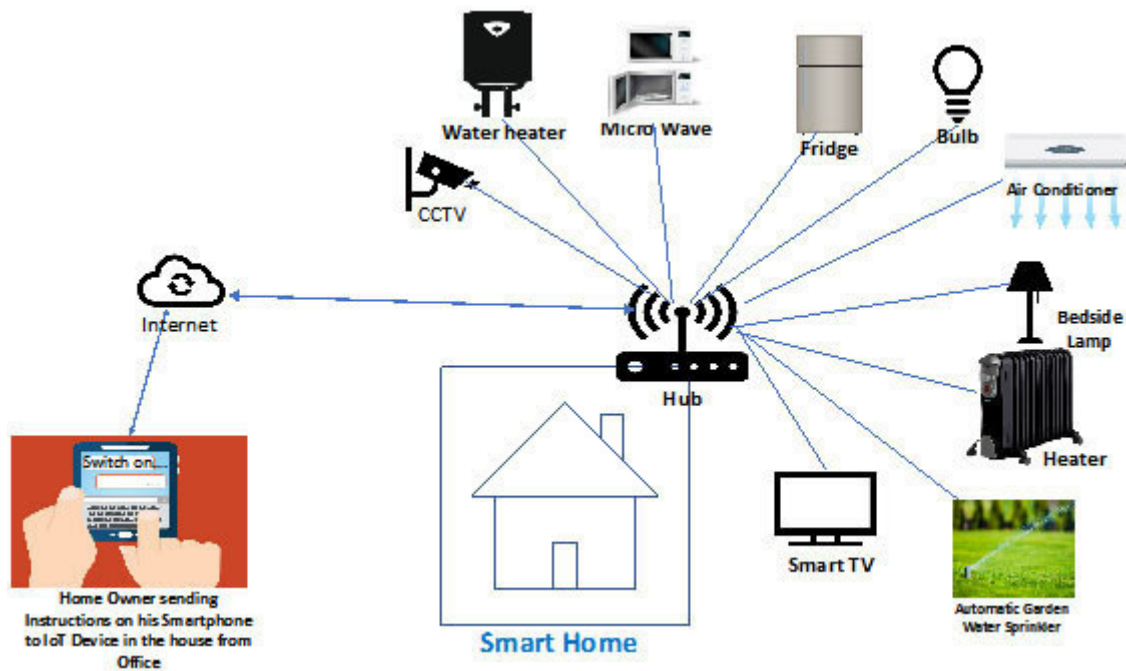
Unfortunately, there are different types of smart home systems due to different manufacturers. This causes incompatibility and integration issues if you don't do your research well.

For smart home systems and devices, compatibility and integration are important. Because without them there is no way the smart home devices can be integrated into the system and work together.

### How does smart home automation work?

Smart home automation or domotics is simply programming your smart devices to undertake certain tasks instead of doing it yourself. This is where making life easier and manageable comes in. It is easier to manage a lot of household tasks, and you save time in the long run. These tasks are done at a faster rate and do not require a lot of effort on your part. Plus, you don't have to hire a maid to do all this. [Figure 6.1](#) below shows some of the daily tasks that we do at home, pretty much on a routine basis that a smart home can do for us if you can integrate IoT devices in your home appliances.

Smart home automation is not limited to indoor home tasks but can be extended to your garden or patio. For example, you are opening your garage door when you pull up or a plant watering kit that waters your flowers at certain times. Some smart home devices are equipped with artificial intelligence that observes your habits. The time you wake up, when you switch on the lights, when you vacuum the living room, restocking of food items, and many more:



**Figure 6.1:** Illustration of a typical Smart Home System

The smart home device observes all these and after a certain time will start to automate these tasks for you. All without you programming the devices or issuing commands. Isn't that great? And these are just a few examples. And because these tasks are executed and done at the right time. It leads to better energy savings and a lower utility bill. Moreover, because more appliances are becoming smart. You can automate the whole house.

### Objective of a smart home

The objective of smart homes is to be able to automate everything in the home and make life easier for the homeowner. Some smart home devices are equipped with artificial intelligence that observes your habits. The time you wake up, when you switch on the lights, when you vacuum the living room, restocking of food items etc. It observes all these and after a certain time will start to automate these tasks for you. All without you programming the devices or issuing commands. Isn't that great? And these are just a few examples. And because these tasks are executed and done at the right time. It leads to better energy savings and a lower utility bill. Moreover, because more appliances are becoming smart. You can automate the whole house.

### Smart home features and functions

There are a lot of features and functions that come with owning a smart home. But, they all fall under major groups. And these groups are what can be automated. They are lighting, security, entertainment, comfort and convenience.



## Lighting

More often than not lighting is the first thing homeowners automate and a popular feature of smart homes. It's the cheapest and easiest to start with when you want to turn your home into a smart home.

The lighting system can either be programmed, voice-controlled or activated through motion detectors. For example, you might step into the living room and the lights turn on automatically. Or you might shout a command, *Alexa, turn on the lights.*

You can also program the lights to turn on or off at certain times in the day. Some lights, for example, the *Philips Hue* can be customized to show different colors. This adds even more customization depending on your mood or tastes.

## Security and safety.

Another great feature and function of a smart home is security. Smart security cameras, smart doorbells, smart alarm systems and smart door locks improve the security and safety of your home.

You can view your home remotely to see what's going on thanks to the smart security camera. Some smart doorbells also come with cameras that automatically start recording when someone is at the door. Smart door locks allow you to lock and unlock your doors from anywhere. And have voice control features.

There are also security modes you can set when you are in or out of your home. This is particularly useful when you have taken a vacation or just leaving your home for a few hours. By enabling the smart security mode, if there is an attempted robbery.

Your smart home can sound an alarm; send a notification alerting you of what's happening while recording what is going on. There are also smart smoke detectors that inform you when it detects a small amount of carbon monoxide.

## Entertainment

You might not believe it. But a smart home takes your entertainment to a whole other level. Now with smart TVs and smart speakers.

You can stream your favorite movies and music. Subscription apps on smart TVs such as Netflix, Hulu, HBO NOW, etc. allow you to stream your favorite movies. Smart TVs and smart speakers also allow you to stream your favorite music from Spotify, SoundCloud, TuneIn, and many more. Then there are the smart home theatres which have some smart TV features and can turn your home into a theatre.

## Comfort

A home wouldn't be a home if it wasn't comfortable. And one of the things that make a home comfortable is the temperature.

Thermostats allow you to regulate the temperature of your home. And there will be times where you forgot to regulate the temperature. Smart thermostats can do this for you automatically. It's able to keep an eye on the weather and adjust the home temperature to comfortable levels.

TheNest smart thermostat does this well. And it even has the artificial intelligence that is to learn what temperatures you like.

There are also smart air conditioners that you can remotely turn on or off. This allows you to put your home at a comfortable temperature before you even get home.

## Convenience

There is nothing more convenient than automating most if not all tasks in your home. And the opening the garage door is one of the things you quickly like to automate. The smart garage door openers have sensors or mobile apps that can automatically open the garage door when you pull up to your home.

There are also smart vacuum cleaners and mops that clean your home around the clock. This keeps your home clean at all times. There are also smart coffee makers that you can program to make coffee for you at certain times. Most if not all are self-cleaning.

These are just a few of several tasks your smart home can do for you.

## Why is a smart home needed? Benefits of a smart home

Do you need to turn your home into a smart home? It depends entirely on you. If you are looking to save money, improve your security, comfort and convenience. Then you can see why smart homes are needed. The benefits of a smart home a lot. Here are a few.

Utility bills have steadily been increasing throughout the years. And one mistake of leaving an electronic appliance on can cost you. By automating a lot of tasks in the house. You can save money when it comes to utility bills.

Your smart devices execute tasks precisely leading to efficiency and savings. You can turn off all devices from your smartphone if not in use. Or the smart device can turn itself off if it has artificial intelligence.

Moreover, if you can view your family or home remotely if you are on a vacation. The security of the home is improved giving you peace of mind when you are away. For the elderly, having a smart home is important, especially if they live alone. Although they might still need someone to look after them. It will still make things easier.

Since most of the tasks are automated, add in smart home security that can detect accidents in the home. It becomes a safe

environment for elders. Although the smart home comes with a lot of benefits, they certainly have their disadvantages.

## How much does a smart home cost?

Smart homes are cheaper than before, but that does not mean they aren't expensive. The cost varies depending on how far and complex you want the home automation to be. It can be anywhere from twenty dollars to thousands of dollars. [Table 6.1](#) below is a cost estimate of a smart home (IoT) devices to expect as you gradually build your smart home:

[illegible]



**Table 6.1:** *Estimates of IoT (smart) devices for a smart home system*

\*These are rough estimates, not absolutes since the pricing of the device depends entirely on the manufacturer and reseller.

Not to mention you need a 24/7 fast internet connection. Plus, if you have no experience in the installation of a smart home system. You have to get an expert. Which is another cost altogether. Here are rough estimates of smart home devices to give you a fair idea.

It can cost you somewhere between USD464.99 – 7041.99. And those are just a few of the devices. Majority of them were left out. For example, the smart refrigerator, smart air-condition, smart coffee maker, and many more.

These are just rough estimates and depend on the price of the device, installation and how far you want to customize your home. It's not cheap but it is worth it. Because it increases the property value of your home while making it easier and comfortable.

### *Are smart homes a smart idea?*

Yes, smart homes are not only a smart idea but a good idea too. If you are looking to increase the comfort, security and convenience of your home. Then turning your home into a smart home is not a bad idea at all.

Permission by author: Kelvin Wireko (blogger and consumer guide advocate, 2019)-

### Multiple choice questions

What is 'IIoT'? (Choose one)

Internet of International organization Technology

Internet of Inter organization Team

Industry Internet of Things

Industry International of Things

Intra Internet of Technology

Which of these can be considered as a 'Smart Home?' (Choose all that is applicable)

Refrigerators that can detect type of items, expiry date and send notification to you

Sending messages to your device at home to adjust the heaters or air conditioners at a certain time before you get home

When you call somebody in your house to switch on the gate light or outside lights before you get home

Getting automatic notification when someone is at your gate/door

When your roommate switches on the TV and sends your notification that your favorite program has started.

Which of these areas do not lend itself to IoT implementation?

Health

Politics

Agriculture

Military

None of the above

Every IoT application is built on an ecosystem of (Choose all that is applicable)

Hardware

Software

Network connectivity

Software and Connectivity only

All of the above

'IoT', also a type of Internet of Things stands for which of the following:

Internet of Body Things

Internet of Battlefield Things

Internet of Biological Things

Internet of Business Things

All of the Above

## Answers

*c*

*a, b and d*

*e*

*a, b and c*

*b*

### Fill in the blanks

Four of the key challenges facing IoT implementation are 1.  
..... 2. .... 3. .... 4.  
.....

Like any other IoT embedded devices system, there is the need  
for high quality data ..... to make sense of the huge  
data collected

The Internet of Things (IoT) application in military become  
possible by integrating systems of sensors, actuators, and control  
systems into existing military infrastructures. Name three of such  
infrastructures that IoT can be incorporated: 1. ....  
2. .... 3. ....

..... farming or agriculture is when farming practices  
are enabled to be more controlled and accurate with respect to  
growing crops and managing livestock.

The principle of ..... is to harness the information  
generated via sensors and networks for military in what is  
considered as "the full realization of pervasive sensing, pervasive  
computing, and pervasive communication."

## Answers

Any four of: Security, Network Connectivity, Infrastructure, IoT Suppliers/Vendors & IT professionals/Experts

Analysts

Military Data War Fare, Military Smart Bases & Military Logistics

Precision

IoT



### Descriptive questions

Why do you think privacy and security are key challenges in IoT implementation especially within the health sector?

Without network connectivity, IoT adoption will have no relevance. Discuss this assertion supporting it with relevant examples.

Briefly describe how IoT in agriculture and IoT in homes can be adopted to improve the life of citizens in developing countries.

Show how IoT can be used for:

Remote operations in construction

Logistics supply in military

political campaigns

Discuss some of the possible cultural practices in your country that militate against the quickuptake and adoption of IoT in the socio-economic life of the people.

## Reference

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O'Maley, D. (2016). The Internet of Things. Journal of Democracy, (3), 176-80.

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Integration of Cloud and IoT

*Today, computing mainly automates things for you, but when we connect all these things, you can truly start assisting people in a more meaningful way. If I go and pick up my kids, it would be good for my car to be aware that my kids have entered the car and change the music to something appropriate for them.*

*-- Sundar Pichai, CEO of Google*

After studying this chapter, you should be able to understand the difference between cloud computing and **Internet of Things (IoT)**, The Integration of both technologies (cloud computing and IoT) which is likely to be call **CloudIoT** pronounced *Cloud IoT*, The significance and challenges face with it application and adoption in the real world, The application of the technology, and the architectural model of CloudIoT.

## Structure

Introduction

Cloud

Internet of Things (IoT)

Difference between cloud computing and IoT

Integration of CloudIoT-the role of cloud computing in the IoT

CloudIoT architecture: service and deployment model

Stages, administrations, and research ventures

Challenges of CloudIoT

Future scope

Conclusion

## Objective

Cloud computing and IoT are two different movements in technology that are both considered part of our everyday life. Their assurance and usage are essential to be logically unavoidable, which make them basic parts of the future internet. An impressive point of view where cloud and IoT are consolidated is predicted as perilous and as a drawing in authority of perpetual application conditions.

## Introduction

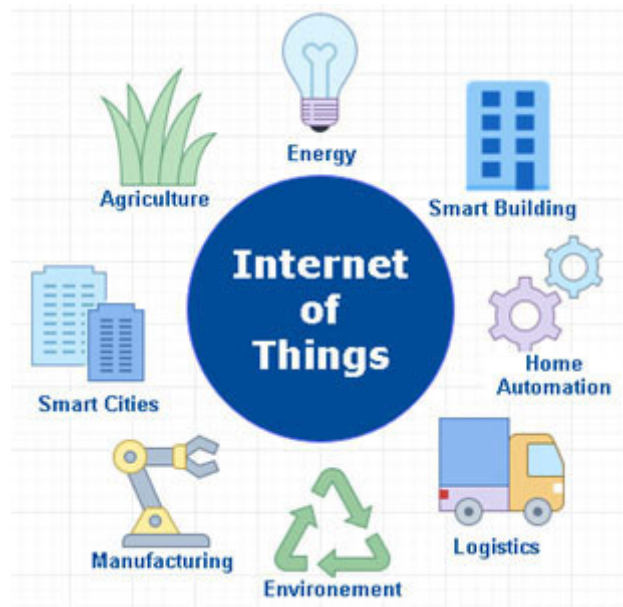
The IoT includes the web associated gadgets use to play out the procedures and administrations that help our lifestyle. Another part set to help IoT succeed cloud computing, which goes about as a kind of front end. Cloud computing is an inexorably famous administration that offers a few focal points to IoT and depends on the idea of enabling clients to perform typical processing activities utilizing administrations conveyed totally over the internet.

A specialist may need to complete a noteworthy venture that must be submitted to a chief, however maybe they experience issues with memory or space imperatives on their figuring gadget.

Memory and space requirements can be limited if an application is rather facilitated on the internet. The laborer can utilize a cloud computing technology to complete their work because the information is overseen remotely by a server. Another example could be that you have an issue with your iPhone, and you have to reset it or reinstall the working framework. You can utilize Apple iCloud to back-up your data from the phone and later restore your backup setting from your iCloud storage.

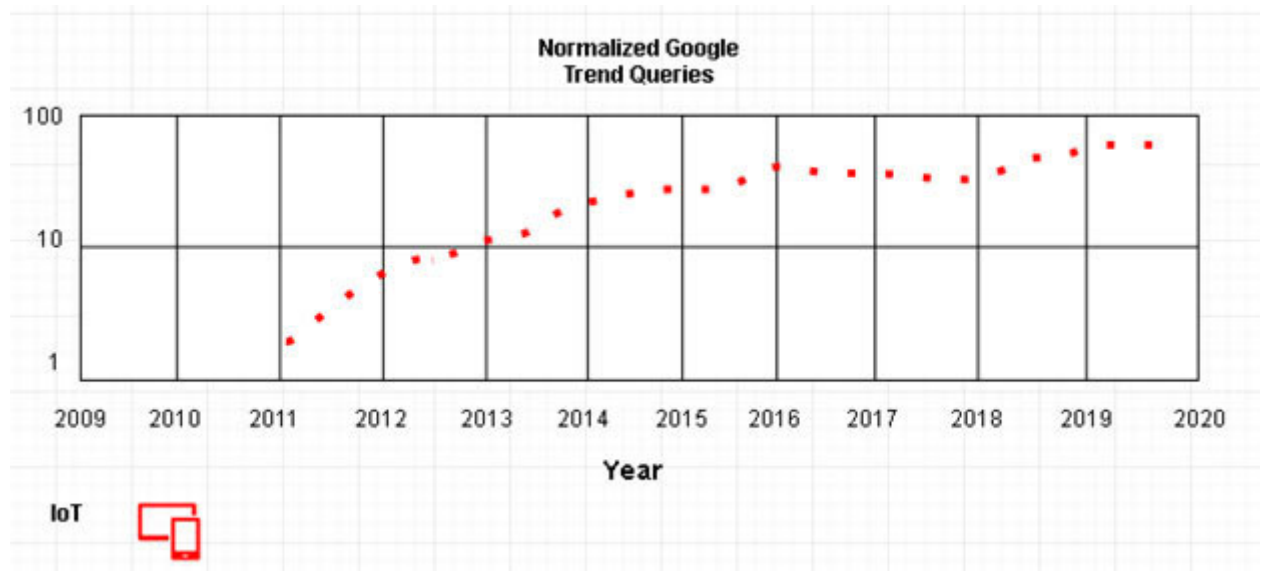
It means these associated gadgets sense parts of this present reality, for example, temperature, lighting, the nearness or nonappearance of individuals or objects, and so on, and afterward, they report that certifiable information or follow up on it. They can transmit data, and even sense, screen, and respond to human conduct. [Figure 7.1](#) below illustrates a diagram of IoT. This

diagram explains about devices that are used to connect to the internet, and how we utilized them for our everyday life:



**Figure 7.1:** Hardware devices and equipment that makes up IoT

As figures assembled by Google appear, the development of the IoT has been developing throughout previous couple of years and is set to soar inside the following year or two, with the two organizations and customers adjusting to shrewd innovation on an enormous scale (See [Figure](#) This diagram shows the research trends since 2011:



**Figure 7.2:** Google research trends



### Concept of cloud computing and IoT

In truth, cloud computing and IoT are firmly coupled. The development of IoT and the quick improvement of related advances make an extensive connection of Things. This has led to the generation of a lot of information, which needs to be kept, process, and circulated. Cloud computing as a paradigm for big data stockpiling and investigation. While IoT is energizing without anyone else, the genuine development will originate from consolidating it with cloud computing.

The blending of cloud computing and IoT will empower new observing administrations and incredible preparation of tactile information streams. For instance, tactile information can be transferred and put away with cloud computing, later to be utilized wisely for smart checking and incitation with other brilliant gadgets. Eventually, the objective is to have the option to change information to knowledge and drive beneficial, smart activity from those bits of knowledge. The cloud successfully fills in as the mind to improved basic leadership and streamlined web-based cooperation. Be that as it may, when IoT meets the cloud, new difficulties emerge. There is a pressing requirement for novel system models that consistently incorporate them. The basic worries during mix are the nature of administration QoS and nature of experience QoE, just as information security, protection, and unwavering quality. The virtual foundation for applied mobile computing and useful, versatile processing, and interfacing

incorporates coordinating applications, storage gadgets, checking gadgets, perception stages, examination apparatuses, and customer conveyance. Cloud computing offers a viable utility-based model that will empower organizations and clients to get to applications on interest whenever and from anyplace.

## Characteristics of cloud computing and IoT

The following are the characteristics of cloud computing and IoT:

Cloud computing of IoT is an on-request self-administration, which means it's there when you need it. Cloud computing is an online administration that can be gotten to with no exceptional help or consent from other individuals; in any case, you need at least a type of web access.

The cloud computing of IoT includes expansive system get to, which means it offers a few network alternatives. Cloud computing assets can be gotten through a wide assortment of associated web gadgets, for example, tablets, cell phones, and workstations. This degree of comfort implies clients can get to those assets in a wide assortment of habits, even from more seasoned gadgets. Once more, however, this highlights the requirement for an internet access system.

Cloud computing takes into account asset pooling, which means data can be imparted to the individuals who know where and how (have consented) to get to the asset, whenever and anyplace. This loan to more extensive coordinated effort or closer associations with different clients. From an IoT point of view, similarly, as I can without much of a stretch relegate an IP address to each *thing* on the planet, I can share the *address* of the cloud-based ensured and put away data with others and pool resources.

Cloud computing highlights fast versatility, which means clients can promptly scale the support of their needs. You can without much of a stretch and rapidly alter your product arrangement, include or expel clients, increment extra room, and so on. This trademark will further enable IoT by giving versatile processing force, stockpiling, and networking.

The Cloud computing of IoT is a deliberate administration, which means you get what you pay for. Suppliers can without much of a stretch measure utilization insight, for example, stockpiling, handling, transmission capacity, and dynamic client accounts inside your cloud occasion. This compensation **pay per use (PPU)** model methods your costs scale with your use. In IoT terms, it's practically identical to the consistently developing system of physical articles that component an IP address for internet network, and the correspondence that happens between these items and other internet empowered gadgets and frameworks; simply like your cloud administration, the administration rates for that IoT foundation may likewise scale with use.

## Cloud

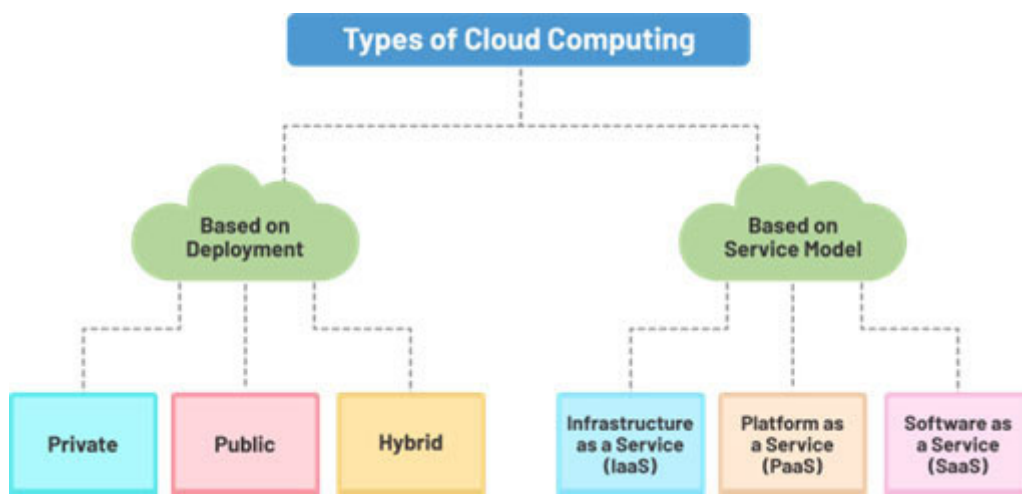
The basic parts of cloud figuring have been accounted for in the definition given by the **National Institute of Standard and Technologies (NIST)**: Cloud registering is a model for empowering universal, advantageous, on-request system access to a common pool of configurable processing assets (systems, servers, stockpiling, applications, and administrations) that can be quickly provisioned and discharged with insignificant administration exertion or specialist organization communication. Although the principle thought behind cloud figuring was not new, the term began to pick up prominence after that Google's *Chief Eric Shmidt* utilized it in 2006, and in the course of the most recent couple of years, the presence of cloud registering has enormously affected IT industry.

The accessibility of for all intents and purposes boundless stockpiling and handling abilities requiring little to no effort empowered the acknowledgement of another registering model, in which virtualized assets can be rented in an on-request style, being given as general utilities. Huge organizations (like Amazon, Google, Facebook, and so on.) broadly embraced this worldview for conveying administrations over the Internet, increasing both practical and specialized advantages.

Cloud computing is a troublesome innovation with significant ramifications for the conveyance of Internet benefits just as for the IT part in general. Be that as it may, a few specialized and business-related issues are yet unsolved. Explicit issues have been recognized for each assistance model, which are, for the most part, identified with security (for example, information security and uprightness, organize security), protection (for example, information secrecy), and administration level understandings, which could drive off a piece of potential clients. Besides, the absence of standard APIs anticipates clients to effortlessly concentrate code and information from a site to keep running on another. More when all is said in done, the re-appropriating framework to a cloud supplier, open cloud clients are fundamentally presented to cost builds, unwavering quality issues, or even to suppliers leaving the business.

## Types of cloud models

Cloud computing models can be categorized into two types, as shown in the figure:



**Figure 7.3:** *Types of cloud computing*

Cloud-based deployment is mainly categorized into three types that are explained in the following sections.

### Public (external) cloud

Public cloud, also known as the external cloud. The data and application in the public cloud are created and retained on third-party servers for general public access and use.

Public cloud subscriber (consumer) is offered a *pay-as-you-go* fine-grained utility pricing model, ensuring that billing is made only for services used and for the period of the usage. In other words, the utility pricing model ensures that the consumer pays only for the number of shared resources used within a stipulated amount of time. This utility pricing model makes the public cloud very economical and easy to use.

Examples of public cloud include Rackspace, Amazon Elastic Compute Cloud (EC2), Sun Cloud, IBM's Blue Cloud, Windows Azure Services Platform, and so on.

The basic characteristics of the public cloud are:

Resources are homogenous.

Operates a common administrative control and privacy policies.

Multi-tenancy and shared resources.



Leased or rented cloud resources.

Economies of scale.

### Private/internal/corporate cloud

The private cloud can also be described as an internal cloud. Private cloud only permits limited access because its highly-virtualized data center architectures are housed by the organization's firewall making it less risky. In the same vein, it also allows consumers to define and set their preferred customized privacy policies and security requirements. These features make it far more secure than the public cloud. However, unlike the economic public cloud, private cloud is capital intensive as it requires some capital to buy, build, and manage.

Examples of private cloud include **Simple Storage Service (S3)**, Red Hat, SaMCloudBOX, and Amazon's **Elastic Compute Cloud (EC2)**.

The general characteristics of private clouds are as follows:

Self-service resource provisioning and compute capability.

Automated and well-managed virtualized environments.

Optimized computing resources and servers' utilization.

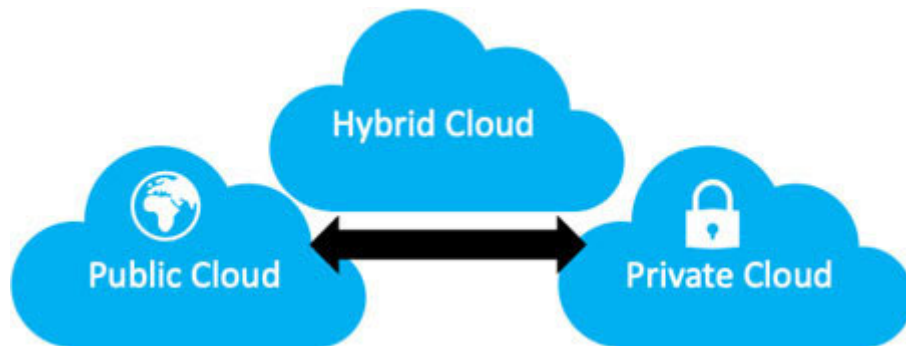
Support specific workloads.

Heterogeneous and dedicated infrastructure.

Customized policies and end-to-end control.

### Hybrid cloud

A hybrid cloud environment integrates the operational characteristics of two or more public and private clouds such that some parts of the organization's resources are partially hosted within the organization and the rest externally in a public cloud. The integration follows a standardized technology that ensures application and data portability. However, with this deployment model, the private cloud could be used to process sensitive (critical) cloud services while other auxiliary cloud service tasks are outsourced to the public cloud. It is often employed for backup purposes and record-keeping due to its high scalability and cost-effectiveness:



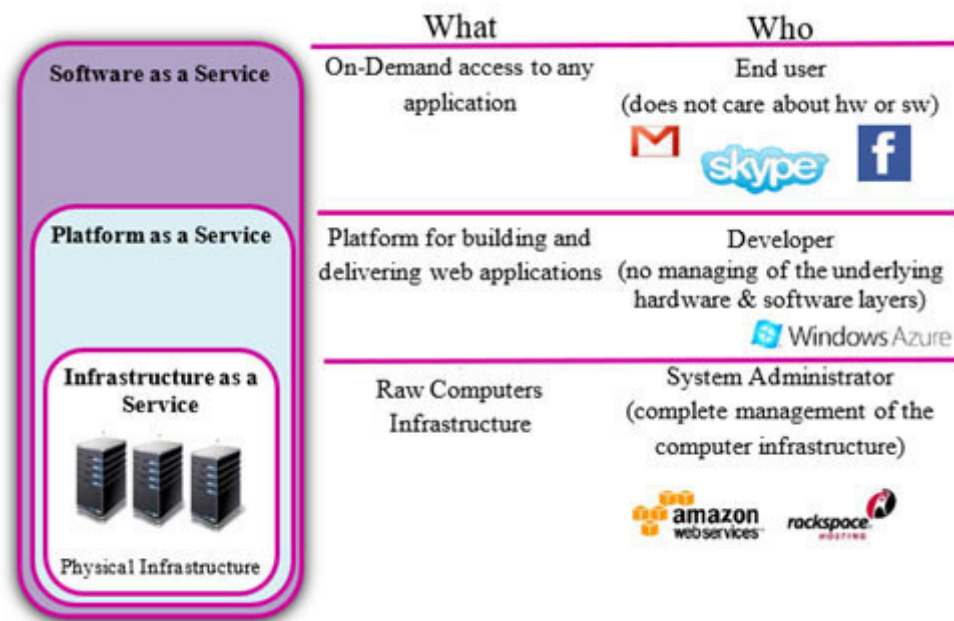
**Figure 7.4:** Cloud types based on deployment

The comparative analysis of cloud deployment models is given below:

below: below:



The service models can be viewed as presented in [Figure](#)



**Figure 7.5:** Types of cloud services

## Infrastructure as a Service (IaaS)

**Infrastructure as a service (IaaS)** is a service-oriented and secure enterprise-level computing infrastructure, including the compute resources and the storage, which can be efficiently scaled and managed to assist organizations in meeting their different business objectives. Technically, it follows from the concept of the virtual private server.

The computing resources (including the CPU hours, bandwidth, storage space, and so on) and infrastructure (data transfers, virtual servers, storage, computing, content distribution networks, and so on) are metered based on a pay-as-you-go utility pricing model.

For example, IaaS cloud storage offers consumers scalable online storage access. Virtual server instances having unique internet protocol addresses and storage blocks are also similarly offered on demand. The unique flexibility of IaaS allows subscribers to use the **application program interface (API)** of service providers to initiate, stop, access, adjust, and setup storage and virtual servers as required.

Examples of IaaS cloud include Eucalyptus, Nimbus, Amazon EC2, S3, and Rackspace.

It is important to note that IaaS can be public, private, and hybrid.

Advantages of IaaS include reliability, almost limitless computing power and storage, cost reduction, business agility, scalability, and privacy.

The characteristics of IaaS are as follows:

Resources are provisioned as a service.

Availability is based on a utility-based pricing model.

Scaling of resources is dynamic and on-demand.

Supports multiple users to concurrently access a single piece of hardware.



### Platform as a Service (PaaS).

PaaS is a product development-oriented cloud service model which provides subscribers (developers) with a virtualized set of the partial or full integrated development environment and tools to create, develop, execute, test, and deploy new applications directly in the cloud with a particular programming language via the internet as a service. It has associated tools and services facilitating efficient coding and deployment of completed applications. The developed applications using PaaS are provisioned as SaaS to the end-users via the internet. With this model, absolute control and maintenance of the key cloud infrastructure, especially the operating systems, storage, and the server, resides with a service provider and not the consumer. With PaaS, collaborative projects can be jointly developed by several members of a project team irrespective of their wide geographical proximities.

Examples of PaaS are Apache's Stratos, Microsoft Azure, IBM BlueMix, Force.com, and GoogleApps.

PaaS is characterized by failover and security capabilities, high server speed, load balancing, availability, uniform authentication, robust scalability, automated backups creation, and high storage capabilities.

The benefits of PaaS are as follows:

**Implements standardization:** PaaS seamlessly allows the developers and the technical operations professionals to access the same services on the same platform.

**Provides ease of service provisioning:** A PaaS offers the easy distribution of development tools and repository services, thereby eliminating interoperability challenges that may be associated with non-standard environments. Consequently, reduced errors, high consistency, and improved efficiency often characterize the development of life cycle management processes. PaaS also offers easy provisioning of scaling applications and runtime services.

## Software as a Service (SaaS).

SaaS is typically the functional layer of the cloud service model. Via a multitenant architecture, SaaS grants multiple users remote and exclusive access to a single application through a web browser.

To the consumer, the SaaS model is more or less a web-based application platform with an interface to present and deliver hosted software applications and services on the internet, which are accessible via a web browser. With SaaS, consumers are relieved of capital investment in servers, software licensing, upgrading, or maintaining software applications on their local systems. However, running and maintenance of the computing resources, the operating system, and the application software are the responsibilities of the cloud service provider.

Characteristics of SaaS include configurability, scalability, high efficiency, and multitenant architecture, among others. Multitenant architecture enables a single software application to be delivered through a web browser to numerous users.

Examples of SaaS hosted applications include Microsoft Office 365, Google Docs, Salesforce, NetSuite, Hotmail, Gmail, WebEx, and Microsoft LiveMeeting, which can be accessed by the client using devices such as smartphones, iPads, and laptops.

In particular, Google docs provide users with interfaces to create, edit, delete and share their documents, presentations, or spreadsheets with anyone while Google retains the responsibility to ensure that the software and hardware supporting the applications are routinely maintained.

The characteristics of SaaS are as follows:

Commercial software gets centralized web-based access.

It provides flexibilities that make the entire business process shift to the cloud.

No burden of software patches and upgrades of some sort.

Integration with different applications is possible via application programming interfaces.

SaaS applications are offered as a service on-demand at no charge via subscription or billed following the *pay-as-you-go* utility pricing model.

### Internet of Things (IoT).

The following wave in the time of registering is anticipated to be outside the domain of customary work area. By this perception, a novel worldview called the Internet of Things quickly made strides over the most recent couple of years. IoT alludes to an overall system of interconnected articles extraordinarily addressable, in light of standard correspondence conventions whose purpose of combination is the Internet. The essential thought behind it is the inescapable nearness around individuals of things, ready to gauge, surmise, comprehend, and even alter the earth. IoT is filled by the ongoing advances of an assortment of gadgets and correspondence innovations, yet things incorporated into IoT are intricate gadgets, for example, cell phones; however they additionally contain regular items, for example, nourishment, dress, furnishings, paper, tourist spots, landmarks, and more.

These objects, acting as sensors or actuators, are ready to associate with one another to arrive at a shared objective. The key component in IoT is, without uncertainty, its effect on the everyday life of potential clients.

IoT has momentous impacts both in work and home situations, where it can assume a main job in the following future (helped living, domestics, e-well-being, shrewd transportation, and so on.). Significant results are additionally expected for business (for

example, strategic, modern computerization, transportation of products, security, and so on.). As indicated by these contemplations, in 2008 IoT has been accounted for by the US National Intelligence Council as one of the six innovations with a potential effect on US interests towards 2025. In reality, in 2011 the number of interconnected gadgets surpassed the number of individuals. In 2012, the number of interconnected gadgets was assessed to be 9 billion, and it was relied upon to arrive at the estimation of 24 billion by 2020. Such numbers propose that IoT will be one of the principle wellsprings of enormous information. In the accompanying, we portray a couple of significant perspectives identified with IoT.

**Radio-Frequency Identification (RFID):** In the IoT situation, a key job is played by RFID frameworks, made out of at least one reader and a few labels. These innovations help in programmed recognizable proof of anything they are connected to and enable items to be doled out one of a kind computerized personality, to be coordinated into a system, and to be related with advanced data and administrations. In a run of the mill use situation, the reader triggers the label transmission by creating a suitable sign, questioning for conceivable nearness of items extraordinarily distinguished by labels. RFID labels are typically detached (they don't require onboard power supply), yet there are additional labels controlled from batteries.

**Remote sensor systems (RSNs):** Another key part in IoT conditions is spoken to by sensor systems. For instance, they can collaborate with RFID frameworks to all the more likely to track the status of things, getting data about position, development, temperature, and

so forth. Sensor systems are ordinarily made out of a conceivably high number of detecting hubs, conveying in a remote multi-bounce design. Exceptional hubs (sinks) are generally utilized to accumulate results.

RSNs may give different valuable information and are being used in a few regions like social insurance, government and ecological administrations (cataclysmic event alleviation), resistance (military objective following and observation), risky condition investigation, seismic detecting, and so on. Nonetheless, sensor systems need to confront numerous issues concerning their correspondences (short correspondence range, security and protection, unwavering quality, portability, and so on.) and assets (control contemplations, stockpiling limit, handling abilities, transmission capacity accessibility, and so forth.). Moreover, WSN has its asset and structure limitations (that are application-and condition explicit) and that vigorously relies upon the size of the observing condition. Mainstream researchers profoundly tended to a few issues identified with sensor systems at various layers (for example, vitality productivity, unwavering quality, strength, versatility, and so forth.).

**Tending to:** On account of remote innovations, for example, RFID and Wi-Fi, the IoT worldview is changing the Internet into a completely incorporated Future Internet. While Internet advancement prompted a phenomenal interconnection of individuals, the current pattern is prompting the interconnection of articles to make a shrewd situation. In this specific circumstance, the capacity to remarkably recognize things is basic for the accomplishment of IoT since this permit to particularly address an

enormous number of gadgets and control them through the Internet. Uniqueness, dependability, ingenuity, and adaptability speak to basic highlights identified with the making of an exceptional tending to blueprint.

Extraordinary recognizable proof issues might be tended to by IPv4 to a degree (generally, a gathering of living together sensor gadgets can be distinguished geologically, yet not separately). IPv6, with its internet mobility characteristics, can relieve a portion of the gadget ID issues and is required to assume a significant job in this field.

**Middleware:** Because of the heterogeneity of the taking an interest objects, to their constrained stockpiling and preparing capacities and to the tremendous assortment of uses included, a key job is played by the middleware between the things and the application layer, whose primary objective is the reflection of the functionalities and correspondence abilities of the gadgets: object abstraction, service management, service composition, and application.



### How IoT functions?

The ecology of IoT comprises of Internet empowered smart gadgets that utilization inserted processors, sensors, and correspondence equipment to gather, send, and follow up on information they obtain from their surroundings.

IoT gadgets share the sensor information they gather by interfacing through an IoT channel or another smart device where information is either sent to the cloud to be dissected or broke down locally. Once in a while, these gadgets speak with other related gadgets and follow up on the data they get from each other. The gadgets do the greater part of the work without human mediation, even though individuals can communicate with the gadgets - for example, to set them up, give them directions or access the information.

The availability, systems administration, and correspondence conventions utilized with these Internet empowered gadgets generally rely upon the particular IoT applications sent.

### Why is IoT significant?

The IoT engages individuals to live and work smartly in corresponding with action in their lives. In spite of offering cunning contraptions to modernize homes, IoT is fundamental to business. IoT gives affiliations a relentless investigation of how their affiliations' structures truly work, passing on bits of data into everything from the acquaintance of machines with the stock framework and coordination endeavours.

IoT connects with relationship to mechanize frameworks and decay work costs. It also abstains from waste and improves association development, making it progressively sensible to fabricate and pass on item comparatively as offering straightforwardness into client exchanges.

IoT contacts each industry, including social insurance, store, retail, and assembling. Sharp urban systems help occupant's abatement waste and significance use, and related sensors are even utilized in creating to help screen accumulate and dairy creature's yields and predict progression structures.

In that farthest point, IoT is one of the most basic movements of standard step by step nearness, and it will keep getting steam as more affiliations fathom the capacity of related device to keep them attentive.

## Benefits of IoT

The IoT offers various advantages to associations, empowering them to:

Screen their general business forms.

Improve client experience.

Set aside time and cash.

Upgrade worker profitability.

Coordinate and adjust plans of action.

Settle on better business choices.

Produce more income.

IoT urges organizations to re-examine the manners in which they approach their organizations, ventures, and markets and gives them the instruments to improve their business systems.

### Advantages and disadvantages

The following are the advantages IoT:

Capacity to get to data from anyplace whenever on any gadget.

Improved correspondence between associated electronic gadgets.

Moving information parcels over an associated system sets aside time and cash.

Robotizing undertakings improves the nature of a business' administrations and lessens the requirement for human intercession.

The disadvantages of IoT include:

As less secure and privacy.

Difficult to manage.

Unemployment issue.

Over-reliance on technology.

### Consumer and enterprise IoT applications

There are various true utilizations of the Internet of Things, going from purchaser IoT and venture IoT to assembling and mechanical IoT. IoT applications length various verticals, including car, telecom, and vitality.

In the buyer portion, for instance, savvy homes that are outfitted with keen indoor regulators, brilliant machines, and associated warming, lighting, and electronic gadgets can be controlled remotely using PCs and cell phones. Smart devices with sensors and programming can gather and dissect client information, sending messages to different advances about the clients with the point of making clients' lives simpler and increasingly agreeable. Smart devices are likewise utilized for open security - for instance, improving people on call's reaction times during crises by giving enhanced courses to an area or by following development laborers' or firemen's fundamental signs at hazardous destinations.

In human services, IoT offers numerous advantages, including the capacity to screen patients all the more near utilizing the information that is created and break down it. Clinics regularly use IoT frameworks to finish undertakings, for example, stock administration, for the two pharmaceuticals and medicinal instruments.

Shrewd structures can, for example, decrease vitality costs utilizing sensors that distinguish what number of inhabitants are in a room. The temperature can alter naturally - for instance, turning the forced air system on if sensors recognize a gathering room is full or turning the warmth down if everybody in the workplace has returned home.

In agriculture, IoT-based smart farming frameworks can help screen, for example, light, temperature, mugginess, and soil dampness of harvest fields utilizing associated sensors. IoT is likewise instrumental in computerizing water system frameworks.

In a smart city, IoT sensors and arrangements, for example, smart streetlights and brilliant meters, can help ease traffic, moderate vitality, screen and address natural concerns, and improve sanitation.

## Difference between cloud computing and IoT

Cloud computing calls *the* includes conveying information, applications, photographs, recordings, and progressively over the Internet to server farms. IBM has accommodatingly separated cloud computing into six distinct classes:

**Software as a Service (SaaS):** Cloud-put together applications keep running concerning PCs off-site (or *in the* Other individuals or organizations possess and work these gadgets, which interface with clients' PCs, ordinarily through an internet browser.

**Platform as a Service (PaaS):** Here, the cloud houses all things required to construct and convey cloud-based applications. The PaaS evacuates the need to buy and look after the equipment, programming, facilitating, and the sky is the limit from there.

**Infrastructure as a Service (IaaS):** Gives organizations servers, stockpiling, systems administration, and server farms on a for every utilization premise.

**Public cloud:** Companies possess and work these spaces and give snappy access to clients over an open system.

**Private cloud:** Similar to an open cloud, with the exception of just a single element (client, association, organization, and so forth.) approaches.

**Hybrid cloud:** This is a private cloud that grants public cloud access.

The IoT, in the interim, alludes to the association of gadgets other than the standard models, for like PCs and cell phones to the Internet. In not too far years to come, other devices such as refrigerators, and heart screens would all be able to be associated through the IoT. What’s more, as the IoT floods in the coming years, more gadgets will join that rundown.

The IoT and cloud computing are unique; however, each will have their activity in handling this new universe of information. [Table 7.1](#) below shows the difference between both cloud and IoT:

|                                                                  |
|------------------------------------------------------------------|
| IoT: IoT:                                                        |
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| IoT: IoT: IoT: IoT:                                              |
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**Table 7.2:** *Difference between Cloud and IoT*

|         |
|---------|
| IoT IoT |
|---------|



|                                             |
|---------------------------------------------|
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| IoT IoT IoT IoT IoT                         |
| IoT IoT IoT                                 |
| IoT IoT IoT IoT IoT IoT IoT IoT             |
| IoT IoT IoT IoT IoT IoT IoT                 |

**Table 7.3:** Similarities between Cloud and IoT

[Table 7.2](#) and [Table 7.3](#) shows the difference between both cloud and IoT, and how they share similar characteristics and the differences in terms of usefulness. It also explains the limitations of IoT when it comes to storage and processing capabilities, as well as connectivity issue.

## *Integration of CloudIoT-the role of cloud computing in the IoT*

Cloud computing and the IoT both serve to build productivity in our ordinary undertakings, and the two have a complementary relationship. The IoT produces monstrous measures of information, and cloud computing gives a pathway to that information to head out to its goal.

AWS one of a few IoT cloud stages at work today, calls attention to six focal points and advantages of distributed computing:

Flexible cost enables you to pay for the registering assets you use, and not more.

Providers, for example, AWS, can accomplish more noteworthy economies of scale, which diminish costs for clients.

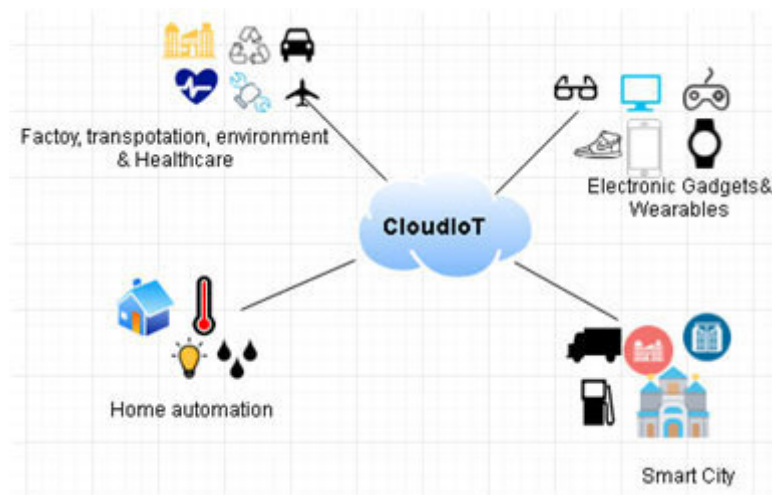
There's no need to figure your Infrastructure needs and wants.

Cloud processing builds speed and nimbleness in making assets accessible to engineers.

You can get a good deal on working server farms.

You can convey your applications worldwide in merely minutes.

A portion of the more prevalent IoT cloud stages available incorporate Amazon web Services, GE Predix, Google Cloud IoT, Microsoft Azure IoT Suite, IBM Watson, and Salesforce IoT Cloud:



**Figure 7.6:** *CloudIoT operation*

[Figure 7.6](#) shows the CloudIoT integration, for example, home automation, smart city, electronic gadgets and wearables and factory, transportation, environment and healthcare.

### *CloudIoT architecture: Service and deployment model*

In order for CloudIoT to succeed, there should be a well-defined and organized structure in place to handle the activities of CloudIoT. Engineers take into consideration the service that will be provided by the CloudIoT and the way in which the services will be deployed.

## Service models

Cloud Computing service conveyance includes three diverse administration models, namely: SaaS, PaaS, and IaaS.

SaaS gives applications to the cloud's end client that are, for the most part, gotten to using an online interface or administration situated design-based web administration technology. These administrations can be viewed as **ASP (Application Service Provider)** on the application layer. Normally, a particular organization that uses the administration would run, keep up, and give support with the goal that it dependably utilized over a significant period.

PaaS comprises of the real condition for creating and provisioning cloud applications. The principal clients of this layer are engineers that need to create and run a cloud application for a specific reason. An exclusive language was reinforced and given by the stage (a lot of significant essential administrations) to ease correspondence, checking, charging, and different viewpoints. For example, the startup just as to guarantee an application's adaptability and adaptability. Constraints concerning the programming dialects upheld, the programming model, the capacity to get to assets, and the long-haul perseverance are conceivable disadvantages.

IaaS gives the vital equipment and programming after that a client can assemble a modified processing environment. Computing assets, storing assets, and the correspondences channel are connected with these basic IT assets to guarantee the soundness of utilizations being utilized on the cloud. Those stack models can be alluded to as the vehicle for IoT, being utilized and passed on by the clients in various strategies for the best possibility of interoperability. This incorporates interfacing vehicles, TVs, cell phones, robots, ATMs, as well as security and expert managements, and examination stages that accompany them.

## Deployment models

Basically, in cloud computing, there are four fundamental models: private cloud, public cloud, network or community cloud, and hybrid or mixed cloud.

A private cloud has foundation that is provisioned for selective use by a solitary association involving numerous shoppers, for example, specialty units. It might be claimed, overseen and worked by the association, an outsider or a blend of them, and it might exist on or off premises.

A public cloud is made for open use by the overall population. Public cloud offers administrations to anybody on the internet. An example of a public cloud could be Microsoft Azure which is a case of an enormous public cloud supplier. This model is reasonable for business necessities that require the executives of burden spikes and the applications utilized by the business, exercises that would somehow or another require more prominent interest in foundation for the business. In that capacity, public cloud additionally lessens capital use and cut down operational IT costs.

A public cloud is made for public use. Public cloud offers computational services to organizations or anybody on the internet that subscribe for such service. A case of a public cloud provider could be **Amazon Web Services (AWS)** which may be a

tremendous public cloud supplier. This is normally deployed over the internet and cut down cost and increase profits for organizations.

A network cloud or community cloud is overseen and utilized by a specific gathering or associations that have shared interests, for example, explicit security necessities or a typical mission.

At last, a hybrid or mixture cloud consolidates at least two unmistakable private, network or public cloud foundations with the end goal that they stay exclusive, yet are bound together by institutionalizing or restrictive innovation that empowers information and application versatility. In a hybrid cloud, data that is not critical are sent to the public whereas confidential information about the organization is kept in the confines of the organization. Since exercises like stockpiling and information handling occur in the cloud instead of on the gadget itself, this has had huge ramifications for IoT. Numerous IoT frameworks utilizes huge quantities of sensors to gather information and afterward making a smart decision.

Utilizing the cloud is significant for amassing information and drawing bits of knowledge from that information. For example, a smart agribusiness organization would have the option to look at soil dampness sensors from Bong and Lofa County subsequent to planting similar seeds. Without the cloud, looking at information across extensive regions is significantly more troublesome, and can led to erroneous decision.



In addition, the usage of cloud can take into account high versatility. When you have hundreds, thousands, or even a great many sensors, putting a lot of computational power on every sensor would be very costly and vitality escalated. Rather, information can be passed to the cloud from every one of these sensors and prepared there in total. The framework for which IoT runs on is in the cloud. Sensors and gadgets gather information and perform activities, yet decision making, and investigation usually occurs in the cloud.

### Stages, administrations, and research ventures

In this area portray open source and exclusive stages for actualizing the new vision and ideal models and the new CloudIoT utilizations. Additionally, the research undertakings concentrated on the CloudIoT worldview, which accept are a case of the exploration endeavors supported in this significant zone.

## Stages

The plan of CloudIoT stages can prompt the creation of wise foundations, empowering savvy applications to profit by cloud-based systems. These stages would empower the new standards. In view of such stages, end-clients would have the option to use keen suppliers' detecting and inciting frameworks, instead of conveying detecting foundations independent from anyone else - which demonstrated to be a tedious and repetitive errand that drastically eases back development. The subsequent virtualization of detecting assets ought to likewise give an intend to modify the virtual detecting framework to adjust to the various applications.

While the writing reports endeavors in characterizing a conventional elevated level engineering to manage the mix of cloud and IoT, Software characterized IoT (SDIoT) – there are a few open source and exclusive stages accessible for cloud and IoT incorporation. Many of them are planned for tackling one of the primary issues in this field is identified with the heterogeneity of things and clouds. These stages attempt to connect this hole actualizing a middleware towards the things and another towards the cloud, and they ordinarily give an API towards the applications. Different stages are rather bound to explicit equipment gadgets or clouds.

CloudIoT is an open source task planned for incorporating the things (advanced mobile phones, tablets, robots, pages, and so on.) with backends for overseeing sensors and their messages and for giving an API to applications inspired by this information. The stage has been displayed with video sensors (that is, IP cameras) on the Future Grid Cloud testbed. The product is accessible online through the code sharing stage GitHub.

OpenIoT is another open source exertion encouraged by an exploration task financed by the EU. The venture targets giving a middleware to design and send calculations for gathering and separating messages by things, while simultaneously creating and preparing occasions for intrigued applications. The creators of talk about the infrastructural practical modules and the plan standards of the middleware for empowering the dynamic, self-sorting out definition of streamlined IoT applications in cloud conditions. Among the principle focal points of OpenIoT are the versatility parts of IoT for vitality proficient organization of information gathering and transmission to the Cloud. The venture site contains various recordings demonstrating potential applications in a few situations. Additionally, this product is accessible online through GitHub.

## Administration

CloudIoT brought forth another arrangement of brilliant administrations and applications that can firmly affect regular day to day existence. A considerable lot of the applications depicted in the accompanying advantage from **Machine-to-Machine correspondences (M2M)** when the things need to trade data among themselves and not just send them towards the cloud. In this area, portray the wide arrangement of uses that are made conceivable or altogether improved gratitude to the CloudIoT worldview.

**Healthcare:** The reception of the CloudIoT worldview in the healthcare field can carry a few chances to restorative IT, and specialists accept that it can altogether improve human services administrations and add to its constant and efficient advancement. For sure, CloudIoT utilized in this situation can streamline social insurance procedures and permits to upgrade the nature of the medicinal administrations by empowering the collaboration among the various substances included. Encompassing helped living (AAL), specifically, targets facilitating the everyday lives of individuals with handicaps and constant ailments.

Through the use of CloudIoT in this field it is conceivable to supply numerous creative administrations, for example, gathering patients' crucial information by means of a system of sensors associated with medicinal gadgets, conveying the information to a medicinal focus' Cloud for capacity and handling, appropriately

overseeing data given by sensors, or ensuring pervasive access to, or sharing of, restorative information as **Electronic Healthcare Records (EHR)**.

CloudIoT empowers savvy and high-caliber, pervasive therapeutic administrations. Inescapable social insurance applications produce a tremendous measure of sensor information that must be overseen appropriately for further investigation and preparing. The selection of cloud speaks to a promising answer for overseeing human services sensor information productively and permits to extract specialized subtleties, taking out the requirement for ability in, or authority over, the innovation foundation. In addition, it prompts the simple computerization of the way toward gathering and conveying information at a decreased expense. It further makes cell phones appropriate for wellbeing data conveyance, access, and correspondence, likewise in a hurry.

Cloud permits to confront basic difficulties of this application situation, for example, security, protection, and unwavering quality, by upgrading therapeutic information security and administration accessibility and repetition. On account of the productive administration of sensor information it is conceivable to give helped living administrations in real-time. Also, cloud selection empowers the execution (in the cloud) of secure sight and sound based wellbeing administrations, beating the issue of running overwhelming interactive media and security calculations on gadgets with restricted computational limit and little batteries, and it gives an adaptable stockpiling and preparing foundation to perform both on the web and disconnected investigations of information streams created in human services **Body Sensor**

**Networks (BSNs).** Because of the utilization of the CloudIoT worldview BSNs can be sent in a network of individuals and can create a lot of logical information that are put away, prepared, and investigated in an adaptable design.

In the social insurance area, normal difficulties are identified with the absence of trust in information security and protection by clients (presentation to programmer assaults, infringement of medicinal information classification, information lock-in and loss of administration, benefit misuse), execution eccentrics (asset fatigue, information move bottlenecks, sway on constant administrations, and gushing QoS), lawful issues (contract law, licensed innovation rights, and information purview) are still object of examination. The absence of explicit research identified with the reception of these innovations with regards to strategic frameworks, of developed unwavering quality examinations and the predetermined number of contextual investigations are regularly characterized as the significant impediments.

Brilliant urban areas and networks. CloudIoT prompts the age of administrations that connect with the encompassing condition, hence making new open doors for contextualization and geo-mindfulness. Practical improvement of urban zones is a test of key significance and requires new, effective, and easy to understand advancements and administrations. The test is to bridle the community-oriented intensity of ICT (systems of individuals, of learning, of sensors) to make group and individual mindfulness about the different manageability dangers which our general public is confronting these days at social, natural, and political levels.

The subsequent aggregate knowledge will prompt better educated basic leadership forms and enable residents, through investment and association, to embrace increasingly maintainable individual and aggregate practices and ways of life.

CloudIoT can give a typical middleware to future-situated savvy city administrations, securing data from various heterogeneous detecting foundations, getting to a wide range of geo-area and IoT innovations (for example, 3D portrayals through RFID sensors and geo-labeling), BSNs have been as of late presented for the remote checking of human exercises in the medicinal services areas yet additionally in other application spaces, for example, crisis the board, wellness, and conduct reconnaissance.

Structures regularly comprise of a sensor stage (with APIs for detecting and activating) and a cloud platform which offers adaptable and seemingly perpetual stockpiling and handling assets for the programmed administration and control of true detecting gadgets in a huge scale sending. Publicly supported and notoriety-based structures additionally exist: creators of propose a system executing the sensing as a service worldview with regards to keen urban communities and went for open security.

Creators of present an environment for versatile group detecting applications which depends on the cloud-based distribute/buy in middleware to get sensor information from cell phones in a setting mindful and vitality productive way. Creators of spotlight on the weight on application designers and last clients made by the need to manage huge scale conditions. Since IoT situation is



profoundly divided, sensor virtualization can be utilized to decrease the hole between existing heterogeneous advances and their potential clients, enabling them to collaborate with sensors at various layers.

Various as of late proposed arrangements recommend to utilize cloud designs to empower the disclosure, the association, and the coordination of sensors and actuators, in this way making stages ready to arrangement and bolster universal availability and ongoing applications for brilliant urban areas. For example, creators of examine an idea towards building up a brilliant city utilizing a savvy, vitality productive, open enlightenment framework, which would likewise offer pervasive correspondence. Additionally, cloud-based stages help to make it simpler for outsiders to create and give IoT modules empowering any gadget to be associated with the cloud. This sort of cutting-edge administration model conceals the unpredictability and the heterogeneity of the basic foundation, while simultaneously meeting complex prerequisites for cloud, for example, high reactivity and practicality, adaptability, security, simple configurability, and adaptability.

Basic difficulties are identified with security, unwavering quality, scale, heterogeneity and practicality. Without a doubt, empowering the essential assets, stockpiling and registering capacities for a lot of heterogeneous and customized information (originating from disseminated sources) in a straightforward and secure way and the improvement and the arrangement of different middleware stages in a such divided situation (in which diverse IoT biological systems are not ready to convey between them) are not trifling

undertakings. The inclusion of various physical sensors in the extent of administration conveyance makes extra difficulties related with continuous collaborations, which forces a requirement for examining augmentations over ongoing working frameworks for inserted gadgets, just as how they could be upheld in the extent of a cloud domain. Additionally, the subsequent framework needs to give a quick arrangement of sent sensors and a simple coordination of new sensors in the detecting condition.

Huge research on detecting, incitation, and IoT is coordinated towards the proficient semantic explanation of sensor information. At long last, while urban communities share regular concerns, for example, the need to adequately share data inside and among urban communities and the craving for upgraded cross-outskirt conventions – they do not have a typical framework and system for teaming up, producing operational and local fracture that at present avert inventive collaborations.

**Shrewd home and savvy metering:** Home systems have been distinguished as nature where clients primarily act: CloudIoT.

### Research ventures

CloudIoT, which means cloud of things, is an exploration undertaking kept running by mechanical and research accomplices just as city organizations from Europe and Japan. The accomplices target creating framework, administrations, devices and applications for regions and their partners (natives, administration designers, and so forth.) to make, convey, and oversee client driven applications dependent on IoT and cloud coordination.

IoT6 is a European research venture on the future Internet of Things. It targets misusing IPv6 and related models (for example, 6LoWPAN, CORE, COAP) to beat current deficiencies (for example in terms of discontinuity) in the zone of IoT innovative work. Its principle destinations are to research, plan, and build up a profoundly scalable IPv6-based help situated engineering to accomplish interoperability, portability, Cloud registering reconciliation, and knowledge conveyance among heterogeneous things, applications, and services.

The OpenIoT venture, referred to before for its open source stage, targets making an open source middleware for getting information from heterogeneous things, concealing the distinctions among these items. The venture investigates productive approaches to utilize and manage cloud situations for things and assets, (for example, sensors, actuators, and shrewd gadgets) and offering

utility-based (that is, pay-as-you-go) IoT administrations. Creators of present a league of **Future Internet of Things** IoT-LAB IoT-LAB) incorporated with OpenIoT, giving an extremely enormous scale framework office suitable for testing little remote sensor gadgets and heterogeneous imparting objects.

A few ventures target research issues identified with IoT and don't expressly make reference to issues identified with their coordination with the cloud. In any case, we report them here in light of the fact that they regularly notice information accumulation and elaboration stages that are likely being clouds (at this moment or in the following years). These undertaking incorporate smart Santander, The Cooperative ITS Corridor from Rotterdam to Vienna, and WISEBED.

### Challenges of CloudIoT

On account of the investigations, continue the primary issues identified with CloudIoT as yet requiring examination activities and call attention to some future headings.

## Open issues

Different open issues described for the CloudIoT are as below:

**Institutionalization:** The absence of gauges is really considered as a major issue towards CloudIoT by countless scientists. At present most things are associated with the Cloud through electronic interfaces, which can diminish the multifaceted nature for developing such applications. In any case, they are not explicitly intended for proficient machine-to-machine interchanges and present overhead as far as system burden, deferral, and information processing. In addition, interoperability is as yet an issue, in light of the fact that both the cloud and the Things execute non-standard heterogeneous interfaces. Despite the fact that mainstream researchers have provided numerous commitments to the organization and standardization of IoT and cloud ideal models, a reasonable need of standard conventions, designs and APIs is being requested so as to encourage the interconnection among heterogeneous keen articles and the making of upgraded administrations, which understand the CloudIoT worldview.

**Power and vitality proficiency:** Late CloudIoT applications involve visit information transmission from the things to the cloud, which, thusly, may depend on cell phones as entryway. Such procedure rapidly depletes battery limit on both the things and the entryway

constraining the nonstop activity to 24hrs or less. The literature demonstrates that, in the field carefully identified with the reconciliation of cloud and IoT, getting vitality proficiency in the two-information handling and transmission is a significant open issue. Be that as it may, noteworthy research exertion has just been spent for what concerns the cloud and IoT independently.

For taking care of such issue, a few bearings have been proposed: progressively proficient information transmission and pressure advancements; information reserving systems for reusing gathered information in time-tolerant applications; middleware to deal with adverse circumstances and to pack information in the event of consistent and long-span observing of information. These open issues have consequently suggestions additionally as far as correspondence technologies, which are depicted later on in this area.

**Enormous information:** In a past area we have portrayed enormous information as a significant research subject, firmly combined with CloudIoT and with a few related difficulties. Regardless of whether a few commitments have been given, we think about enormous information as a significant open issue, where research is still unequivocally vital. CloudIoT includes the administration and handling of enormous measures of information and occasions originating from various areas and heterogeneous source types, where most applications require complex tasks to be performed continuously.

From one viewpoint, this implies appropriately synchronizing occasions originating from remote sources and reproducing and

corresponding their semantics so as to derive the circumstance significant for the particular application. Then again, it intends to process progressively tremendous measures of sight and sound information and to infer in time the data important to trigger pertinent administrations and help the client in his present area.

**Security and protection:** With respect to the past case, we believe security and security to be both an exploration challenge that is accepting a great deal of consideration just as an open issue where more exertion is still required. While numerous clients are as of now worried about protection and security in cloud-based applications, since CloudIoT brings information originating from this present reality into the cloud and empowers activating activities into this present reality, such concerns are significantly more pertinent.

With respect to security, giving appropriately planned approval jobs and approaches while straightforwardly ensuring that lone approved individuals approach touchy information is as yet a test, particularly when information uprightness must be guaranteed in light of approved changes. As to, it stays testing to adapt to various dangers from programmers: malware can be infused into physical sensors to create altered information; crude or handled information can be taken/altered on the cloud; bargained entryways can cause security breaks to the CloudIoT framework; the communication channels are helpless against side-channel data spill.

**Insight:** The centralization into the cloud of ongoing information originating from heterogeneous things empowers upgraded basic



leadership abilities by utilizing refined data choice and combination components. In spite of the fact that examination endeavors have been spent in this field, amplifying the insight in this setting is as yet an open test.

**Incorporation philosophy:** While CloudIoT arrangements have been as of now worked around explicit applications, little exertion has been spent to determine a typical procedure to coordinate cloud and IoT frameworks. Since sets of utilizations have regular necessities, a few institutionalized work processes could be characterized. In addition, a conventional and adaptable stage could be the beginning stage for actualizing such work processes all the more effectively.

**Valuing and charging:** Various elements engaged with CloudIoT frameworks have their own client and administration the executives, and strategies for installments and evaluating. Additionally, the expense for conveying things is diminishing, while the expense to keep them associated with the cloud increments exponentially. Subsequently, setting the cost for coordinated administrations, circulating it among the various elements, and dealing with the installment procedure are as yet open issues in the CloudIoT incorporated situation.

**System interchanges:** CloudIoT includes a few heterogeneous arrange advances, where numerous applications require progression in the transmission of information and generally speaking utilization of transfer speed increments drastically. On the one hand, the productivity of the entrance the executives for empowering progression and for optimizing the data transmission

use is as yet an open issue. Then again, ebb and flow data transfer capacity restriction can't bolster the increasing pattern, and extra research work is important to improve the distribution strategies at immense scales.

Additionally, numerous CloudIoT applications (for example, human services) require issue tolerant and solid constant exchange of information from the things to the Cloud. For example, a patient wearing body sensors might be out of coverage zone from the passage (for example, a cell phone). Along these lines, situations naturally inclined to association disappointments require explicit help so as to dodge collection of mistakes.

Novel answers for system interchanges are of fundamental significance in both human-driven and M2M settings. The proliferation of cell phones, and the expanding utilization of cloud computing, alongside the expanding interest for media administrations, are changing the way of life of clients and making new opportunities to suppliers and customers. Without a doubt, sight and sound information will account for up to 90% of all the Internet traffic in a couple of years, and a large portion of the substance will be made and gotten to by versatile things (for example, cell phones or tablets) conveyed by people or put in vehicles.

The joining of human-driven mixed media arranging into Internet cloud registering situations enables portable clients to have new mixed media encounters not beforehand available. For example, the cloud can be designed to play out a lot of significant undertakings and administrations for portable mixed media clients

and systems, extending from surveying the video quality level and burden adjusting to sight and sound transcoding and excess/blunder correction plans. This tale portable interactive media period forces new difficulties for systems, substance, terminals, and people, and should defeat issues related, for example, with high congestion, low versatility, quick battery utilization, and poor client experience.

CloudIoT includes M2M interchanges among numerous heterogeneous gadgets with various conventions, which rely upon the particular application situation. M2M correspondence can be viewed as a propelled type of sensor systems, where a definitive objective is to give thorough associations among all shrewd gadgets. Be that as it may, the correspondence system from sensor systems faces challenges in fulfilling the prerequisites of ongoing situations, for example, ITS, where each savvy gadget can play more than one of the potential jobs: sensor, chief, and activity executor. M2M to a great extent profited by the advances in remote correspondence advances, for example, wearable and implantable biosensors, alongside ongoing improvements in the installed computing, canny frameworks, and cloud registering territories.

In the writing, a couple of acknowledge of M2M interchanges have likewise been proposed, utilizing for instance, Bluetooth (IEEE 802.15.1), ZigBee (IEEE 802.15.4), or WIFI (IEEE 802.11b and p) innovations. Notwithstanding, there is still no accord on the system engineering of a general situation for M2M interchanges. Dealing with the things in a uniform manner in a heterogeneous situation, while providing required execution still speaks to an

open issue. Most of uses don't include versatility: in stationary situations, IoT regularly receives IEEE 802.15.4/6LoWPAN arrangements.

Then again, situations, for example, vehicular systems generally embrace IEEE 802.11p. Be that as it may, being WIFI and Bluetooth the most broadly utilized radio advances for remote systems, their adoption for IoT applications is expanding: they speak to a less expensive arrangement, most cell phones as of now bolster them (for example, PDAs), and the two norms are winding up increasingly more low power. In some different cases, when power limitations are less critical, GPRS is as yet utilized for Internet network, however it brings about an exorbitant arrangement (for example, numerous SIM cards are vital).

As of late, genuine consideration was pulled in by the institutionalization progress of LTE-advanced, and the effects of bringing M2M correspondences into LTE-advanced are currently under significant investigation in 3GPP. We accept that examination on system communications for CloudIoT is still required so as to give compelling and productive arrangements.

**SLA requirement:** CloudIoT clients require things-produced information to be moved and prepared by application ward imperatives, which can be exacting if there should arise an occurrence of basic situations. Ensuring a certain QoS level about cloud assets probably won't be constantly workable for a solitary supplier, in this way depending on various cloud suppliers may be important to maintain a strategic distance from SLA infringement. Be that as it may, progressively choosing the best blend of cloud

suppliers still speaks to an open issue as a result of costs, time and heterogeneity of QoS the executives support.

**Capacity:** Capacity arrangements have been much of the time considered in this paper. For instance, we have just thought about them as a driver for the coordination of cloud and IoT. Nonetheless, the writing considers this as a still open issue as present arrangements may not give the vital help to future applications. For instance, the capacity of information moved from the things to the cloud includes some building issues as yet requiring examination endeavors. While information must be appropriately time stamped to empower server-side reproduction and handling, moves require legitimate planning so as to stay away from over the top burstiness of system and preparing load. One conceivable heading to address such issues includes the presentation of prescient stockpiling and storing.

**Adaptability and adaptability:** CloudIoT requires effective instruments to match gathered information and occasions to suitable applications and administrations. Giving adaptable membership outlines and occasions the executives while ensuring adaptability regarding things and clients is as yet viewed as an open issue.

### Future scope

So as to empower the maximum capacity of CloudIoT, extra research exertion is normal in a few ways:

Properly distinguishing, naming, and tending to things will be important to help both the gigantic number of things and their portability. While IPv6 could be the best possible arrangement, its enormous scale selection is as yet a continuous procedure and extra research is important to both accelerate this moderate procedure in explicit situations (for example, access systems) and to adapt to new versatility and adaptability necessities.

Solutions for recognizing natural changes dependent on IoT information will empower the conveyance of upgraded setting-based administrations, giving the best assistance relying upon the circumstance. Such open door will be motivating force the examination of progressively viable calculations for conveying customized substance and promotions.

Large scale support for multi-organizing (for example, multihoming, multipath, and multicast), association handover and wandering will be obligatory for improving system unwavering quality and ensuring ceaseless network, QoS, excess, and adaptation to non-critical failure. In this setting arrangements dependent on software defined networking are additionally visualized.

Many uses of CloudIoT would profit by productive and adaptable components for making sensibly secluded system segments over comprehensively dispersed system foundations, which could be another significant driver for research in system virtualization and programming characterized systems administration fields.

Converging towards a typical open help stage environment for giving APIs to grow outsider CloudIoT-based applications will empower new business openings and drive investigate endeavors toward characterizing standard conventions, libraries, dialects, and systems for CloudIoT.

## Conclusion

The integration of cloud computing and IoT speaks to the following huge jump ahead in the future internet. The new applications emerging from this combination called CloudIoT - Open up new energizing bearings for business and research. In this chapter, we overviewed the writing so as to recognize the integral parts of cloud and IoT and the fundamental drivers for incorporating them into an exceptional situation. Since the appropriation of the CloudIoT worldview empowered a few new applications, inferred the fundamental research difficulties of enthusiasm for every one of them. Further broke down such difficulties so as to distinguish current look into bearings. At last, we studied accessible stages and extends by looking at their principle angles and distinguished open issues and future research headings in this field.

On account of the CloudIoT worldview regular day to day existence and exercises will be possibly improved for everybody: shrewd urban communities will empower progressively effective open administrations and advance new business openings, pervasive medicinal services applications will improve the personal satisfaction for some patients, and so forth. These new application situations present significant research difficulties, for example, the heterogeneity of included gadgets and innovations; the required presentation, dependability, adaptability and security; protection safeguarding; lawful and social angles. The open issues of



CloudIoT worldview relate essentially power and vitality productivity, SLA requirement, estimating also, charging, security and protection. The imagined future bearings incorporate the ID of the conclusive answer for naming and tending to things, the huge scale support for multi-organizing, what's more, the union toward a typical open help stage condition.

The integration of cloud computing and IoT speaks to the following huge jump ahead in the Future Internet. The new applications emerging from this combination called CloudIoT - Open up new energizing bearings for business and research. On account of the CloudIoT paradigm regular day to day existence and exercises will be possibly improved for everybody.

By reading this chapter, one will understand what is meant by cloud computing, and how one can benefit from it. They will also understand the importance of IoT in cloud computing, as well as the types of devices, use in accessing the cloud resources in cloud computing. The integration of both technologies will see the emergence of an almost new ways of life and solutions. Also, the reader will also understand that one utilizing CloudIoT will get to know how easy his/her life will become in terms of having access to things they use on a daily basis. Discerning urban communities will empower progressively effective open administrations and advance new business openings, pervasive medicinal services applications will improve the personal satisfaction for some patients, and so forth.

### Exercise

What is role of cloud computing in IoT?

Difference between cloud versus IoT.

What are the advantages and disadvantages of the integrated CloudIoT?

Explain the CloudIoT architecture in detail.

What are the challenges of CloudIoT?

### Fill in the blanks

The Integration of both technologies cloud computing \_\_\_\_\_ and which is likely to be call CloudIoT pronounced Cloud IoT.

In IoT situation, a key job is played by \_\_\_\_\_ frameworks, made out of at least one reader and a few labels.

The IoTs, in the interim, alludes to the association of gadgets other than the \_\_\_\_\_, for like PCs and cell phones to the Internet.

The open issues of CloudIoT worldview relate essentially power and vitality productivity, \_\_\_\_\_, estimating also, charging, security and protection.

SDIoT acronym stands for\_\_\_\_\_.

## Answers

Internet of Things (IoT)

Radio-Frequency Identification (RFID)

Standard models

SLA requirement

Software characterized IoT

### Smart City Using IoT Integration

With the help of **Internet of Things (IoT)**, we will be looking forward towards the modern technology that we have adopted or the future ideas that will be transforming our daily city lives into a smart city living. Since its origination, the smart city thought has developed from the execution of explicit undertakings to the usage of worldwide procedures to handle more extensive city challenges. So we need to study the concept of a smart city with various issues, its challenges and different application areas are of it.

## Structure

Concept of smart city

The emergence and definition of a smart city

Dimensions and components of smart city initiatives

Adoption of IoT in automation

Design strategies

Issues and challenges

Factors affecting automation using IoT

Existing application areas

IoT applications in smart cities

Education

Healthcare

E-governance

Industry

### Objective

After studying this chapter, you should be able to understand the concept of a smart city, its emergence and definition, dimensions and components of it in details. Adoption of IoT in automation is also explained in detail with design strategies. In the last various issues and challenges of smart city and different application, areas are also discussed in detail.



## Introduction

With the help of IoT, we will be looking forward towards the modern technology that we have adopted or the future ideas that will be transforming our daily city lives into a smart city living. IoT is finished utilizing various sorts of gadgets, made by various sellers and having various details. Since its origination, the smart city thought has developed from the execution of explicit undertakings to the usage of worldwide procedures to handle more extensive city challenges. So we need to study the concept of a smart city with various issues, its challenges and different application areas are of it.

### Concept of smart city

We are living in a modern society in which our needs and demands frequently change with changes in lifestyle. Are food, cloth and shelter still our basic needs or we have uplifted from these needs and put yourself into a comfortable zone in which every task should be performed or initiated with a snap of fingers. We will be answering these questions and will be looking forward towards the modern technology that we have adopted or the future ideas that will be transforming our daily city lives into a smart city living.

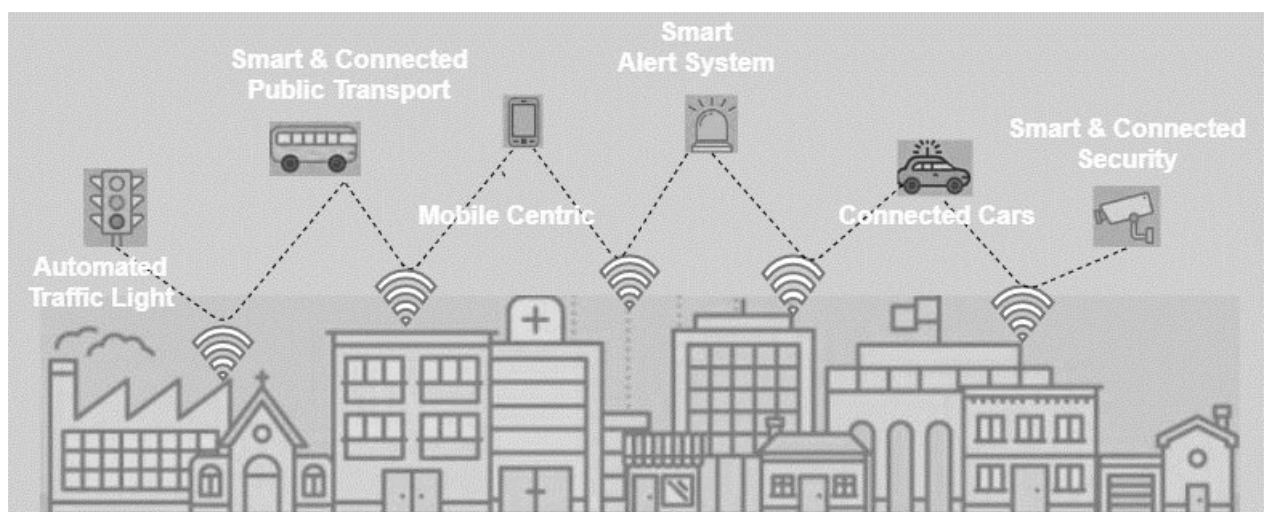
So what is a smart city? What features does it possess, and how is it different from our normal existing cities? Does everyone see the smart city from his/her viewpoint? What essential features do a city possess to declare itself a smart city? Here is the answer to the above-mentioned questions.

The meaning of smart city or what is the *smart city* has been in conflict as far back as the term was begotten. The appropriate response is there is no all-around acknowledged meaning of a smart city. It implies various things to various individuals. The conceptualization of smart city, consequently, differs from city to city and nation to nation, possible upon the degree of advancement, eagerness to change assets and desires of the city occupants.

A smart city would have an alternate meaning in India or Ghana than, a state in Europe. Some definitional limits are required to manage urban areas in the mission. In the mind of any resident, the

image of a smart city contains a list of things to get of framework and administrations that portrays their degree of desire.

To accommodate the yearnings and requirements of the natives, urban organizers in a perfect world target building up the whole urban eco-framework, which is spoken to by the four mainstays of exhaustive advancement-institutional, physical, social, and financial foundation. This can be a long haul objective and urban areas can work towards growing such far-reaching framework steadily, including layers of smartness. The focal point of any smart city activity is on a practical and comprehensive improvement plan:



**Figure 8.1:** Smart city

The term smart city means a city where people use the human resources optimally for gaining better services in the field of transportation, telecommunication, infrastructure, environment, and many more which leads to a better future. The main objective is to increase the city dwellers' quality of life. The idea of a smart city will fully come into existence when it is connected with IoT. It connects

the residents to the services provided by the city so that all can enjoy the services securely without any difficulty. On a daily basis, citizens come face to face with so many problems and challenges as they move about to do their business and household chores.

Amongst these myriads of problems are sanitation, education, transportation, infrastructure and many more. Governments spend millions of money in an attempt to improve the quality of life by instituting or embracing the smart city concept in its developmental agenda.

A lot of people think that these smart cities are useless if we are unable to provide the basic facilities to all corners of our country. They think that initially, government should uplift the poorer section by providing the basic human facilities and then we can move towards the concept of smart cities. We all know that most countries in developing economies lack the provision of basic facilities such as clean drinking water, electricity (in rural areas), health, and education facilities. Thus, some have argued and criticized the focus on providing high-end state of technology such as the bullet trains, technology framework and many more. According to such critics, there will be an injustice if more than half of the population is not getting the basic resources while the few and a richer section of the community is enjoying the quality life with the smart city.

### *The emergence and definition of a smart city.*

Since its origination, the smart city thought has developed from the execution of explicit undertakings to the usage of worldwide procedures to handle more extensive city challenges. In this manner, there is the requirement for a complete review of the accessible potential outcomes and relate them to the particular city challenges close by.

These days, numerous activities planned for examining the origination procedure, organization strategies for smart city ventures are being created in various fields. In any case, there is an absence of institutionalized measurements and approaches to help evaluate, organize, fund, actualize, oversee and repeat these tasks the nation over or the globe as there is a wide assortment of meanings of what a smart city could be.

For the most part, two ways to deal with characterizing the smart city idea can be obviously recognized and distinguished in connection to the fundamental perspectives or center thought of that smart cities. On the one hand, there is a lot of definitions that put accentuation on only one urban angle (innovative, natural, and so on.) advancement or issue hence, leaving the remainder of the city's difficulties.

This perspective on methodology is considered as a mono-theme representation that demonstrates a misconception that the last objective of a smart city is to give another way to deal with urban administration without reference to the different parts of the connected life that are between associated and occur in the actual life of the city. Improving only one piece of an urban environment doesn't infer that the issues of the entire environment are being separated. Then again, there are a few creators that bring out distinctly how the fundamental contrast of the smart city idea is the interconnection of all the urban viewpoints. As it were, the smart city idea ought to become nearer from an incorporated and interconnected methodology or point of view. The city's issues don't occur in separate and in detachment however have in reverse and forward linkages that require a helicopter perspective on gratefulness while simultaneously applying arrangement of single activities analysis and answer for effectiveness. The tangled issues between urbanization are infrastructural, social and institutional that is simultaneously interlaced are reflected in the smart city idea.

The smart city idea speaks to a convincing stage for IT-empowered assistance advancement. It offers a perspective on the city where specialist organizations use data advancements to connect with residents to make frameworks that can improve the personal satisfaction to the point that suburbanites never again need to possess a vehicle to have one available to them.

Such newness is made possible as ICTs (Internet) are quickly embraced and embedded within the transport system that creates flexibility, multitasking, multi-nodal, multi-modal, and ease of

mobility of the citizens within the city physical space (Ropke, 2015).

Numerous meanings of the smart city center only around the key job of ICT in connecting city-wide administrations. Some attention on ICT as an innovation driver and empowering influence, while more extensive definitions incorporate financial, administration and multi-partner angles, for example, the utilization of social support to upgrade manageability, personal satisfaction and urban welfare.

While a few people keep on taking a tight perspective on smart urban communities by considering them to be places that utilize **Information and Communication Technology (ICT)**, others view smart cities as an expansive, coordinated way to deal with improving the effectiveness of city tasks, the personal satisfaction for its residents, and developing the neighborhood economy *al.*, 2016). The emphasis is on manageable and comprehensive advancement, and the thought is to take a look on conservative zones, make a copy model which will act as a beacon to other striving urban cities.

*Catriona et. al.*, (2014) have inventoried over ten unique meanings of the smart city by different analysts since 2009. For instance, while, *Caragliu et. al.*, (2009) have characterized smart city from an expansive based point of view as interests in human and social capital and conventional and present-day correspondence framework fuel practical financial development and a high calibre of life, with a hard administration of common assets, through participatory administration. *Caragliu et. al.*, (2009) *Catriona et. al.*; (2014) has additionally characterized Smart City from innovation

perspective as: .a city trying to address open issues through ICT-put together arrangements with respect to the premise of a multi-partner, municipally based organization. *et. al*; 2014).

This definition tries to inspect the utilization of ICT in the everyday exercises of the populace as they move about in the city. In like manner, the reason for the Smart City's crucial to drive financial development and improve the personal satisfaction of individuals by empowering neighborhood and saddling innovation that empowers urban communities to utilize data and information to improve foundation and administrations and 2016; Skouby et al., 2014).

Smart gadgets are structured so that they catch and use all of the information which you offer or use in regular daily existence. What's more, these gadgets will utilize this information to interface with you on a consistent schedule and complete assignments.

From the definitions, it is seen that frameworks are a focal bit of the Smart City and that innovation is the empowering influence that makes it conceivable however, it is the blend, association and incorporation of all frameworks what winds up principal for a city being really smart *and* 2011). From these definitions, it should be surmised that the smart city idea suggests an exhaustive way to deal with city executives.

These definitions demonstrate an equalization of the mechanical, financial and social components engaged with an urban biological system. The definitions mirror an all-encompassing way to deal



with the urban issues exploiting the new advances so the urban model and the connections among the partners can be re-imagined *Del Bo*, and *Nijkamp*. 2009; *Harrison et al.*, 2010; *Nam* and *Pardo*, 2011; *Batty et al.*, 2012 and *EC*, 2014).

### Dimensions and components of smart city initiatives

Catriona et. al., (2014), expanding on crafted by the *European Smart City Project* and other research contemplates; have proposed six smart city attributes or measurements (highlights). These are smart governance, smart economy, smart mobility, smart environment, smart people, and smart living.

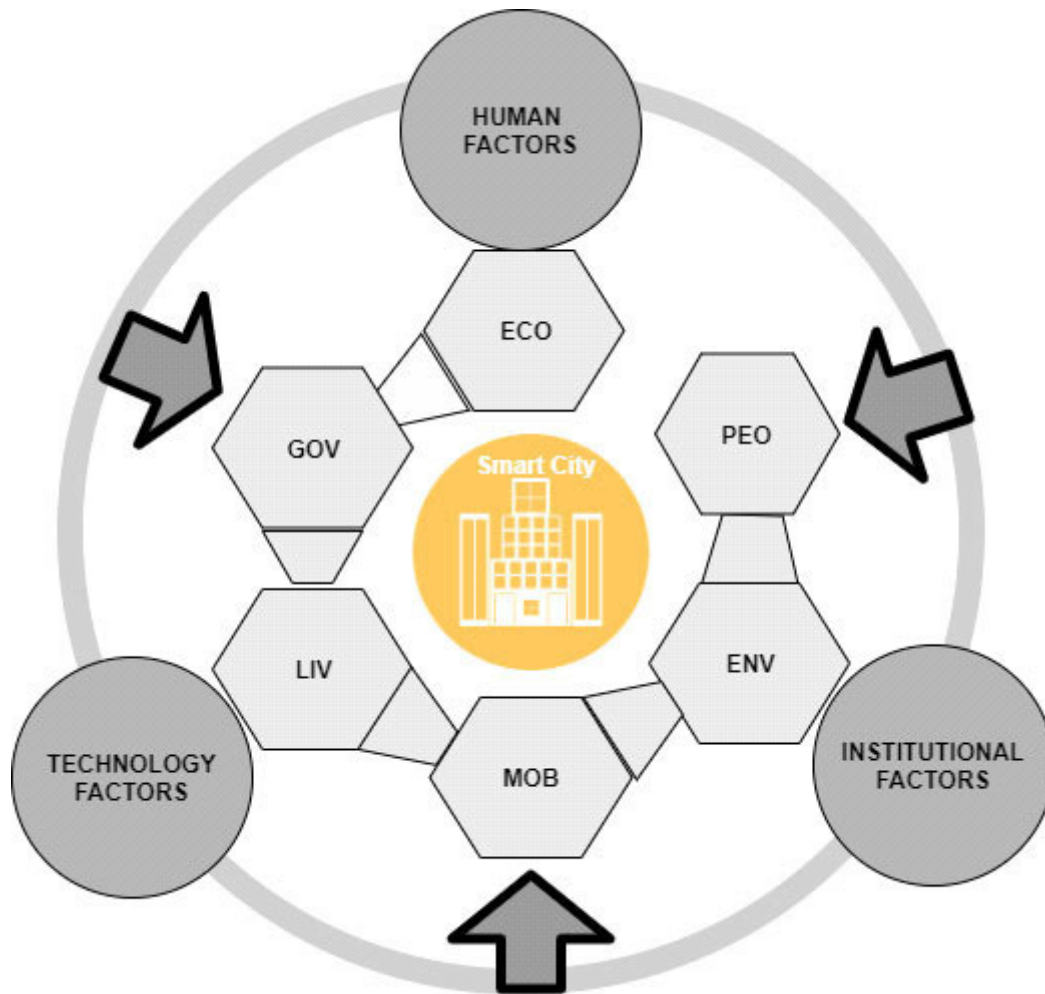
Components are the variety of exercises, advances, standards or measures, assets and techniques, some of which are prior, while others been gathered in a large group or even made explicitly for accomplishing the end. Smart city components are considered as key drivers of explicit qualities, in light of the particular difficulties and requirements, a city faces regarding that trademark 2007; Peek and 2014).

There is a very close connection among components and qualities or measurements. While the measurements delineate the assortments of ventures or activities of smart cities (the finishes), the segments are the methods by which the aim is accomplished.

Because of the way that smart city activities go past the advancement and utilization of innovation, there is the need to consider the human and social factors, for example, training and social capital just as institutional components with respect to the job of partners. Nam and Pardo (2011) have displayed a comprehensive methodology, sorting smart city components inside

three center factors to be specific, human, institutional, and innovation; and the connection between these components as qualities.

A *smart city* is a complex ecosystem, consisting of government policies, process, people, technology and many other factors that work together to provide a set of outcomes. Human Factor, Technological Factor, and Institutional factors are the major and important consideration in designing an ecosystem for the smart city, as shown below in [\*Figure\*](#). Other factors that can also create some values are involved in designing an ecosystem for a smart city. Some of the factors work in collaboration, and some components work individually. Across the whole ecosystem, engagement of stakeholders is the necessity for successful and sustainable smart cities. Engagement of stakeholders for sustainable smart city requires to follow an efficient programmatic approach:



**Figure 8.2:** *Smart city ecosystem*

A well-functioning infrastructure is absolutely necessary but not enough to be deemed to be called as a smart city. IT infrastructure and applications are pre-requisites, but without real engagement and willingness to collaborate and cooperate between public institutions, private sector, voluntary organizations, schools and the citizenry (multi-stakeholder) there is no smart city.

There are hundreds of services that smart cities should possess, but the core features/services must include sustainable infrastructure, digitalization and IT connectivity that should drive

services like e-governance, health and education, safety and security, sanitation and waste management.

### Adoption of IoT in automation

IoT guarantees far-reaching computerization and information trade, which will change the manner in which we work and live. The IoT is going to change the next decade. Actually, everything will be connected to everything. Some gauge that 50 billion gadgets will be IoT-associated by 2020. The clock on the connected device progress is ticking very loud. Some time ago during industrial revolutions of the eighteenth and nineteenth hundreds of years, new innovation, for example, steam-fueled transportation and machines changed the economy and the structure of society yet now the IoT reach incorporates transportation and machines too however it is propelling this hardware with new innovation as very little sensors that gather and hand-off data. The essential drive for robotization IoT is to lessen altogether working uses when mechanization gadgets, sensors and actuators become Internet-empowered gadgets. It's the following enormous jump in profitability on the grounds that there are significant focal points to be gotten from the management and association of already unbelievable measures of data. The IoT is a huge system of equipment, programming, and applications that cooperate with each other. Notwithstanding innovation like sensors and ecological meters, things can incorporate regular articles like coffee pots and autos. These items are outfitted with sensors to accumulate information and web ability to share the data. The figure below shows how IoT integration makes gadgets smart.



**Figure 8.3:** *Adoption of IoT in automation*

Computerization is a key rising subject for undertakings of different sorts. Today we see a flood of advanced change clearing businesses, for example, fabricating, vitality, transportation and that's only the tip of the iceberg. As urban communities hope to end up smarter, many are investigating how they can exploit these equivalent abilities to all the more likely deal with their advantages and activities. For instance: how might they best apply the standards behind computerized mechanization and the omnipresent network of the web of things (IoT) to address urban administration challenges - from the upkeep of city assets to sanitation to open safety?

In straightforward terms, IoT interfaces various gadgets through the web and in this manner; we can share the information which

robotizes the business forms completely. Robotization ventures particularly the appropriation and assembling businesses need to utilize enormous measures of sensors into gadgets, so interfacing them with the web helps in diminishing the generation cost, time, endeavors, and upgrading lean assembling. IoT structure gives the discussion a chance to happen straightforwardly between interfaces of M2M, which facilitates the procedure of information move. Quality checking has turned out to be simpler than before in light of the fact that hardware and processors can screen gadgets. The effect of IoT has turned out to be significant on account of its capacity to utilize gadgets to their fullest capacity at a remarkable level. The objective of utilizing IoT is to cause gadgets to convey easily and without any stops.

In specific manners, urban communities look like modern conditions (though on a lot bigger scale), and face a significant number of similar difficulties. Thus, they should have the option to foresee mechanical issues inside the civil foundation of different types, from waste-water handling plants to traffic the executive's frameworks to road lighting establishments. They additionally should have the option to convey a wide scope of city administrations to inhabitants and guests the same; much of the time, as their populaces and geographic impression grow quickly too.

Speaking especially about the smart city, some city governments have perceived the potential advantages of smart innovation and some have begun incorporating it into an existing framework. One driving smart city, Barcelona, has been working with Cisco since 2012 to actualize IoT over its district.



News outlets have revealed that Barcelona utilized its effectively broad system of fiber-optic link to introduce about 20,000 ecological meters that screen vitality utilization over the city. Barcelona's lampposts help decrease vitality utilization through LED lights and sensors that change brilliance when walkers are close by, yet the streetlights additionally give different capacities.

They measure traffic, air quality, noise, and crowds, and even offer free web access through an association with the city's Wi-Fi arrange. Refreshing the lampposts alone produced vitality reserve funds of 30% over Barcelona's lighting framework. IoT innovation additionally changed transportation in Barcelona; sensors inserted in parking spaces enable drivers to effortlessly discover open spaces and pay for leaving on the web, while computerized transport stops offer free Wi-Fi, USB charging, and transport refreshes.

The outcome was that residents and guests of smart urban communities experience improved personal satisfaction. Seeing genuine advantages of smart innovation, an expanding number of metropolitan territories have put resources into their own IoT systems. This speculation is relied upon to keep on developing quickly.

In the zone of co-ordinations, for example, shipping organizations are as of now expanding their utilization of scanners and sensors to improve tracking of containers, equipment and vehicles. Online vehicle inspections are enabling technicians to perform proactive maintenance, helping diminish armada personal time.

The analysis offers the possibility to figure and give an account of traffic clog and gauge conveyance times. Essentially, organizations in regular asset businesses are now utilizing automations that help constant video nourishes - to examine pipelines, keep an eye on the advancement of yields and scope of equivalent administrations. These equivalent methods can be utilized to improve situational mindfulness for open well-being associations and salvage groups just as for the visual assessment of city foundation, for example, bridges.

The present mechanization frameworks applications are awkward and complex to create. Each new gadget requires a lot of customization and support just to play out the fundamental assignments. Interfacing unique gadgets and frameworks involve a significant level of designing unpredictability, due to various information groups, differing systems, inconsistent IP tending to plans, many working frameworks, and more.

Altogether, for the market to truly take off, a few key innovation obstacles should be tended to: genuinely pervasive gadget interchanges principles and new programming design to help hugely distributed, complex occasion drove information the board. The present-day absence of interoperability with disseminated, constant gadget information stays a tremendous obstacle.

As a general public, we are truly at the earliest reference point of this procedure. A little, yet developing number of forward-looking urban communities have started to send the sorts of abilities depicted above; commonly in generally constrained geographic zones, and frequently cited in and around specialized colleges and modern grounds.

## Design strategies

Urbanization today comes with a lot of pointers attached to it, adding various constraints and requirements according to the changing times. Cities should be suitable and ready for large growth in populations, and also these cities should bear the capability of withholding and managing the ever-increasing carbon footprints, three-fourth of which is produced from these cities only (reference?). As changes in climate are threatening many cities through various dangers like rising sea levels due to melting icebergs, increasingly changing weather patterns as well as resources depleting. The government in both developing and developed nations are facing the scrutiny for not only improving the social conditions or changing economic prospects, but also, drastic steps that are needed to be taken to get environmentally-sound cities.

The present reality needs affirmation that urban communities are moving towards more vitality productive and low contamination. Today the interest is to make layers of smartness, by simply utilizing assets productively as well as get the monetary soundness required, get the necessary well-being principles, and the capacity to confront any future difficulties that may emerge. Or more this, we have to get a connection between these layers so we can bridge information which can upgrade the efficiency of city activities and to raise the urban expectation for everyday comforts more.

New approach towards these smart cities can go from getting a full city-wide arrangement, for example, covering framework, financial, business, administration strategies, transport and well-being, utilizing the strategy for computerized ace wanting to every one of the undertakings independently, in the case of structure, focuses, grounds or be an open spot. The primary string in this is a minor comprehension about the interrelation among the expert administrations and joining them together to work all the more adequately while at a similar occasion looking for greater chances to get efficiency with the utilization of innovation.

The capital expenses and ROI can be a significant part of basic leadership in the city administration. An incredible model is the high line venture in New York. Here one anticipate has changed a scourged, and a molding zone of the city and has been the hotspot for \$2bn of private ventures (reference?), including a huge number of new private spots, a huge number of new openings, inns, new eateries, displays and shops. The examples below illustrate how cities are being re-designed with IoT integration.

**Existing design:** In London, there is a traffic board framework known as SCOOT, which improves green light term time at traffic crossing points by nourishing back to magnetometer and preparatory information which is in a circle to a supercomputer, which can help in the coordination of traffic lights over the city. The city of *Northern* has around 20,000 sensors interfacing structures, transport, foundation, systems and utilities, which offers a physical space for experimentation and approval of the IoT capacities. In Santander, the sensors can screen the degrees of contamination, commotion, traffic, and parking.

Kyiv has advanced vehicle dispatch framework. It contains GPS trackers, introduced on each open vehicle, just as 6,100 video reconnaissance cameras which throughout the day screen the traffic. The accumulated information through these sensors is utilized by neighborhood traffic management service and transport application engineers.

There are various other methods that can be used to transform a common city into a smart city. Application of these methods in real life depends on the investment to be made in the project. Some of the major innovations that we will see in the future of smart cities are described in the following sections.

### Smart parking management systems

Smart parking management systems are the needs of the future. There is a lot of fuel and energy wasted by the individuals in the cities just for finding a parking spot. Wasting time on such low work could affect the efficiency of the individual in his life and work. Innovation is used to control the traffic of the city in many ways and up to a great extent. In the past, not a lot of efforts and investments were made in finding the solution to this particular problem. But the future has more to offer, and investment on large scales is now being made to find the solution to this problem. Latest innovations suggest that there is a system being developed which allows you to not only see the available parking spots but also book them for a particular time period, just before leaving home ahead of time.

### Smart transportation

Automobile manufacturers are designing and producing vehicles that they believe are going to be an essential element in the smart city agenda. Companies are working towards and have showcased fully autonomous vehicles, with cars capable of understanding their surrounding environment and make decisions. People might not even drive in the smart city. Just step into a smart vehicle that communicates with the machine learning and other surrounding vehicles to ensure that you reach point X from point Y, both ASAP and as efficiently as possible.

### Smart energy.

The idea of having zero-emission in the city is about to become a reality wherein the future smart city must and will work on clean energy sources to meet its power needs. Energy production and supply in smart cities are efficient and wasting less energy because of the real-time data collection and analysis. solar and wind energy will be two major sources of energy and will be collected all through the day and sent to the required areas. Integration of solar energy with the roads, buildings, and residential areas will be one key characteristic of a smart city.



### Smart infrastructure

As indicated over, the examination of information will assume a greater job in the arranging of increasingly smart urban areas down what's to come. With the information gathered, city organizers and designers would have the option to make structures that are streamlined and appropriate for individuals dependent on past information. Innovations that were non-existent before can be tried on residents to guarantee that they are really profiting the natives.

Likewise, a smart foundation takes into account the avoidance of numerous general medical problems, possibly halting airborne ailments or water pollutants before they even occur.

### Issues and challenges

When we hear the term smart city, we usually think about all those cities which we see in science fiction movies, where there are flying cars, smart robots picking trash and garbage and keeping the city clean but the reality is far from it. Although we are moving closer to those futuristic cities as many smart devices are used such as IR scanner, face recognition, fingerprint scanners for public records and security cameras for public and traffic safety. Transparency may be good, but it is something we need to be careful because it touches on the ultimate nerve of what makes us human. They are in our hands to shape the future but not the other way. That's machines shaping our future.

Despite the fact that a significant number of people are now partaking in the IoT upheaval; they do so without realizing their involvement and acceptance. The most mainstream items in this class are web empowered TVs, phones and tablets. When arranged, these associate every day with cloud administrations and specialist organizations' servers with no mediation from us. Because of this smart innovation improvement and IT, issues keep on developing, although numerous engineers and different pioneers handling these issues yet at the same time there stay some basic difficulties and issues looked by some smart cities even today.

IoT devices with constrained usefulness have been around for, at any rate, 10 years. What has changed as of late is the state or quality of being present in all parts of a particular thing or place of network alternatives (WIFI, 3G, and Bluetooth and so on.), cloud administrations and investigation, which are incredible empowering influences for IoT. The cloud gives a stage to facilitating savvy programming, organizing countless IoT devices and provisioning them with a lot of information. This empowers smart choices to be made without human mediation. In any case, there are still some present difficulties restricting the appropriation of IoT:

### **Challenge no.1: Power and technological advancement**

The smart city utilizes various types of sensor in huge numbers, which required an exceptionally great or large amount of electrical power and technical staff to establish smart circuit and connections and all these required a huge amount of capital, which any government does not want to invest. The figure below shows some of the electrical power challenges in Smart City initiatives.



**Figure 8.4:** *Challenges in smart city*

### **Challenge no.2: Hackers attacks and security leak threads**

As a smart city utilize many sensors and cameras such that these are interlinked with each other through internet connections, so there is always the threat of Hackers who try to sabotage these systems or steal confidential information through system loopholes.

### **Challenge no.3: Policy development**

Another roadblock for smart cities is policy development, although it is good that many cities are moving towards technological advancement and innovations however policies and governments are

not always quick to these changes especially when it comes to city building code and road policies such as smart parking system and smart traffic system.

#### **Challenge no.4: Inclusive development**

As the smart city develops, it has to make sure innovation also benefits marginalized people such as farmers, informal workers, micro-entrepreneurs, indigenous people and the poor in order to close the digital divide amongst the citizens.

#### **Challenge no.5: Long term sustainability**

Final challenge which these smart cities face is long term sustainability, which is not just in terms of the environment but in power consumption and regards to the amount of infrastructure that is needed to be maintained. Today smart cities move towards short term benefits rather than long term sustainability with technological advancement, which adds a very fast route requiring a continuous update in order to stay current for the future.

### Factors affecting automation using IoT

This phase gives the demanding situations confronted all through automation the usage of IoT:

**Privacy and safety:** when the numerous varieties of facts are collected and evaluated within the IoT platform, the system may be at risk of several attacks. Furthermore, the gadget may be subjected to much vulnerability due to its multi-tenancy nature. Consequently, smart cities need to adopt very severe measures to ensure the privacy and security of 'their citizens' information. Without such assurance, residents may not believe the authorities and the collection of data or information may be difficult. Consequently, all structures and devices must be made resilient and resistant to such cyber-assaults.

**Heterogeneity:** IoT structures are usually evolved with precise solutions in which every element of the principal device is joined, retaining in mind the special application context. On this foundation, the government need to study the goal scenarios, make a list of the wished hardware/software program and in a while combination, those couple of heterogeneous subsystems to form the complete system. Offering such substructures and the guarantee of a right cooperating working amongst them is going to be a primary undertaking.

**Reliability:** IoT-based systems can motive some reliability problems. As an example, because of an automobile's mobility in special orientation, the interconnection among them may be very unreliable. Furthermore, the participation of huge numbers of clever technologies might lead to a few reliability demanding situations, mainly regarding their failure.

**Large scale:** Scalability is going to be a difficult as running on an extensive scale requires the mixing of sensors on a large scale. Making the sensors and subsystem occur at the same time or with coordinated timing is a major challenge.

**Legal and social components:** The IoT machine is specifically provided according to statistics supplied by the customers. For such structures, the provider (vendors) needs to be primarily based on numerous nearby and international regulations. Likewise, the candidates can also face enough incentives to take part in a certain situation and data gathering. it would be extra powerful if people were to determine the software and participation in these activities. Regulations of the city ought to be gentle closer to such facts collection and helpful closer to the software element.

**Big Data:** Considering round 90,000,000,000 devices, its miles truly going to be essential to be aware of data transfer, garage and recall as well as the analysis of the massive amount of information generated by means of them. It should be clear that the IoT substructures might be among these good-sized resources of massive statistics. In massive facts troubles, 3 principal highlights are present, including the range, velocity in addition to variance.

### Existing application areas

IoT can possibly tame the weight of urbanization, make new understanding for city inhabitants, and make everyday living increasingly agreeable and secure. IoT-empowered brilliant city use cases range numerous regions: from adding to a more beneficial condition and improving traffic to upgrading open security and streamlining road lighting. Beneath, we have given some cases of the most mainstream use cases that are as of now executed in shrewd urban communities over the globe.

The key motivation for IoT applications in smart cities

**Cost** is key in the choice to change over from wired answers for remote, as it is enormously costly to introduce and look after landlines.

**Remote control:** Most IoT devices can be controlled by their user through something as simple as a smartphone. In fact, 76% of smart devices are currently controlled by smartphones since IoT technology enables its users to control their smart devices remotely. A greater sense of control and influence is available to the people and computers using IoT technology. Managing processes and control systems are much easier now. You can log onto your connected devices and systems using IoT technology from anywhere in the world. The devices will even work on their



own accord according to the mandate they've been given. This is probably the greatest benefit of driving IoT growth in 2019.

**Automation:** The ability to control things remotely also opens new doors. Automating systems and processes are among the top benefits of owning IoT-capable devices. You can program a computer nowadays to do almost anything. So giving it the capability to connect to other devices worldwide will help automate businesses, services, and many more industries.

Smart home products love to market home automation as a major benefit. Automation-based smart home technology leads to four specific benefits for its users:

Saved time

Saved money

Saved effort

Improved personal security (smart security products)

Automation will be one of the key value propositions to grow the consumer IoT market in the coming years.

**AI integration:** Interestingly enough, artificial intelligence is older than the internet. *John McCarthy* is widely referred to as the father of AI. The term *artificial intelligence* was adopted back in the 1950s

at a Dartmouth Conference. Research centers all across the US, and subsequently, the world, started looking into the nature and uses of AI. Today we find traces of AI within some of the most basic smartphone apps, like predictive text. But generally speaking, a computer will only have AI capabilities if the software is installed directly onto the device. Thanks to the global IoT market, you don't need to download AI software to gain its benefits. AI can be integrated into any system or device that logs onto IoT networks.

Some brief AI benefits:

Saving humans from monotonous tasks

Eliminating human error

Enhancing automation

Combining technology and creativity

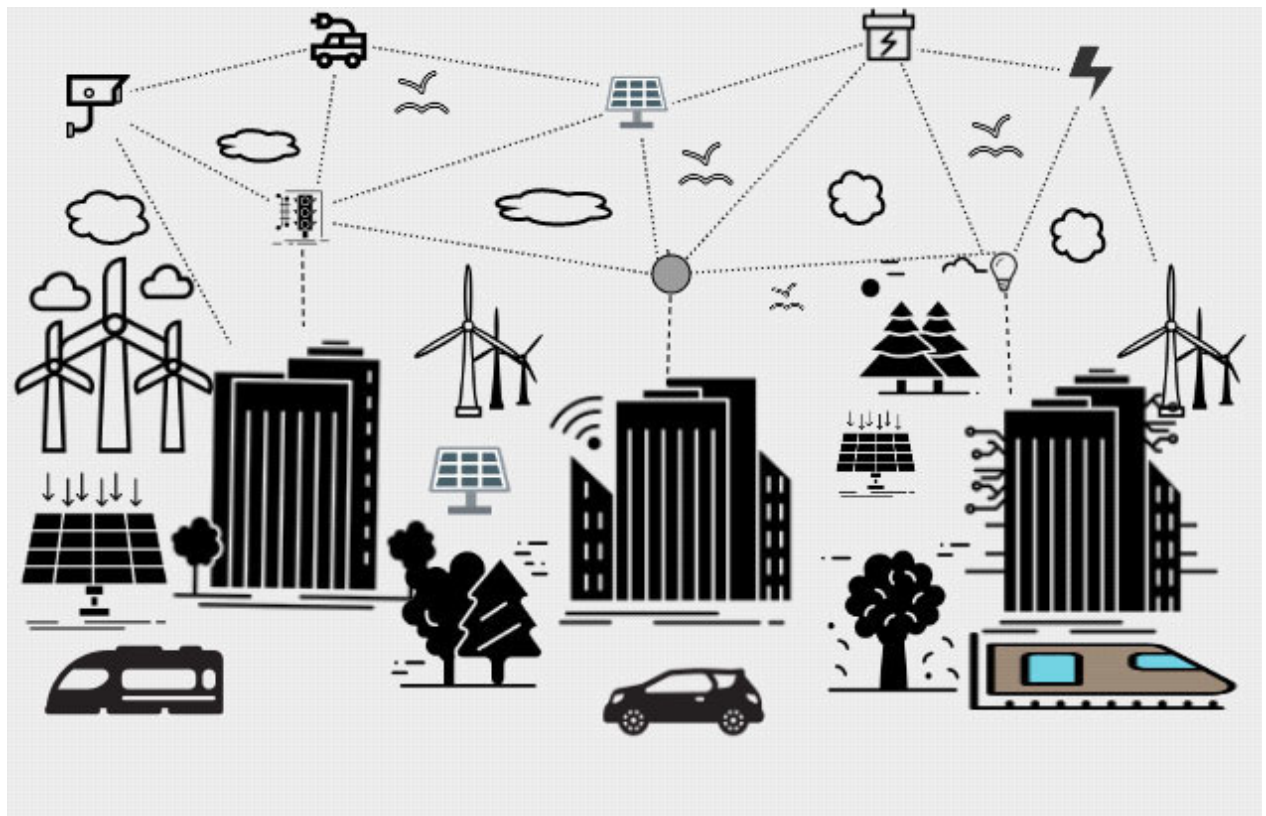
The expanding pace of reception for IoT and robotization frameworks in ventures is relied upon to drive the development of worldwide AI as a help (Malays) advertise. Improving client encounters by fortifying deals and promoting with more prominent experiences is one of the essential forces, either internal or external, that impels driving the AI appropriation today. A few mechanical mechanizations are driven by billions of associated gadgets. Attributable to IoT, smart and associated advances have expanded the route of information that is being made, exponentially. Accordingly, enterprises are discovering chances to

investigate this information and devise new arrangements so as to improve the current foundation and working efficiencies.

Subsequently, the expanded selection of computerization in all business portions and consistent development in mechanized innovation that guides to diminish the time and overhead costs supports interest for AI as assistance over the globe.

**Efficiency:** This is another significant motivating force. In most wired arrangements, administration faculty should physically go to the establishment site to review and administer the interchanges infrastructure. By differentiate; remote correspondences empower remote checking and the executives of IoT organizations. This empowers chairmen to perform firmware updates and security fixes over the whole organization, and get mechanized notices in case of any issues.

**Asset control:** Asset decrease is frequently a driver also, especially being used cases, for example, brilliant road lighting and observing resources. The figure below shows some of the areas where IoT can be applied to create Smart City.



*Figure 8.5: Application areas of smart city*

### IoT applications in smart cities

Integration of IoT in day to day processes and life of a person is one feature of a smart city. The air conditioning system in your house could receive commands from your smartphone, or take cues from the motion sensors in the house, that help to reduce heating or cooling efforts when no one at home. Smart assistant software running on our smartphones is now imaginable, sending notifications to the house, which activates the heating or cooling start when one is thirty minutes from home.

Similarly, a diagnostic machine or medical device in a doctor's office or lab hooked up to a private cloud can trigger a technical support call, or the shipment of new consumables from the supplier when required. Other existing examples that have been implemented in some form include having a manufacturing execution system trigger the **ERP (Enterprise Resource Planning)** to order new parts for a manufacturing plant or the reading of electricity consumption and automatic billing of customers.

It is not even a far-fetched idea to have your refrigerator trigger a service call or order milk automatically from a nearby shop for delivery when you use the last drop in your morning cereal. The only limit is your imagination.

## Road traffic

Smart urban ensure that their locals get from one point to another as safely and viably as possible. In fulfillment of ensuring safety, governments have taken the steps towards IoT development and to execute wise traffic game plans. Smart traffic feature makes use of different sorts of sensors, getting GPS data from drivers' PDAs to collect data and then choose the number, region and the speed of vehicles based off on this information. Meanwhile, smart traffic lights related to a cloud the officials stage grant checking green light timings and normally adjust the lights reliant on current traffic situation to stay away from congestion. Besides, using recorded data, adroit responses for traffic, and the administrators can expect where the traffic could continue to take measures to turn away potential blockage.

For instance, being, a standout amongst the most traffic-influenced urban communities on the planet, Los Angeles has executed a keen traffic answer for control traffic stream. Street surface sensors and closed-circuit TV cameras send constant updates about the traffic stream to focal traffic the board stage.

### Smart parking

With the help of GPS in users mobile phones (or street surface sensors installed in the ground on parking spaces), one can decide if the parking spaces are engaged or accessible and make a constant stopping map. The moment, parking spaces are available, people get a notification regarding the same with a guide that will lead them to the vacant space. This helps in making the whole process of parking, streamlined and quicker, bypassing the episode of indiscriminately moving around for parking.

## Waste management

Many waste accumulators void their containers as per fixed schedule. However, this is not an effective methodology since it prompts the inefficient utilization of waste containers and enormous fuel utilization by waste gathering trucks.

A smart city arrangement powered by IoT helps in upgrading this whole system by providing it with a planned enhancement through operational analysis. Every garbage container will be provided with a sensor with edge detection. When the amount of garbage crosses a certain edge, the waste management department is informed. Similarly, this can be done for other containers as well and then a collection vehicle can be deployed. Along these lines, the truck driver discharges a full holder, abstaining from cleansing half-full ones.



## Environment

IoT powers city allow tracking parameter crucial for a healthy environment and thus maintaining them at an optimal level.

For instance, in the case of water quality, we can lay down a system of sensors over the water matrix and associate them to a cloud platform. Sensors measure pH level, the measure of broken up oxygen and disintegrated particles, and if any changes occurred in any of the fields, the cloud stage triggers are suit characterized by the clients. For instance, if a Nitrate ( $\text{NO}_3^-$ ) level surpasses 1mg/L, a water quality administration arrangement arm support keep groups of sullying and consequently makes a case for field laborers, who at that point begin fixing the issue.

Another utilization case is of checking air quality. For that, a connection of sensors is laid along occupied streets and plants. They accumulate information on the measure of CO, nitrogen, and sculpture oxides, while the cloud stage investigates and pictures sensor readings, so which users with cloud platform can see the guide of air quality and utilize this information to bring up regions where air contamination is basic and workout proposals for natives.

### Public safety.

For improving open well-being, IoT-based keen city advancements offer continuous checking, investigation, and basic leadership apparatus uses. Joining information from acoustic sensors and CCTV cameras conveyed all through the city with the information from online networking feed and investigating it, open security arrangements can foresee potential wrong doing scenes. This will enable the police to stop potential culprits or effectively track them. For instance, in excess of 90 urban communities over the United States utilize an acoustic gun fire identification arrangement. The arrangement uses associated amplifiers introduced all through a city. The information from microphone pieces is sent to the cloud stage, which breaks down the sounds and identifies gun fire. The stage estimates the time it took for the sound to achieve the amplifier and assesses the location of the weapon.

This is just a beginning, the reamorous amount of existing applications of IoT in smart-cities but here we have explained only a few. Also, it is predicted that in the coming time, we all will be surrounded by living and non-living things connected to the internet, and we will use the major part of daily routine.

## Education

Education has an important role to play in sustainable development and increasing the literacy rate. It uplifts the economy of society, which helps in improving the quality of life. Education contributes to the nation's economy. Now the question rises again, which is to be given more importance the primary education in every corner of the country or improving the education system in a smart city. The new-age schools and colleges in smart cities are equipped with data-driven and software-mediated infrastructure, which will help students learn through digital media.

Digital education will surely help students in enhancing their skills. Moreover, every college in the smart city should possess a college app from which student can pull out the maximum data, that is, when they are on campus, distance to reach their destiny or a particular place in college and all the notice displayed by department heads and hostel, wardens, timetable, and more. This will surely help new students on the first day of their college.

### Case study: How students benefit from smart technology in education

It is the first day of Amit at university, and it's his first time he is away from his home. He is very excited and enthusiastic, but as he enters the vast campus, he seems to be lost as he is experiencing this scenario first time in life. He pulls out his smartphone from his pocket and opens the university app, sign in using the profile and comes across various options. He tries to open his current location, and he is successful at this attempt. The app showed the notification that there is fresher's meeting in 15 minutes at the east wing of the university and the rest of his university day schedule.

In turn, the app also shows the direction to reach the east wing on foot or using public transport. After attending the fresher's meeting, he gets the direction of his first lecture and the details, that is, name of the lecturer, course name and today's topic. This technology helped Amit all day in attending lectures and other important notifications. The latest feature added to university app is that it tells you the distance covered, and calories burned in a day if you carry your smartphone all day along with you.

As the days pass, Amit gets his attendance status on it and also the dates of the upcoming mid-semester exams and assignments that he has to submit. The contact number of group members in the project task is also updated with their hostel room number so

that they do not feel any difficulty in contacting them. Additionally, it also shows Amit that 'Elements of chemistry' book is due, and he needs to return it by tomorrow. He is also notified of the need to pick up future books as the syllabus proceeds. Smart technology does make a difference in a student's life and will continue to help the student to evolve.

The reforms that are going to happen in the schools of a smart city are as follows:

Constantly imparting real-time knowledge and learning, at par with the changing world.

Experimentation, analysis and implementation of the learning.

Students' overall development through advanced technology.

Guiding of individuals at every phase and providing everyone with equal opportunities to excel.

Imparting qualities like responsibility and inspiring them to contribute towards the development of society and humanity.

## Healthcare

Healthcare is one of the most important facilities which every smart city required. Smart healthcare combines smart technology and the latest mobile devices with health. Let's have a look at how healthcare looks like in a smart city.

### Smart healthcare for smart cities

As cities population are increasing day by day, but due to this there is also an increase in number of various diseases and number of patients, so it is very difficult for administration to cope up with the healthcare issues, but for smart cities, health is one of the most important aspects, so that is why there is a need for smart healthcare system. Smart healthcare combines smart technology and the latest mobile devices with health. Numerous ways have been designed to increase or encourage healthy habits, and well-being due to the smart variable devices such as fitness trackers and fitness bands have been invented which can help in monitoring a person's health condition through apps in smartphones. These devices not only monitor health conditions but also provide a solution if needed at the right time. Through these devices, doctors, researchers, and healthcare professionals can provide better-personalized healthcare through the data which these devices collect.

### Technology roles in smart healthcare

As connectivity is the basis of the smart city, therefore smart Healthcare is also based on connectivity as it provides better connectivity of a patient with his doctor through real-time data collection and monitoring ability, thus improving the effectiveness of the overall healthcare system. These data need to be collected and managed; therefore, AI needed to be applied. All these data can be properly recorded through blockchain technology; this method not only maintained electronic health record but also can link to other services such as payment and Insurance claim. This technology is not only critical for smart cities but can overall improve the living and health conditions of citizens.



## E-governance

E-governance is one of the applications of ICT for turning in diverse authorities offerings, exchanges in records, communication transactions, and integration of diverse systems and services between **government and citizens (G2C)**, **government and business (G2B)**, **government and government (G2G)**, **government and employees (G2E)** as well as various back-workplace approaches and interactions within or outdoor the complete government framework. With the help of e-governance, numerous governments' services are given to the citizens in an efficient, handy, and obvious manner. There are 3 essential goal organizations inside the system that can be distinguished in governance principles. Those are authorities, residents, and groups/hobby organizations. In e-governance, distinct limitations are not there in the previous years, and e-governance has won plenty greater recognition in making unnecessarily complex enterprise global. Among the control scholars have defined this idea of e-governance, that's rising as a crucial and crucial pastime inside the business subject. It's far installed that e-governance transforms the effectiveness, performance, transparency and accountability of transactional and informational exchanges inside the government, among diverse government and one of a kind government. Companies on countrywide, state, municipal and local stages. The arena bank explains that E-governance is the use by means of government businesses of IT(such as WAN, internet, and mobile), which have the ability to talk records and services, in addition, to rework

relations with citizens, businesses, and other parts of presidency. These technologies can serve as a ramification of different ends: higher delivery of numerous authorities' offerings to residents, improvement in the interactions with business and enterprise, fundamental citizen empowerment thru getting entry to facts, and extra efficient authorities' management. The resulting advantages from such an era can be much less corruption, greater convenience, elevated transparency, sales boom, and or cost discounts.

### Features of e-governance

E-governance and with the involvement of the public in selection making manner will be an essential issue of clever governance.

The equipment used to gain them are as follows:

**Use of ICT:** This implies that using computers, the internet, and different digital gadget for accumulating, processing, sharing, and retrieving of statistics. Such channels might be used to convert the effectiveness, efficiency, transparency, and responsibility of transactional and informational exchanges inside authorities.

**E-session:** Human beings' participation will be the principal feature of E-governance. There can be a right channel of the interplay among government and citizens. Residents have to be empowered to elevate voice for his or her reviews, thoughts approximately government applications and schemes, and so on. Their remarks concerning various guidelines have to attain without delay to the leaders, counselors, city managers, or nearby head.

## Industry.

The most important aspect for the smart industrial city is economic growth, to achieve this, the first step is to engage with the stakeholder then the required administrator should be diligently thinking about public-private partnership, policies and citizen partnership. It is a well-established fact that today most of the world population is concentrated in urban areas, approximately 55%, and this number is still increasing.

The major benefits smart city provide to industries are as follows:

**Smart infrastructure:** In smart cities, houses and office spaces are installed with IoT devices. These devices provide useful streamline information about everything in an office building. All the industries usually which are installed with IoT devices see an increase in productivity and an increase in efficiency of employees. Smart buildings also provide better security through CCTV cameras and also provide protection against cybersecurity and data leakage risks.

**Increase efficiency in travelling:** In smart cities travelling from one place to another is very convenient as there is a proper traffic control system which utilizes IoT devices and AI technology to optimize traffic control of the city due to this there are no traffic jams which saves time in travelling. Thus employee can reach their offices on time, and there is public transport such as metro

and buses and taxi services like Uber and Ola which make travelling more convenient.

**A smart economy:** Smart cities provide a huge economic boost for industries. Business growth will increase exponentially because, now, with the help of the data collected, they can better understand their target buyers and can focus their products towards the same providing a more comprehensive service. For example, travel industries use the data on how many tourists come to visit the city yearly and which tourist spots is most famous so that better planning could be done for the tourist.

**Increase sustainability:** As the population in the cities grows, we require sustainability. As these smart cities use resources which impact surrounding environments; therefore, the smart city seeks to reduce the damage, it caused by looking for sustainable solutions. Those industries which are situated in smart cities also come under sustainability programmer. Due to these problems such as disposing of waste and energy consumption will automatically be solved by these cities. With constant demand for new technology, industries will also be developed; this will become a powerful source of new innovation. New technology and business will rise up, and more jobs will be created, which will provide good profit.

## Conclusion

Smart cities use information and innovation to make efficient, improved maintainability, make monetary advancement, and upgrade personal satisfaction factors for individuals living and working in the city. IoT, in the form of interconnected sensors, collect data that help the cities to improve performance of physical infrastructure in main socio-economic areas of the country and therefore this is what it takes to build a smart city. The focus of any smart city should be its people, providing benefits like Better quality of life and sustainable livelihood for residents and visitors. A smart city is a position given to any city that uses ICT to improve the quality and performance of services such as natural resources, transportation and utilities to ultimately reduce resource usage, wastage and overall cost concluding, IoT enables businesses to achieve more visibility, collaboration, and smoothens it; thus, helping the business to flourish.

## Exercise

### Multiple choice questions

What are the important components in IoT?

Hardware

Software

Verbal Exchange Infrastructure

a and b both

In order to promote \_\_\_\_\_, the government should employ more management.

eGovernment related

Enterprise-based

Company based

Business-oriented platform

The \_\_\_\_\_ may become an important facilitator and stimulate for the modern economy.



eGovernment related

Enterprise-based

Company based

Business-oriented platform

The core element of architecture of smart city is \_\_\_\_\_

Mobile Unified Service

Urban Application Platform

Management center

Integrated Information Provider

The core element is operated by \_\_\_\_\_

PaaS

IoT service Provider

SaaS

IaaS



## Answers

A and B Both

eGovernment related

Business-oriented platform

Integrated Information Provider

IoT service Provider

### Descriptive questions

What role does governance play in developing smart cities?

What can be various perspectives on *Definition of Smart City*?

Are there any models or frameworks available for assessing (the state of development of) smart cities?

Is there any connection between smart cities and resilient cities?

Is there any difference between compact city and smart city?

What is a smart city? How do you define it?

How can measure the *smartness* of smart cities?

What could be similarities and differences between smart city and non-smart one?

Internet of Things (IoT) Use Cases

To the average person, IoT simply means the smartphone. This is because it is going to be everywhere and carried by everyone as a mobile phone. IoT is gradually being used in large numbers on a daily basis across the globe giving users (industry and consumers) more options and efficiency.

Therefore, understanding the role and the extent of IoT deployment and adoption is very important. The future of IoT applications in our lives now appears to be limitless as it is envisaged to provide solutions in all sectors including education, healthcare, fashion, military, transportation, manufacturing, fashion, and more. Due to the low cost of IoT and its intelligent connectivity abilities, it is now being deployed in an unprecedented number and in an unimaginable, non-traditional way. This chapter is intended to present some of the ways that IoT is being deployed and throw more light on the emerging trends and the possible future direction of IoT applications.

## Structure

In this chapter, we will focus on the following use cases of IoT:

Industrial Internet of Things (IIoT)

Smart energy

Smart transportation

Smart health

Smart buildings

Smart education system

Smart cities

## Objectives

At the end of this chapter, you will be able to:

Identify the various ways, in particular, the three key areas of industry, consumer and governance categories in which IoT concept has been applied.

Understand how IoT devices work in concert to ensure end-to-end delivery of projects, ideas and initiatives.

Expand your creativity beyond the current traditional approach by industries in using IoT to improve profitability and return on capital.

Appreciate the fact that the concept and application of IoT in addressing contemporary problems in our society and community have no boundary but only subject to the imagination.

## Introduction

**Internet of Things (IoT)**, in simple terms, means connecting all the things in the world to the Internet. Soon, IoT will become IoE, the Internet of Everything! Generally, in the domain of IoT, all the things that can be connected to the Internet can be categorized into three, namely:

Things that collect and send information. For example, devices and sensors such as motion sensors, temperatures & moisture sensors, and more automatically collect information from the environment and send to a central point (server) via the Internet in most cases.

Things that receive the information and act upon it. This is where IoT devices or sensors add value to the entire concept of IoT and smart ideas. IoT devices at this point are not just passive but active whereby the devices initiate commands or actually act on commands to execute activities to give owners the desired or anticipated objective or goal. For instance, when your door sensor *sees* or *feels* your presence and then it automatically opens the main gate or door for you to enter without you having to turn the doorknob or push the gate to open. Things happen in a seamless and natural way to your convenience.



Things that do both, collect, send, receive and act on the information. The epitome of IoT device in the value-adding process is when such devices are able to do the two activities above. For instance, using the example of agriculture (farming), sensors, or devices that are attached to animals or planted in the soil are able to collect the movement of the animals or the moisture in the soil respectively while at the same time they are able to water the crop and even to certain extent identify the moisture level and automatically turns off or on all thanks to the simultaneous analysis of its own vast data that it has been collected and sent.

In this chapter, we will cover three broad categories of use cases pertaining to the IoT ecosystem, namely industrial, consumer, and governance use cases. Under the industrial use cases, we will focus mainly on two broad use cases of smart energy and smart transportation systems. The consumer use cases will feature Smart buildings and smart education.

Finally, we will show use cases on how citizens can achieve a high quality of life and engaged in conversations that impact directly and indirectly on their lives by looking at how IoT is used in governance via the implementation of Smart City projects.

### Industrial Internet of Things (IIoT) use case

Technological innovations and grand break-through in industrial operations have become possible because some companies are taking advantage of IoT in their setups. IIoT innovations such as edge, cloud, advanced 3D, and digital visualizations are but a few examples that are turning abstract industrial concepts into reality, a phenomenon that is being considered as the frontier of the 4th Industrial revolution.

Businesses around the globe must address and face global competition in the areas of geopolitical uncertainties, commodity prices, supply fluctuations, demographic and generational changes, environmental, quality, and safety enforcements.

Implementing strategies that incorporate IoT in a diverse and non-traditional way or approach could give companies a significant advantage, leverage, and a competitive edge in the market.

Sensors embedded in manufacturing equipment that are located across the factory floor help in detecting errors and faults and hence reduction in time and waste in the manufacturing process.

The following examples are being presented to illustrate what value of digital transformation can bring to industries.

### IoT and smart energy.

Construction of digital buildings is now becoming the future of buildings in the real estate and property market. Digital building is where IoT devices and sensors are embedded in buildings at several locations to help monitor and identify fail points for correction, efficiency and optimization of energy resources.

Hitherto, when a power outage or failure occurs, it takes several hours and physical leg work to determine and verify the point of failure for correction.

By embedding IoT devices in the building, operation centers are able to determine and restore services based on the analytics of data generated by these IoT devices and sensors. A key feature of IoT power management is its ability to recycle input power remotely to correct faults without the need for human (manual) intervention.

Thus, managing the power usage resources by way of IoT contributes to the operational budget of many organizations and industries. Certainly, this gives an increased customer (user) satisfaction as the portal provides high levels of availability and reliable service-level reporting. IoT becomes the digital building enabler turning old building and high-power consuming office into energy-efficient and cost-saving edifice.

Making changes to a building to make it smart over-night is not a matter of one single item (device) change or implant but one that is carefully based on unique and desired features (requirements). This makes it difficult for users to immediately appreciate and understand the value of smart building unless there is a significant leap to implement low-capital cost, non-invasive IoT application in key lifestyle and economic areas.

For a successful *smart building* to be achieved there is the need for three key elements to be present.

The first is what *Matt Ernst* (2018) call **Occupant** For some-time now and in the past, occupant experience and comfort had really not been a major part of the engineering conversations. Engineers had focused on designing and building systems in order to achieve energy efficiency, reliability, and other metrics goals but lesser on attention to occupant experience.

With IIoT, it is now possible and in fact, a compelling approach for engineers to design and build homes or offices or factories to incorporate those features that guarantee good user or occupant experience

Apart from meeting and even exceeding user (occupant) experience, engineers (property owners) will save or make money by embedding such devices for users. For example, occupants of buildings could save money if the water supply to their premises is fitted with an IoT device which is able to shut off automatically the water supply in the house whenever there is pipe burst and/or

send notification or alerts to his or her phone of any leakage detection

Smoke detectors do not only set off the alarm but also able to send alerts on the specific location of the origin of the fire to both the occupant and to the fire service station. Keyless entry, auto adjustment of lighting and bulbs, heaters and air conditioners according to the number of occupants in room or the complete shutdown or closure of these utilities when everyone has left are some of the smart ways that companies can achieve efficiency and high profitability with the use of IoT devices and sensors

Providing basic home automation and providing the ability for landlords to provision apartments can drastically simplify property management by reducing the cost of operations and save money.

The second is the Renaissance in the construction industry (Matt E, 2018). The rise in energy demand will be driven by developed countries retrofitting their buildings and in developing countries through the production of new building stocks. In order to address these increases in energy demands, developing countries can focus on passive approaches to maximize the scalable energy benefits while developed countries turn to active technology for retrofits.

In this respect, IoT can be used to optimize energy consumptions in buildings, both in commercial/factory and residential/private domains. Connected heating, water, cooling and electricity systems can lead to significant productivity gains when attached to smart

grids that provide automatic, scalable electricity pricing and supply to meet demands.

In a study at Fortaleza, Brazil focusing on right-sizing pumps in the municipal water supply and automatic control system was reported to have cost the government US\$1.1m but led to savings of US\$2.5 within four years of operations. In Aarhus, Denmark, the introduction of IoT into its wastewater electricity generation yielded about 90% more than what was originally needed (References).

The adoption of IoT in Energy efficiencies is not only limited to large scale infrastructural development. It can also be extended to cover private homes and domains where water heaters or washing machines are made to run automatically at off-peak periods when energy is cheap. No more forgetting to turn off the lights when leaving the room—let IoT solutions streamline your energy bill and your carbon footprint.

The third element to be present for a successful IoT implementation is architects and planners' collaboration in deploying various and diverse IoT devices. Such IoT collaboration and applications could include the monitoring of greenhouse gas emissions across the city and key industrial areas of the region.

For instance, by planting sensors across the city in places such as signposts and traffic lights, municipal authorities would have continuous and up to date access of the ecological health status of the city for the necessary intervention before things get out of hand.

Generally, therefore, IoT makes it possible for consumers to manage their own personal energy consumption with greater access to information while at the same time commercial suppliers or providers also have the opportunity to operate at an efficient level by leveraging IoT in the production and monitoring of energy for the citizenry.

By processing these sensor data, intelligent algorithms can micromanage energy usage in real-time far more effectively than we ever could, saving money and the planet in one swoop.

## Smart transportation

Transportation is very important in every country of the world and represents a large part of energy consumption and economic activities 2019). Transportation is also one of the major contributors to global pollution and carbon dioxide emissions. It is estimated that through the activities of **Light-Duty Vehicle (LDV)**, global energy consumption from transportation will continue to rise significantly if no action is taken.

In the light of this, solutions that leverage on the **International Energy Agency (IEA)** proposed **Avoid-Shift-Improve (ASI)** paradigm is urgently needed to avert any future catastrophe such as the much talked about global warming.

The avoid (avoidance) aspect of the IEA proposal or framework deals with methods that seek to decrease the need for transportation by leveraging passive solutions that make it possible for people to be close to where they need to be. Due to the fact that energy consumption per capita decreases as density goes up, so will the amount of transportation needed for people to get to their workplaces in urban mixed-use zones.

This will allow commuters to rely on walking or cycling, which are all part of the smart city agenda. Freight and logistics also benefit in megacities as goods take shorter distances to reach a larger part of consumers-something being practiced by Amazon!



The shift strategy in the ASI proposal is to encourage users to move away from the use of private modes of transportation and patronize the use of public modes of transportation, especially within the core urban areas.

For example, the use of public buses especially in developing countries where traffic congestion has become a chronic feature of the anatomy of the city can be encouraged by aligning road incentives to increase the use of public modes at a cost or tax to private modes. One such approach could be limiting parking slots to discourage driving to the **central business district (CBD)** or impose congestion pricing regime.

The *improved* methods are more dependent on vehicle-level efficiency. It is based on strategies that tend to mitigate or minimize fuel consumption of the specific vehicle of transportation. For example, by tracking the average gas consumption of a bus and taking remedial actions to improve efficiency/consumption per trip.

The key question here is what is the role of IoT in all these **Avoid-Shift-Improve** paradigms? In other words, what are the various ways in which the concept of IoT can be applied in the socio-economic life of the people to provide that *smart* transportation for their convenience, safety and improved quality of life?

In the area of transportation, IoT is being adopted across all the various facets that leverage the ASI framework towards unprecedented levels.

For instance, while encouraging people to avoid using private means of transportation to the CBD in favor of public transportation by limiting parking spaces; efforts or strategies must also be put in place to ensure that the search for these parking lots do not end up creating inconvenience to the motorist as such situations become counterproductive and antagonistic.

Thus, the use of IoT devices and sensors in the smart parking system where approaching motorist could be notified on their mobile phones or car dashcam about the parking availability (capacity) status and other related traffic information. More importantly, through mobile apps, the parking space, toll fee, and more, can be purchased and reserved in advance, providing convenience to users.

IoT in the transportation industry could lead to a rise in **Shared Autonomous Vehicle (SAV)**. This is because, IoT will make it possible to know various aspects of one's journey such as the arrival, routes, passenger load, location, price, and many more, of the vehicle ahead of time for timely decision making.

There is efficiency everywhere! Studies have shown that rideshare reduces the driving distance by 27% and also replaces up to 12 regular cars on the average on the road (reference). The shift from motorized to non-motorized such as scooters especially for last-mile commuting will require the embedding of IoT to enable

users to locate free scooters, including other parameters such as electric power level (indicator), and more.

In the not too distant years to come, **Artificial Intelligence and Machine Learning (AIML)** will dominate and shape the transportation sector. This will enable automated and predictive analytics for better decision-making and road safety leading to lowering of costs and fewer disruptions to transporters.

## Smart health

Healthcare institutions are beginning to adopt IoT technologies into their operations at a very fast and lightning speed. The desire to make their health outfits smart is to enable them to achieve excellence and efficiency across all aspects of health care; from remote monitoring to health care delivery and in-home diagnostics.

Leveraging on machine learning and IoT capabilities, healthcare institutions are striving to provide around the clock real-time data streaming, medication alerts or reminders and real-time medical diagnosis, all with the view to reducing the need for hospital visits.

IoT in the healthcare industry, which is also sometimes referred to as the **Internet of Medical Things (IoMT)** consists of all and any medical devices, patient monitoring tools, wearables and sensors that send and receive signals to and from other devices via the Internet.

These devices store a massive amount of data that are analyzed for actionable insights so as to prevent and/or manage diseases in the form of patient care needs. For example, predicting patient's deteriorating illness or condition requires continuous feedback be it from the bedside or home through monitoring tools such as Bluetooth enabled blood pressure machines, scales, thermometers, pill bottles, and many more. Thus, machine

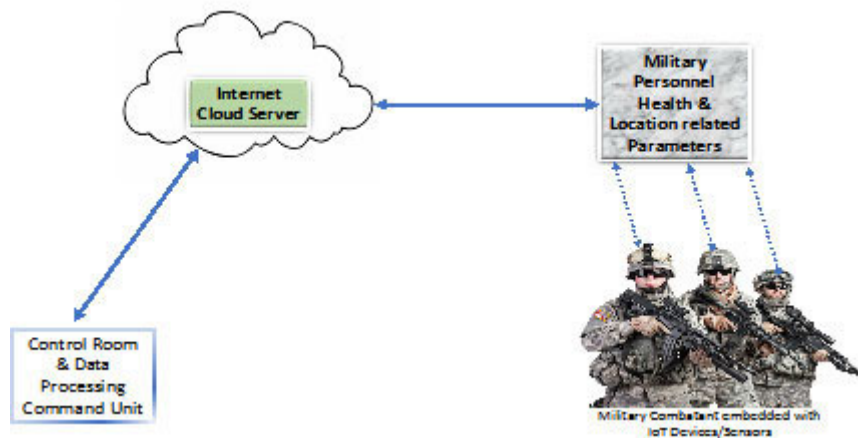
learning, blockchain and big data analytics are all essential elements of the usefulness of IoMT.

### Wearable technologies

One of the key advantages of IoT in healthcare is the use of wearable technology in gaining insights and access to patient's information for underlying causes and behaviors. The wearable technologies are causing significant disruption in the healthcare industry due to its rapid applications.

As a result of the rapid development in IoMT, many wearable devices are now working in a real-time patient monitoring mode, especially from the fitness and wellness segment. Healthcare wearable can be defined as a non-invasive and autonomous device that has the ability to perform a certain medical function.

As wearables are gradually becoming indispensable tools in healthcare, their greatest contribution is their ability to generate massively personalized (individual) data required to gain better control and visibility over one's health outcomes or status. Some examples of assistive wearables or IoMT tools are Google's Smart Contact Lenses, Cloud DX's Vitaliti, Chrono Therapeutics' Smart Stop, and many more:



**Figure 9.1:** *Use of IoT in military and combat positions from remote locations*

In summary, IoT is being used in the healthcare industry from patient's medical diagnosis and management (in-patient and outpatient), Health care facility (asset) and institution management to healthcare logistics delivery and management point of view.

Let us now look at some of the ways medical practitioners and the industry is adopting IoMT sensors and devices.

### Monitoring *patient's medicine in-take administration*

Administering medication to a patient's sometimes become problematic. For examples, it has been observed that some patients stop taking their medicine when they begin to feel better or more often than not fail to complete the dosage or in some terminal cases, adopt erratic behavior, and skip medication without any reason.

While in some cases, skipping or completely abandoning the medication could lead to recurrence of illness or prolonged recovery, it may in some situation become critical and cause irreparable damage. To ensure that patients do not renege on such behavior, some medical practitioners have introduced an IoMT tool in the form of a wearable device to remind, alert and record patient's medication routine.

*Porteus Digital Health* solution is one such firm providing a wearable device to address patient's reneged behavior when it comes to drug medication in treating patients with severe mental health disorders.

The patient's pill is equipped with a sensor that sends a message to a smartphone application which is then accessed via a web-based portal over the internet by health professionals.



*Zephyr Anywhere's BioPatch* is also a device that tracks a patient's health condition on a continuous basis by attaching the device to the chest. Messages sent to health professionals via the smartphone's application provide the patient's near-real-time health status.

**Patient Identity Management (PIM)** is one of the crucial activities within the healthcare system. In the USA, about 200,000 people die every year as a result of medical errors, and 58% of such deaths are attributed to identification

In dealing with the above problem, *Zebra Technologies* proposed a printed wristband with **radio frequency identification (RFID)** tags that contained a patient's health records linked to hospitals management software.

This made it easier for doctors just to scan the wristbands using PDAs for pre-surgery checks and administration of medication. More importantly, movement of patients and even assets such as beds, equipment, and more, are tracked and monitored on a continuous basis for effectiveness, efficiency and transparency.

### *The fetal heart monitor and AnteNatal healthcare*

Somewhere in Africa, within the healthcare system, an IoT enthusiast has tried to intervene positively in the antenatal health care of vulnerable and deprived people in remote villages by introducing fetal monitoring device based on a smartphone.

In most places in developing countries, pregnant women are usually confronted with both lacks of financial resources and access to antenatal clinics or services and end up with several complications during pregnancy etc. A simple novelty of turning one's smart mobile phone into a listening device to monitor the heart of the fetus and data sent to a medical center many miles away via the mobile app has had a major intervention of the health of the rural people (Grand Challenges Canada). Health care management should be leveraged to prevent needless death during labor and childbirth.

### Healthcare logistic delivery\_(drone delivery in Ghana)

From the healthcare logistic delivery perspective, Ghana has launched an ambitious and aggressive program to address the challenges in the sector by introducing the use of drone in the delivery of mission-critical drugs and equipment.

In most developing countries (including Ghana) severe bleeding during or after childbirth has been identified as the commonest cause of maternal mortality. Timely access to blood in such situations has been a major challenge, a situation that has needlessly sent many mothers to their untimely death.

This, amongst others, prompted the Ghana government to initiate this novelty, yet life-saving project in 2018. Within a year, the medical drone program was launched by Ghana's vice president *Dr Mahamudu Bawumia* and the first medical assignment by the drone in the form of yellow fever vaccines were dropped at the *New Tafo Government Hospital* (located in the eastern part of the country) that had run out of vaccines during immunization campaign.

Through integrated health management software, an order was placed, and it took only 21 minutes for the drone to drop off a parachuted box of vaccines from a height of 80m and without stopping, the drone returned to its base. Normally, it would have taken several hours for this order to be accomplished.

From the first drop off, the drone has since delivered various medical supplies notable amongst them blood to hospitals in various parts of the country. More *Zipline* drone operational centers are being opening to cover the entire country:



**Figure 9.2:** Map of Ghana showing Zipline's four operation centers (Source: Zipline, 2019) & Drone dropping-off medical Item in New Tafo, Ghana (Credit: Kwasi Gyamfi Asiedu, 2019)

The health care industry is now witnessing a massive adoption of IoT devices, especially in the wearable segment. The wearables enable users to connect to the cloud and remain accessible for data transmission and feedback and thereby making it possible for both operators and patients to reduce costs and improve efficiency.

Wearables are now empowering both patients and doctors to take control of their respective responsibilities and roles in the health care industry and that the sky is only the limit.

## Smart home and smart buildings

Although the concept of *smart buildings* is still vague in the real estate, it is nonetheless becoming a disruptive technology to reckon with. This is because; the adoption of IoT in creating smart building is enormous and provides many benefits.

In the realm of smart buildings, the real estate industry is able to incorporate new capabilities and user experiences arising from solutions communicating directly to buildings via IoT devices; to the extent that HVAC system, temperature and proximity-based sensors work to optimize both energy consumption and workflow within the building to the comfort, convenience and safety of both occupants and visitors.

To leverage the adoption of IoT in smart buildings, let us look at a simple yet high volume and routine flow of visitors to an organization (office building); visitor management.

As an employee of a large and busy organization, you invite a friend or business associate to your office. You do this by sending an invitation via email that has a QR code embedded. The QR code contains certain essential information such as names (host and visitor's name), office number or location, a designation of an employee, phone numbers, and more.

The visitor arrives at the office (reception) scans the badge (QR code sent by an employee) using the smartphone at the front desk/elevator to announce his/her presence or arrival. At this point, the security unit is alerted with the relevant details, the employee (host) also receives alert or notification immediately via email and/or SMS text messages.

The host either goes to the front desk to receive a visitor or simply asks the front desk to direct visitor to the office room. As the visitor moves through the various lounge and large corridors towards the host, the QR code populates the visitor management software with all the touchpoints, thus leaving a trail.

The tracking case is closed when the visitor leaves or exits the office and depending on the nature of configuration, relevant or interested parties are alerted. In certain organizations where there are large numbers of visits in a day, managing visitors can be extremely chaotic, challenging and disruptive to otherwise a more efficient and well-organized outfit.

Subsequently, visitor data can be analyzed to predict peak visit times and days and combined with other data such as tenant occupancy, certain pro-active and pre-emptive activities initiated automatically. For example, the building's HVAC system can automatically start the heating or cooling appliances based on the analytic insights of the data regarding the peak and low visitor periods of the day.

Additionally, the security staff can be optimally deployed or scheduled and even alert the maintenance team when to begin

work. Visitor management is just one of the many activities in a building that can leverage on IoT device to create a smart building.

Notwithstanding the above benefits, a building is not a smart building if it fails to change the way an employee or occupants do their jobs for better in a more quantifiable and measurable way. In other words, it should be possible to measure the gains or otherwise of the application of IoT in buildings in order to designate them as smart. Thus, building technology (IoT) must leverage the method to achieve the goals of the organization.



### Smart education system

The role of education is to disseminate knowledge through various means and platforms. Typically, and traditionally, the delivery of (formal) education had been through face-to-face format and not too long ago also incorporated a hybrid type of long-distance approach. However, in recent times, the model of education is going through an accelerated pace of conversion and delivery modes in which the different learning styles/modes of students are to be met. In this perspective, the smart education system becomes a viable route.

Smart education can be defined as the use of different information and communication technologies in delivering or disseminating knowledge while at the same time adjusting to the different learning needs, styles and requirements of students.

The application of IoT in education is to ensure that the quality of the learning process is improved on a continuous basis as a result of the ability to monitor, analyze and obtain feedback of the state and activities of students by means of sensing devices and information processing platforms.

The Internet of Things in education, which could also rightly be said to be the **Internet of Education Things (IoET)**, proposes to take the delivery of education to a new level that has already

started to break down teacher-student relationship barriers to transnational (borderless) education.

The adoption of IoT concept in education will no doubt help students to learn at their best level while at the same time helps teachers also fulfill their job efficiently. IoT in education makes it possible for educational institutions to deliver a high level of personalized learning modes and styles to students, keep track of important resources, enhance access to information, design secure campuses, and many more.

Essentially, like a smart school, IoT in education can be considered as a novel approach in classroom administration with new tools. Amongst such smart tools are Smartphones, smart digital boards, Interactive learning environment (digital classrooms), Google apps that enable students and teacher interaction, wireless with swipe locks (and wearable devices) for managing attendance and advanced security measures, temperature sensors, **Learning Management System (LMS)** and other cloud-based platforms, and more.

The benefits of IoT in education are enormous. One can immediately think of the cost advantages recruiting teachers. Although hiring more teachers may lead to increased access to education and improve the teacher-student ratio, there are always budgetary and funding issues and constraints associated with the recruitment.

The key question is, when or how do you achieve optimum teacher-student (including challenges with adequate physical

classroom space) population ratio for an effective learning environment? The adoption of IoT in education at this point may be strategically used to create and maintain a good balance in quality education. Again, purchasing and access to new or even old textbooks both by teachers and students are also constraints in providing quality education.

Thus, access to a digital library, eBooks, online educational resources, and more are all part of the IoET agenda. Today, as a student you can take a class from a prestigious university located thousands of miles away from your living room while teachers could get their students assignments or exams graded immediately after submission with ease and less marking stress.

The education sector is currently under profuse injection of IoT devices and sensors in countless and diverse ways. Below is some of the overview of the different use cases.

### *The medium of instruction (language).*

One of the important means of learning is language. This, at the early formative ages, is done in your native language (in most cases). However, as you progress and grow in your life and academic endeavor, the chances are that you will have to learn new things in new or different languages.

To speed up this process, language experts say one needs to immerse in the language. This is best done when you are not afraid to speak or talk and also ask for continuous feedback. Such environments are difficult to create outside one's country where the language is spoken.

It is at this point that IoT in education comes in handy. Thus, through platforms such as podcasts, YouTube, Google translator, TV/Film subtitle setting based on geographical location settings etc. the immersion processes are deepened.

There are also numerous software/applications that automatically translate reading/educational materials into the preferred language. All these take some form of link or connectivity from the IoT domain.

### Smart classrooms

The classroom, which epitomizes the classical teaching and educational delivery and experience is gradually changing and soon will become truly technologically enabled. **Augmented Reality (AR)** or **Virtual Reality (VR)** is replacing the dissection biology laboratory and history classes, respectively.

With your wearables, the course instructor gets notification and update on all students seated in class as well as any relevant stakeholders. Whiteboards will record all notes taken in class and along with video clips and sent to students automatically via their emails immediately after class.

### *Task-based learning*

Knowledge transfer in Education is gradually taking a new dimension in the form of collaboration. This model, which is also known as an information-sharing system, is a task-based learning environment where the use of IoT frees up teachers to concentrate on their core work as students learn-by-doing. The IoT system will provide feedback as well as assistance automatically.

Using cloud-based shared drives and other collaborative software like Microsoft SharePoint, Google Drive, and many more, students are able to work in a group from a remote location. This helps in inclusive education and provides convenience and diversity through global and transnational participation in education.

### Disability accommodation

Inclusive education requires that all persons irrespective of their socio-economic and physical condition get equal access to education. In this respect, IoT immediately comes to mind regarding its capability to empower the physically challenged to access quality education. Devices such as hearing aids, connected gloves and tablets that help translate sign language to speeches, convert sounds into written languages. Using IoT devices this way gives a new life and enabling capability in providing quality education to the physically challenged.

In conclusion, one can argue that IoT in education has gathered unprecedented momentum; however, it could take some time before IoT applications enter mainstream education.

### School security.

Most schools generally have a large population and progressively increase year by year. Monitoring students, including teachers and other staff members of their activities each day, is near an impossible task.

With IoT such as wearables, RFID tagged devices and even smartphones; the movement of students and teachers can be monitored and prompted of any security (including physical security) threats and dangers. For example, parents or guardians are automatically notified when their wards enter the school premises are in class.

It also helps to prevent unwanted or unauthorized persons from entering the school premises. With the recent spate of people and child kidnappings, IoT devices can help authorities and parents monitor their wards at anytime and anywhere. IoT devices and sensors certainly add value in areas of security, safety, and protection.



### Challenges of IoT in education

Notwithstanding the significant benefits of IoT in education, this novelty has, however, started to attract some criticisms from some section in the society. This includes the fact that IoT is making students lazy and have become 'surface' or passive learners rather than deep learners.

There is also the fear of teacher delinquency and complete reliance on online and distance education which are seen as suspects in quality education. Ethical implications and privacy issues are gradually rearing its ugly head in the IoT application in education as they inevitably create risks in their adoption and unknowable challenges.

### Governance use case

Governance is *the process of decision-making and the process by which decisions are implemented or not implemented* (Pereira et al., 2018). Smart governance is, therefore, about the use of technology in delivering better decision making and planning, especially within the public sector space.

Smart governance help improves the democratic process by facilitating and supporting the ways that government and municipal authorities interact and engage with the citizenry and thereby ensuring convenience, transparency, responsiveness, innovation and openness to change, ethical conduct, and accountability.

The other characteristics or elements that ensure good governance are participatory, consensus-oriented, effectiveness and efficiency, equitability and inclusiveness, and the rule of law. It is important to note that governance is not only confined within the political and public domain but also a key feature within the private arena, especially within the corporate and business world.

So, the key question here is what the relevance of IoT in governance is? IoT helps in the automation of key characteristics or elements of governance that operate throughout the numerous departments of government infrastructure and set up. It achieves this while enhancing accuracy, speed, and convenience.

For example, in the area of National defense, IoT improves and supports military systems, operations and services by providing certain important technology needed to control the numerous and arrays of infrastructure.

By way of City planning and control, IoT enables governing bodies and engineers to analyze complex issues while at the same time simplifying citizen's access, control and compliance. Areas such as zoning, mapping, water and food deliveries, waste management, transportation and mobility patterns, social services, land use, and more, within the city or the country, come under the purview of smart governance.

The whole idea of adopting or incorporating IoT in governance is to ensure that towns and cities become smart and better-living places for all, most especially for the aged, elderly and the disadvantaged or vulnerable members of the society and to the extent that no one is denied, restricted or constrained in their inalienable rights as citizens!

## Smart cities

As you have by now been introduced to the concept of smart cities in the previous chapters, we will only discuss how IoT is being used in key areas of the city's infrastructure to make the city smart and resilient. The central benefit of the concept of a smart city is the number of citizens who are engaged and empowered by technology to seek more fulfilling lives!

In building a simple understanding of a use case for a smart city, let us use the analogy of corporeal metaphor in demonstrating the relevance of IoT in governance (smart city). Thus, assuming that the smart city was a human body, then technology (IoT) would be its nervous system taking in and reacting to our information needs while at the same time moderating our long term physical and intellectual resources.

This *nervous system* of the city requires sensory inputs to function well. IoT constitutes these sensory inputs of nose, eyes and ears that take in information (big data) from the environment (ecosystem) for the central nervous system to process. To make cities efficient, liveable and sustainable, they must be made smarter, and the use of IoT is a sine qua non.

Thus, in the world of smart city, data become the bloodline that nourishes the body and keep it alive and fresh (smart). This means that as more IoT devices and sensors are embedded and

become ubiquitous, data become massively available for analysis and used by relevant stakeholders. Smart cities use IoT to improve infrastructure, utilities, buildings, and more.

The following are some brief overviews of how IoT devices are being used to disrupt positively in the cities across the world.

### Automatic car parking notification

Santander, a city in the northern part of Spain, considered to be the first city in the world to go 'smart' has embedded thousands of IoT devices and sensors to manage its public urban landscapes and parks. For example, special sensors that measure and switch on sprinklers in dry periods have been installed. Public paths lamps are only lit in the night when there is a person approaching or nearby, while trash cans automatically send status messages for removal to be emptied.

One of the impressive smart cities' agenda by Santander is the introduction of the smart mobility solutions that monitors available car parking spaces in the CBD. Sensors detect free parking spaces and automatically re-route traffic, relieving the pressure on drivers to spend time and fuel going about hunting for a free parking spot to park.

### Louisville 'If this then that' (IFTTT) smart city platform

Although the city of Louisville is not the most famous metropolis in the USA, the implementation of smart home *applets* that simplify citizen's daily activities has given it high profile amongst the high-end American cities. The smart applet puts the control of one's home into their hands by allowing them to set up the air filters to activate when the digital health IoT detects changes in air quality. The applet also hooks them onto the city's large event organization (calendar), and citizens are automatically invited to open social activities with no administrative effort required.

### *District of Songdo new resident integration*

In the district of Songdo where there are over 3 million people, new residents that connect immediately to the Internet on their arrival are enabled by sensors and IoT devices to communicate with neighbors or businesses by way of introduction.

Residents can even borrow salt from a neighbor without leaving the house or seek grooming help from salons via a video conference from the comfort of their homes. Residents never miss appointments as their calendar is synchronized with the city's centralized digital assistant software.



### Copenhagen zero carbon emission agenda

In Copenhagen, Denmark the city has incorporated IoT in several areas of the city's administration. Denmark's agenda to be carbon neutral by 2025 is encouraging them to implement several smart city initiatives ranging from smart lighting, smart traffic management, waste management, and smart building management.

More importantly, the city's policy to introduce the free exchange of data allowing everyone (both public and private & businesses) to explore, consume and publish data has suddenly opened up for massive development of smart city products and services.

## Conclusion

IoT is envisaged to be deployed and permeate in all aspects of our lives in the coming years. At the heart of the IoT ecosystem and accessibility is the issue of connectivity. Businesses and individuals want smart technology at their beck and call.

Telecommunications are indispensable to the successful application of IoT and must be able to connect all manner of things including human-to-human, human-to-machine and machine-to-machine. From high-speed Internet data, VPN, wireless, video, and audio conference as well as voice communications to interactive media and digital signage is expected to be simple, resilient, seamless, plug, and play.

However, as IoT technologies continue to spread, and even become embedded in the human body, questions about data ownership, accuracy and privacy protection will assume greater importance. IoT vulnerability to cyberattacks and the potential for security breaches to cause serious harm will also assume an unprecedented rate.

In the coming years, IoT will become invisible as insights and automation will be enabled in a real-time access mode. It will be uncommon to check for anything manually when all that is needed is to press a button. The stage is set for an explosion of IoT!

### Multiple choice questions

What is IIoT?

Internet Industrial of Things

Industrial Internet of Things

Inter Industrial of Things

Industrial International of Things

International Industrial of Things

What is IoMT?

Internet of Ministry Things

Internet of Military Things

Internet of Medical Things

Industrial of Medical Things

Internet of Management Things

Wearable Technology is a key feature of

Smart education

Smart health

Smart mobility

Smart transport

Smart governance

In the concept of smart transportation, IoT will make it possible for commuters to check these various aspects of their journey:

Routes

Location

Price

Arrival

All of the above

IoT plays a key role in governance. Which of the following are elements of smart governance?

Convenience

Transparency

Responsiveness

Ethics and accountability

All of the above

## Answers

b. Industrial Internet of Things

c. Internet of Medical Things

b. Smart health

e. All of the above

e. All of the above

Fill in the blanks

Name the three essential characteristics (categories) of the Internet of Things (IoT) things that can be connected to the Internet:

.....

.....

.....

Digital building is where ..... devices and ..... are embedded in buildings to help monitor and identify faults

In the analogy of corporeal metaphor, Data is the ‘.....  
.....’ That nourishes the ‘body’ of the Smart city concept.

The IFTTT is an applet launched by the city of Louisville and it stands for .....

A key feature of IoT power management is its ability to ..... input power remotely to correct faults without the need for ..... intervention.

## Answers

Things that can collect and send information

Things that can receive and act upon the information

Things that can do both

IoT and Sensors

‘Blood Line’

‘if this then that’

recycle ..... human (manual)



### Descriptive questions

Briefly describe two (2) ways that IoT can be used to create smart education system.

What is smart governance?

With examples, briefly explain how IoT can be used in the **Avoid-Shift-Improve (ASI)** framework proposed by the **International Energy Agency (IEA)** to improve transportation in your country.

Briefly imagine and describe a futuristic situation where you will use IoT to provide solution to a problem (either an old or emerging problem) within your community or country.

Identify and describe three key areas where you think IoT implementation will create problem rather than benefits when introduced in your community or country.

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Zipline 2018:

**Multiple Choice Questions (MCQ)**

IoT can be defined as?

The day that computer take control of the world

The intelligent connection of people, process, data, and things

Both of these

None of these

IIOT stands for what?

Informational Internet of Thing

Industrial Internet of Thing

Industrial Internet of Things

Innovative Idiocy of Technology

Which protocol is most common for short range low power communication?

Bluetooth

RFID (Radio Frequency Identification)

ZigBee

Wi-Fi

Zigbee is based on which protocol?

IEEE 802.15.2

IEEE 802.15.3

IEEE 802.15.4

IEEE 802.15.5

What is the minimum size of the IPv6 MTU (total transmission).

1280 bytes

1680 bytes

2460 bytes

3620 bytes

MQTT is \_\_\_\_\_ protocol.

Machine to Machine

Internet of Things

Machine to Machine and Internet of Things

Machine Things

Which one out of these is not a data link layer technology?

Bluetooth

UART

WiFi

HTTP

A closely detached software device (software version of actual hardware) that could run its own operating systems and application as if it is running on a physical computer, is called\_\_\_\_\_.

Guest OS

Host OS

Virtual Machine

Physical Machine

In which attack the malicious node manipulates its own name by offering sensitive recommendations?

On - off attacks (OOA)

Attacks by ballot-stuffing (BSA)

Self-promotion attacks (SPA)

None of above

‘IoBT’, also a type of Internet of Things stands for which of the following:

Internet of Body Things

Internet of Battlefield Things

Internet of Biological Things

Internet of Business Things

All of the Above

### **Fill in the Blanks**

Near Field Communication (NFC) is a \_\_\_\_\_ wireless connectivity standard that uses magnetic field induction to enable communication between devices.

REST Services operates over \_\_\_\_\_ Protocol.

\_\_\_\_\_ IoT layer protocol determines how data is physically sent over the physical medium in the network.

MQTT is a \_\_\_\_\_ oriented protocol.

Zigbee specifications are based on \_\_\_\_\_ protocol.

\_\_\_\_\_ Sensors are used in touch screen device.

The Internet of Things, in the interim, alludes to the association of gadgets other than the \_\_\_\_\_, for like PCs and cell phones to the Internet.

SDIoT acronym stands for \_\_\_\_\_.

Digital building is where \_\_\_\_\_ evices and \_\_\_\_\_ are embedded in buildings to help monitor and identify faults

The Integration of both technologies Cloud Computing \_\_\_\_\_ and which is likely to be call CloudIoT pronounced “Cloud IoT”.

### **Short Answer Questions**

Name Raspberry pi's components.

Which type of RAM is used in the Arduino Board?

What is Zigbee?

What is the Role of “Things” in IoT?

Distinguish between “Web of Things” and “Internet of Things”.

### **Descriptive Questions**

Explain the difference between Request-Response and Publisher-Subscriber communication Model. Draw a neat and clean diagram to explain the working of both of the model.

Show the working of a Push-Pull Communication model also explain how it differs from Exclusive pair model of communication.

What are the different layers present in five-layer structure of IoT?



Describe the three basic components of SLoT and justify with example?

Draw and explain privilege levels of x86 hardware architecture, also explore the various techniques of x86 hardware virtualization.

## Model Question Paper 2

### Multiple Choice Questions (MCQ)

IoT is built on\_\_\_\_\_

Networks of data gathering sensors

Cloud computing

Both of these

None of these

Which layer is not present in three-layer structure of IoT?

Application Layer

Perception Layer

Business Layer

Network Layer

Which of the following language is preferred for IoT analytics?

Python

S

R

All of the mentioned

A software, that is used to create and run virtual machines called  
\_\_\_\_\_.

Guest OS

Hypervisor

Virtual Machine

Physical Machine

What are the important components in IOT?

Hardware

Software

Verbal Exchange Infrastructure

A and B Both

Every IoT application is built on an ecosystem of (Choose all that is applicable)

Hardware

Software

Network connectivity

Software and Connectivity only

All of the above

The core element of architecture of smart city is \_\_\_\_\_

Mobile Unified Service

Urban Application Platform

Management center

Integrated Information Provider

What is IoMT?

Internet of Ministry Things

Internet of Military Things

Internet of Medical Things

Industrial of Medical Things

Internet of Management Things

**Fill in the Blanks.**

Physical devices include different types of sensors like the ones based on \_\_\_\_\_ technology.

Hardware assisted virtualization technique allows to run VMM (Virtual machine monitor) in \_\_\_\_\_ privilege mode.

\_\_\_\_\_ is one of the major problem, i.e. a way to connect computers to the internet as well as the platform for cloud computing.

The Integration of both technologies Cloud Computing \_\_\_\_\_ and which is likely to be call CloudIoT pronounced “Cloud IoT”.

SDIoT acronym stands for \_\_\_\_\_.

The principle of \_\_\_\_\_ is to harness the information generated via sensors and networks for military in what is considered as "the full realization of pervasive sensing, pervasive computing, and pervasive communication."

The IFTTT is an applet launched by the city of Louisville and it stands for '\_\_\_\_\_'

### **Short Answer Questions**

Define the term Internet of Things; and how it works? Explain with the help of an example?

What are the different layers present in five-layer structure of IoT?

How are the various python libraries useful for IOT?

Describe the three basic components of SIOT and justify with example?

What are the Advantages and disadvantages of the Integrated CloudIoT?

What is a smart city? How do you define it?

### **Descriptive Questions**

What are the various characteristics of IoT? Explain in detail.

Explain the difference between REST-based and Web Socket Based API.

What are different layers in Zigbee architecture? Explain all layers.

How Virtualization technique is useful for IoT resource management?

What are the Security and Privacy challenges faced in IoT devices?

Explain the CloudIoT architecture in detail.

Why do you think privacy and security are key challenges in IoT implementation especially within the Health sector?

With examples, briefly explain how IoT can be used in the 'Avoid-Shift-Improve' (ASI) framework proposed by the International Energy Agency (IEA) to improve transportation in your country.

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